

Day	Time	Topic	Detailed Description
1	1	Dynamo & Generative Design Trends	활용 사례 소개 및 목표
	2	Dynamo 개념 및 인터페이스	Dynamo, Generative Design 개념, 작동 방식 및 UI 확인
	3	Dynamo 기본 Geometry 활용	Point, Line, PolyCurve, Surface, Solid 순환
	4	Geometry 응용	응용 및 Revit 정보 및 형상 연결
2	1	공간 배치 자동화 개념 소개	다양한 Space Planning, Space Layout 사례 및 방식 설명
	2	공간 배치 자동화 예제 1	공간 배치를 활용한 형상 생성, 고정된 형상의 공간 배치
	3	공간 배치 자동화 예제 2	Space Planning - 주거 공간 배치 예제
	4	공간 배치 자동화 예제 3	형상 생성을 통한 1~4개 건물 규모 검토 (건폐율, 용적률)
3	1	주차장 배치 및 검토 자동화 개념 소개	주차장 배치 자동화 개념 및 사례 소개
	2	주차장 배치 검토 조건 확인	Revit 패밀리 활용 / Dynamo Geometry 활용 비교
	3	외부 주차 배치 방식 개념	정해진 공간의 외부 주차 배치 방식 개념 확인
	4	내부 주차 배치 방식 개념	정해진 공간의 외부를 제외한 주차 배치 개념 확인
4	1	공원 구성요소 배치 자동화 개념 소개	공원 진입로의 변경과 휴식 공간 검토 사례
	2	공원 구성요소 배치 자동화 실습	정해진 공간의 구성요소를 변경하며 배치 검토 실습
	3	외피 패턴 형성 자동화 개념 소개	Facade 패턴 변경 사례
	4	외피 패턴 형성 자동화 실습	다양한 노드를 활용한 패턴 변경 방식 실습
	5	Dynamo 별도 실행	Sandbox 설치 및 Curve Mapper Node 확인
			Curve Mapper Example
			Geometry SAT 내보내기
			SAT Revit 형상 적용

Day Time Topic

Detailed Description

2

1 공간 배치 자동화 개념 소개

다양한 Space Planning, Space Layout 사례 및 방식 설명

2 공간 배치 자동화 예제 1

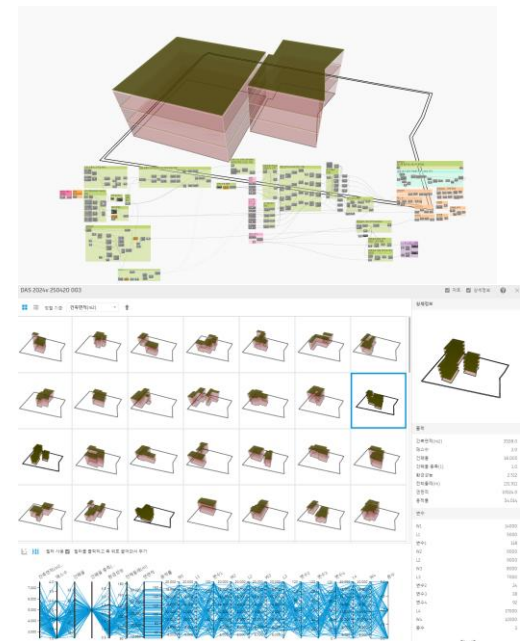
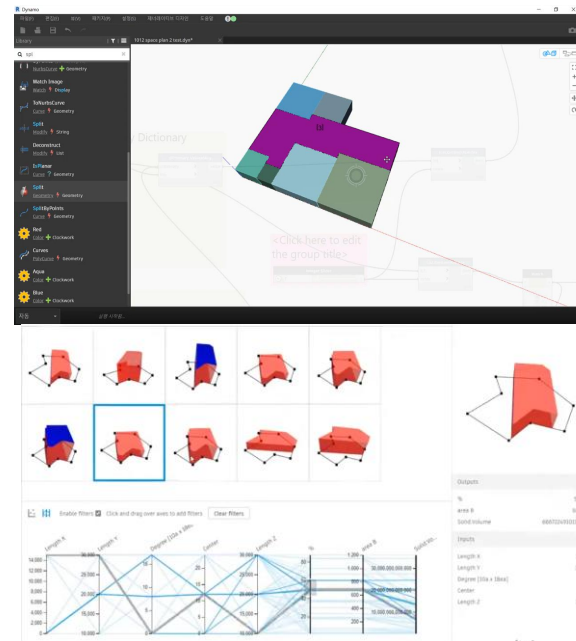
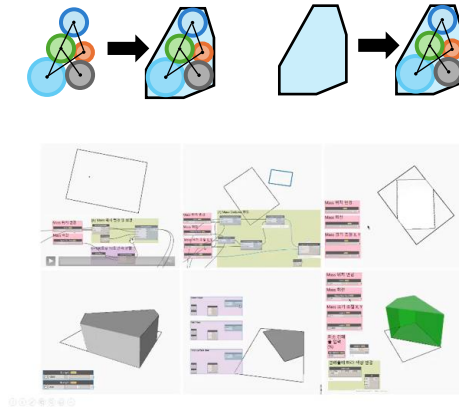
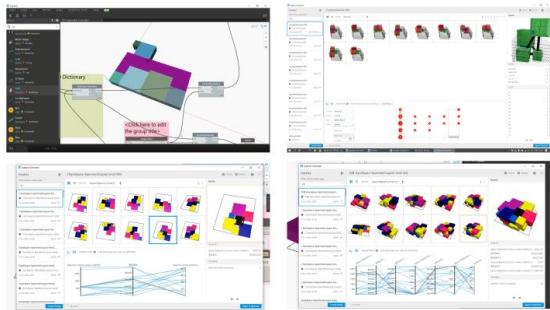
공간 배치를 활용한 형상 생성, 고정된 형상의 공간 배치

3 공간 배치 자동화 예제 2

Space Planning - 주거 공간 배치 예제

4 공간 배치 자동화 예제 3

형상 생성을 통한 1~4개 건물 규모 검토 (건폐율, 용적률)



About _ Space Planning
Space Layout

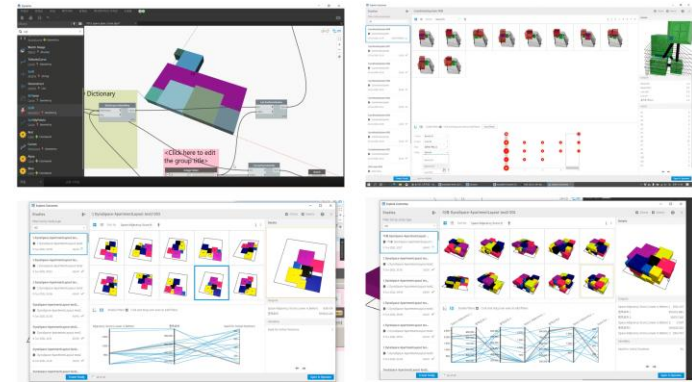
Space – Out line

Space Planning 01

Space Planning 02

01 About _ Space Planning, Space Layout

다양한 Space Planning, Space Layout 사례 및 방식 설명

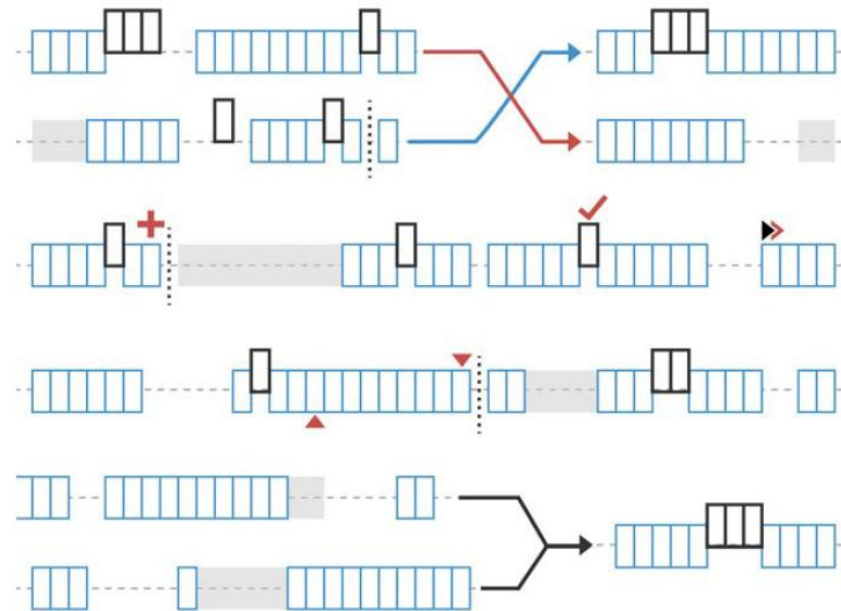




Designing with Lists

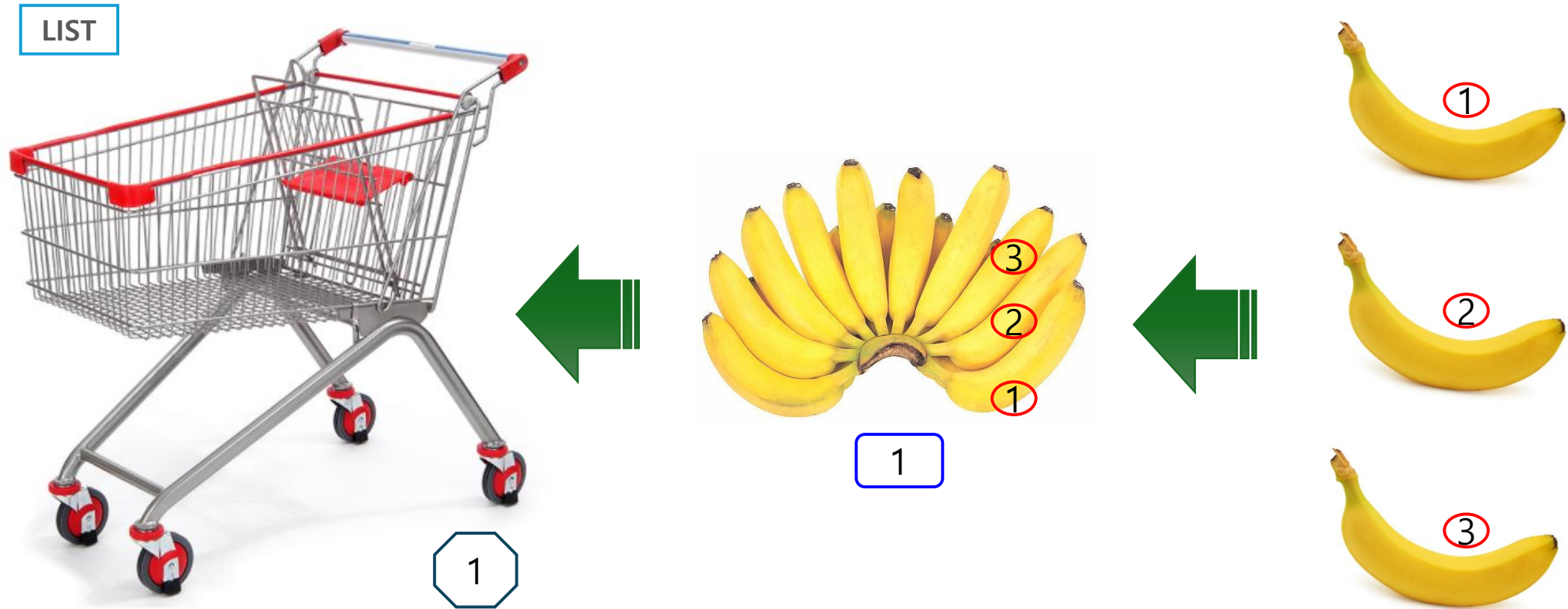
List

리스트를 이용한 다이나모의 데이터 관리 / 정리 방식



List

단일 요소 또는 그룹화된 요소의 모음



The screenshot shows a software interface with three main components: a Code Block, a Point.ByCoordinates block, and a Watch window.

- Code Block:** Contains the text `0..10..1;` followed by a right-pointing arrow.
- Point.ByCoordinates:** A block with three input fields labeled `x`, `y`, and `z`, each with a right-pointing arrow. To its right is a block labeled `Point` with a right-pointing arrow.
- Watch:** A window titled "Watch" showing a list of 11 points. The list is as follows:

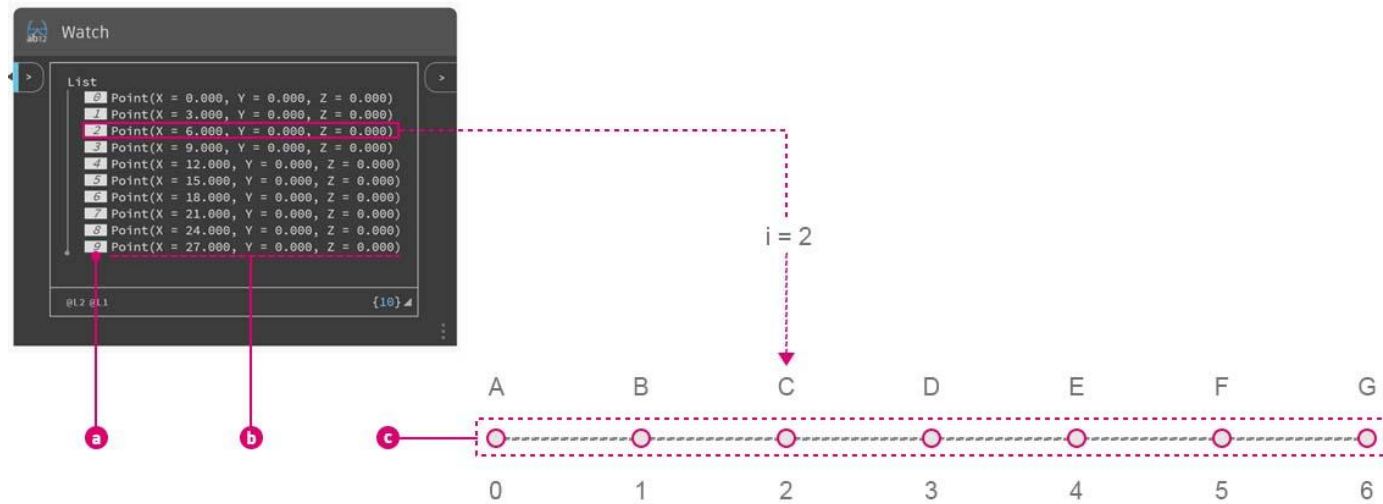
Index	Point(X, Y, Z)
0	Point(X = 0.000, Y = 0.000, Z = 0
1	Point(X = 1.000, Y = 0.000, Z = 0
2	Point(X = 2.000, Y = 0.000, Z = 0
3	Point(X = 3.000, Y = 0.000, Z = 0
4	Point(X = 4.000, Y = 0.000, Z = 0
5	Point(X = 5.000, Y = 0.000, Z = 0
6	Point(X = 6.000, Y = 0.000, Z = 0
7	Point(X = 7.000, Y = 0.000, Z = 0
8	Point(X = 8.000, Y = 0.000, Z = 0
9	Point(X = 9.000, Y = 0.000, Z = 0
10	Point(X = 10.000, Y = 0.000, Z =

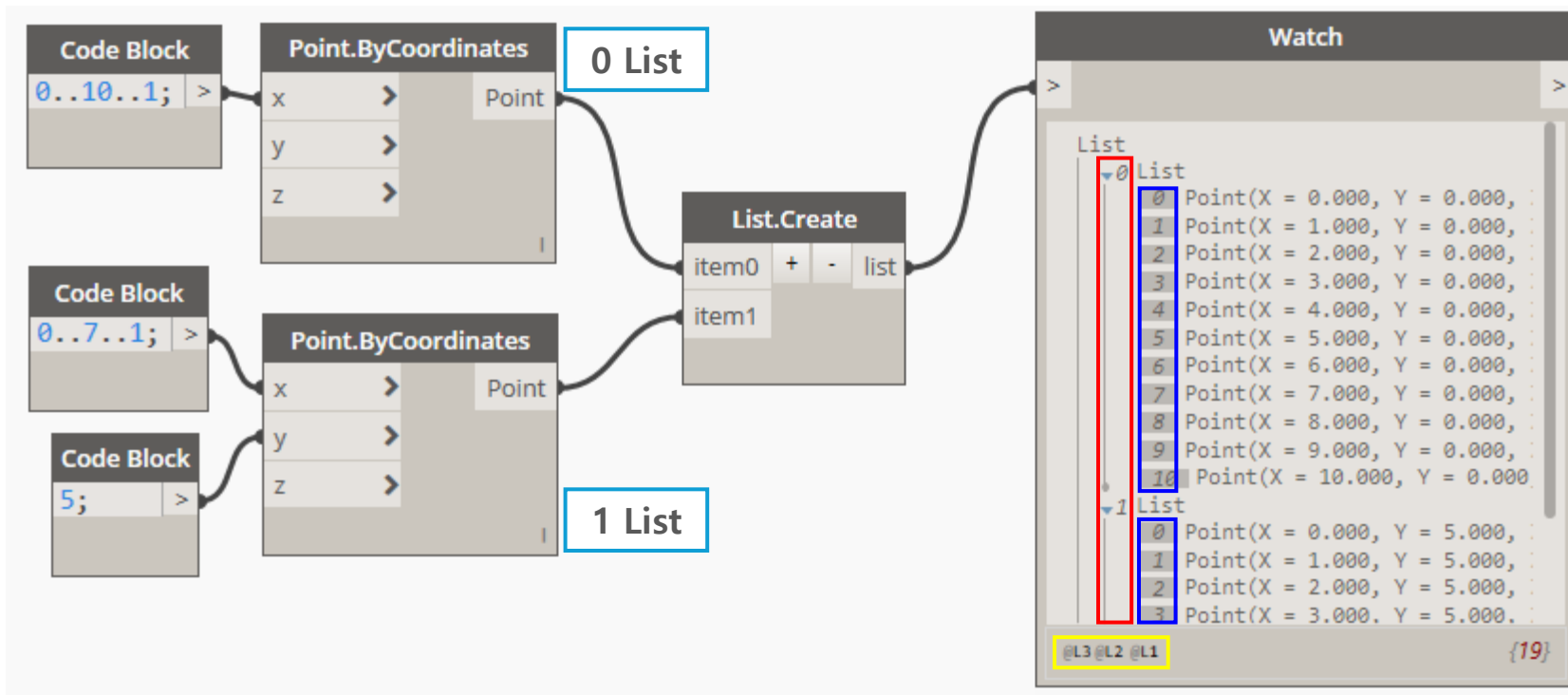
Annotations in the image:

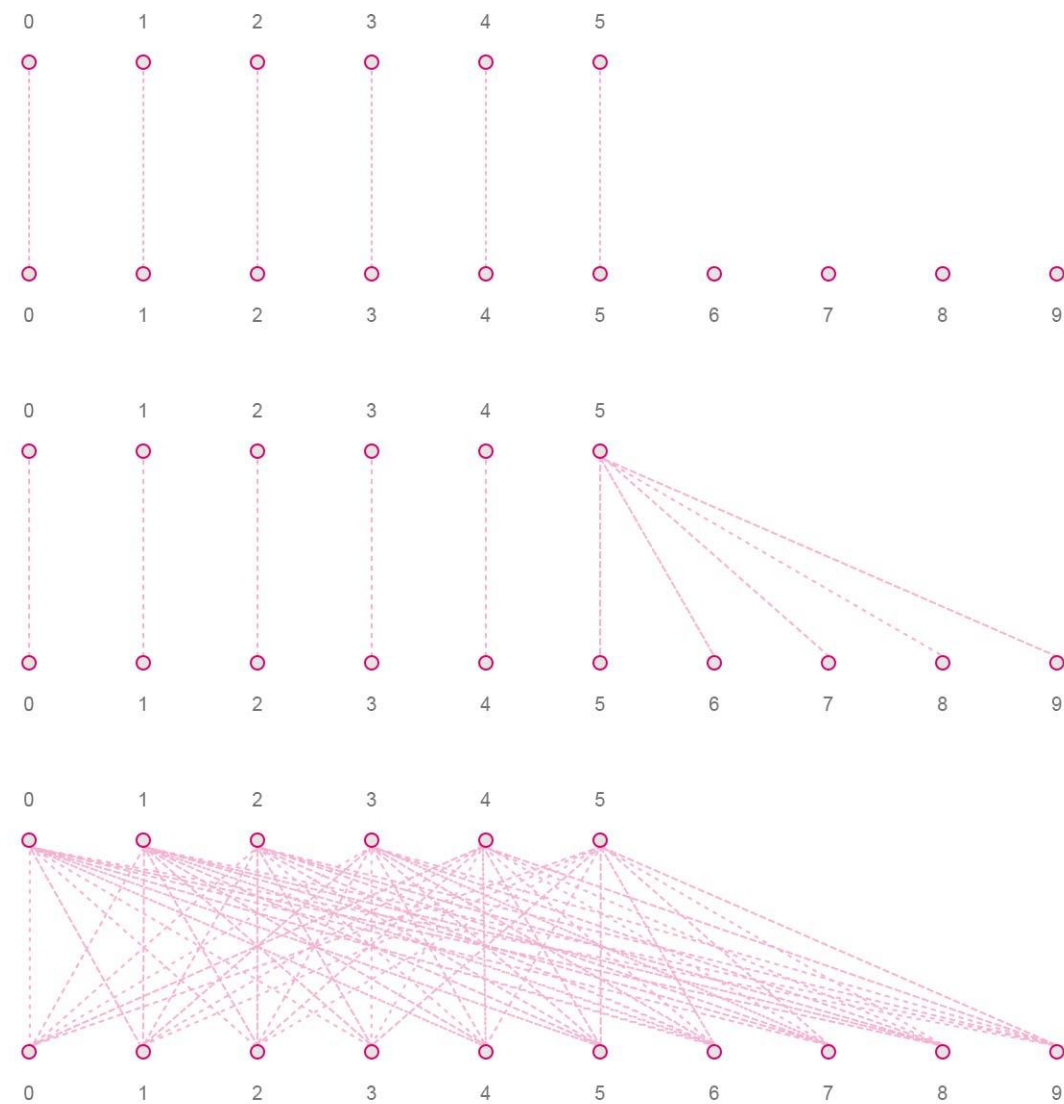
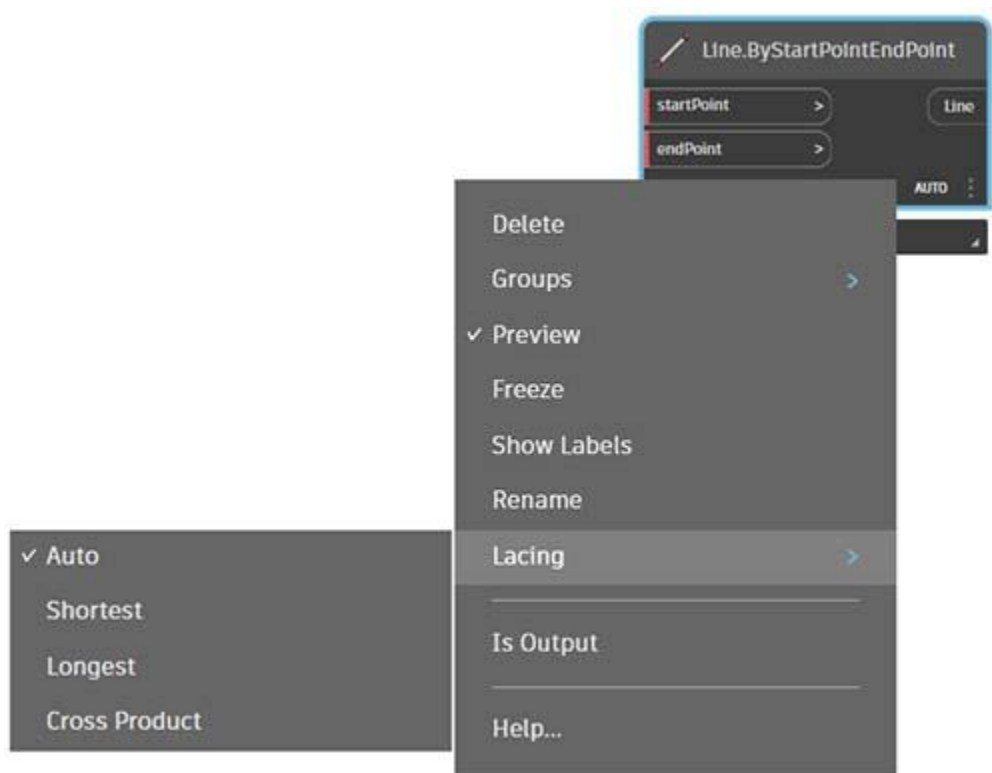
- A red box highlights the index column (0 to 10) in the Watch window. A circled number **1** points to this box.
- A purple box highlights the data for index 10. A circled number **2** points to this box.

At the bottom of the Watch window, it shows `@L2 @L1` and `{11}`.

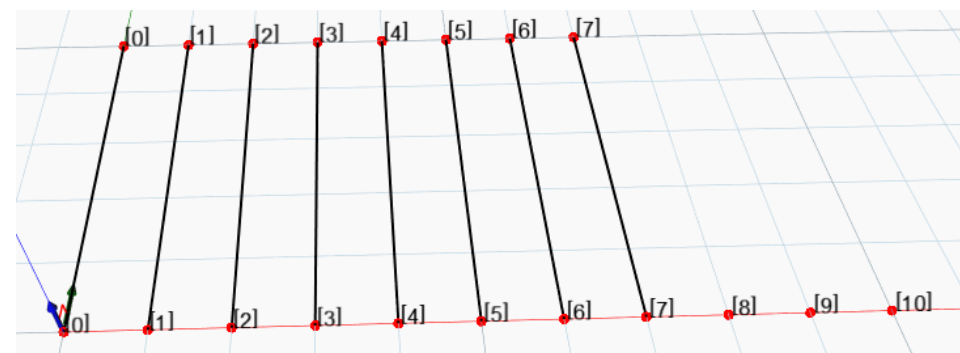
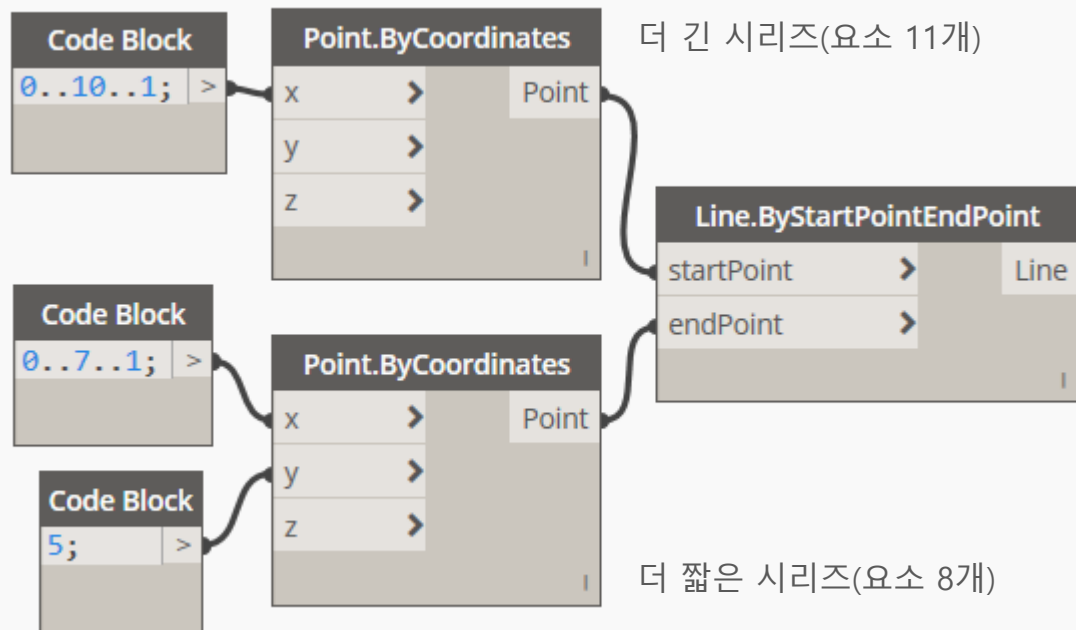
1. Index : 항목의 색인 / 0 부터 시작
2. 해당 Index의 데이터





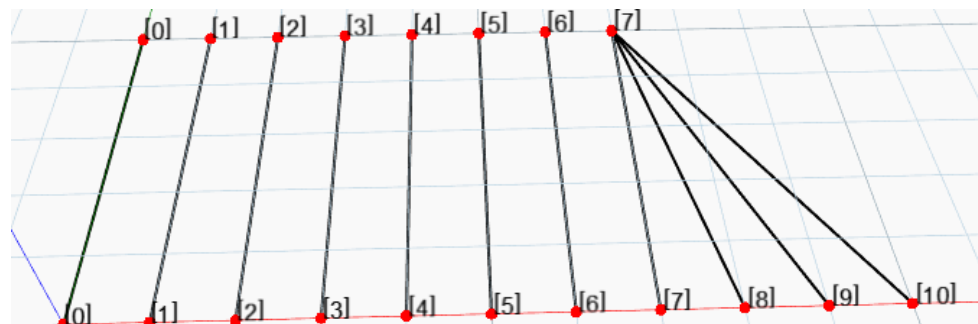
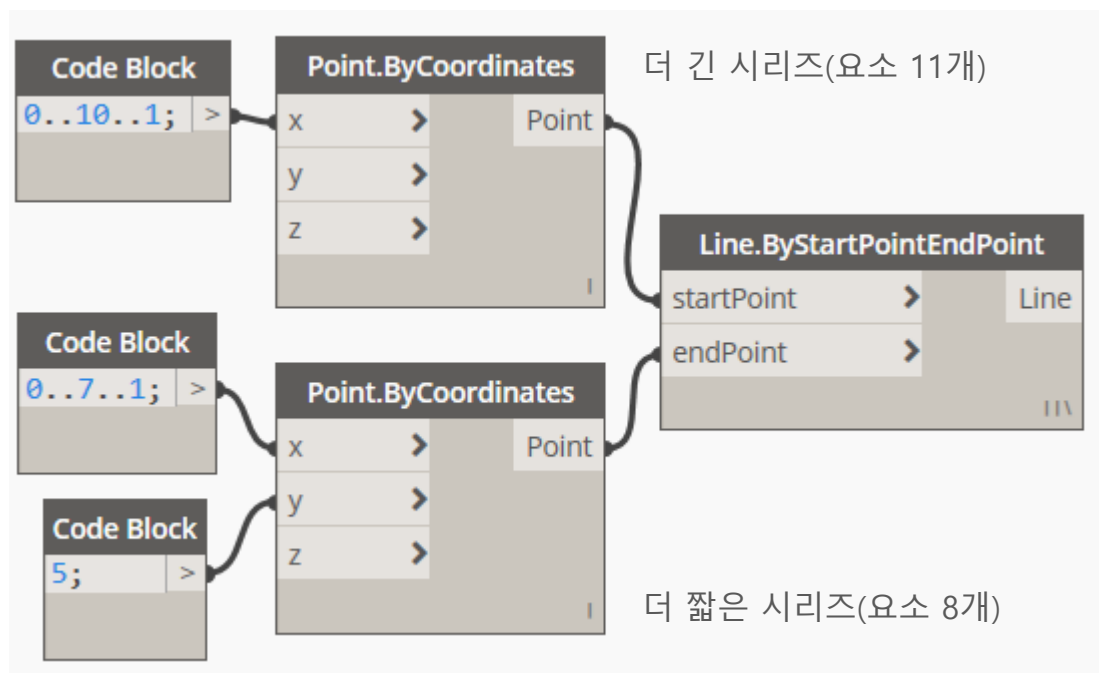


최단 레이싱



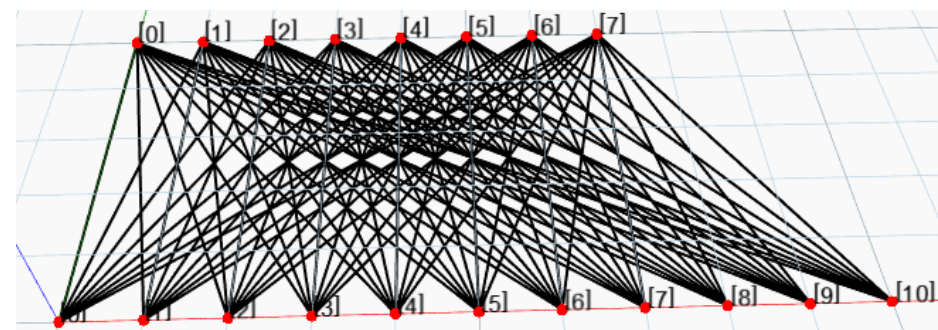
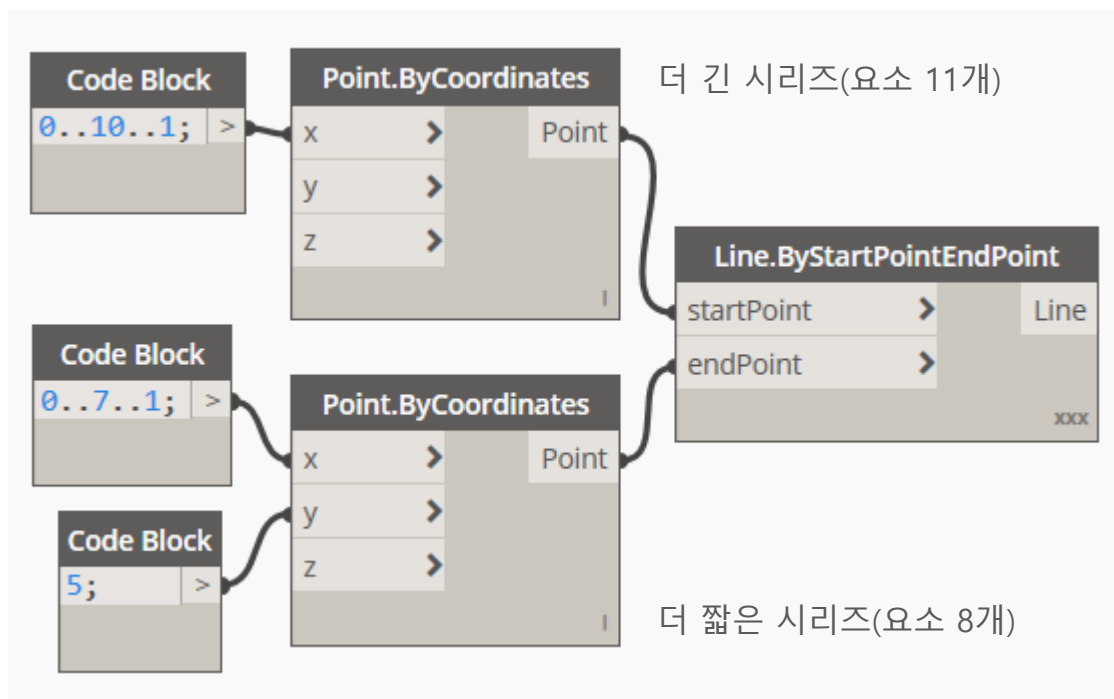
최단 레이싱선은 가장 짧은 입력 리스트의 항목 수만큼만 그려집니다. 더 긴 리스트에 "남아 있는" 점은 무시됩니다.

최장 레이싱



최장 레이싱 더 긴 입력 점 리스트의 항목 수만큼 선이 그려집니다. 더 짧은 리스트에 더 이상 점이 없을 경우 노드에서는 가장 긴 리스트의 나머지 항목을 일치시키기 위해 필요한 수만큼 마지막 항목을 복사합니다.

외적 레이싱



외적 레이싱가능한 모든 쌍의 조합이 작성됩니다. 긴 리스트에 11개의 점이 있고 짧은 리스트에 8개의 점이 있는 경우 $11 \times 8 = 88$ 개의 선이 생성됩니다.

□ 자동화란?

어학사전

[국어사전]

자동화 (自動化) 

다른 힘을 빌리지 아니하고 스스로 움직이거나 작용하게 됨. 또는 그렇게 되게 함.

[국어사전 다른 뜻 1](#)

[영어사전]

자동화 (自動化)

automation

[영어사전 결과 더보기](#)

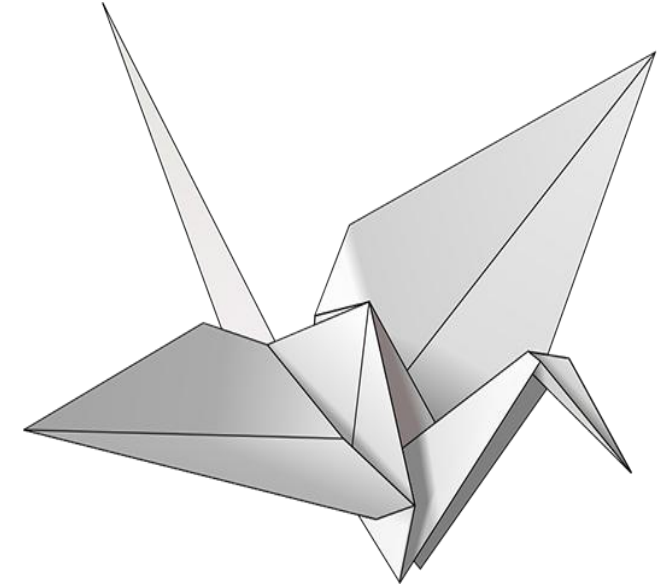
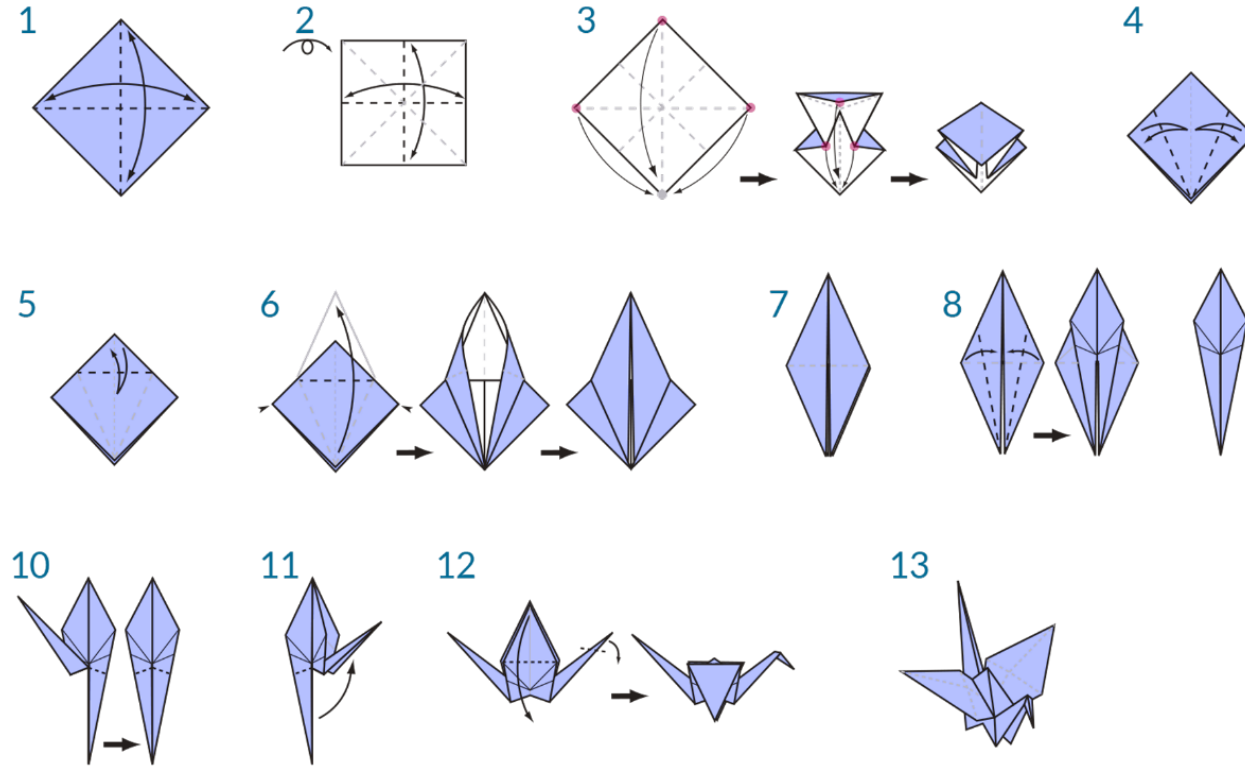
□ Target / Key Words

시간 절약

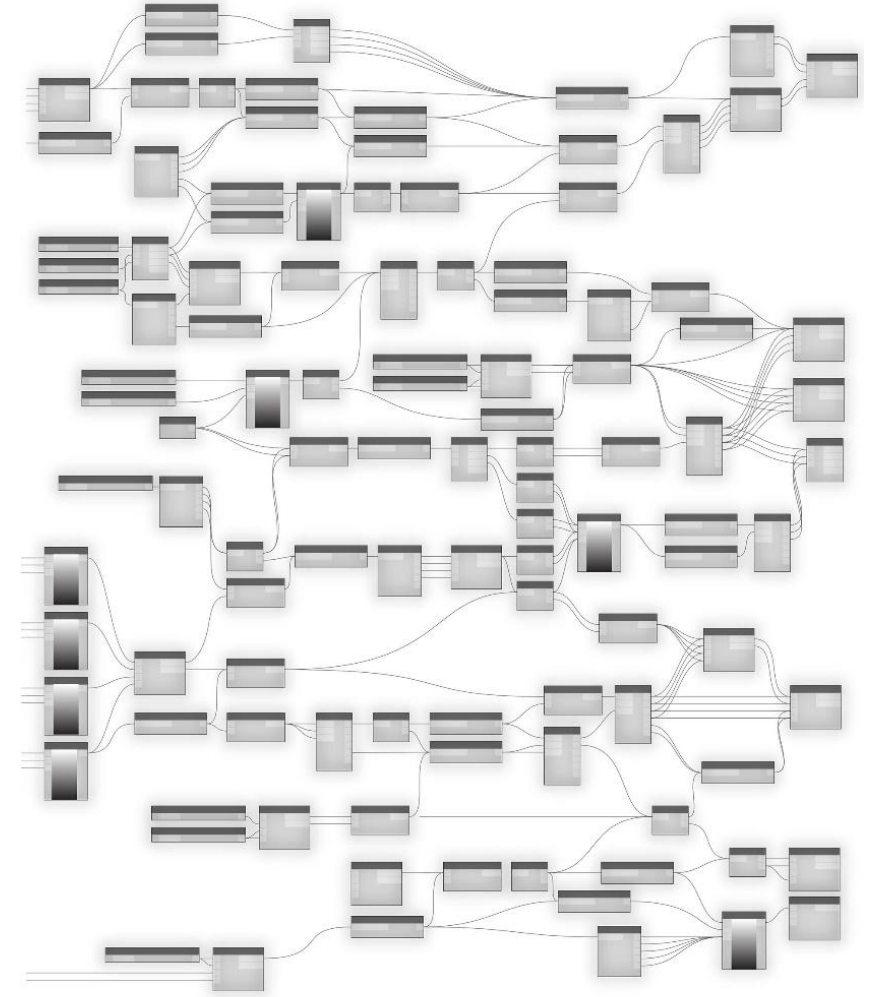
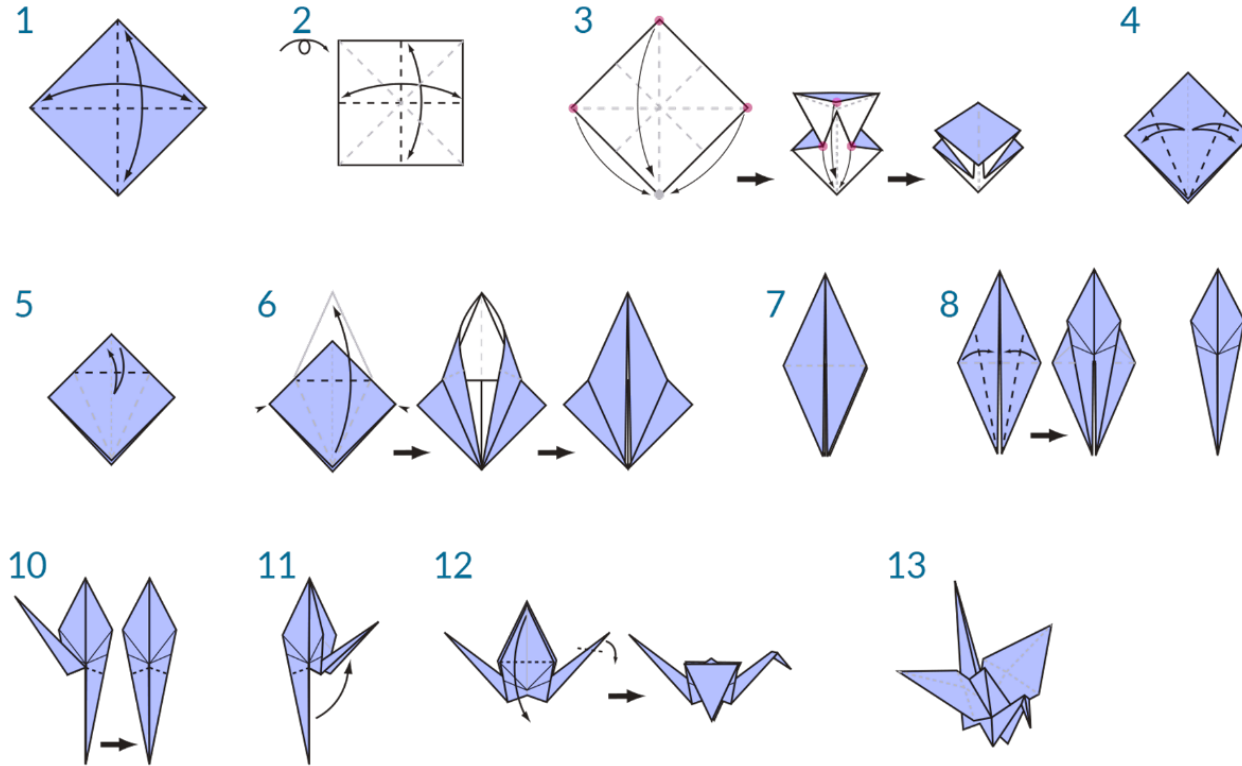
반복 작업

어려운 일

Automation _ Dynamo _ Visual Programing Tool



Automation _ Dynamo _ Visual Programing Tool



Item Issue

시간 절약

반복 작업

어려운 일

Item Issue

시간 절약

반복 작업

어려운 일



자동화

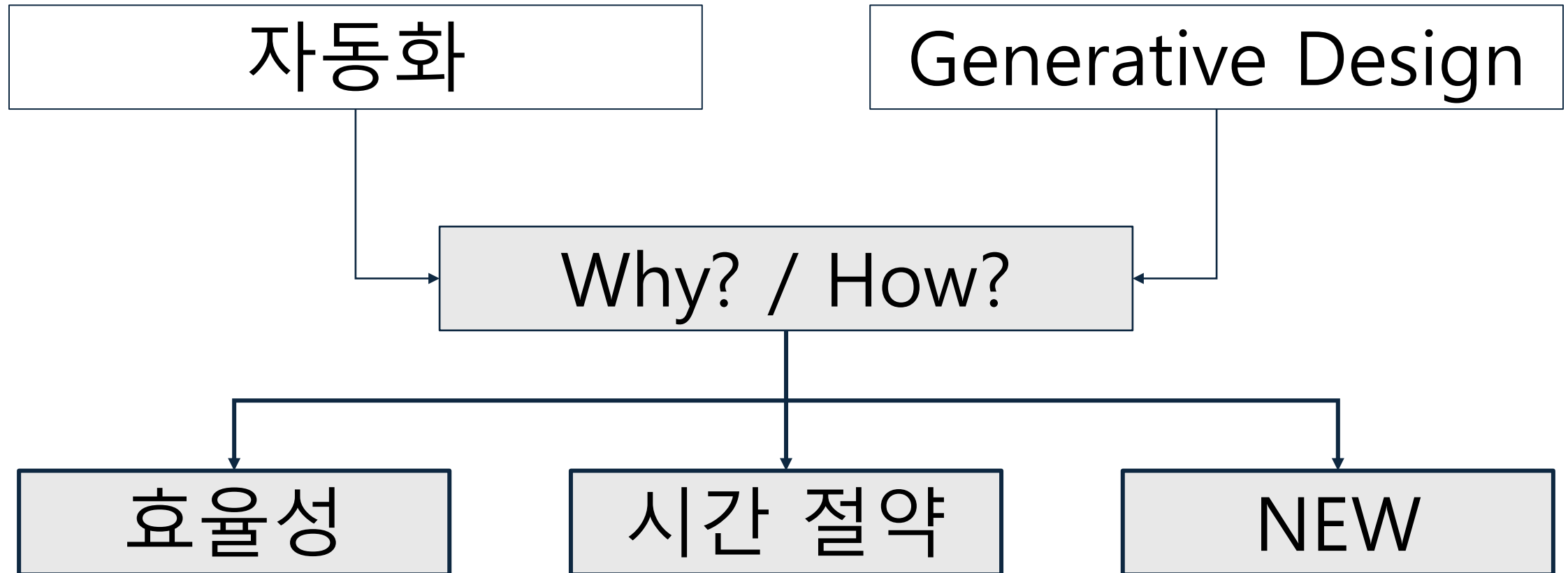
Item
Issue

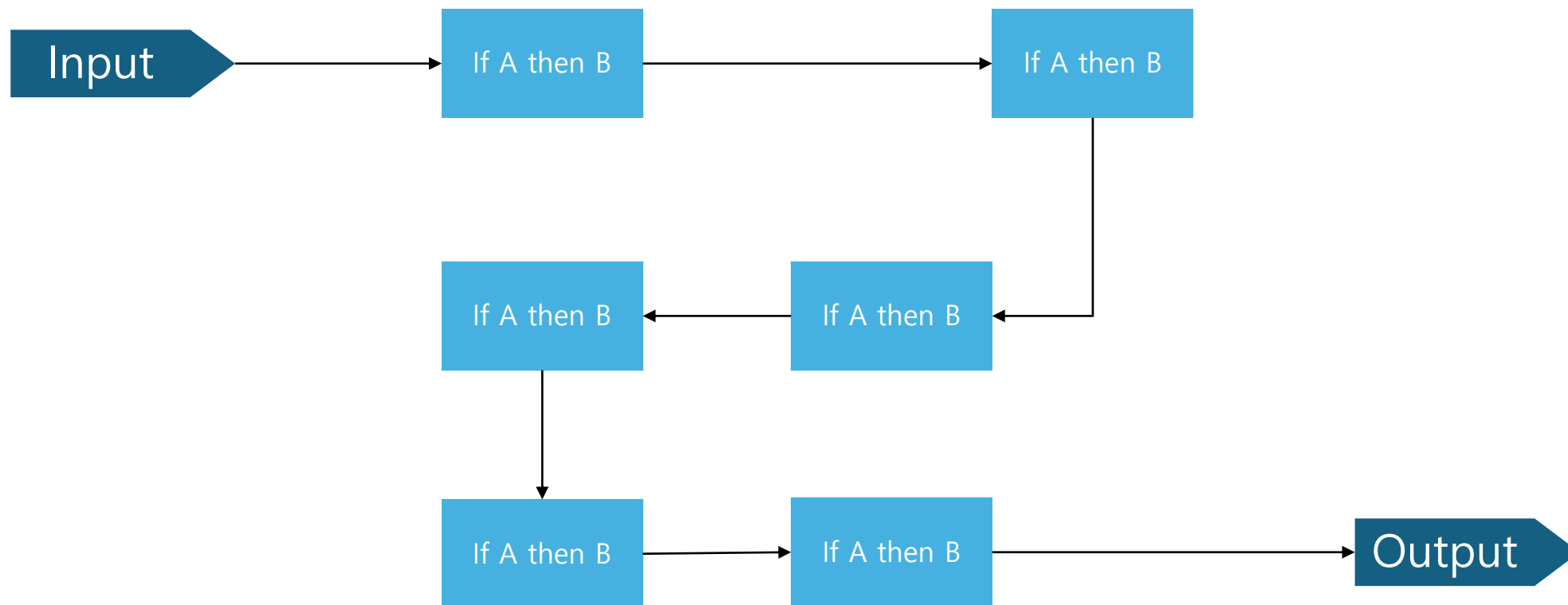
시간 절약
반복 작업
어려운 일

HOW

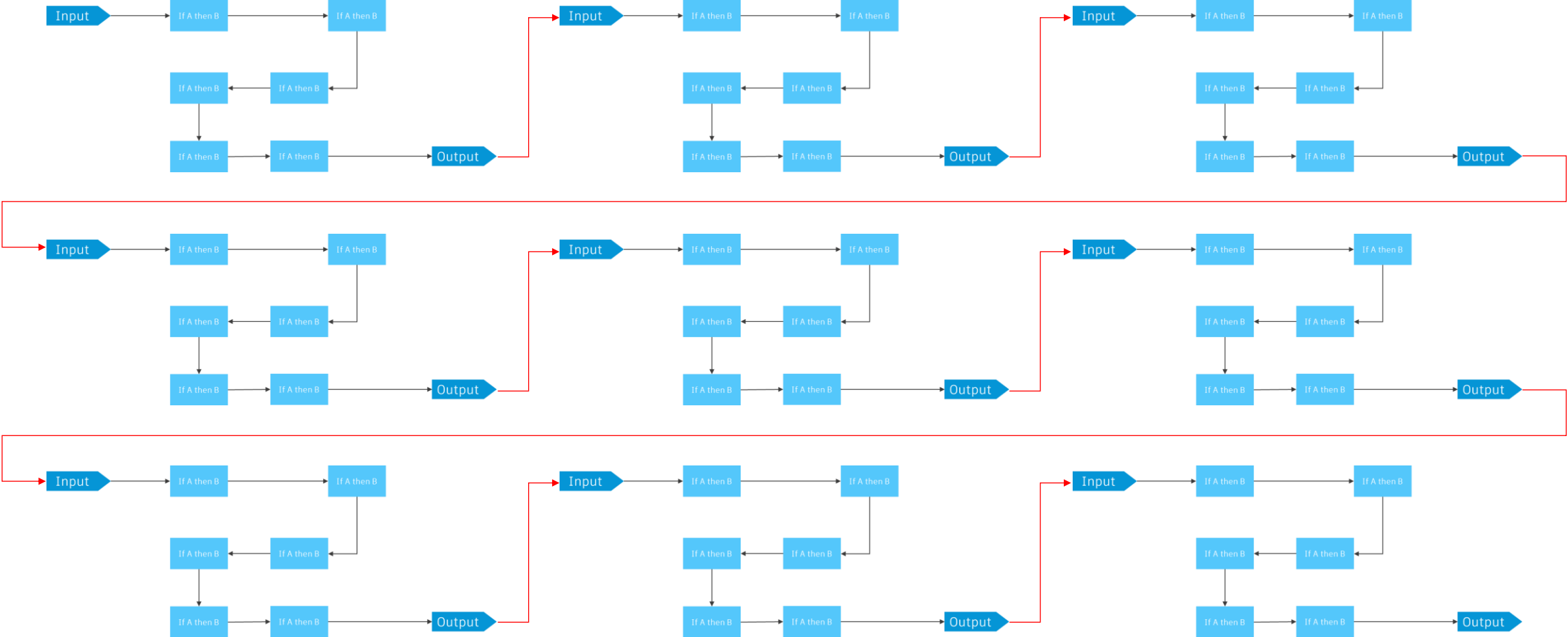


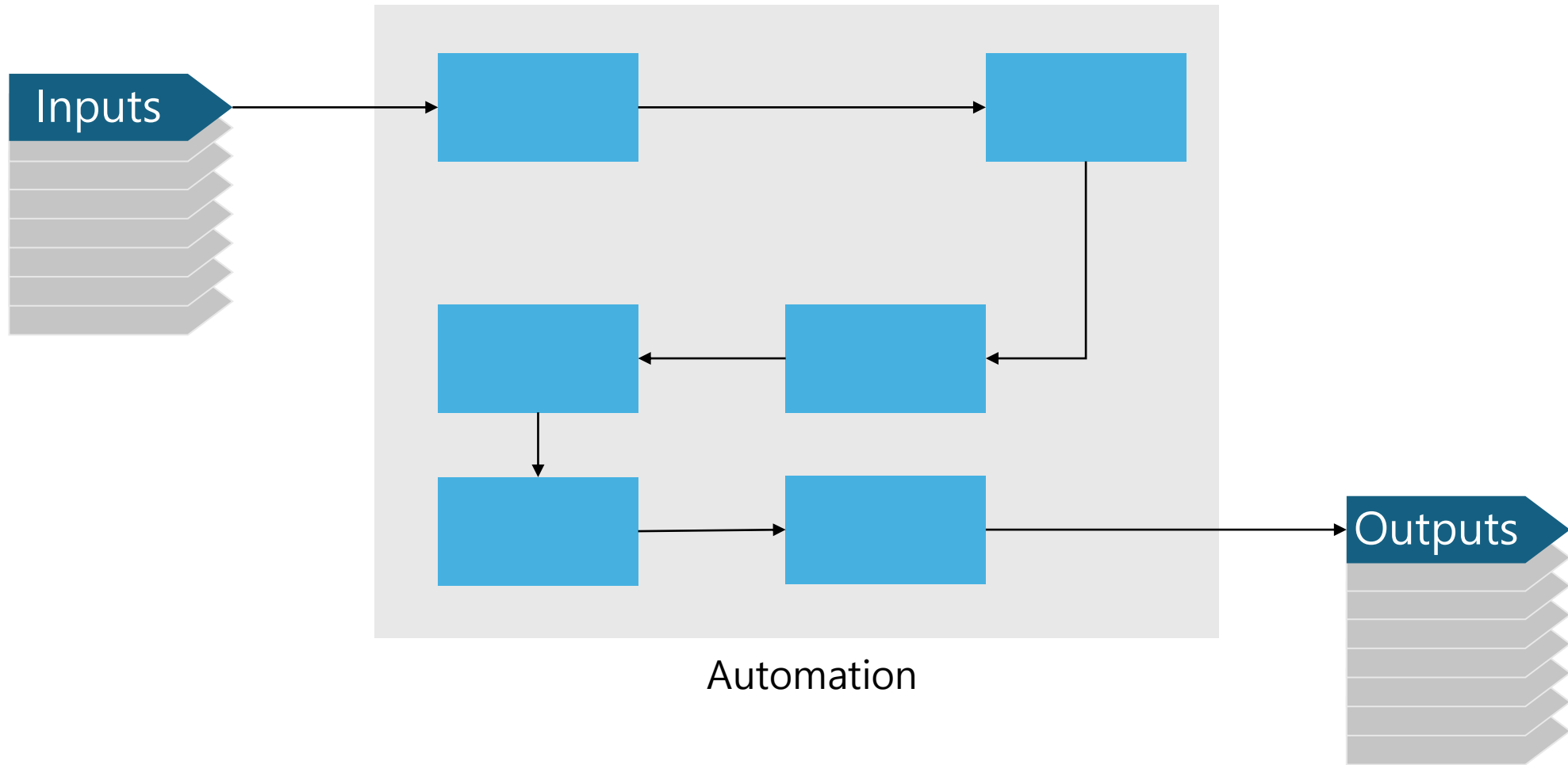
자동화



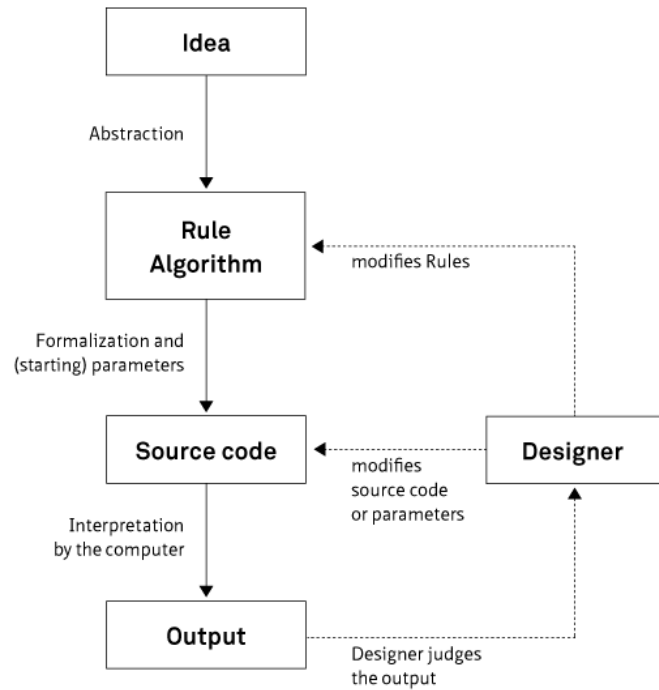


Process _ 자동화 프로세스





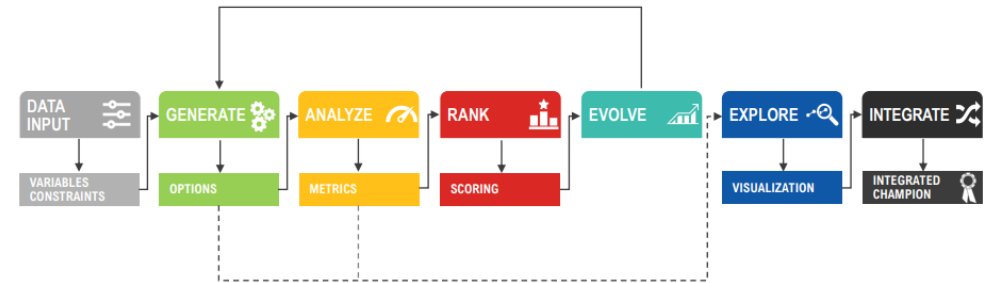
자동화



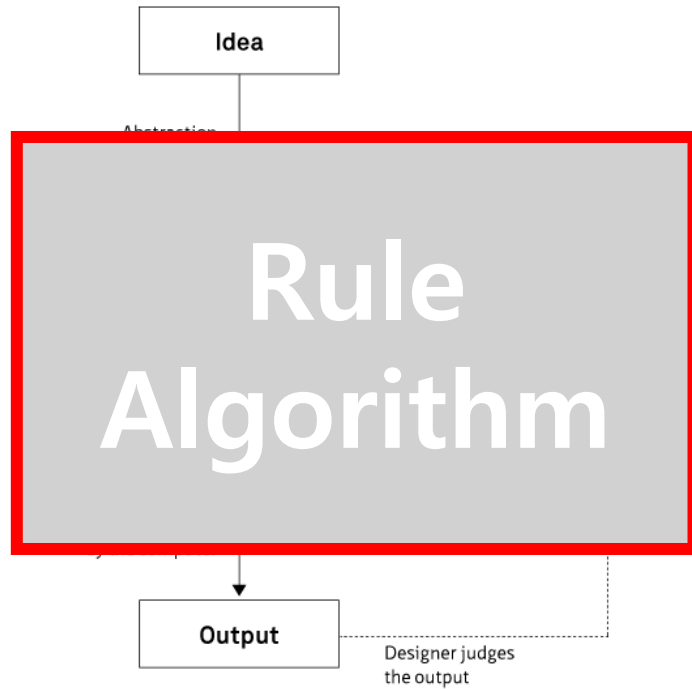
Copyright Hartmut Bohnacker, Julia Laub, Benedikt Groß, Claudius Lazzaroni (2009)
Book „Generative Gestaltung“, www.generative-gestaltung.de

Generative Design

Generative Design Process

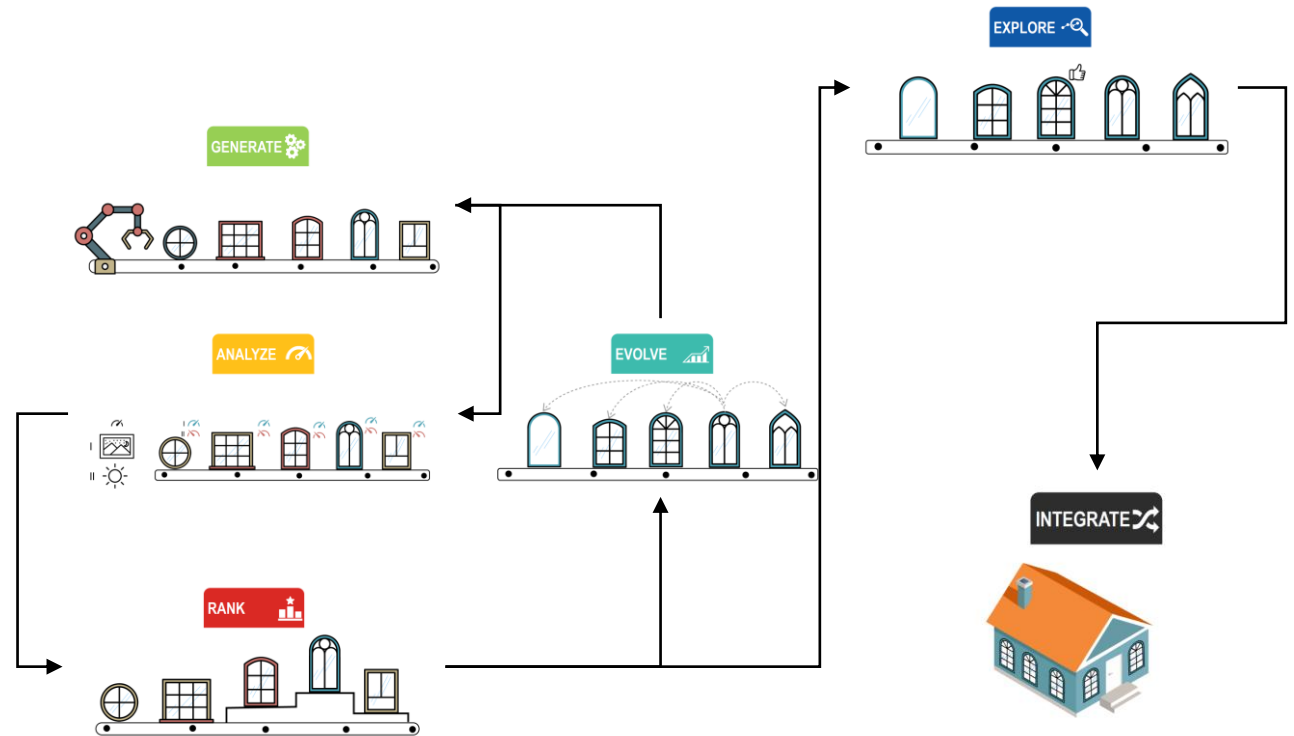


자동화

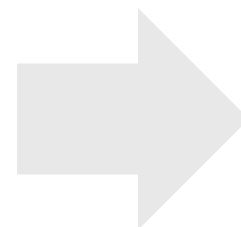
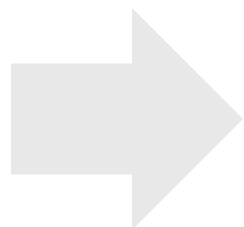


Copyright Hartmut Bohnacker, Julia Laub, Benedikt Groß, Claudius Lazzeroni (2009)
Book „Generative Gestaltung“, www.generative-gestaltung.de

Generative Design

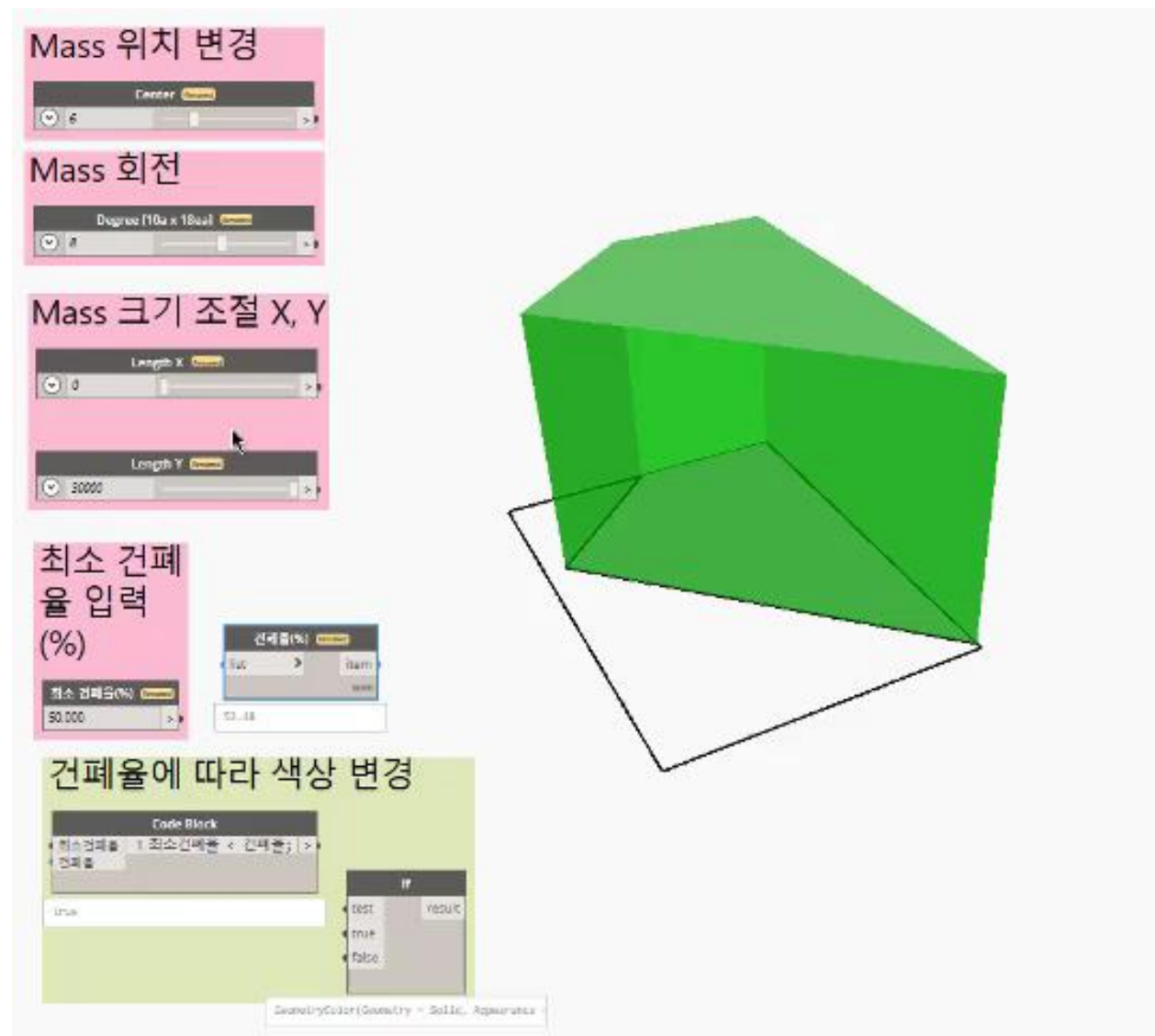


Item
Issue



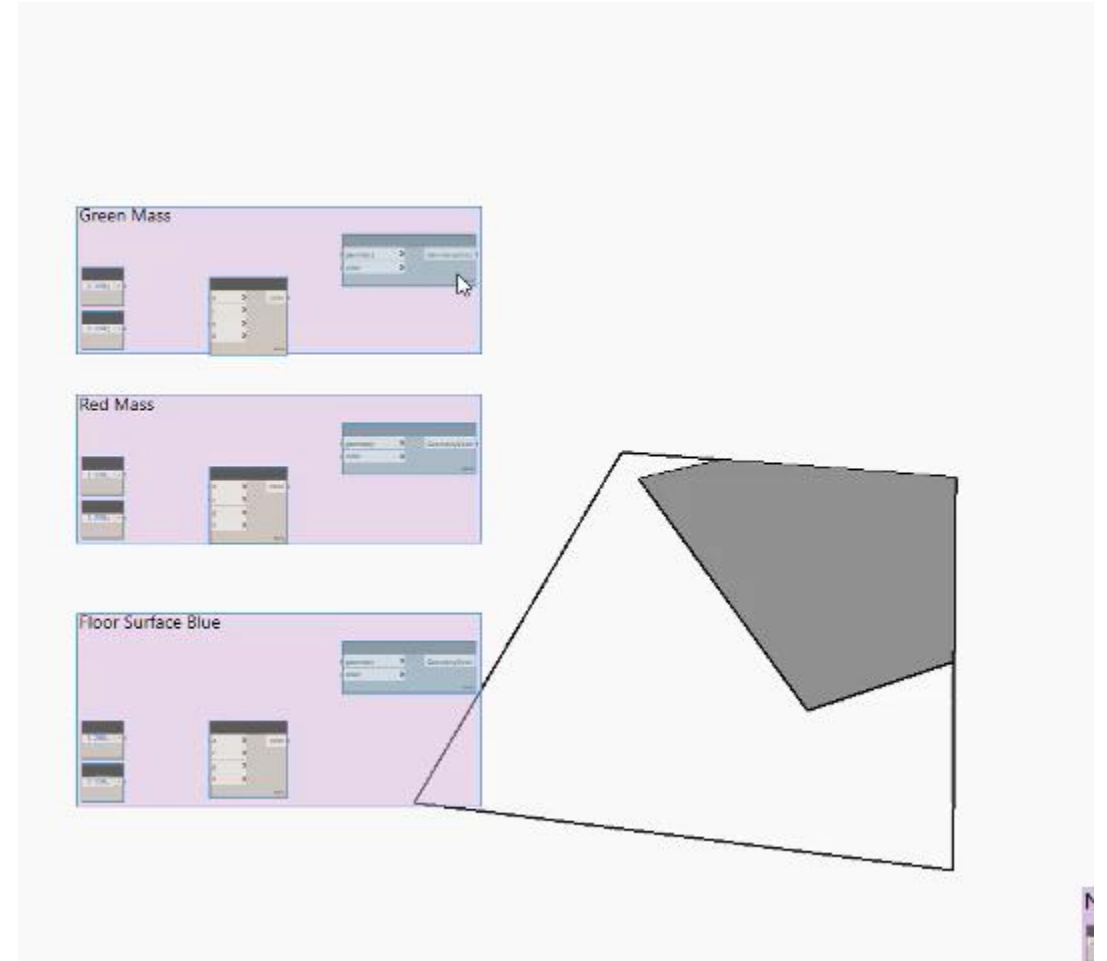
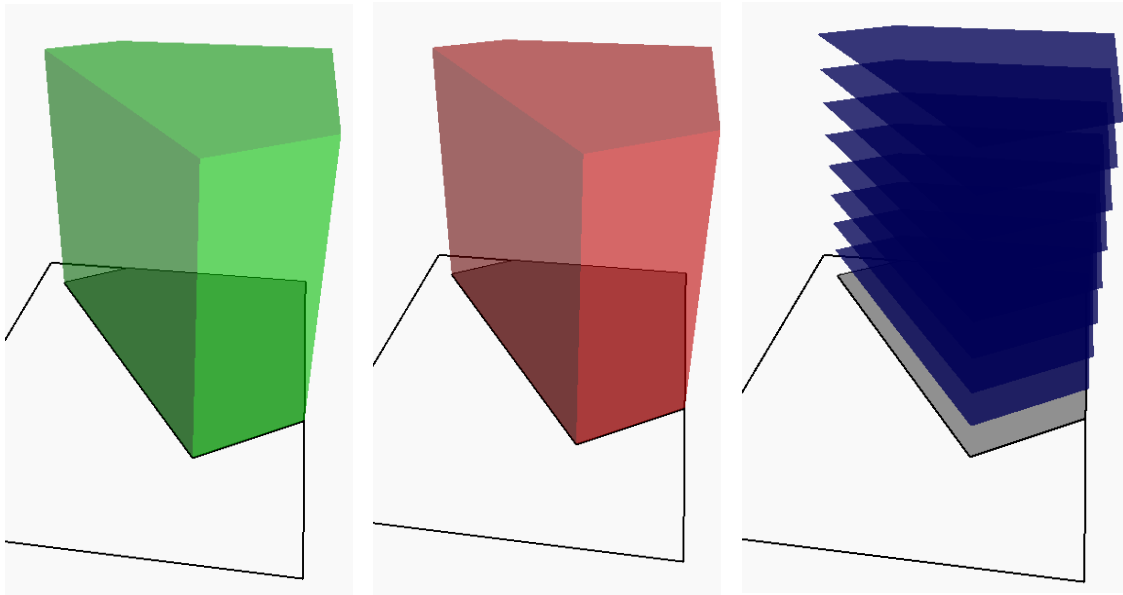
OUT
PUT

용적률
건폐율
법적 제약
조건
등



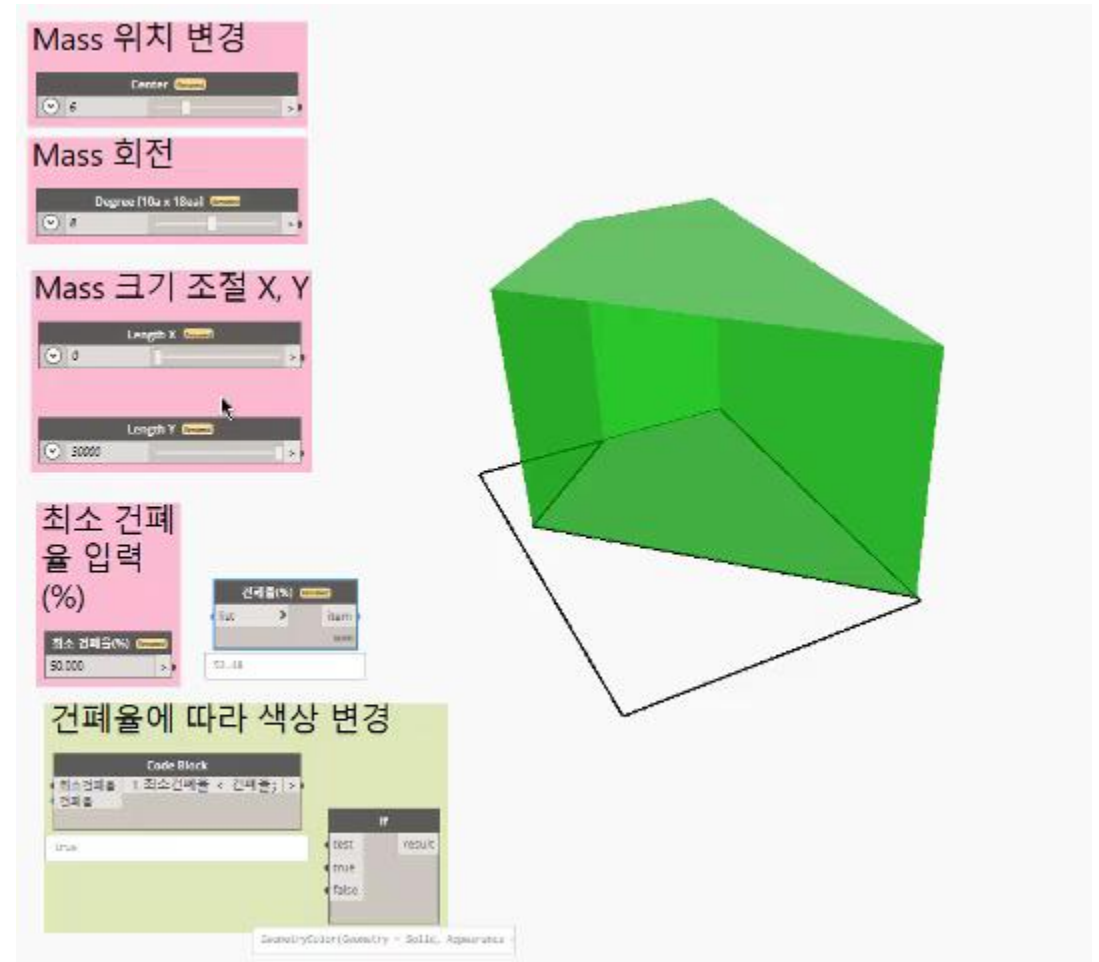
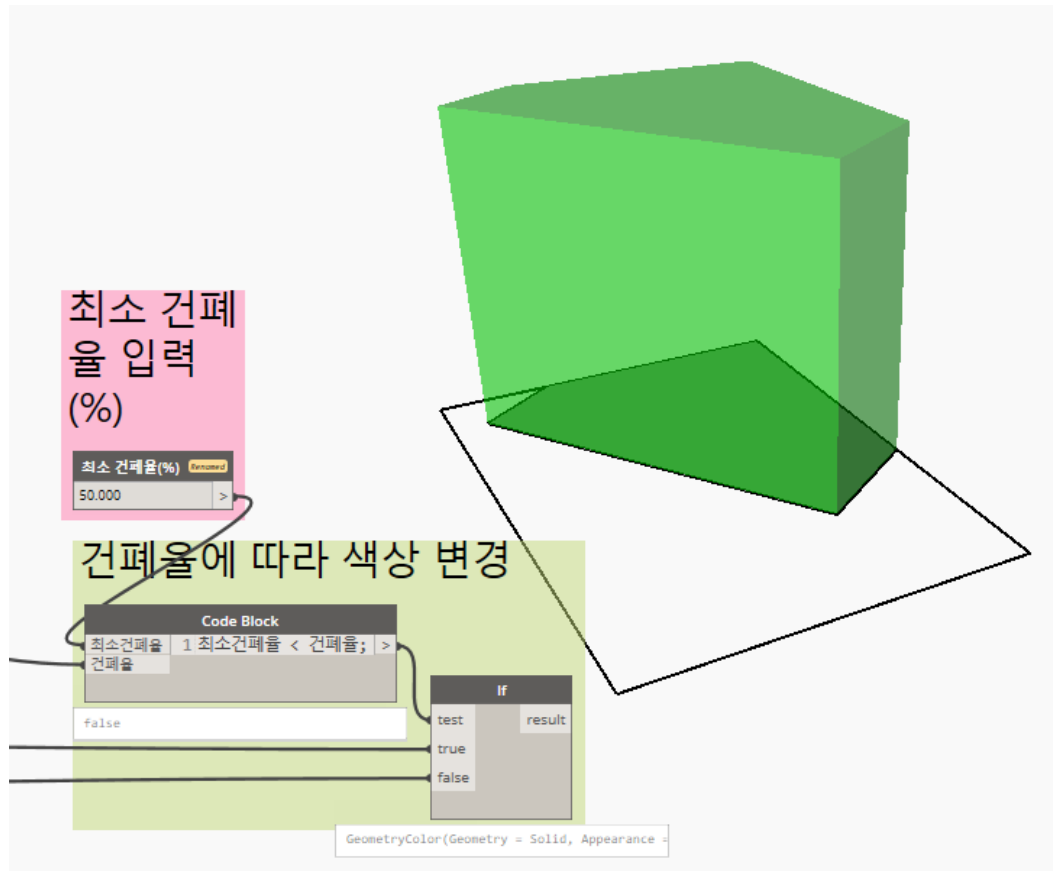
생성된 Solid, Surface의 Color 작업

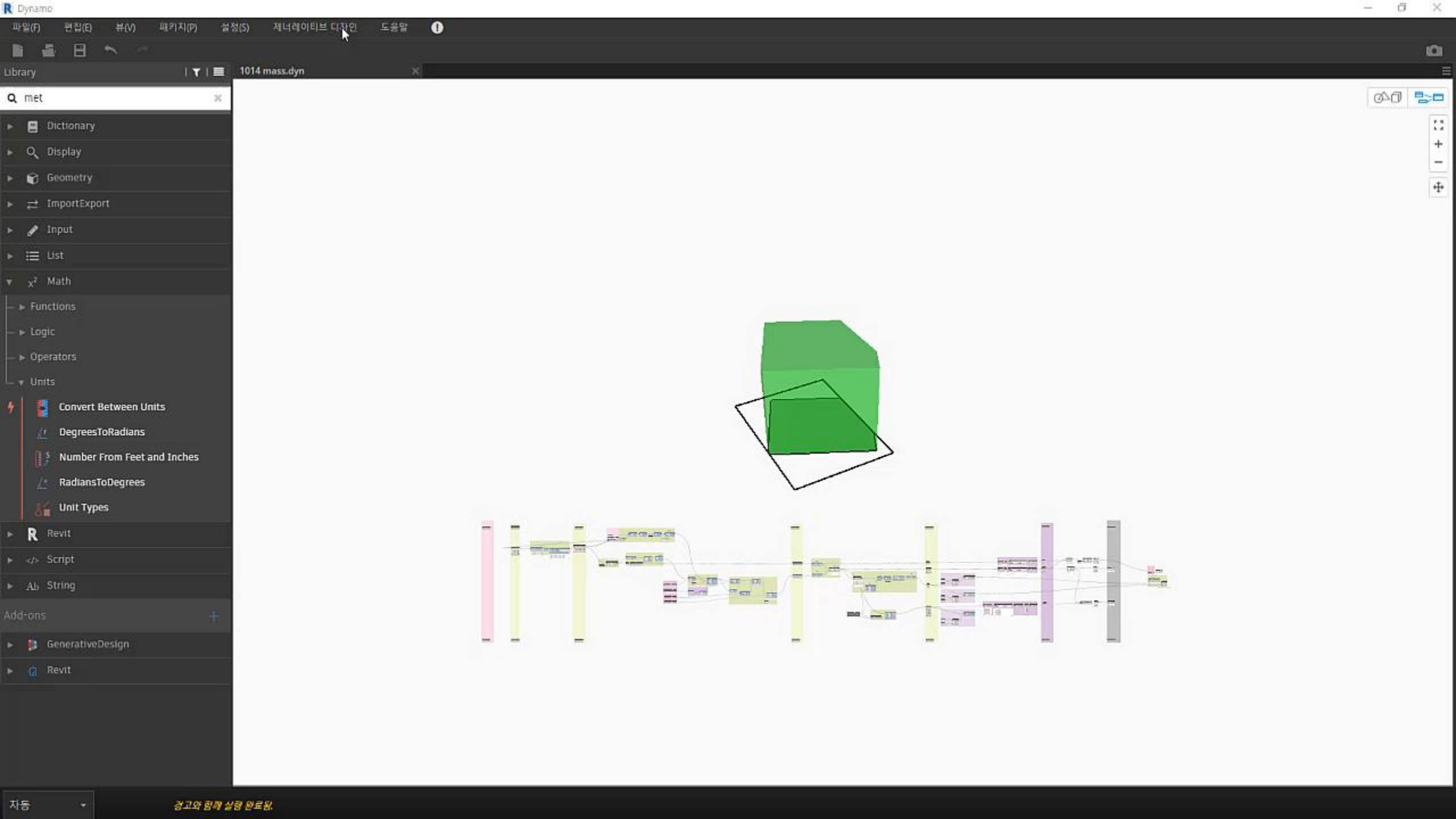
뒤에서 사용될 규칙에 따라 색상을 변경하기 위한 사전 작업

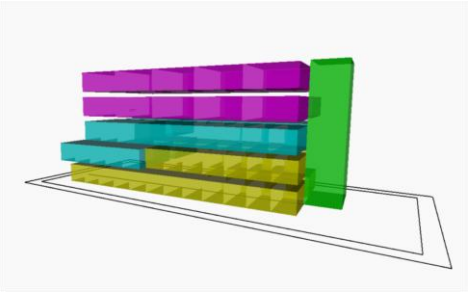
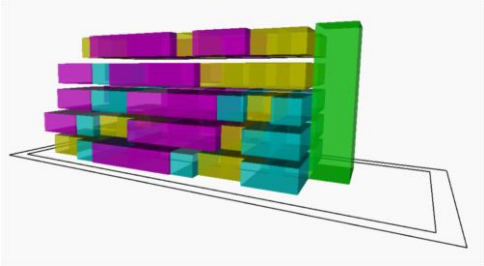
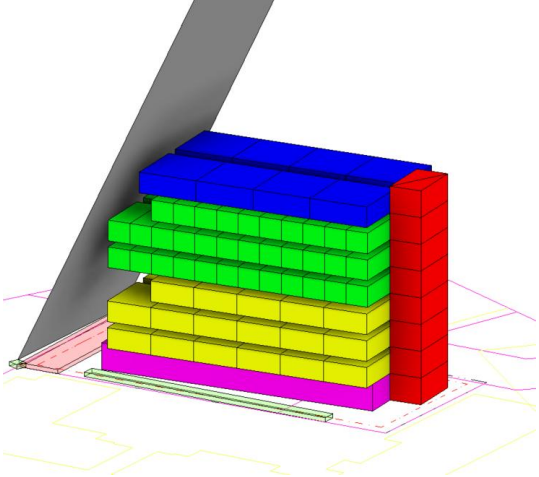
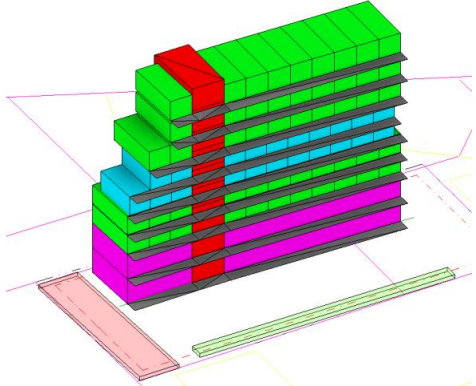


규칙 설정

건폐율 50%가 넘을시 Green / 미만 Red (상황에 따라 변경 가능)

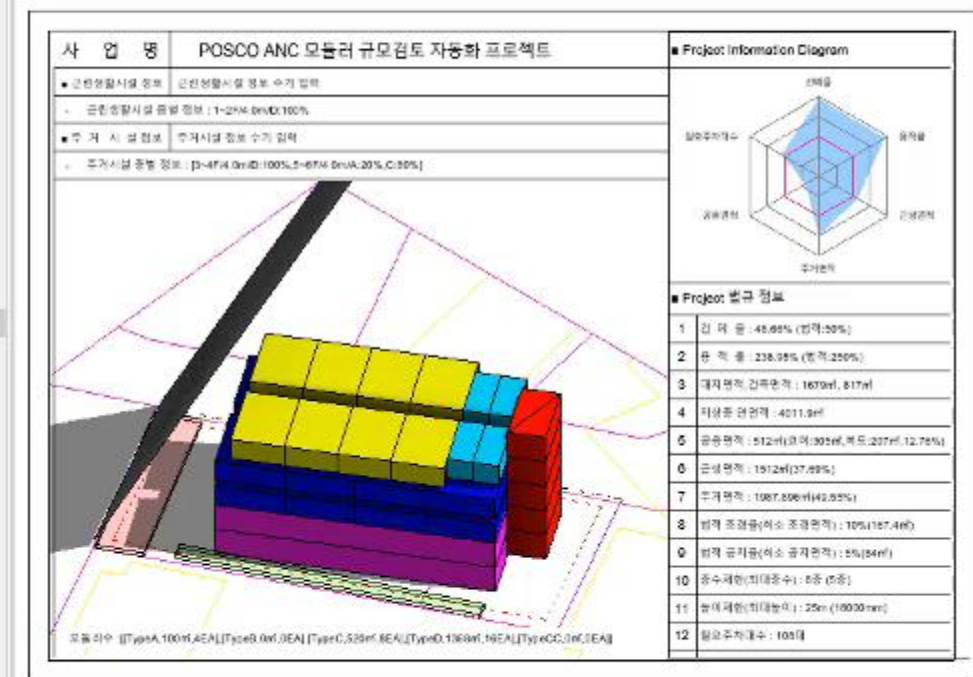
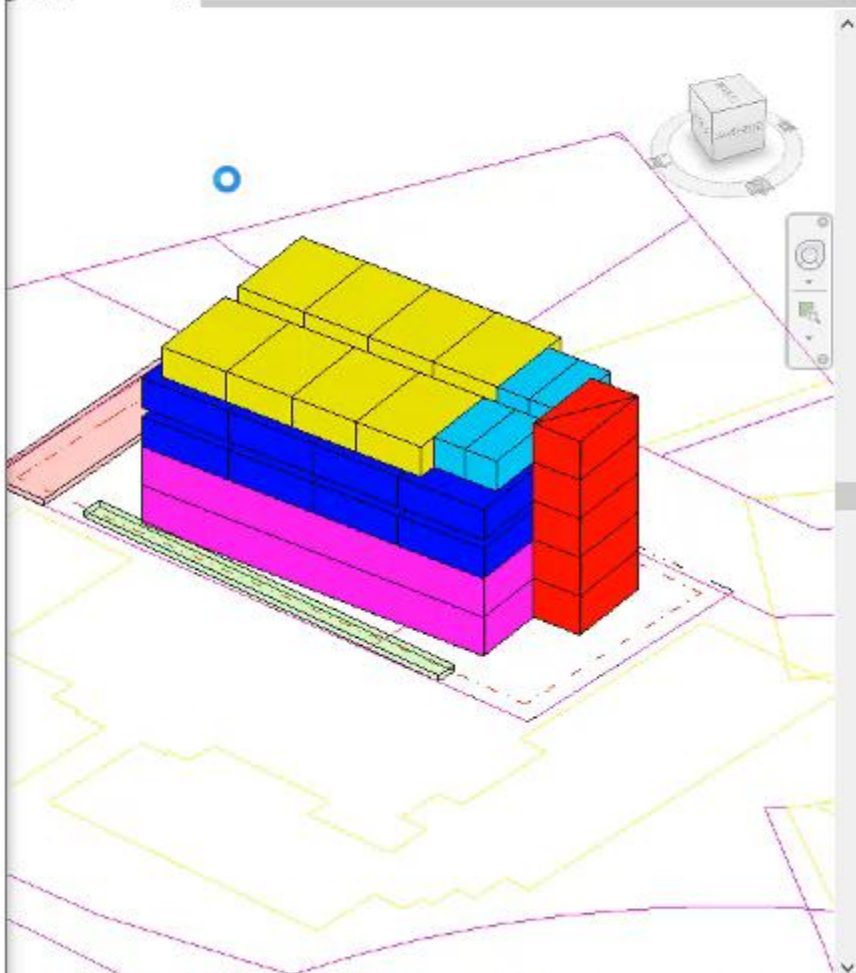




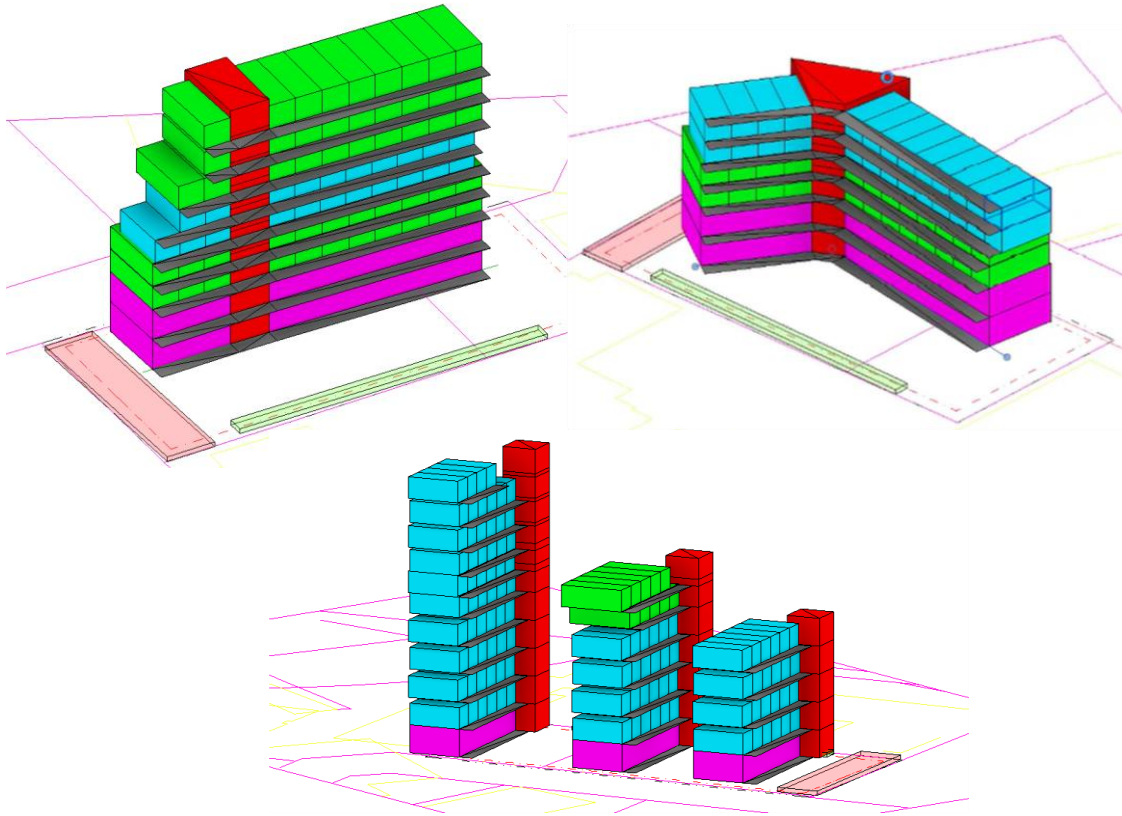
Group 배치	Random 배치	층별 비율(%) 배치 (수정)	선 선택
			
전체 Mass에 같은 유닛 그룹으로 순차적 모듈러 배치	전체 Mass에 지정된 비율로 무작위 모듈러 배치	[저층부, 중층부, 상층부] 3개의 구역을 층별로 나눈 후 모듈러를 배치	Model Line 작성 후 모듈러 패밀리 배치

test X

A102 - 모듈러 배치 (Group, Rand...



선 선택 모듈러 배치



Dynamo 플레이어 List



Dynamo 플레이어

(모듈러 배치 모듈러 배치) POSCO ANC 모듈러 규모검토

준비

[1] 선 선택 (Model Line) :

선택

Elements : 230228 230312

[2] T:층별비율, F:선선택 :

☐

False

[3] T: 중복도, F: 편복도 :

☐

False

[4] 복도 위치 변경 (편복도) :

☐

False

[5] 구역별 층수 입력 :

1,2,2

[6] 층수별 높이 입력 :

4000,4000,4000

[7] 모듈러 타입 배열 입력 :

A,A,A

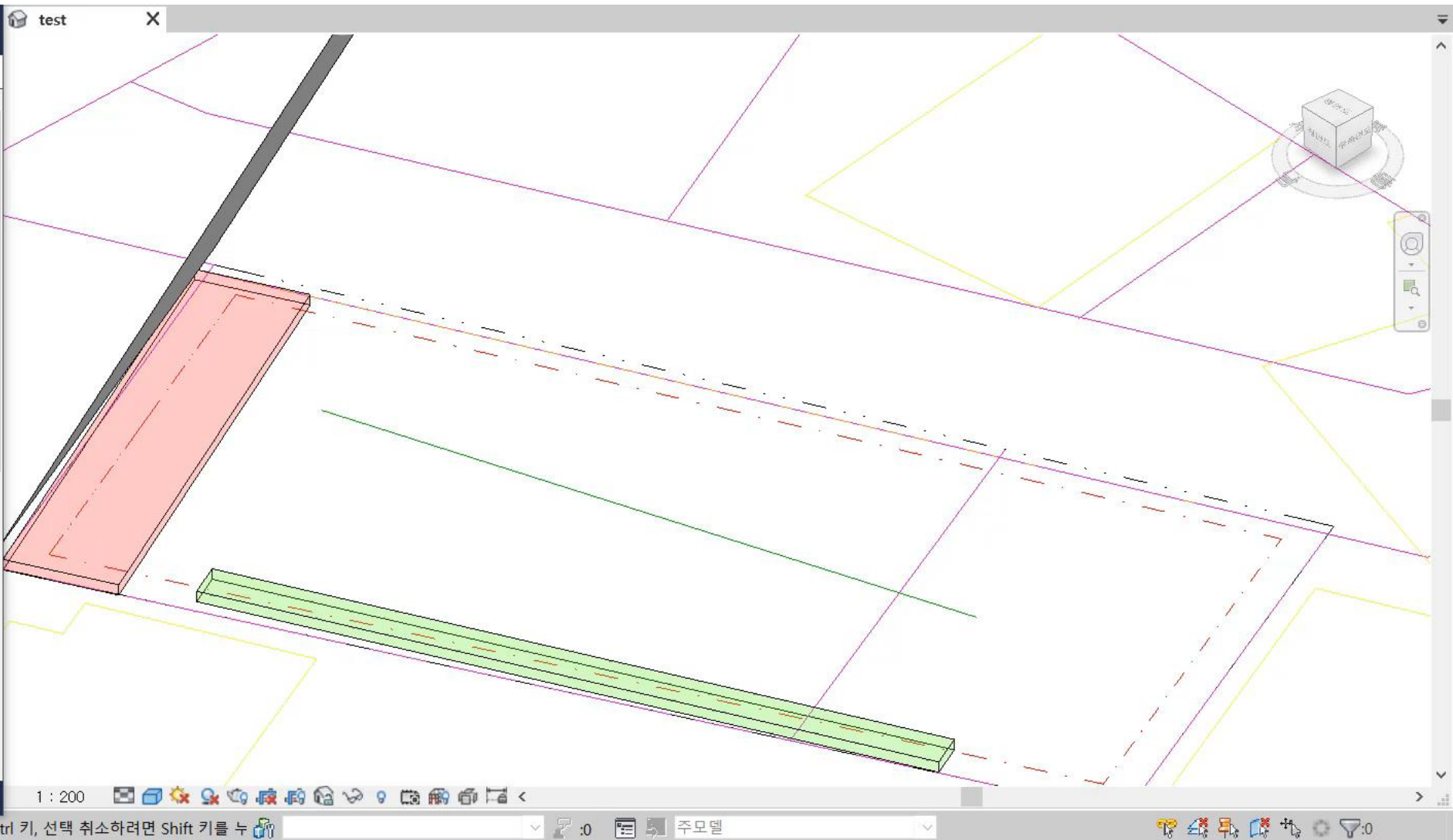
[8] 모듈러 타입 비율 입력 :

100,100,100

[9] 공지율 입력(%) :

60

POSCO ANC 모듈러 규모검토 자동화 Revit 파일.rvt



test

A102 - 모듈러 배치 (Group, Rand...



1 : 200



선택하려면 항목을 클릭하고, 다른 항목을 선택하려면 Tab 키, 추가하려면 Ctrl 키, 선택 취소하려면 Shift 키를 누르세요

0

주모델

Dynamo 플레이어



name

[3] T: 중복도, F: 편복도:

☒ True

[4] 복도 위치 변경 (편복도):

☐ False

[5] 구역별 층수 입력:

1,3,1,2

[6] 층수별 높이 입력:

4000,3100,6000,3100

[7] 모듈러 타입 배열 입력:

C,A@B,B,A

[8] 모듈러 타입 비율 입력:

100,50@50,100,100

[9] 공지를 입력(%):

30

[10] 조경률 입력(%):

5

[11] 건축 한계선 선택:

개수: 37 ... 자세히 표시

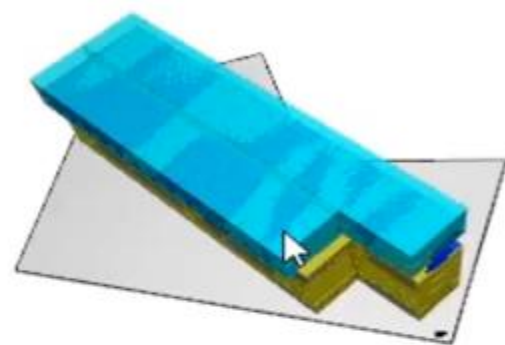
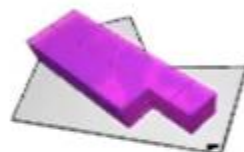
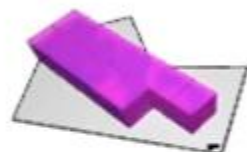
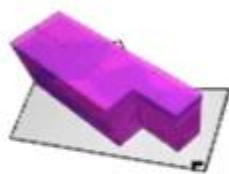
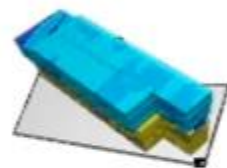
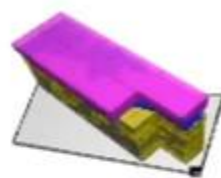
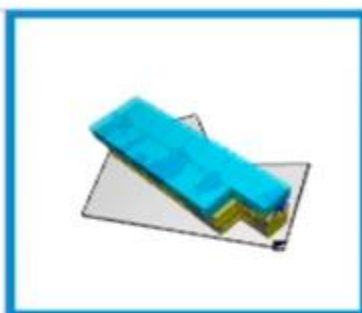
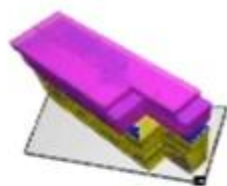
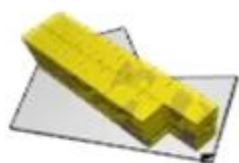
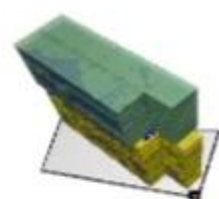
[12] 일조,채광 라인 선택 (전체):

Elements: 129817

[13] 최고 높이 (m):

55

[14] 공지를,조경률 과잉의 생성:



Details

Outputs

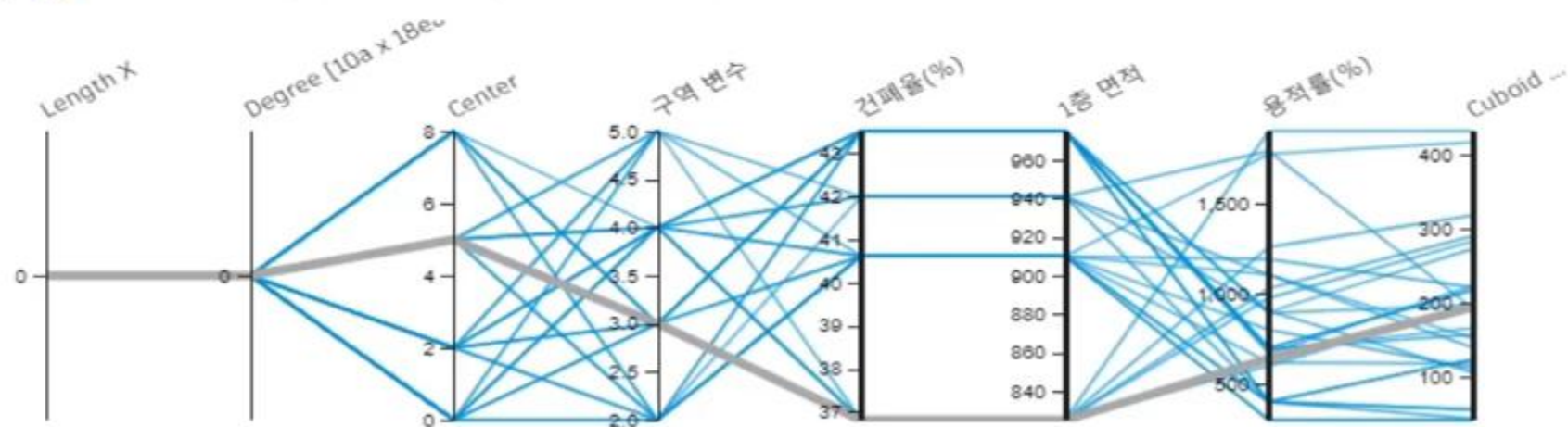
건폐율(%)	36.810
1층 면적	825.0
용적률(%)	639.030
Cuboid Count	194.0

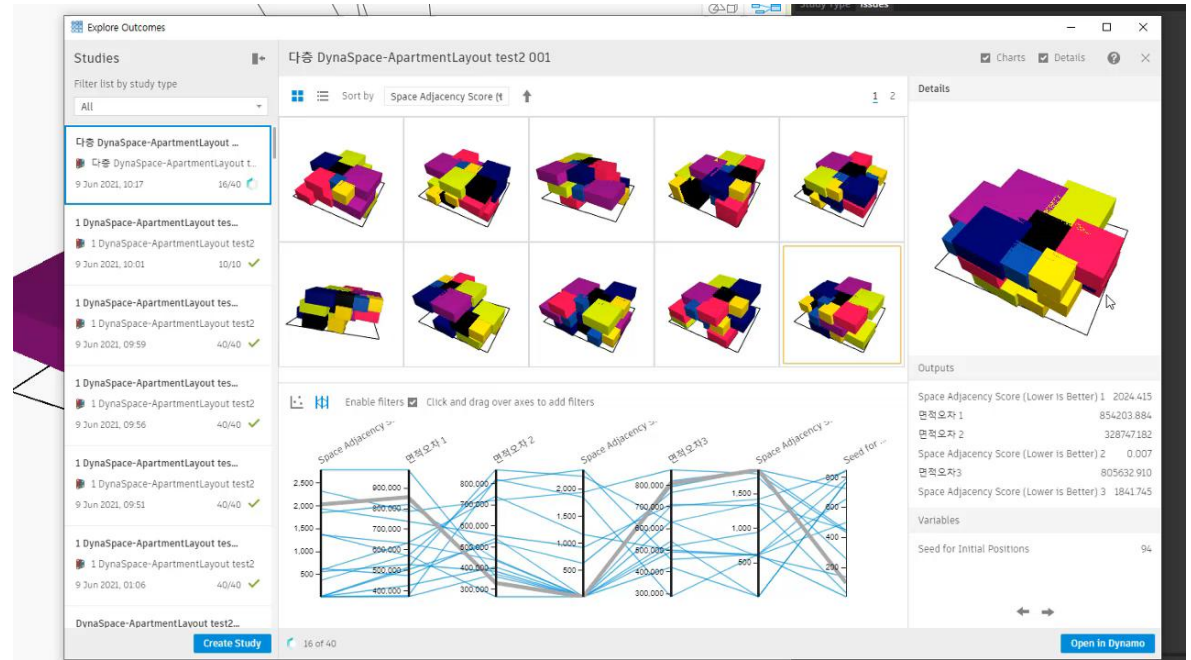
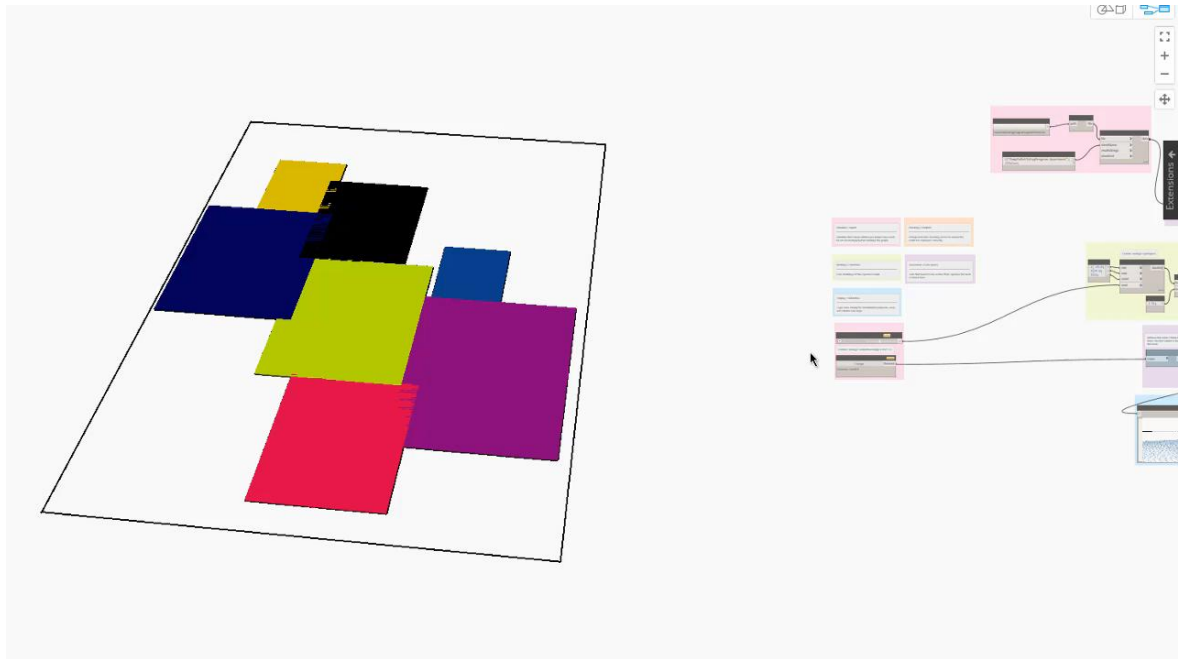
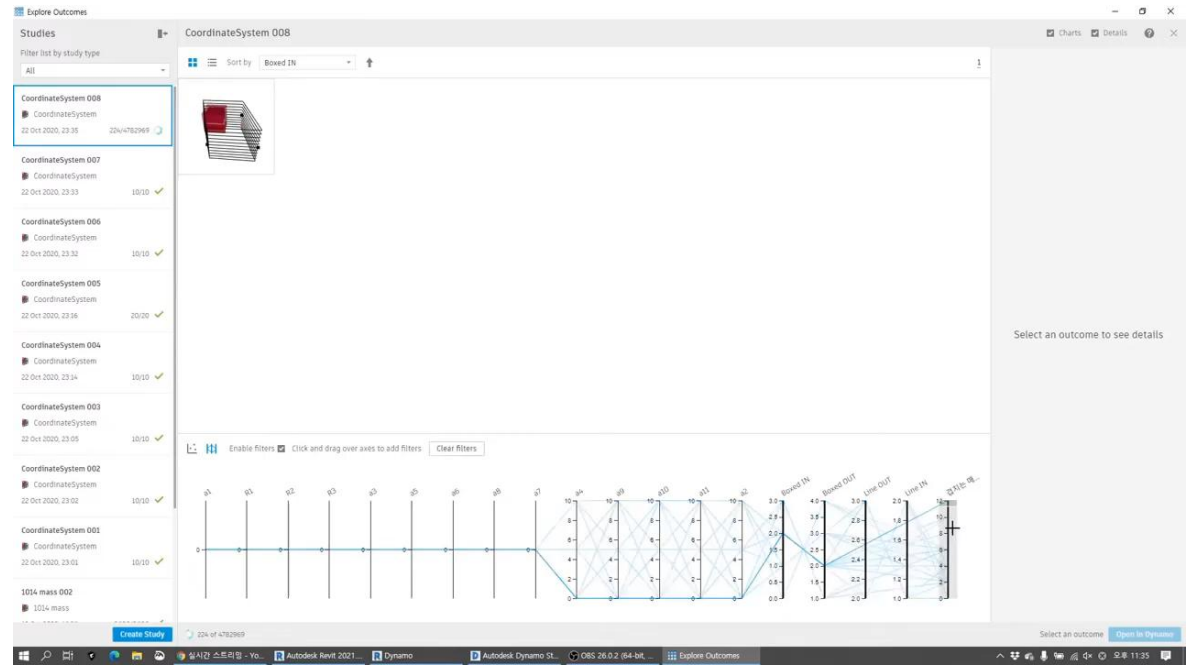
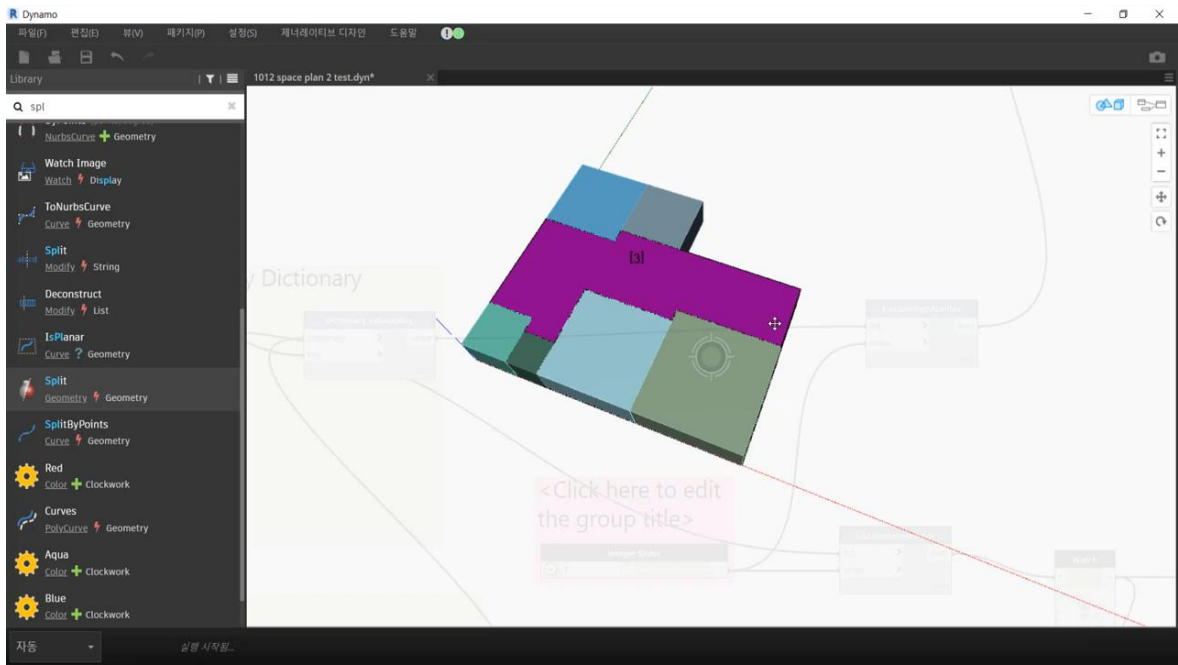
Inputs

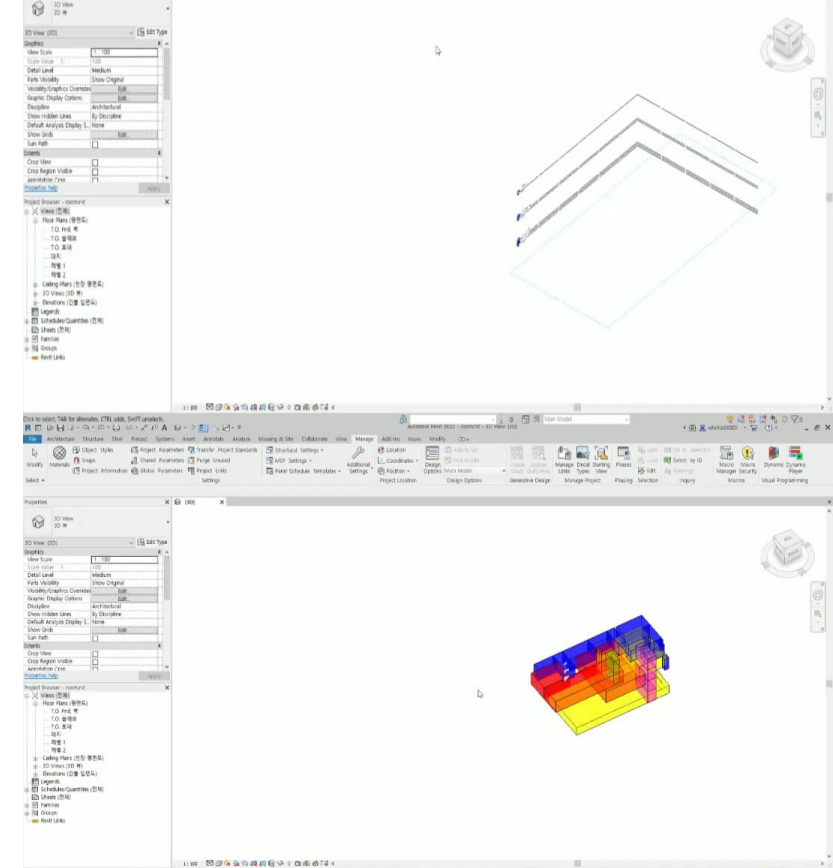
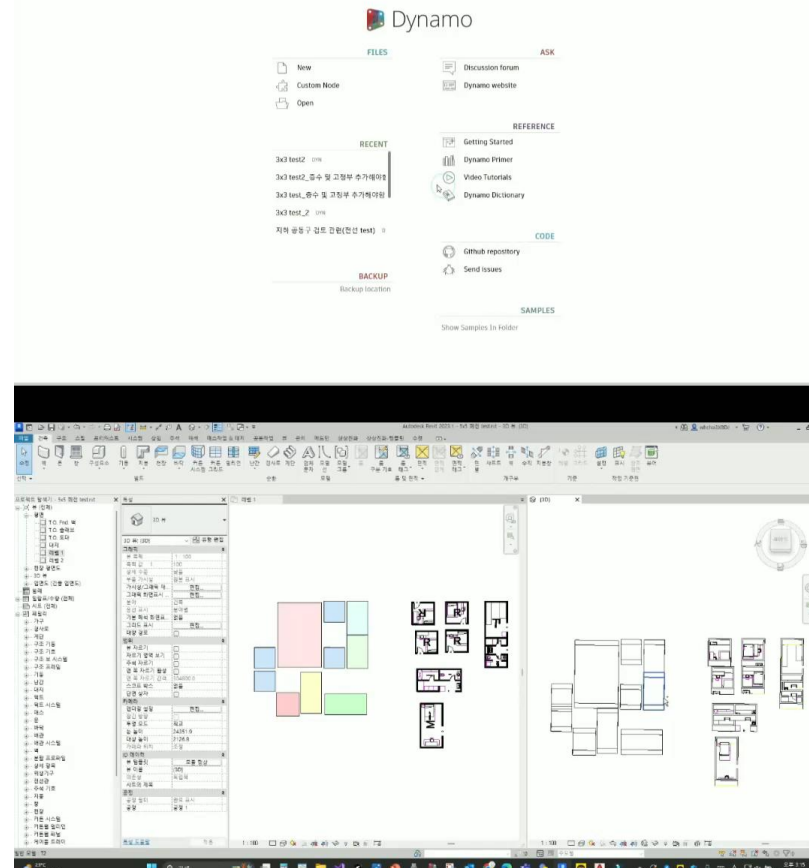
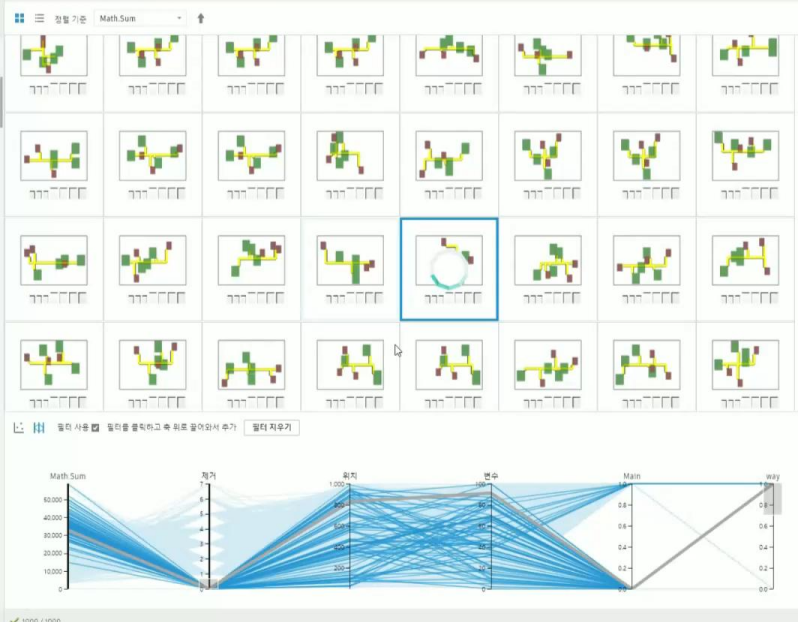
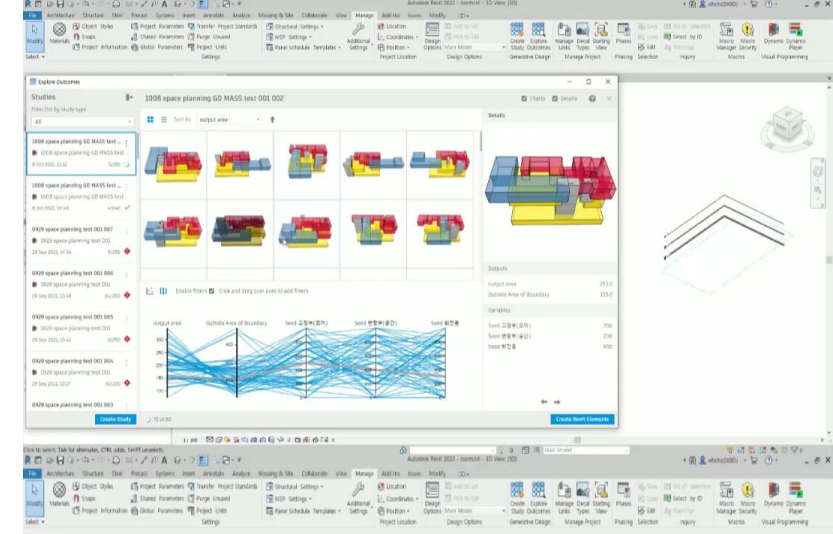
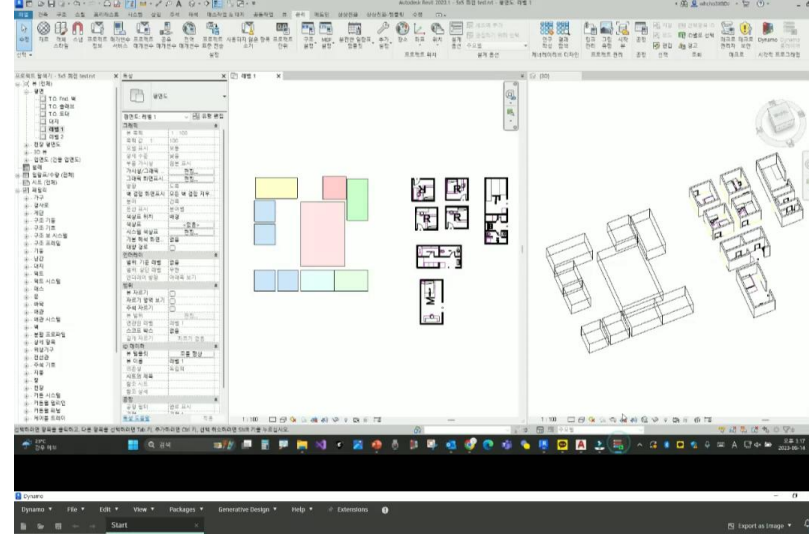
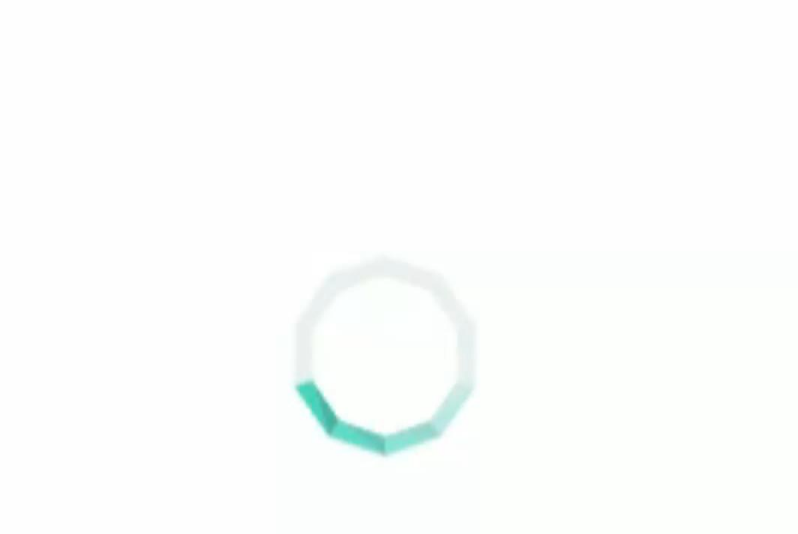
Length X	0
Degree [10a x 18ea]	0
Center	5
구역 변수	3

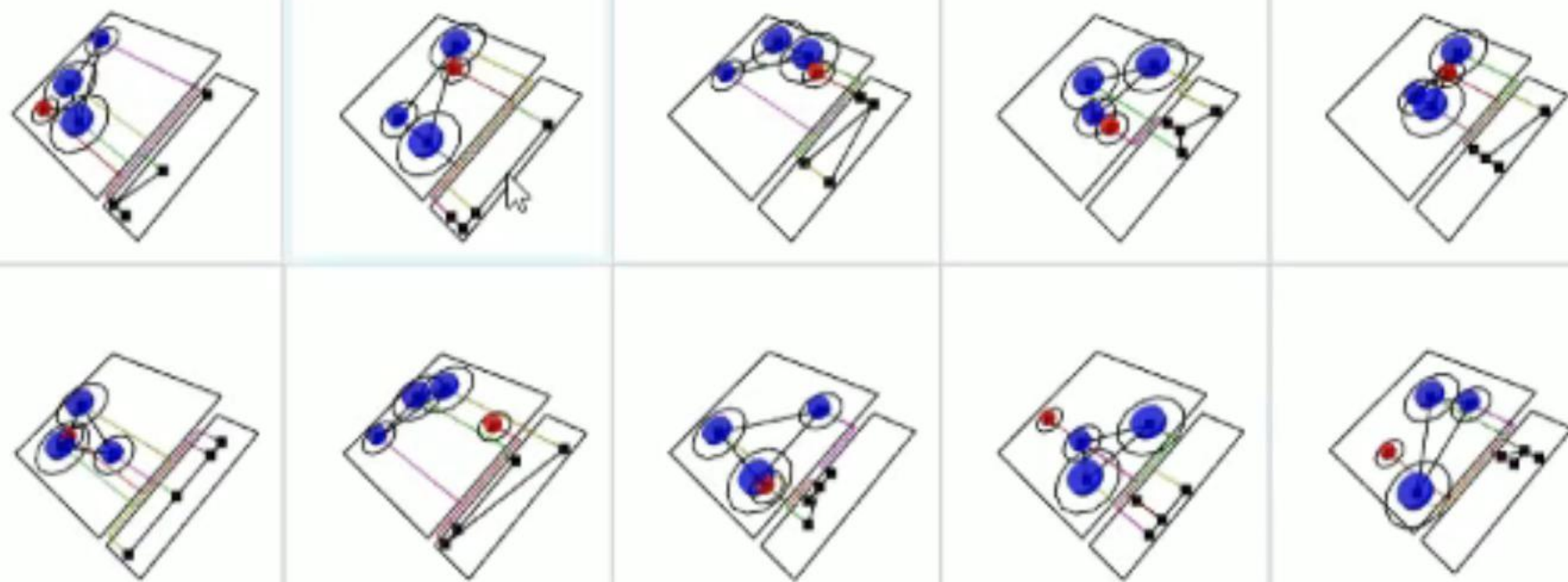


Enable filters ☒ Click and drag over axes to add filters



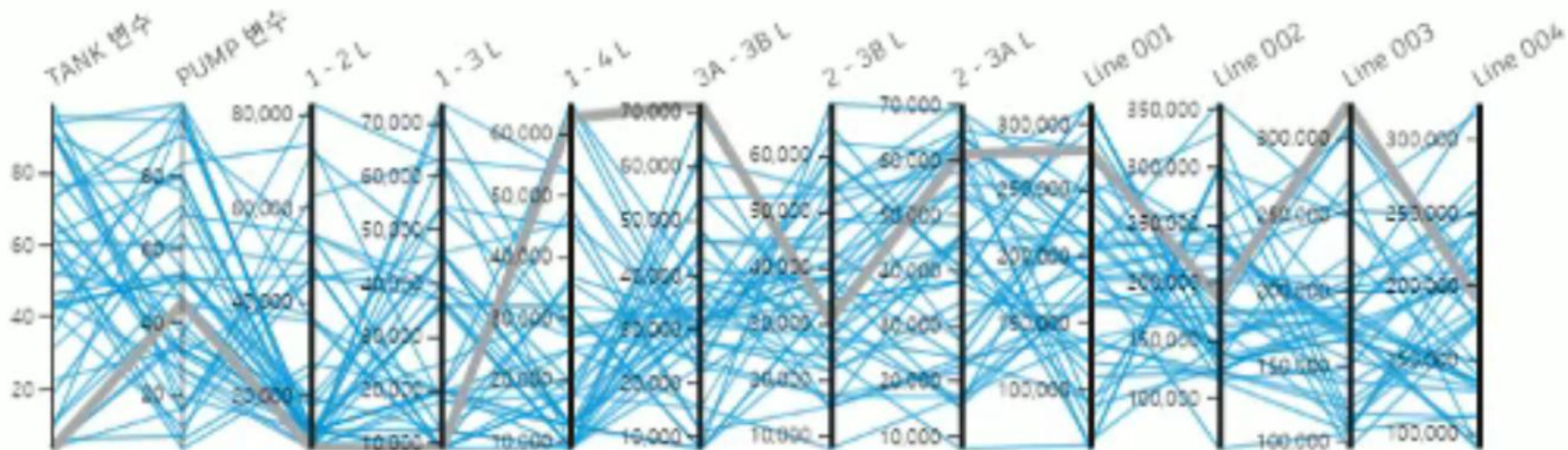






결과를 선택하여 상세정보 보기

필터 사용 ☒ 필터를 클릭하고 축 위로 끌어와서 추가



연구 유형을 기준으로 리스트 필터링

전체

20231101 Space layout test(v5) 복도 test

2023년 11월 1일 오후 1:26 40/40

20231101 Space layout test(v5) 복...

20231101 Space layout test(v5) 복도 test

2023년 11월 1일 오전 11:27 100/100

20231101 Space layout test(v5) 복...

20231101 Space layout test(v5) 복도 test

2023년 11월 1일 오전 11:21 100/100

20231101 Space layout test(v5) 복...

20231101 Space layout test(v5) 복도 test

2023년 11월 1일 오전 10:18 200/200

20231101 Space layout test(v5) 복...

20231101 Space layout test(v5) 복도 test

2023년 11월 1일 오전 10:12 100/100

파이프랙 자동화 test 1030 003

파이프랙 자동화 test 1030

2023년 10월 30일 오후 2:10 1000/1000

파이프랙 자동화 test 1030 002

파이프랙 자동화 test 1030

2023년 10월 30일 오후 2:09 40/40

파이프랙 자동화 test 1030 001

파이프랙 자동화 test 1030

2023년 10월 30일 오후 2:07 500/500

파이프랙 자동화 test 0920 006

파이프랙 자동화 test 0920

2023년 10월 27일 오후 12:01 100/100

파이프랙 자동화 test 0920 005

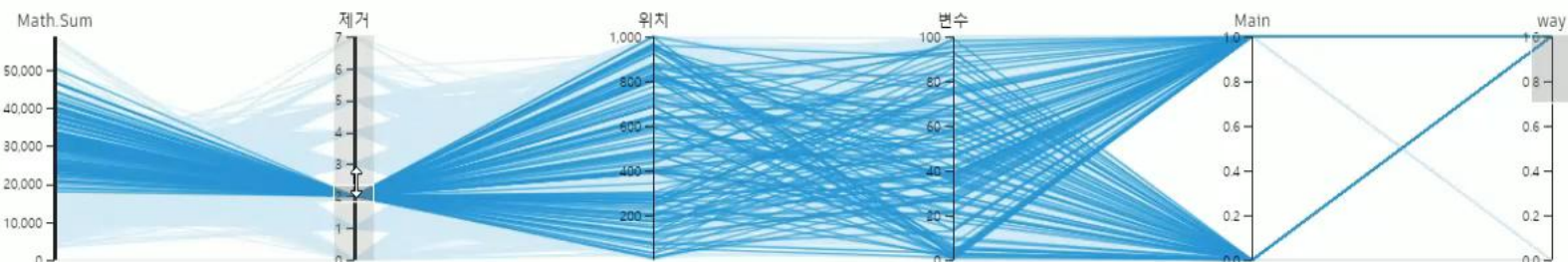
파이프랙 자동화 test 0920

2023년 10월 27일 오전 11:49 100/100

연구 작성

파이프랙 자동화 test 1030 003

정렬 기준 Math.Sum

필터 사용 ☒ 필터를 클릭하고 축 위로 끌어와서 추가 필터 지우기

1000 / 1000

결과를 선택하여 상세정보 보기

결과 선택

Revit 요소 작성



공간(시설) 배치 방법론

REC 움직이기

Point 제외시키기

REC 다중 배치

REC 크기 변경

겹침 피하기

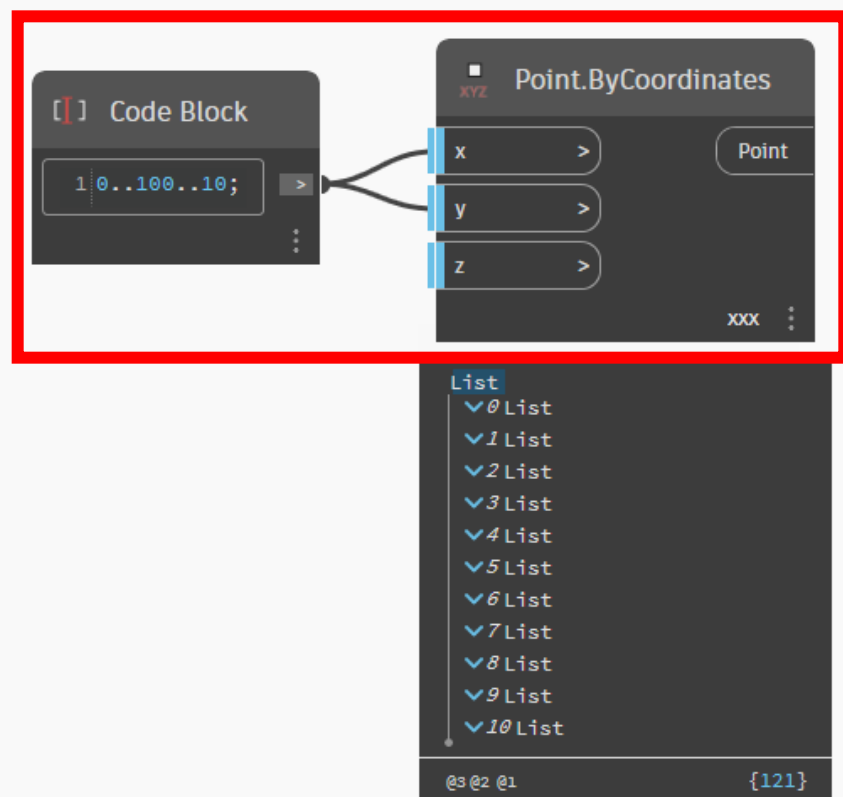
REC Rotate 후 배치

겹치는 REC 제외

겹치는 REC 제외 중심점

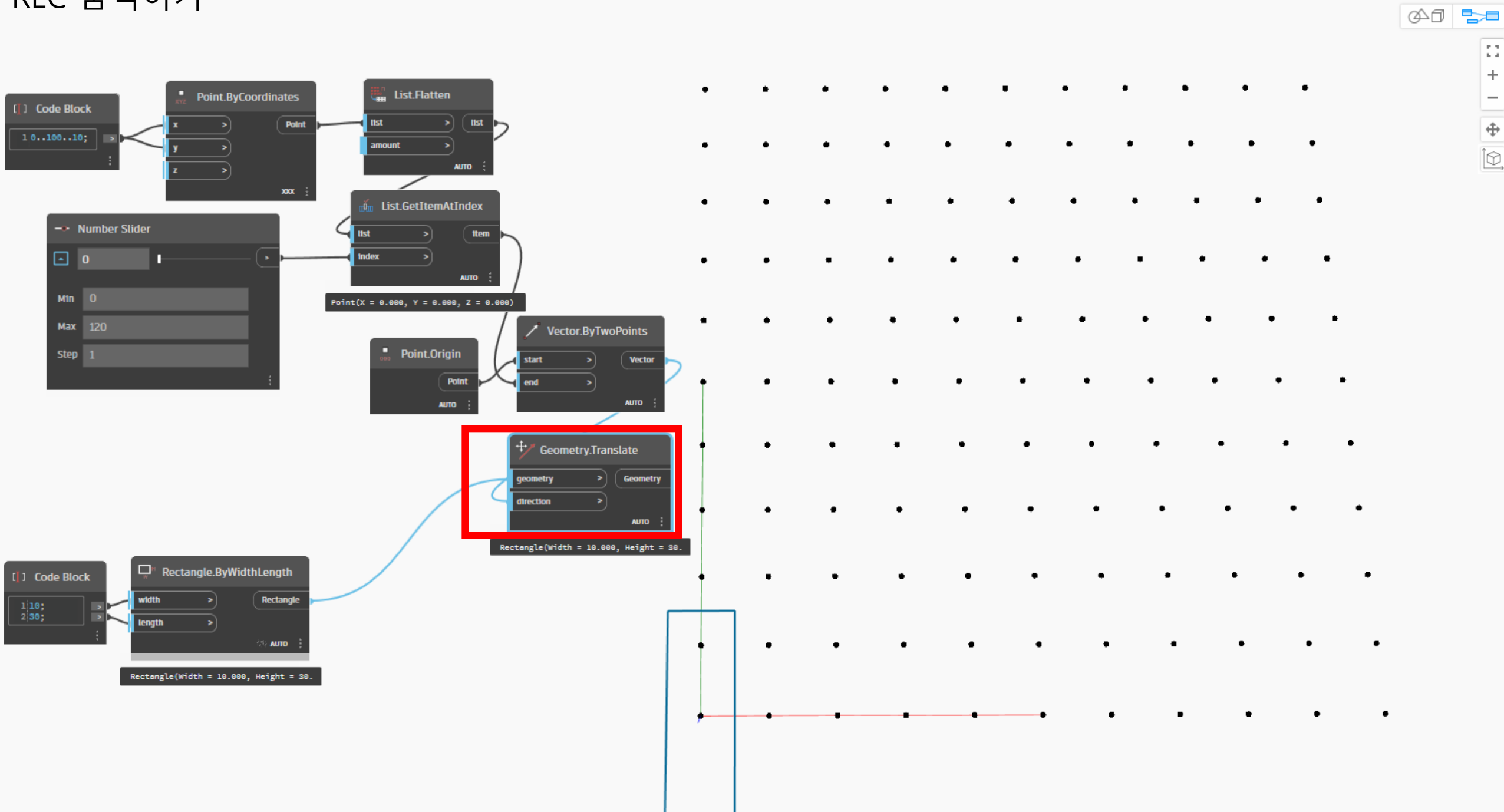
안겹치는 경우의 수 REC

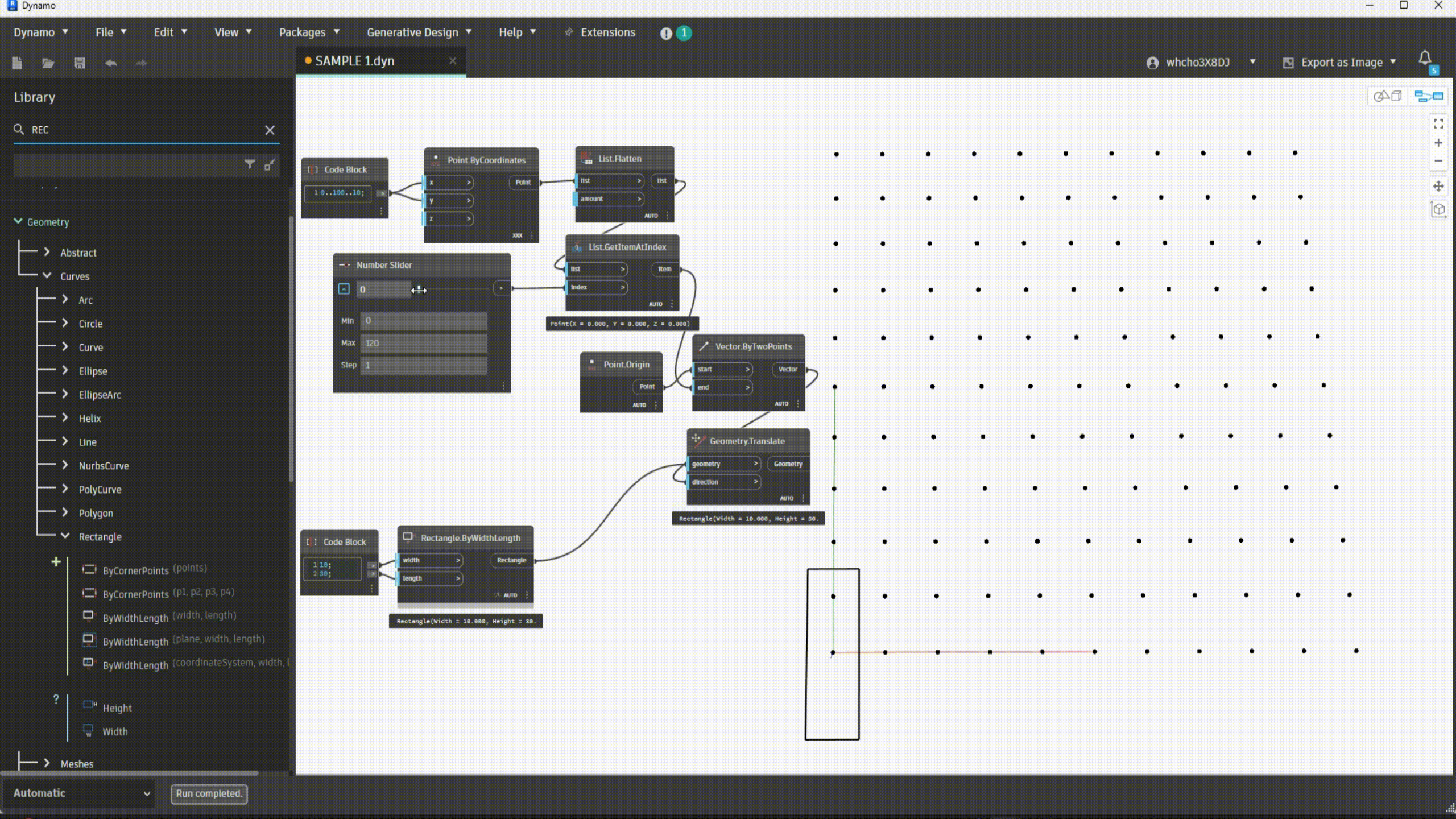
REC 움직이기



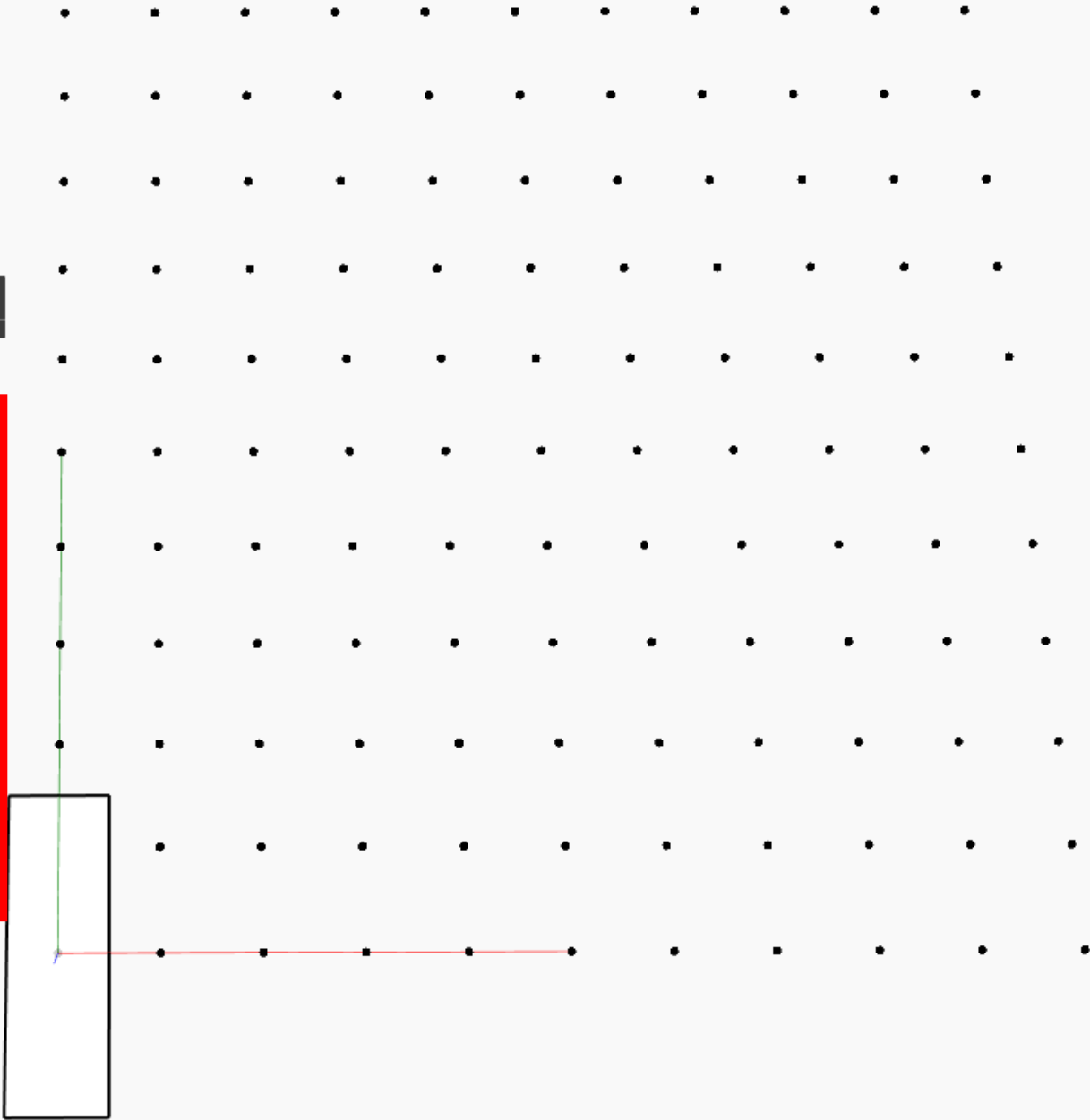
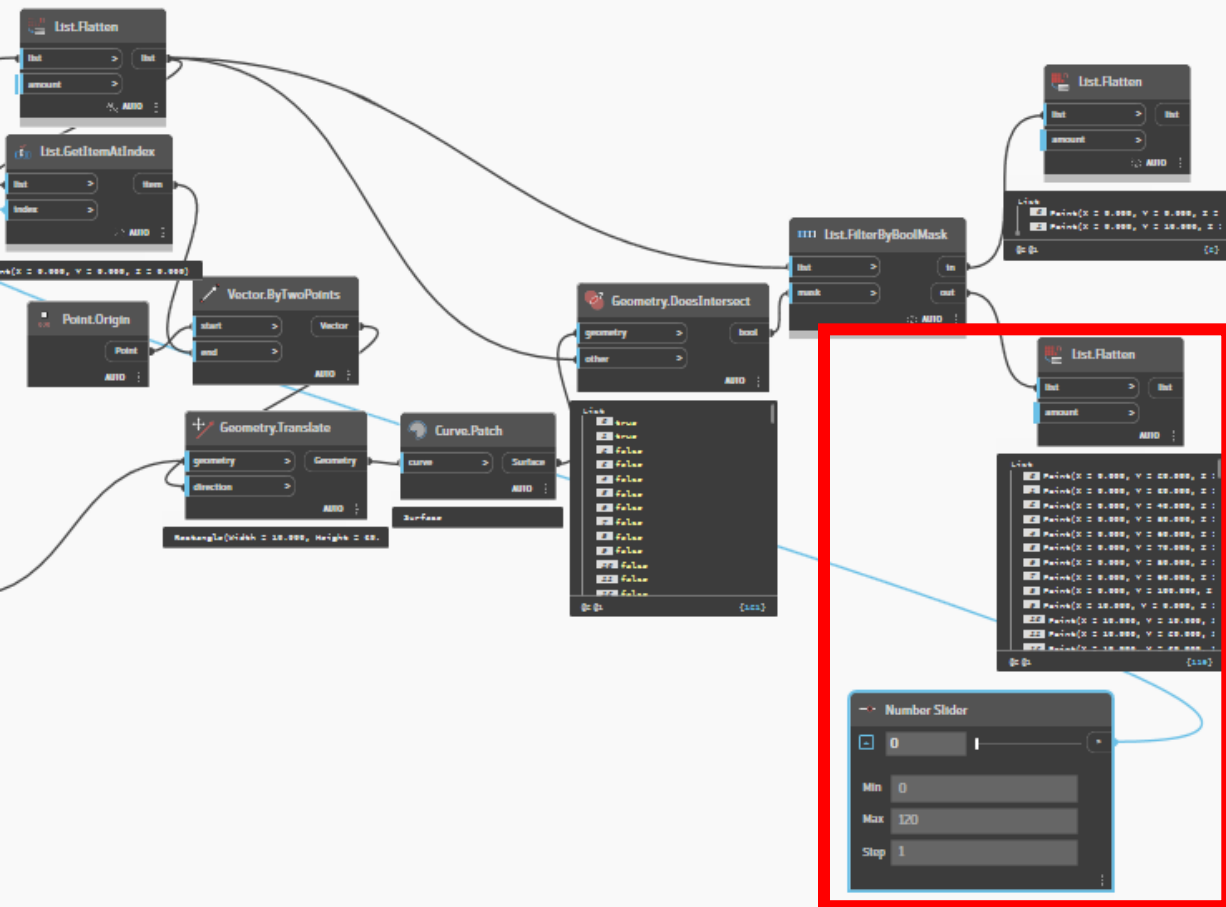
[0,10][1,10][2,10][3,10][4,10][5,10][6,10][7,10][8,10][9,10][10,10]
[0,9][1,9][2,9][3,9][4,9][5,9][6,9][7,9][8,9][9,9][10,9]
[0,8][1,8][2,8][3,8][4,8][5,8][6,8][7,8][8,8][9,8][10,8]
[0,7][1,7][2,7][3,7][4,7][5,7][6,7][7,7][8,7][9,7][10,7]
[0,6][1,6][2,6][3,6][4,6][5,6][6,6][7,6][8,6][9,6][10,6]
[0,5][1,5][2,5][3,5][4,5][5,5][6,5][7,5][8,5][9,5][10,5]
[0,4][1,4][2,4][3,4][4,4][5,4][6,4][7,4][8,4][9,4][10,4]
[0,3][1,3][2,3][3,3][4,3][5,3][6,3][7,3][8,3][9,3][10,3]
[0,2][1,2][2,2][3,2][4,2][5,2][6,2][7,2][8,2][9,2][10,2]
[0,1][1,1][2,1][3,1][4,1][5,1][6,1][7,1][8,1][9,1][10,1]
[0,0][1,0][2,0][3,0][4,0][5,0][6,0][7,0][8,0][9,0][10,0]

REC 움직이기





Point 제외시키기



Library

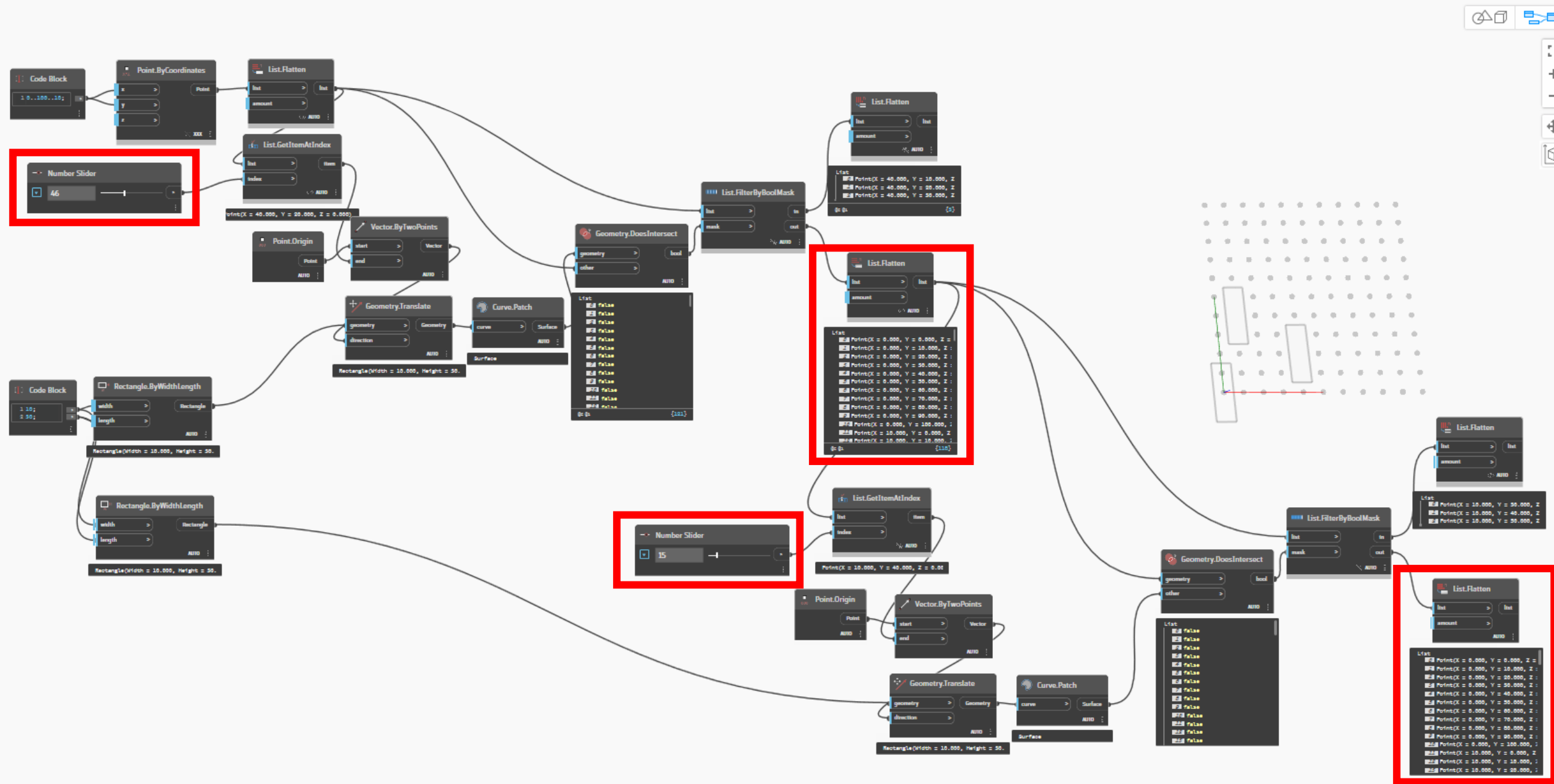
Abstract

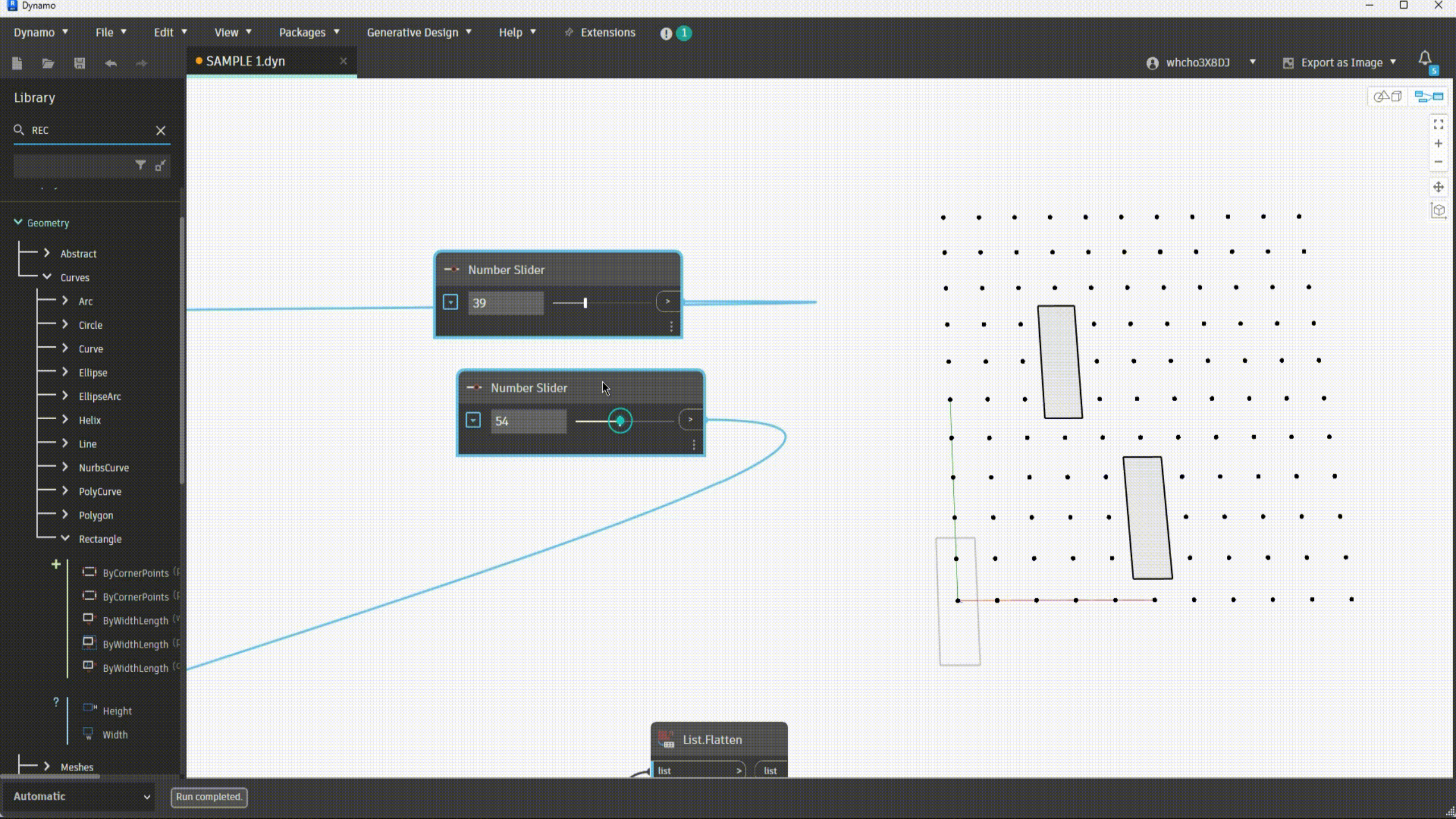
- ByCornerPoints
- ByCornerPoints
- ByWidthLength
- ByWidthLength
- ByWidthLength

- Meshes

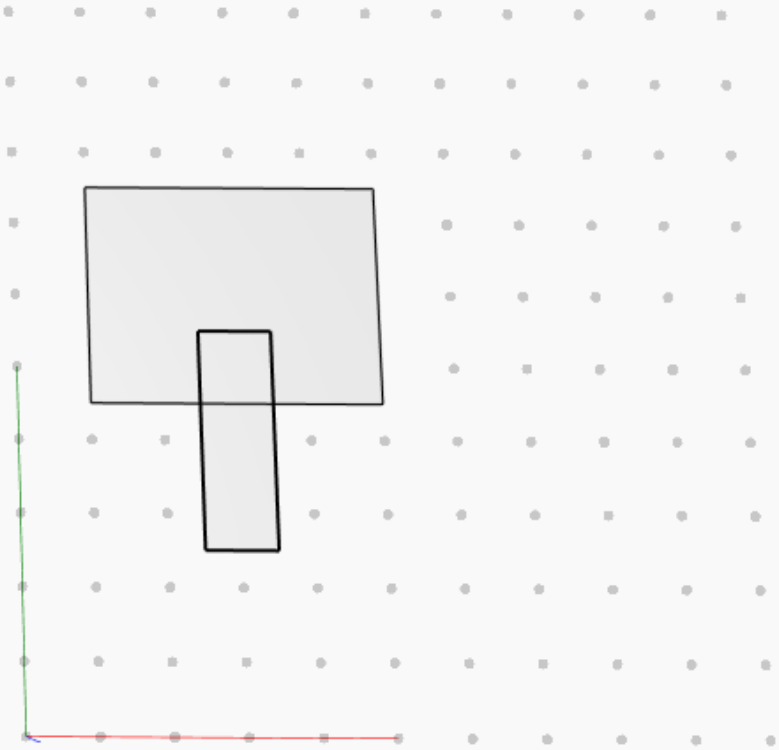
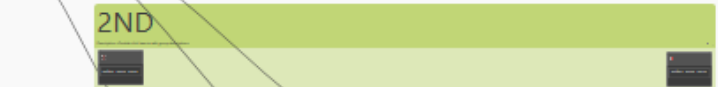
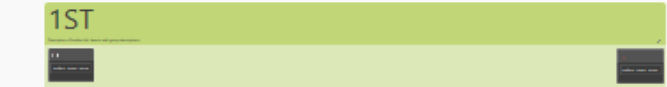
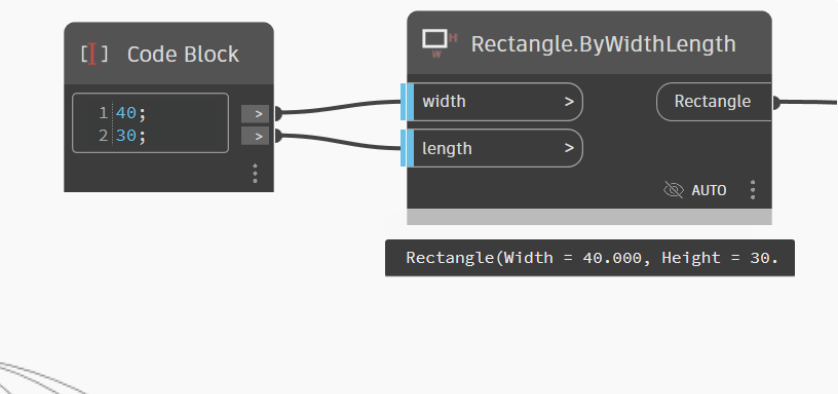
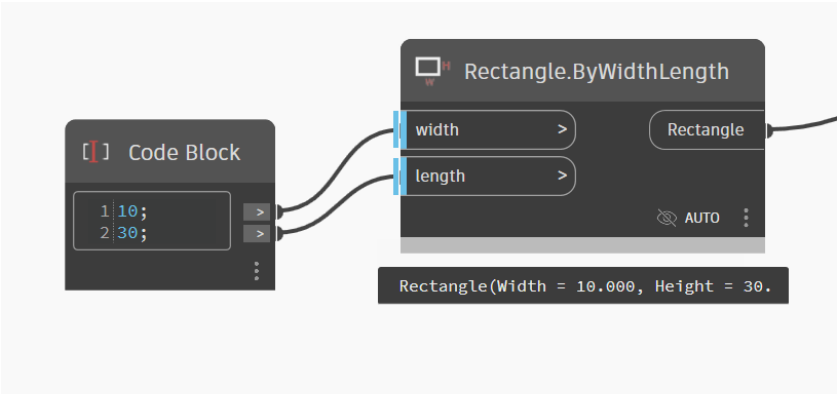
Run completed.

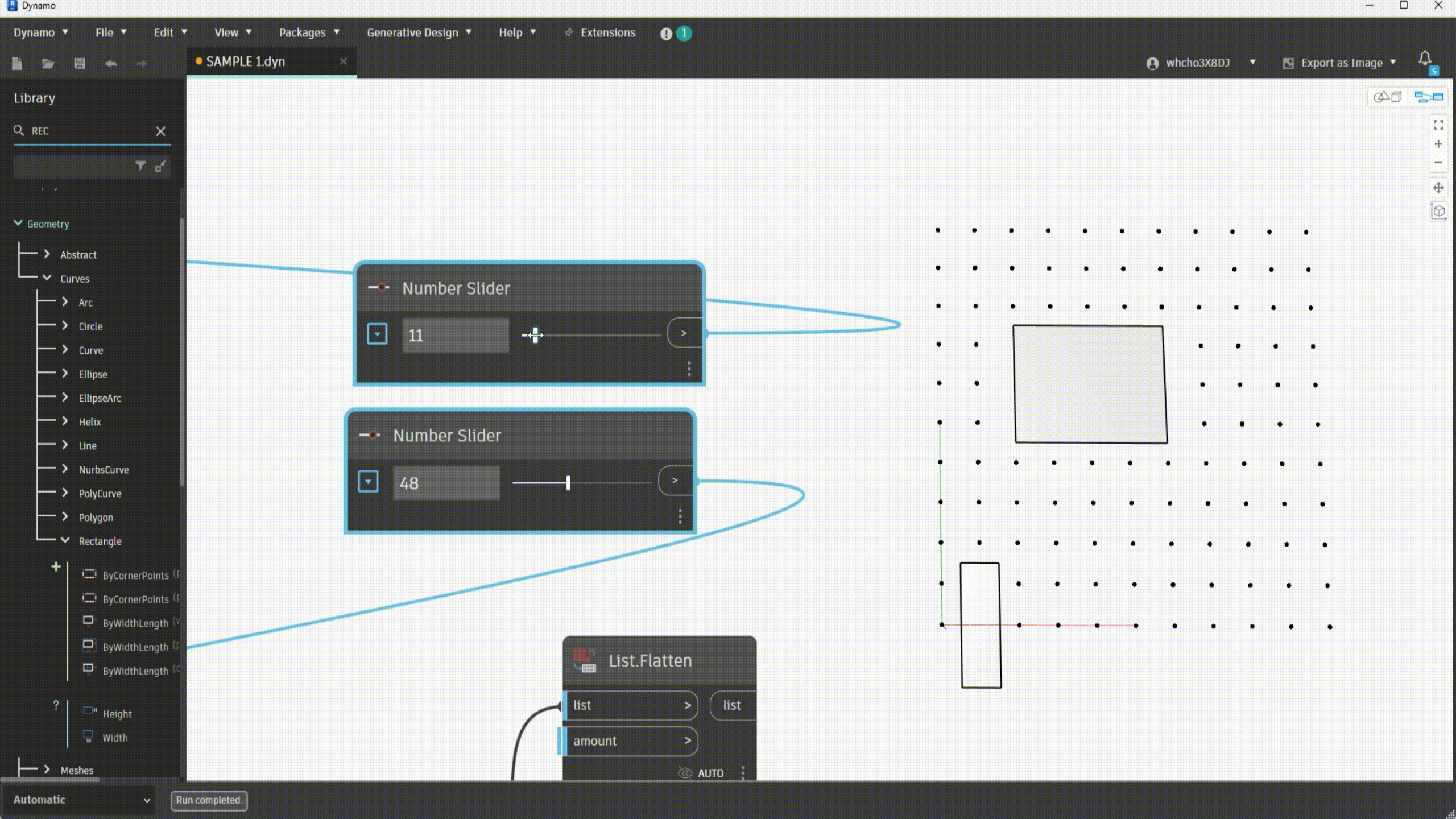
REC 다중 배치



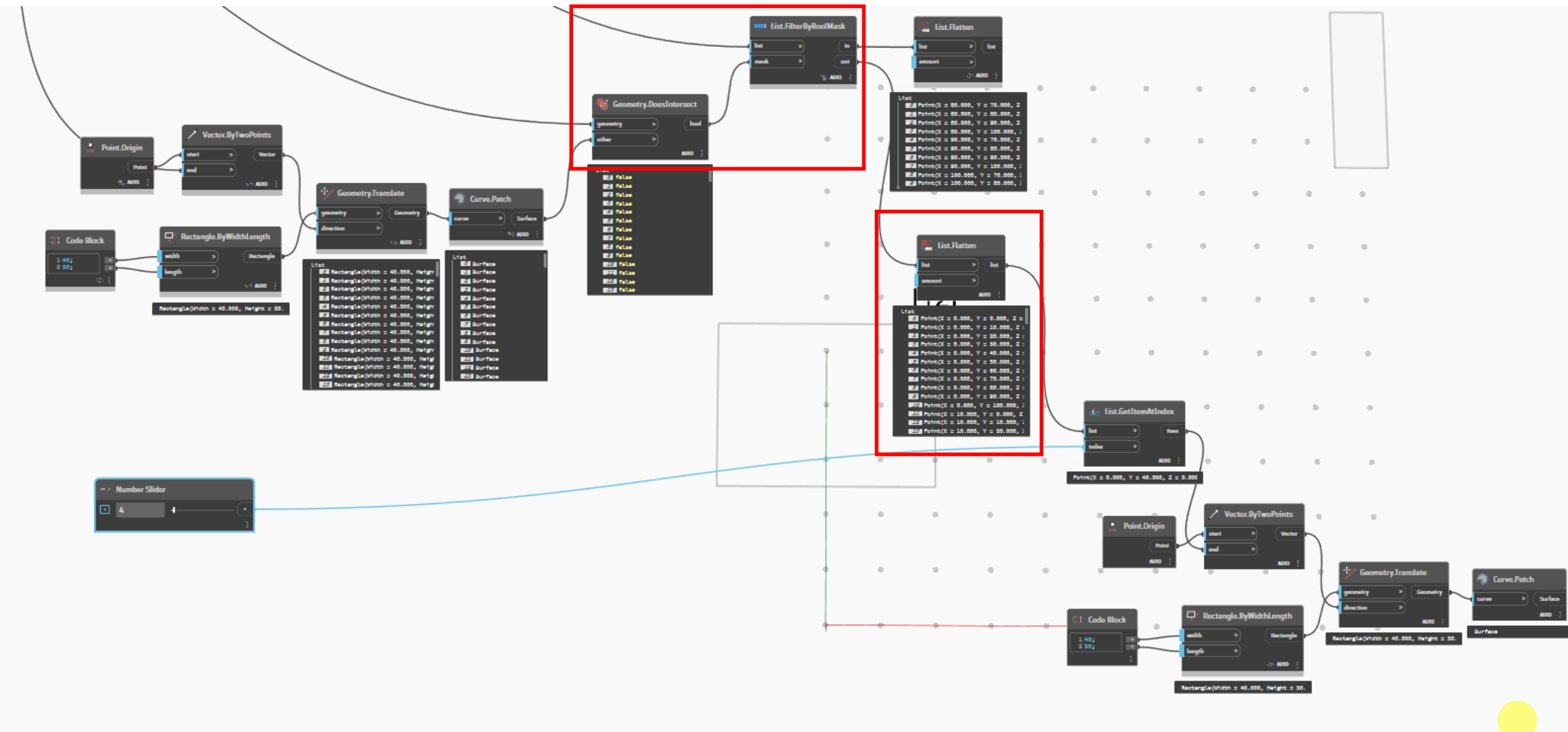


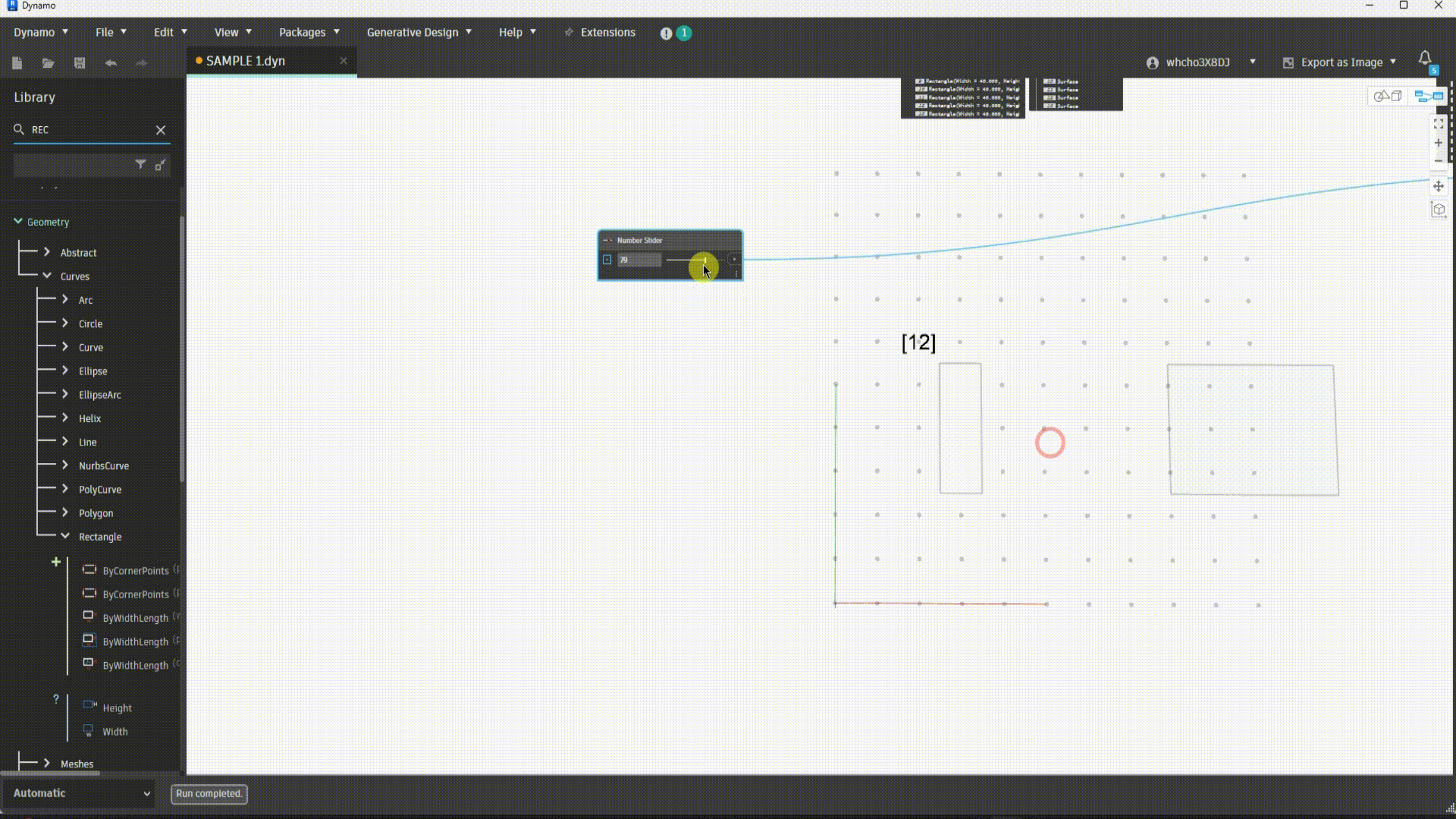
REC 크기 변경



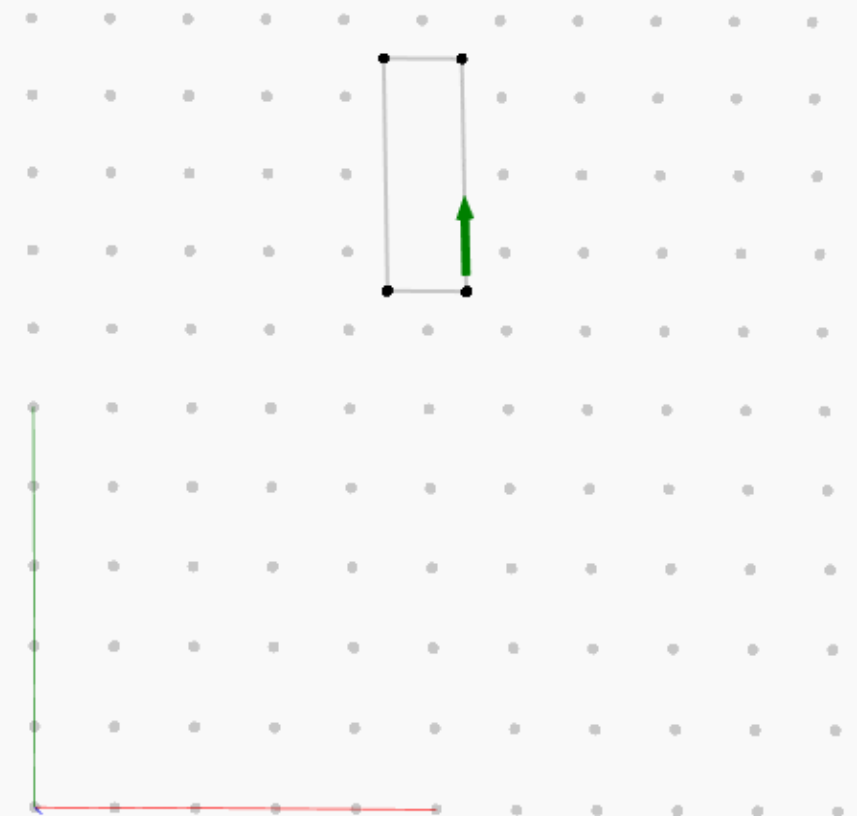
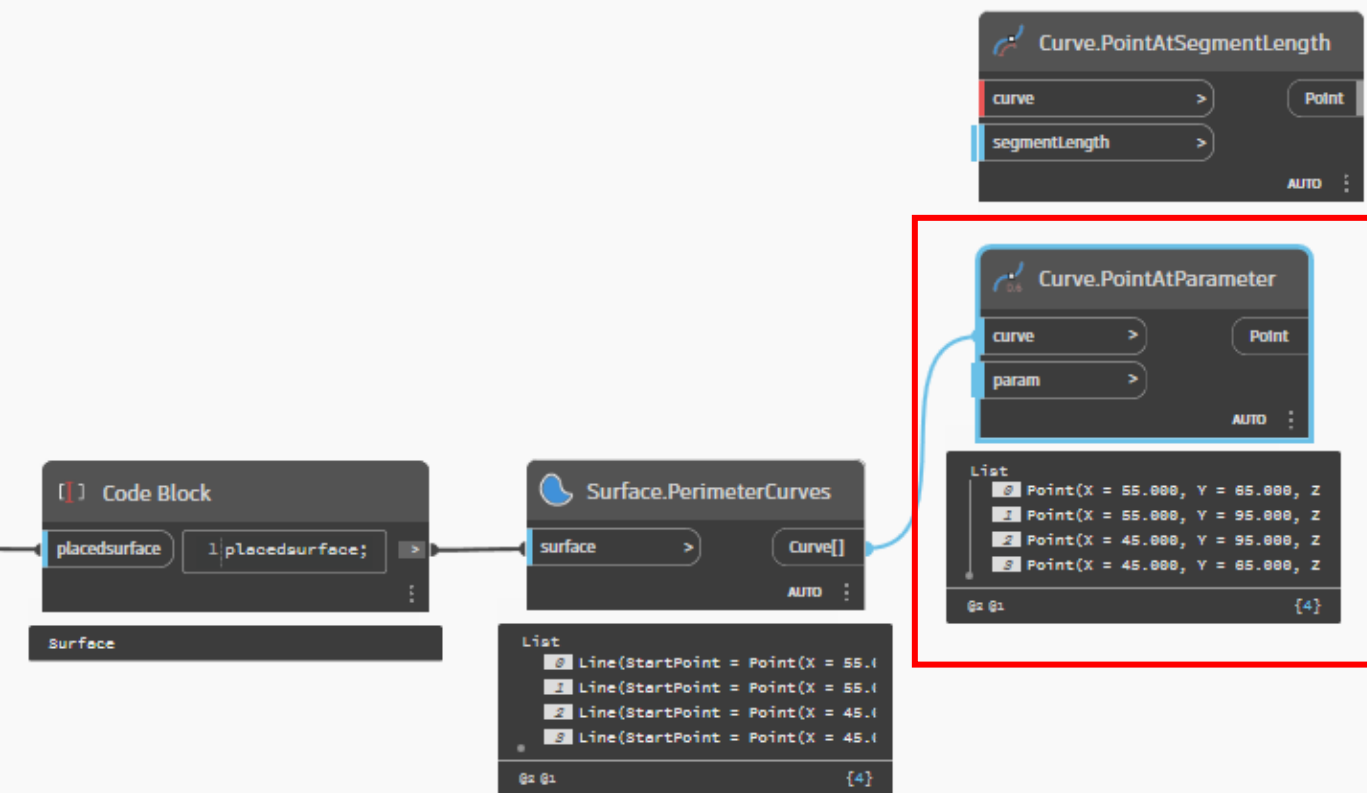


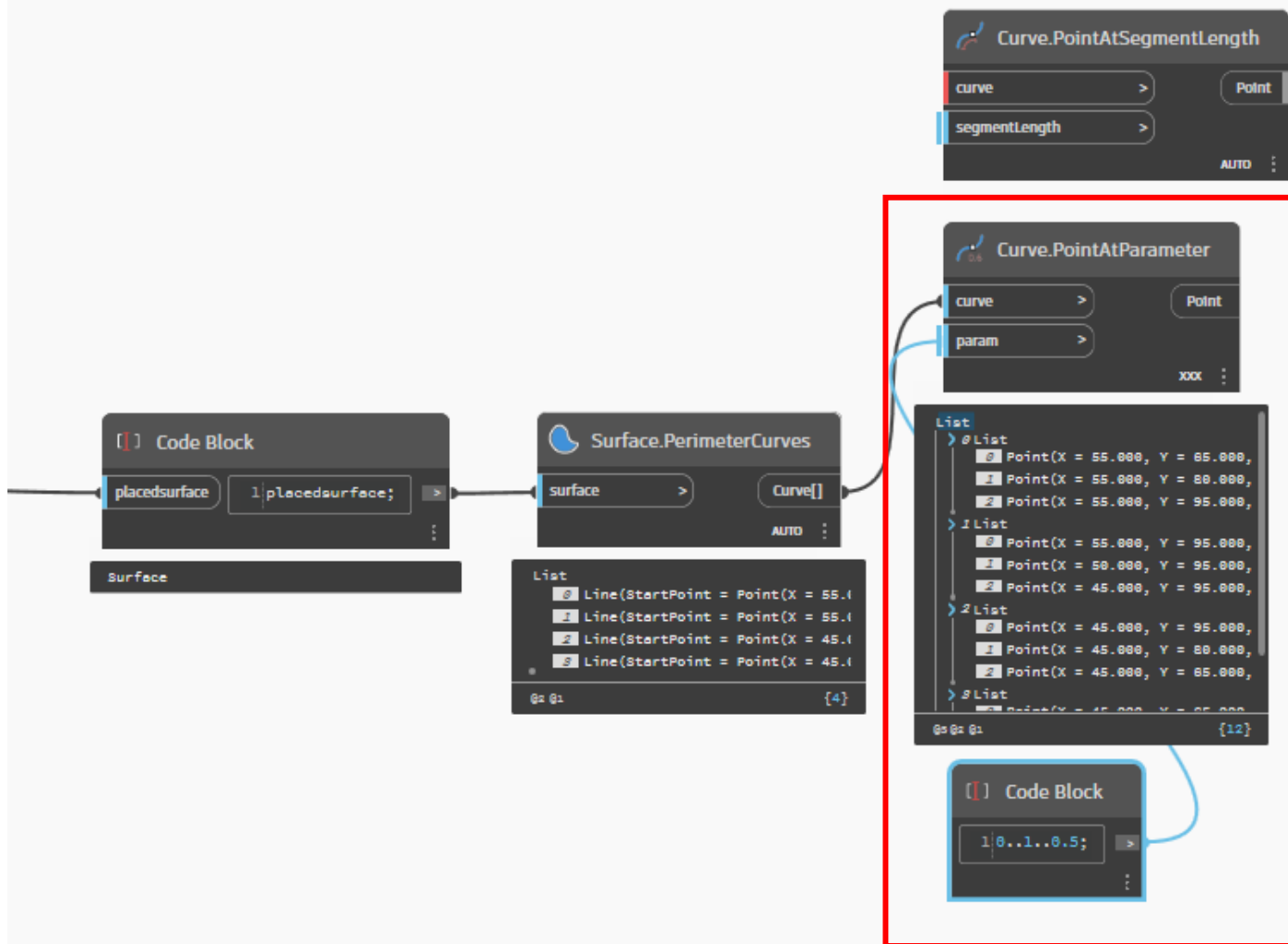
겹침 피하기

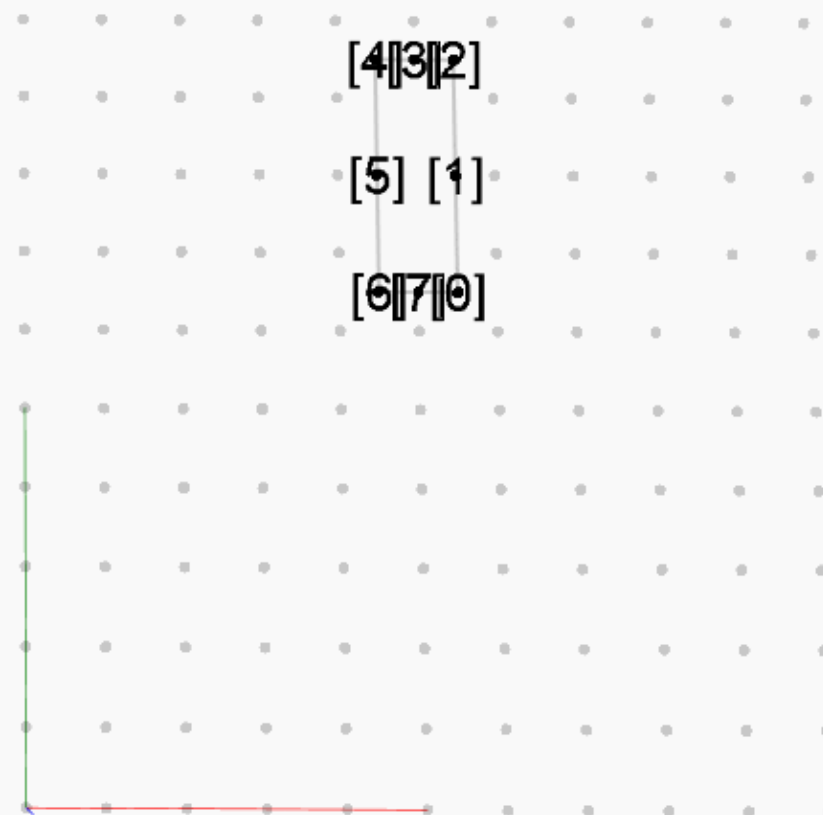
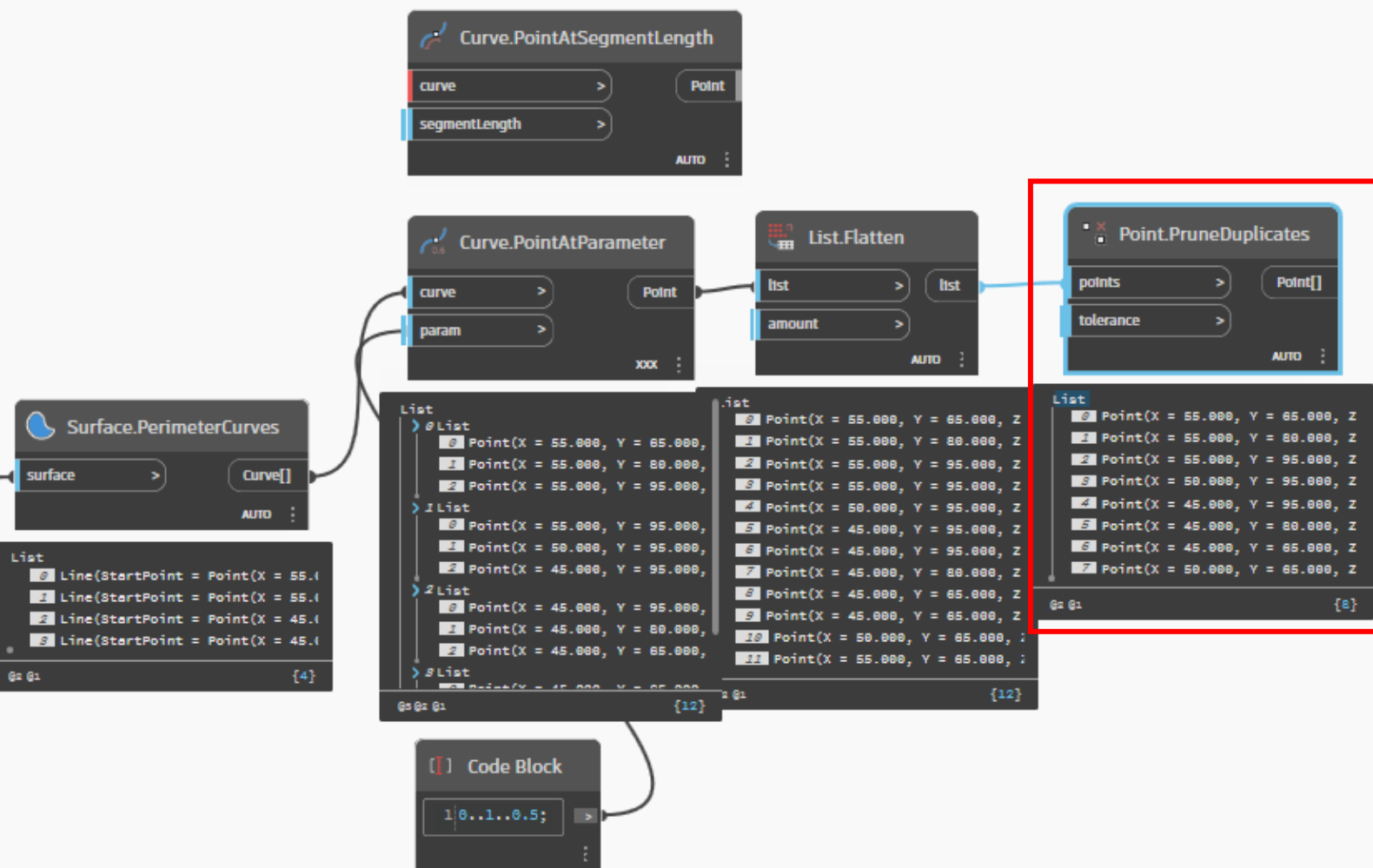


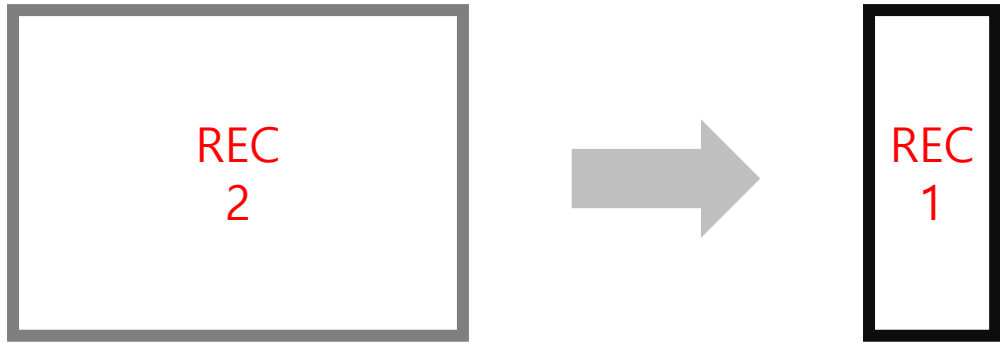


REC Rotate 후 배치

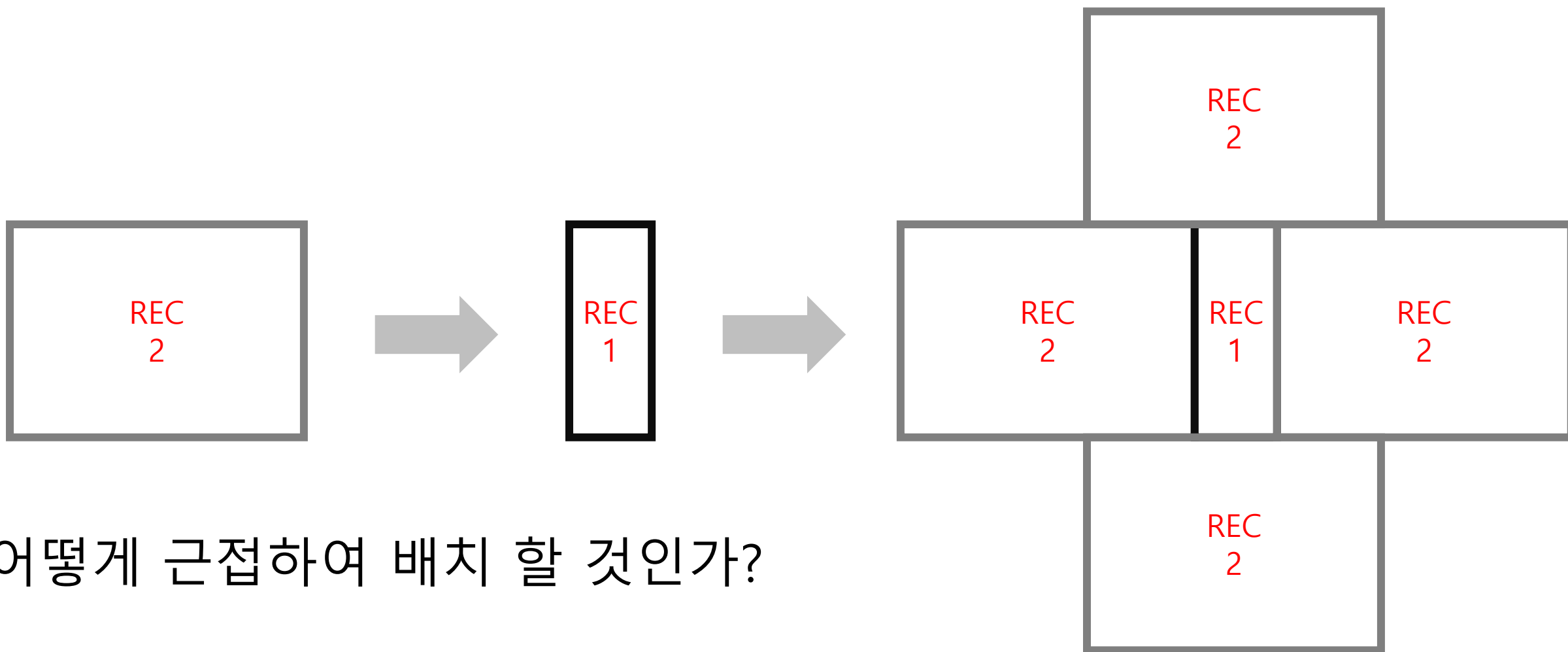




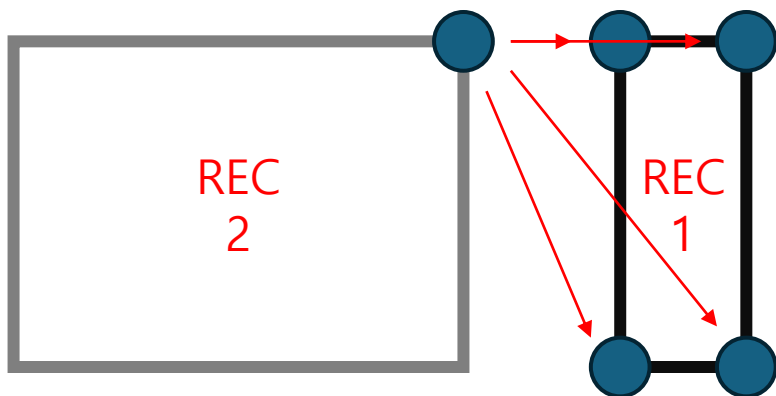


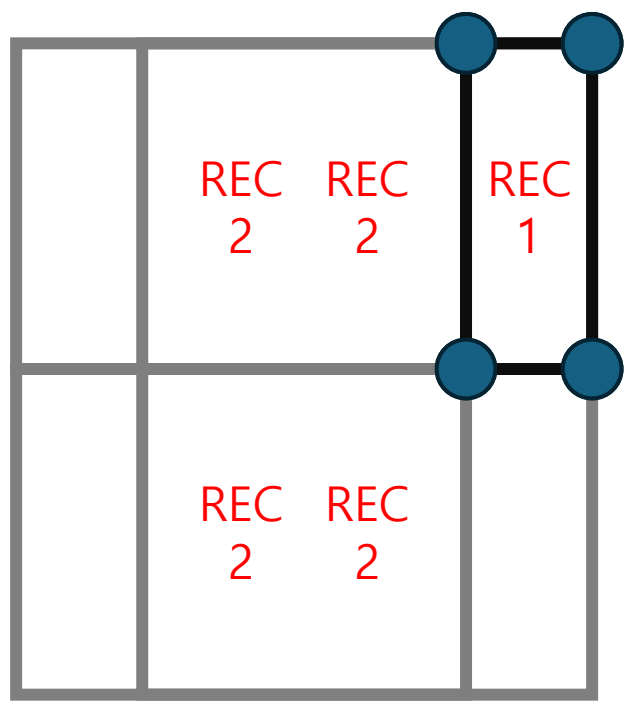
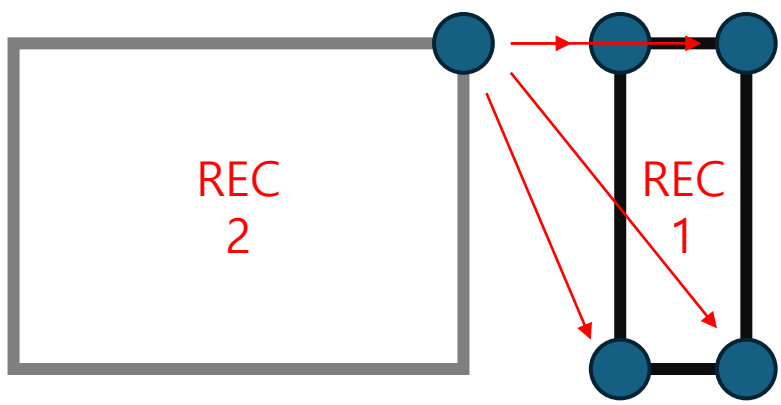


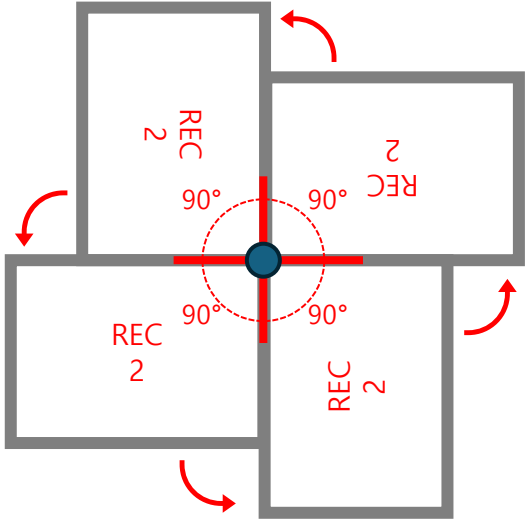
어떻게 근접하여 배치 할 것인가?

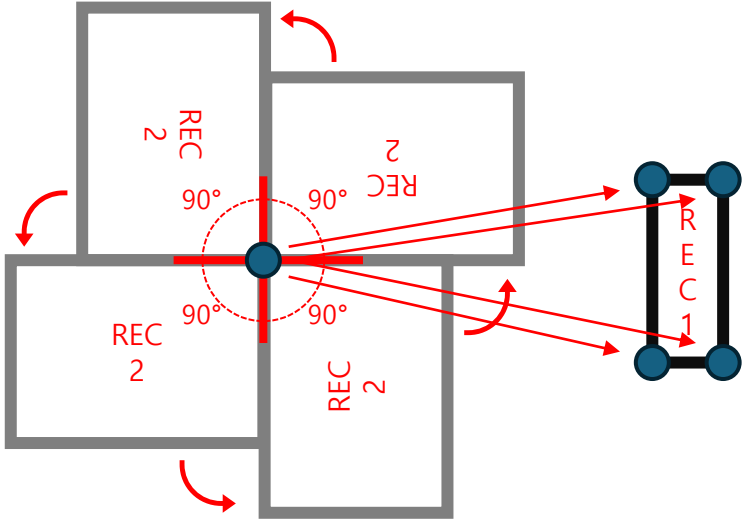


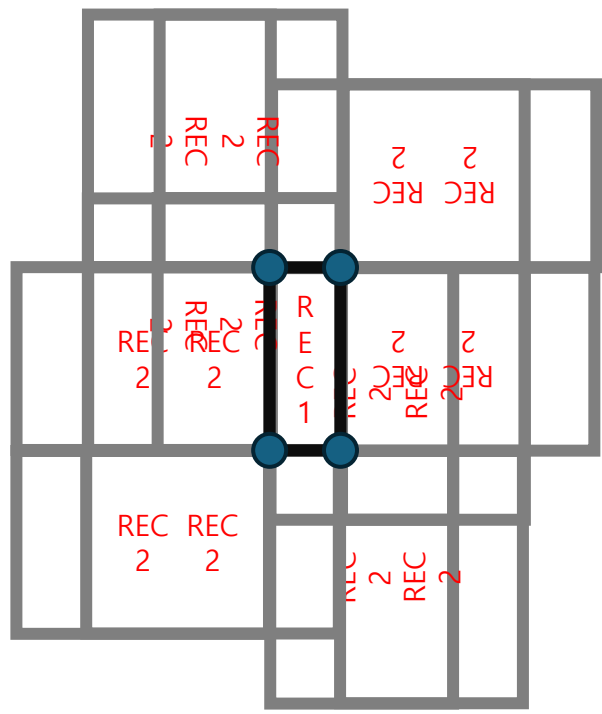
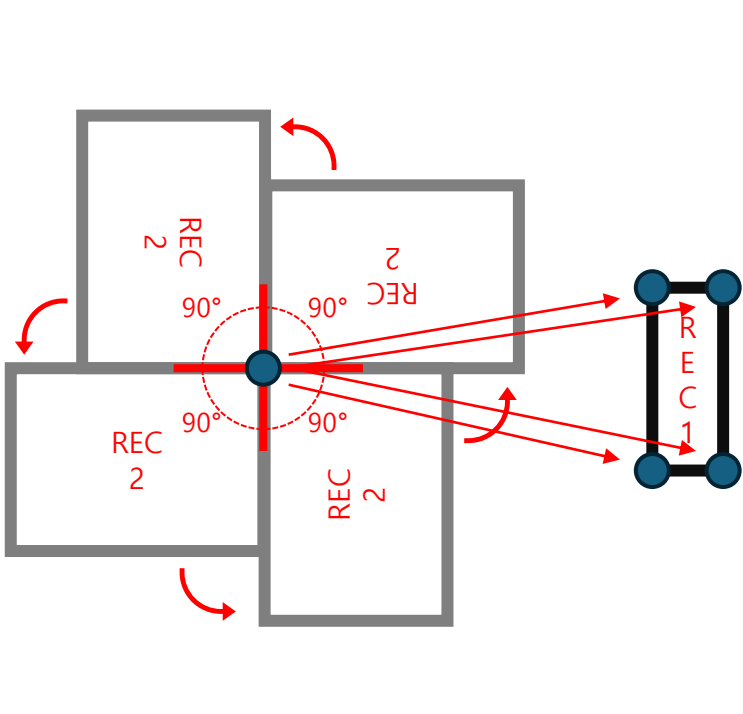
어떻게 근접하여 배치 할 것인가?

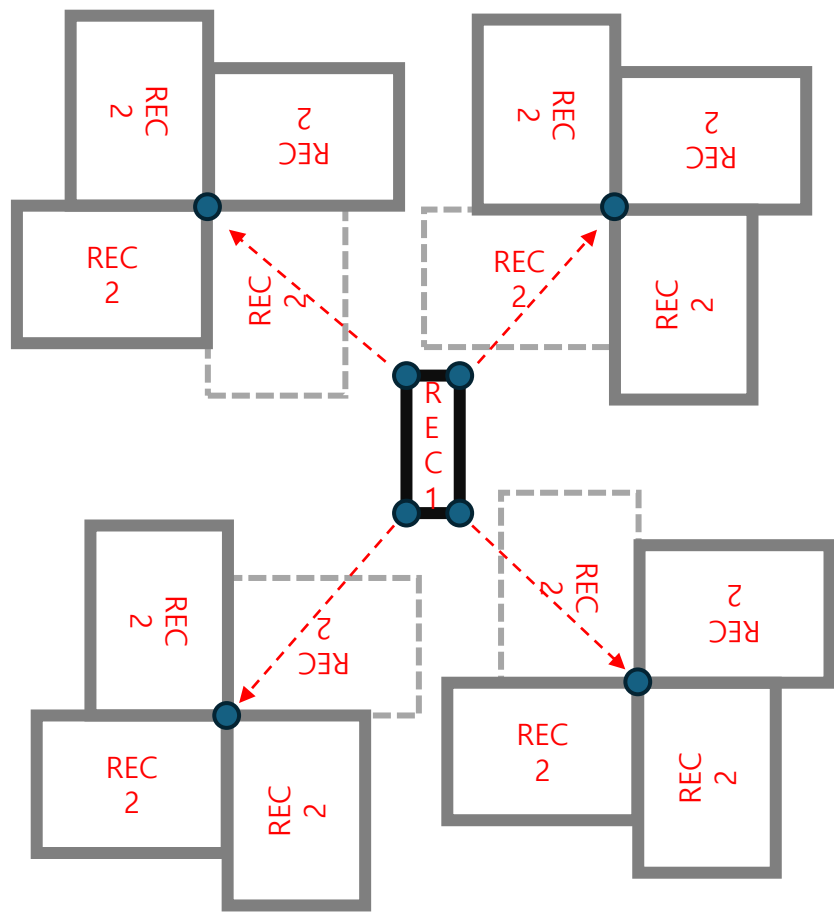
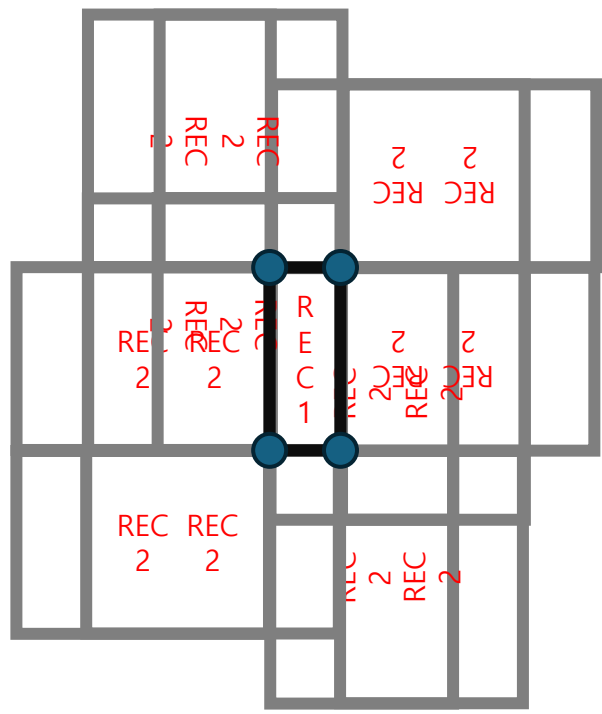
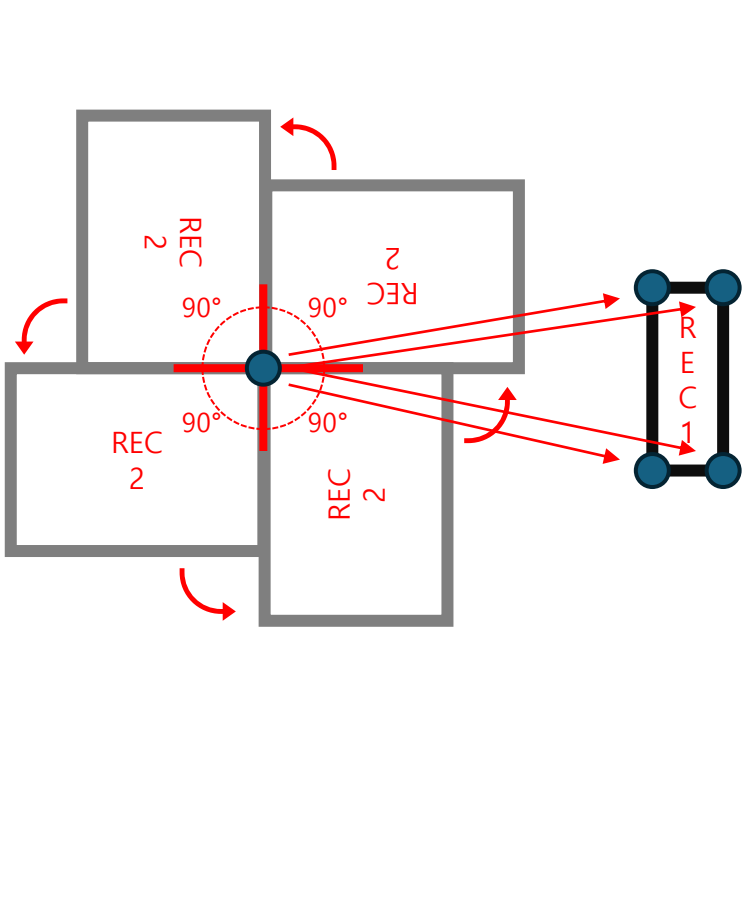


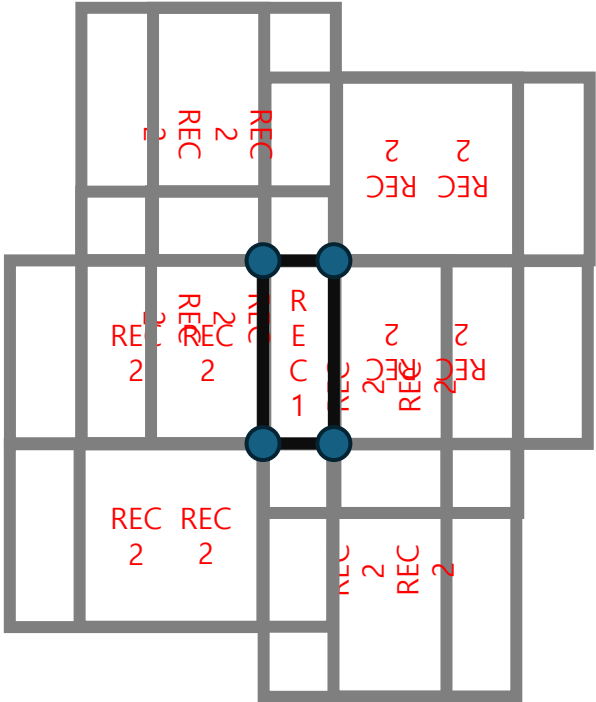
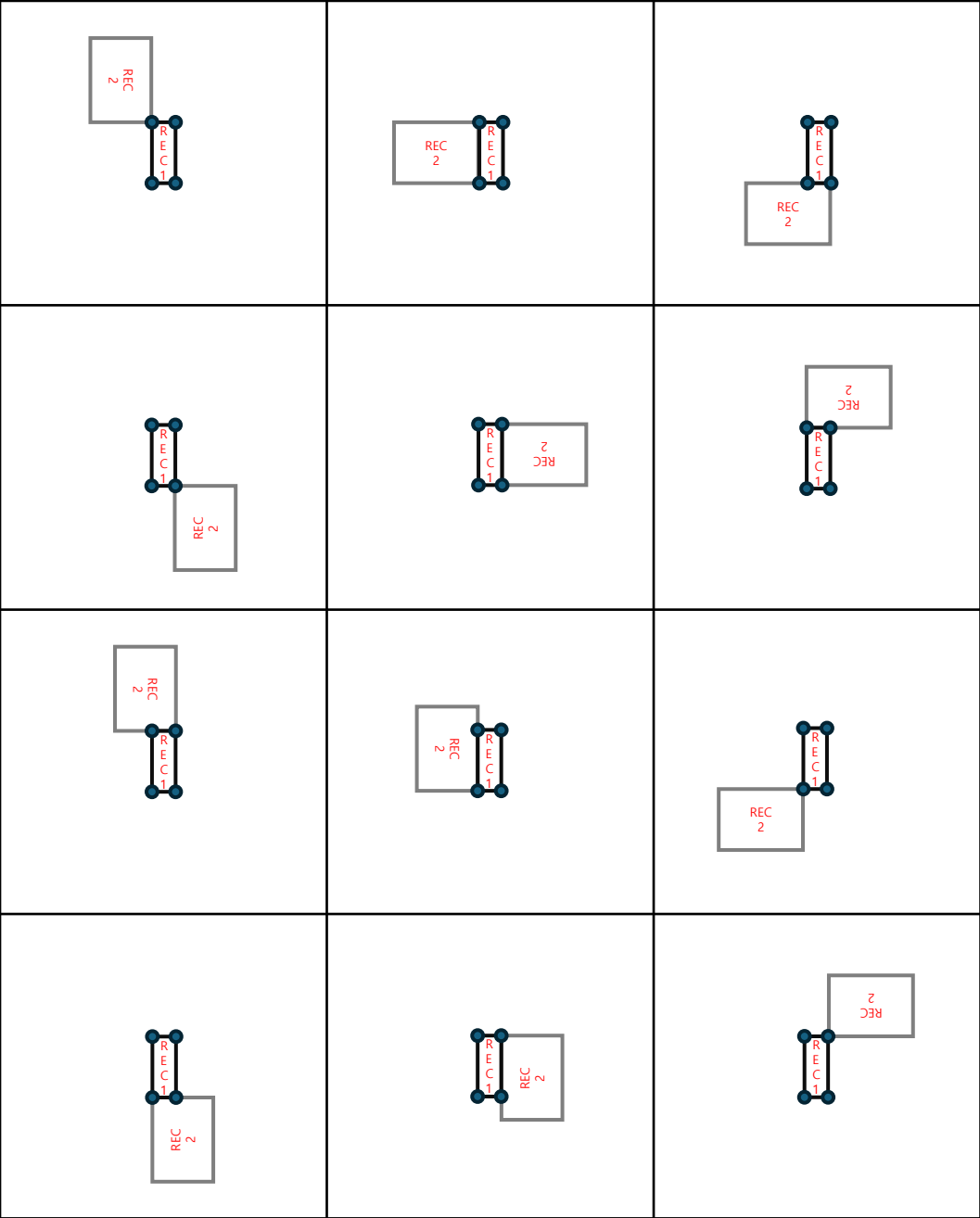


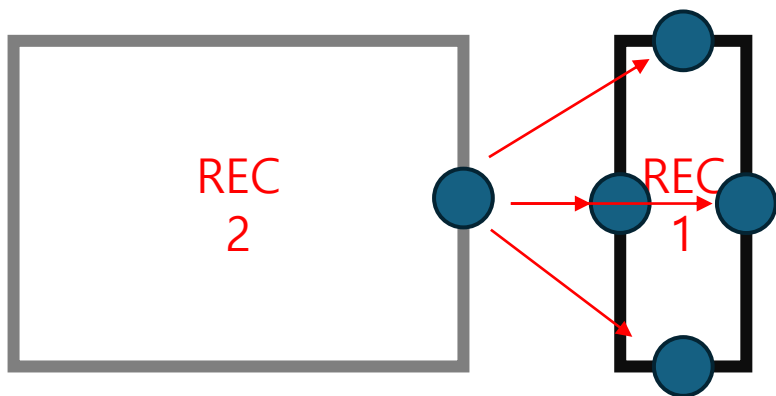
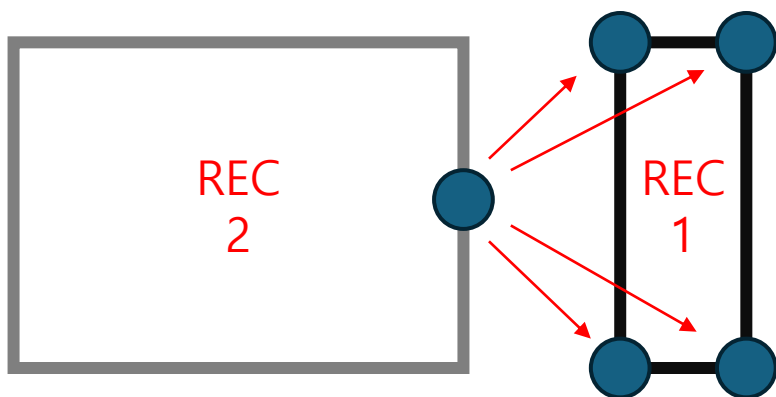


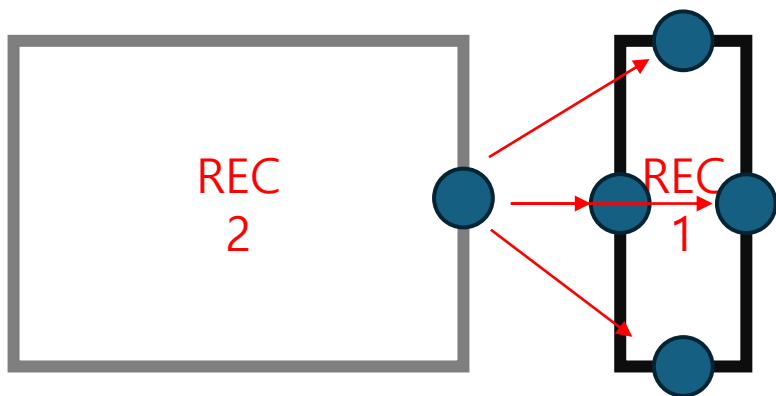
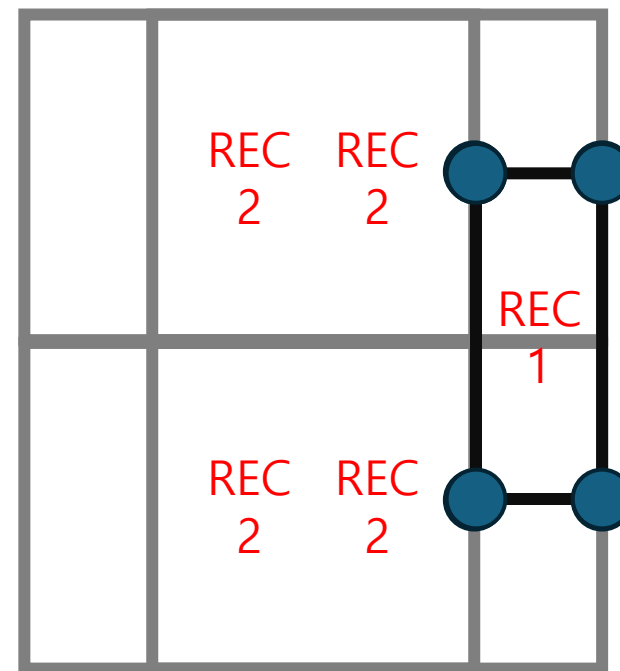
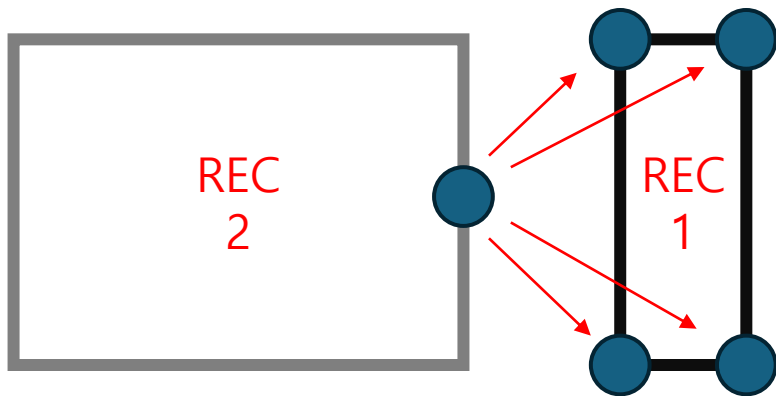


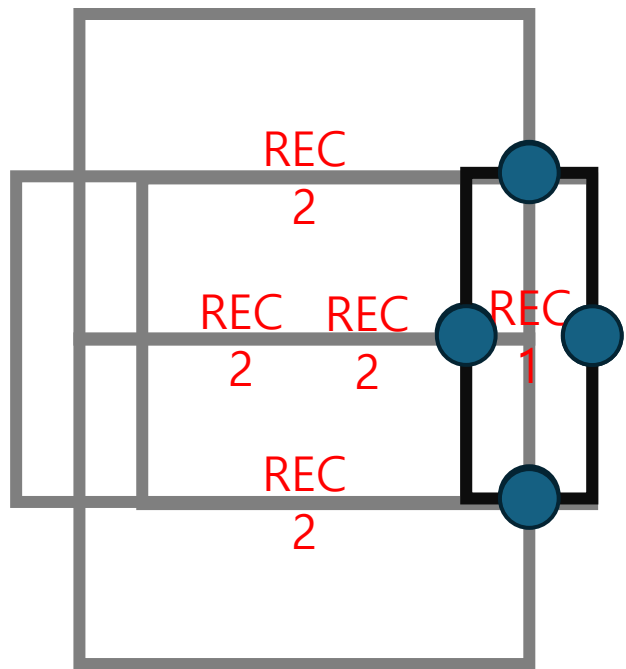
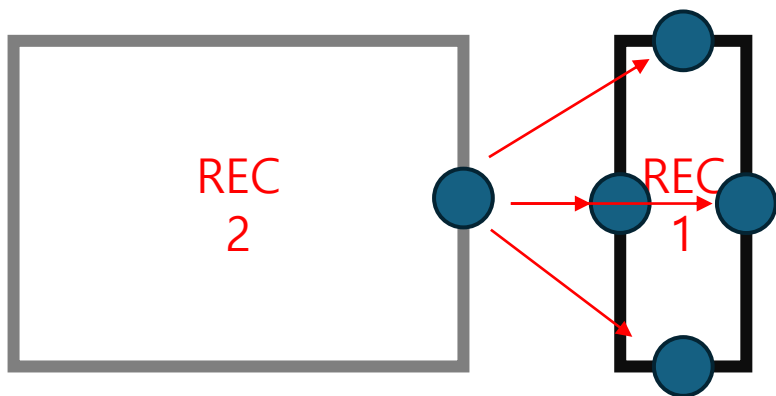
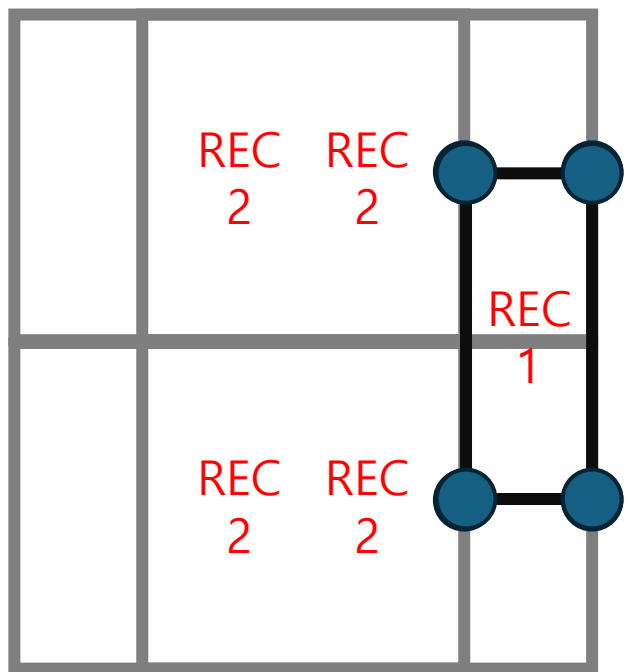
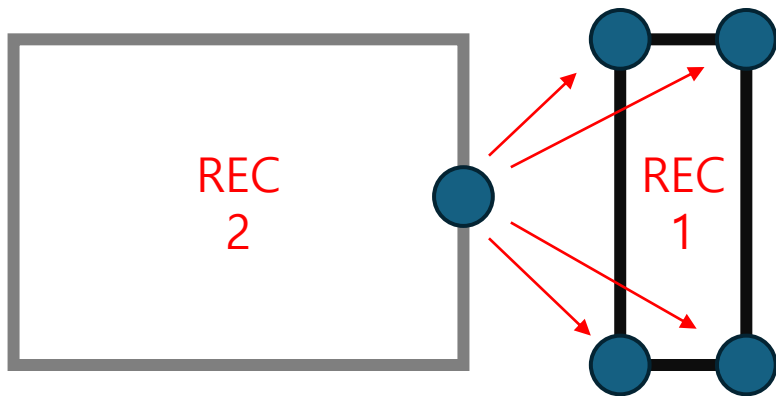


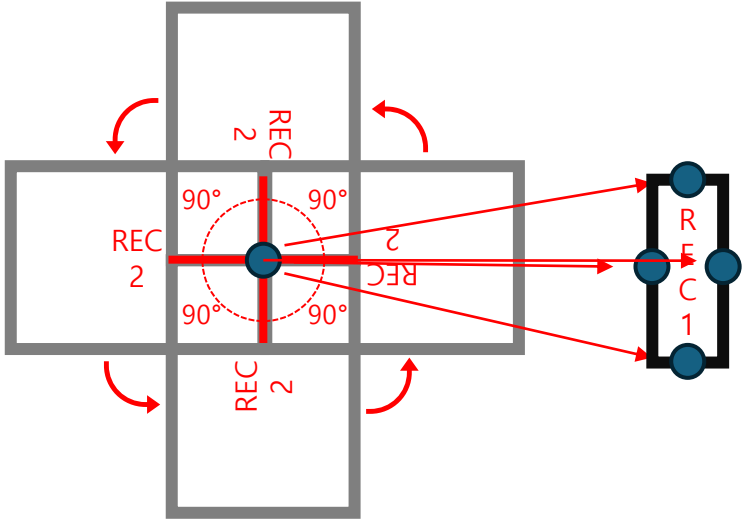


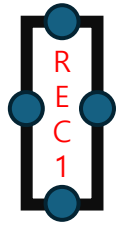
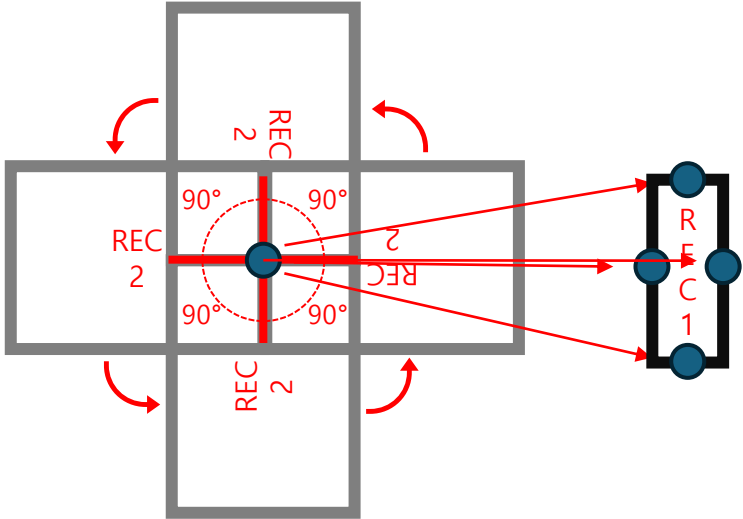


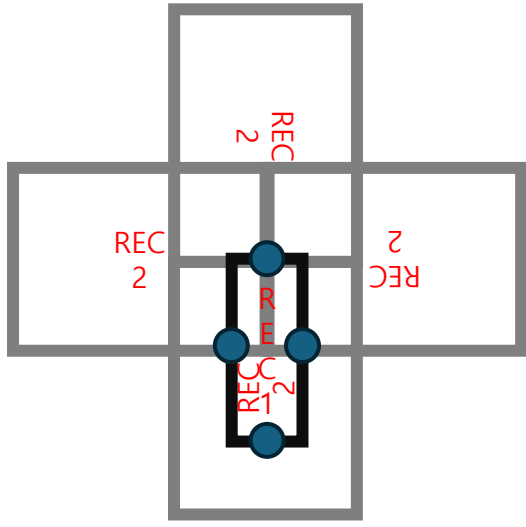
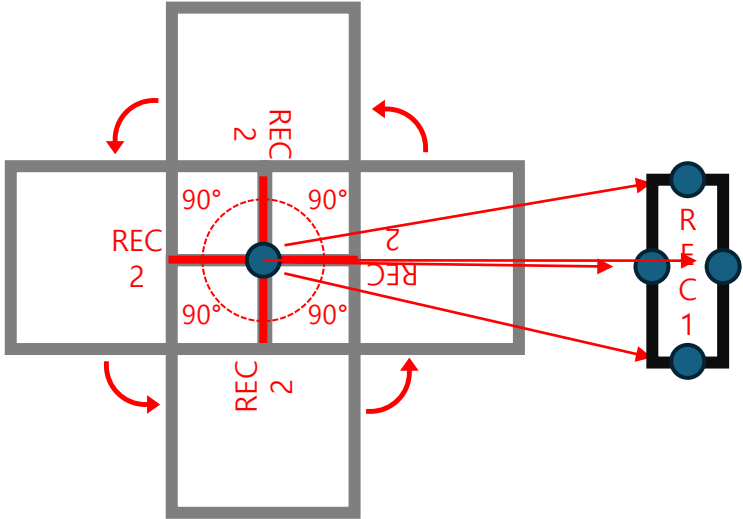


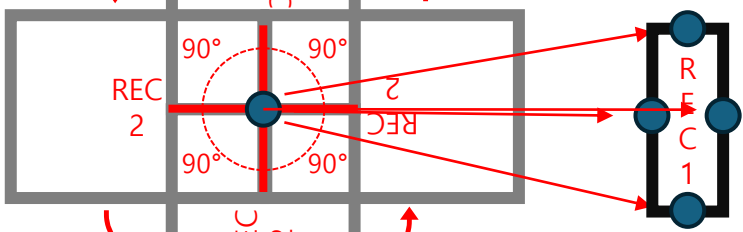
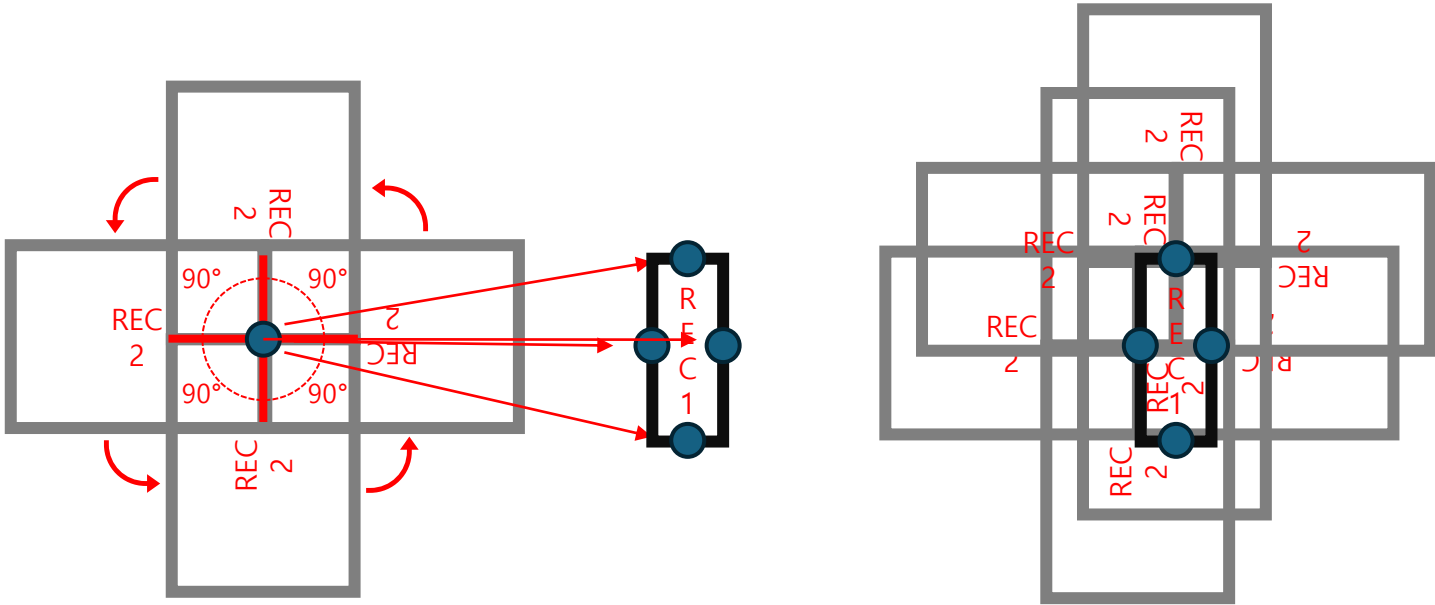


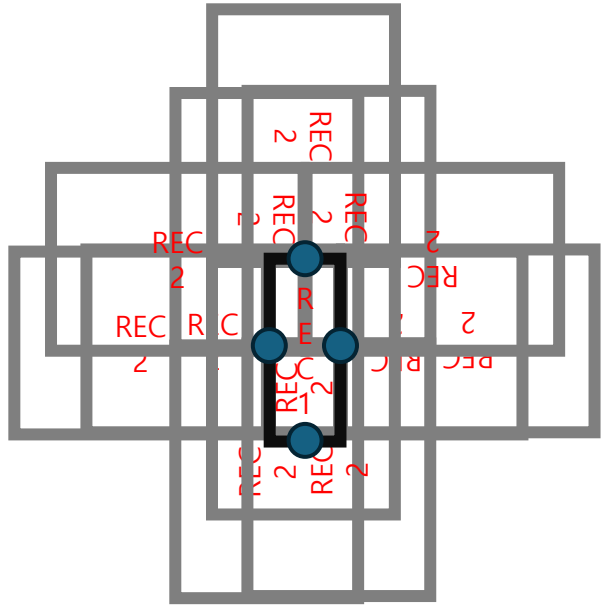
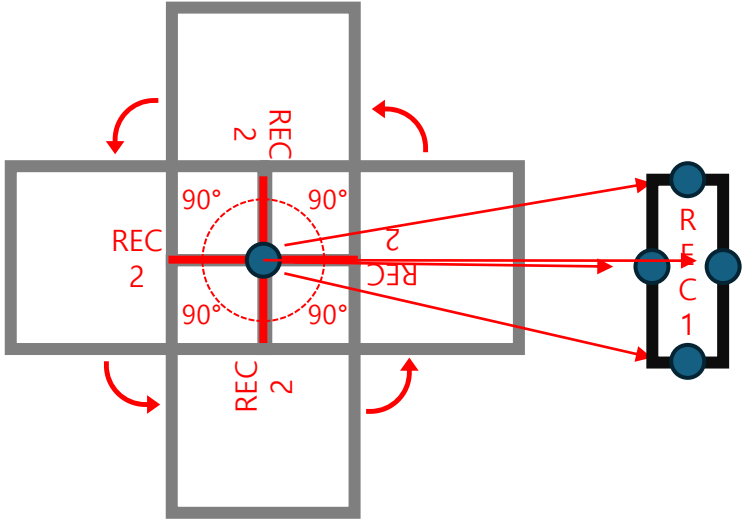


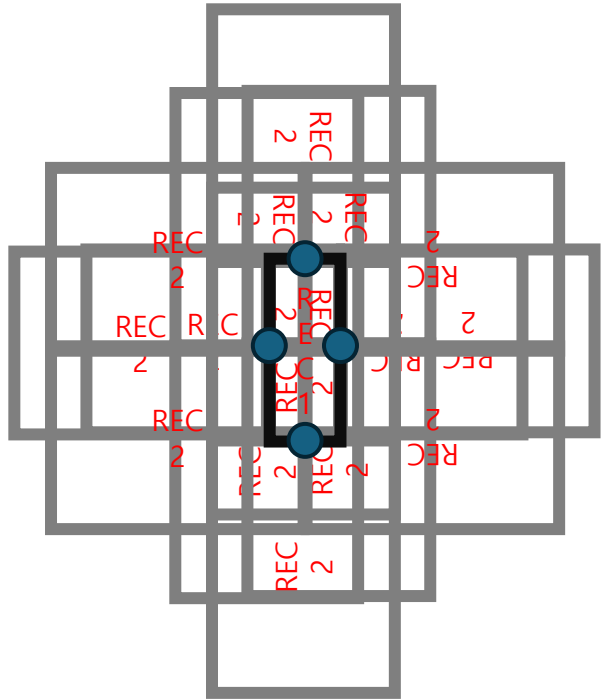
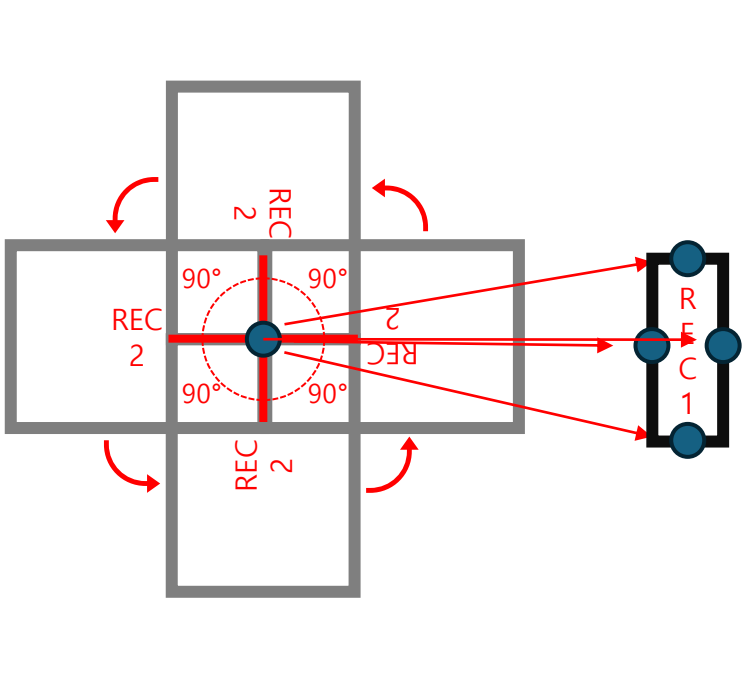


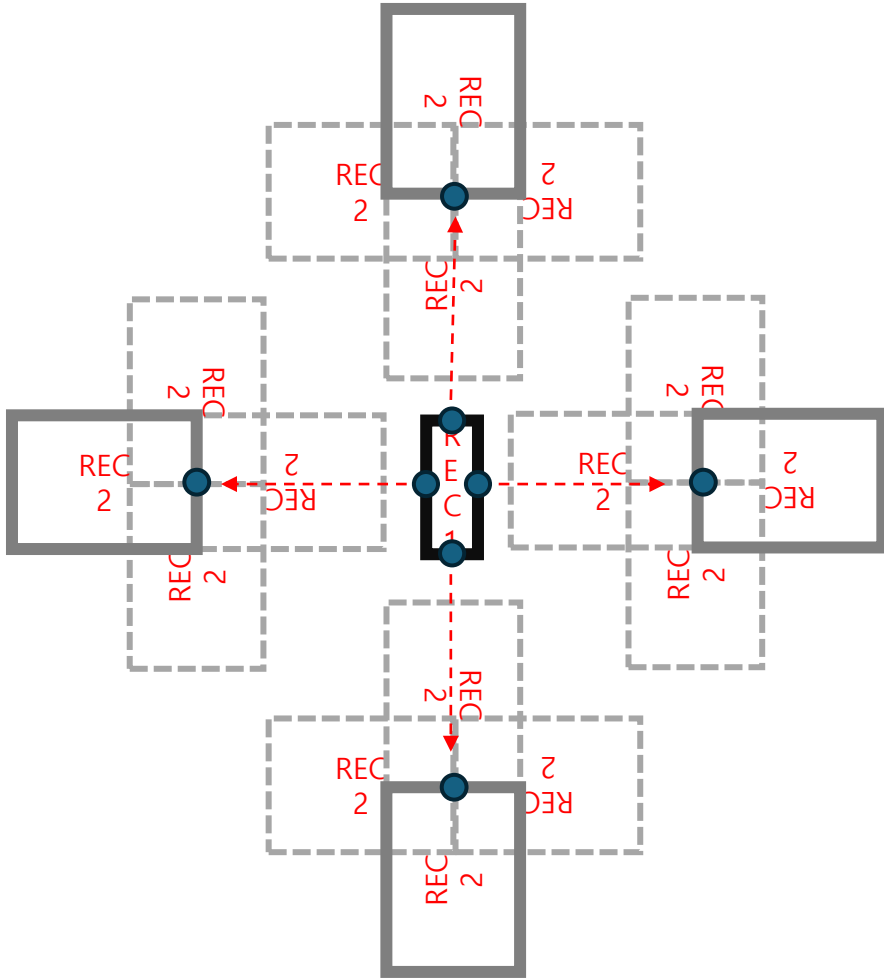
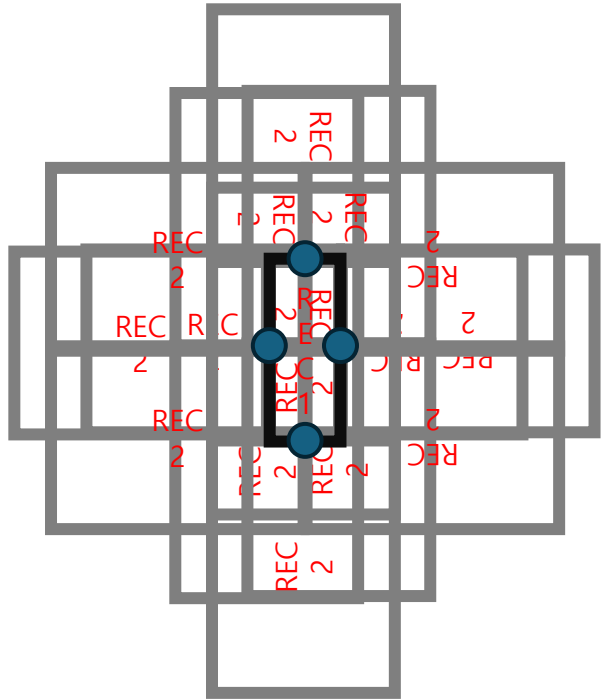
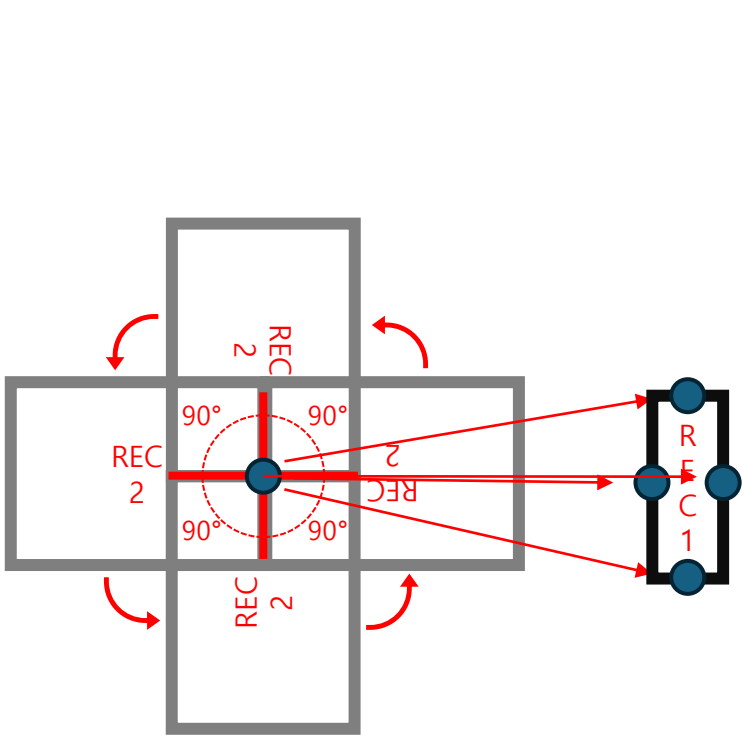


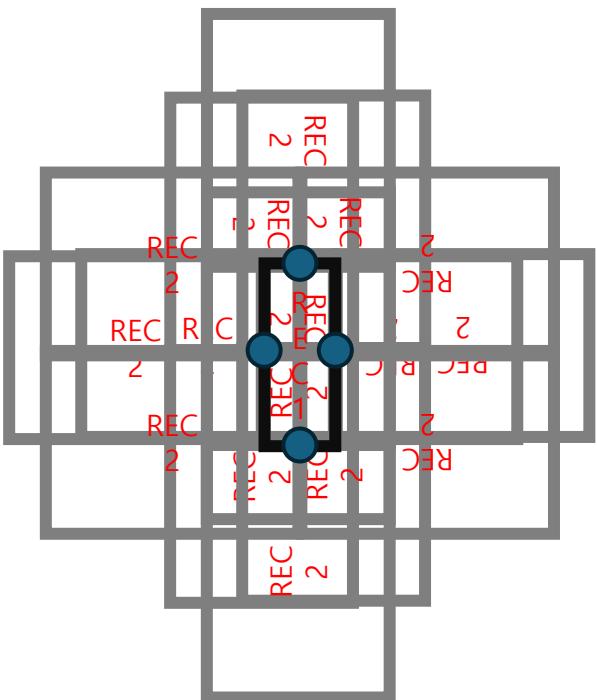


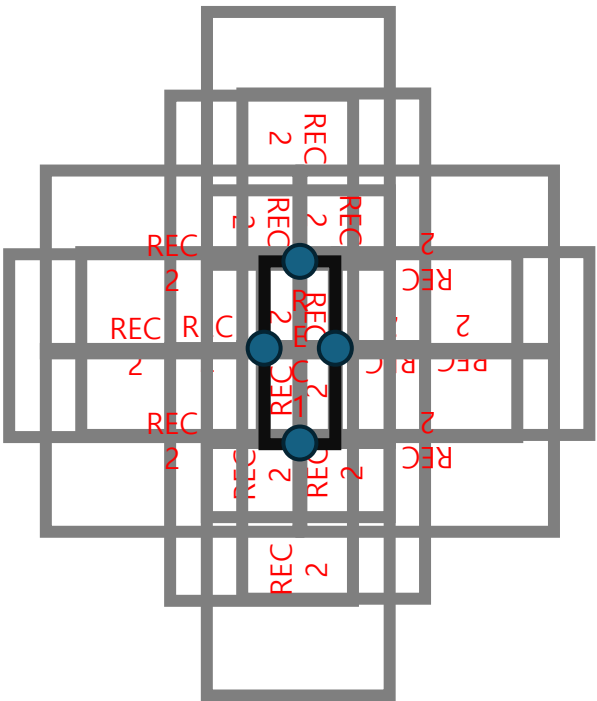
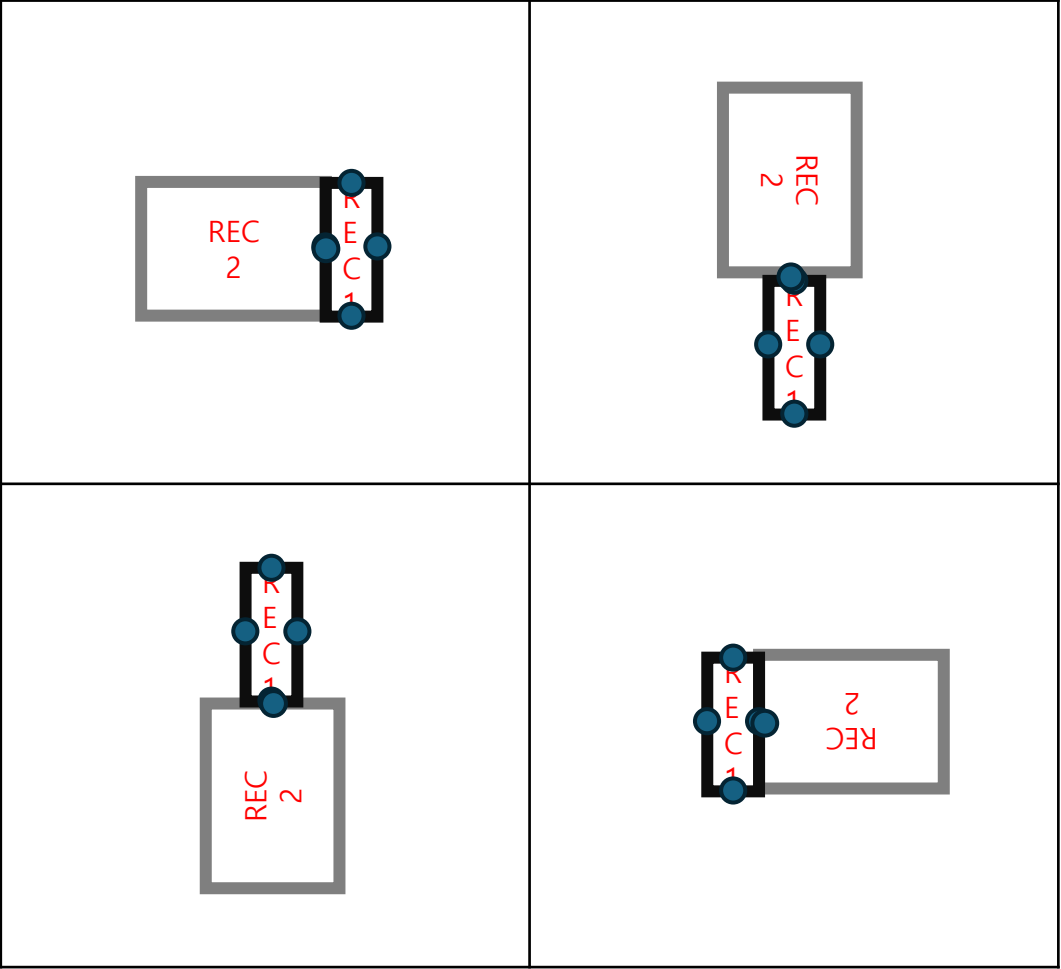


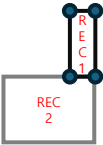
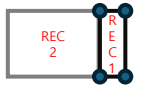
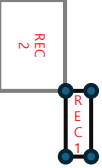
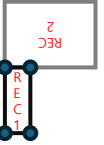
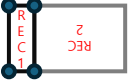
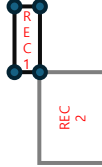
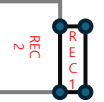
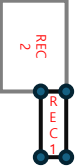
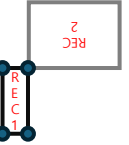
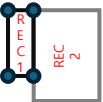
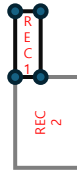


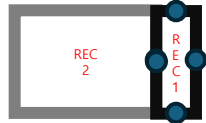
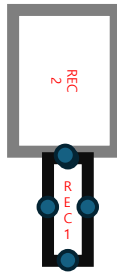
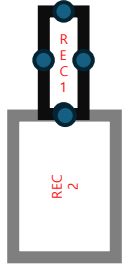
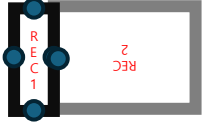




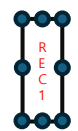




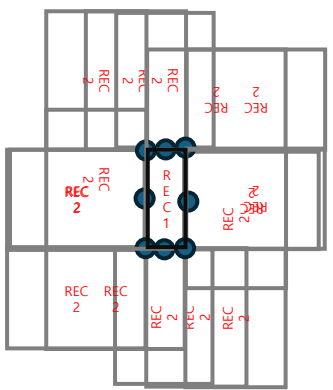
		
		
		
		

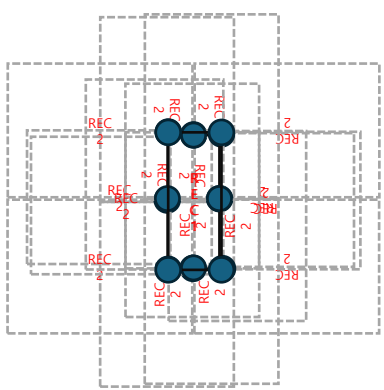
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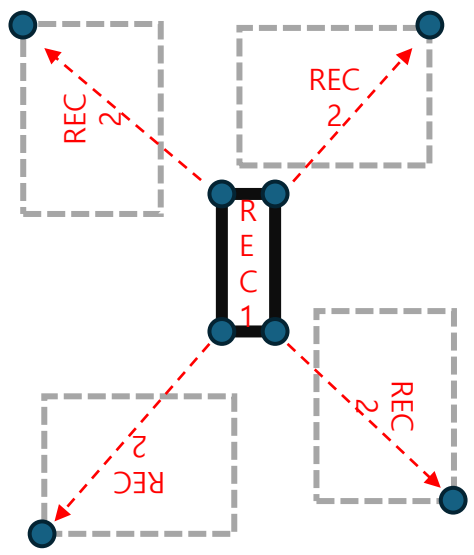
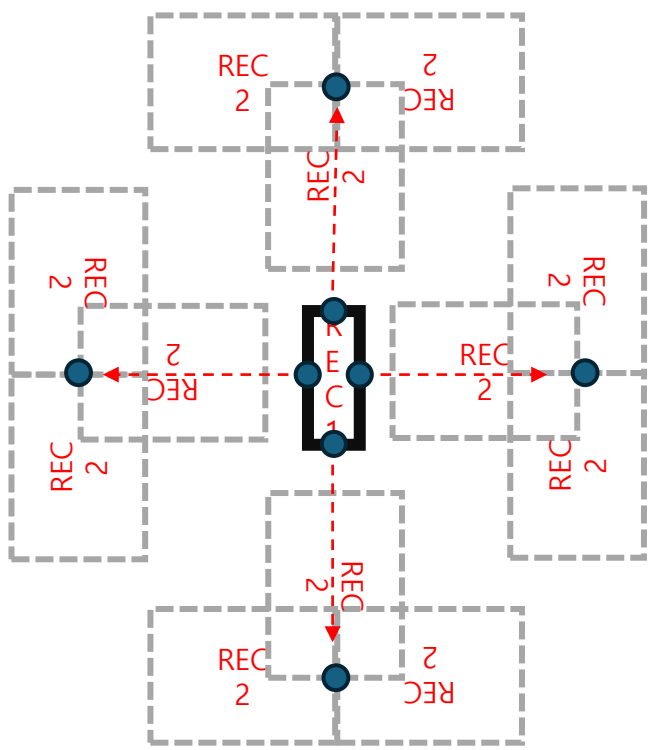


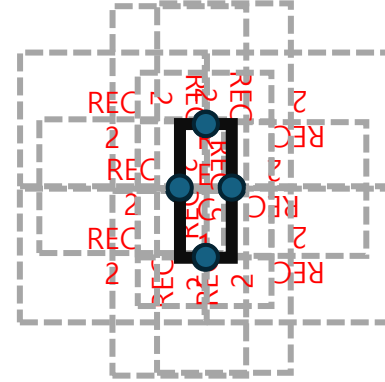
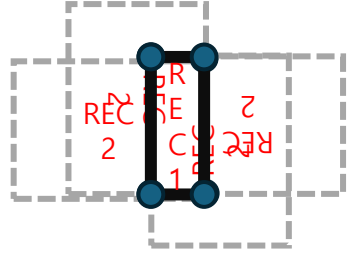
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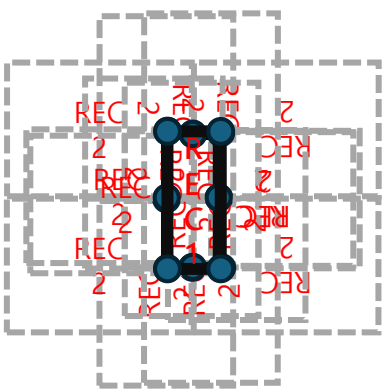


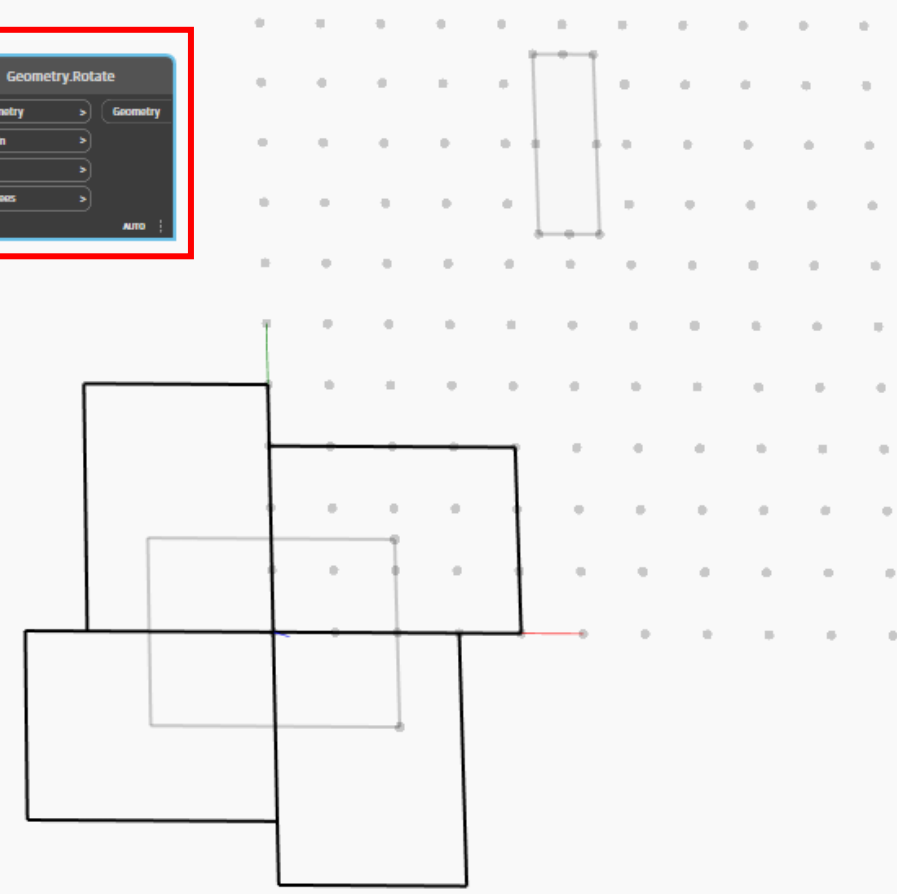
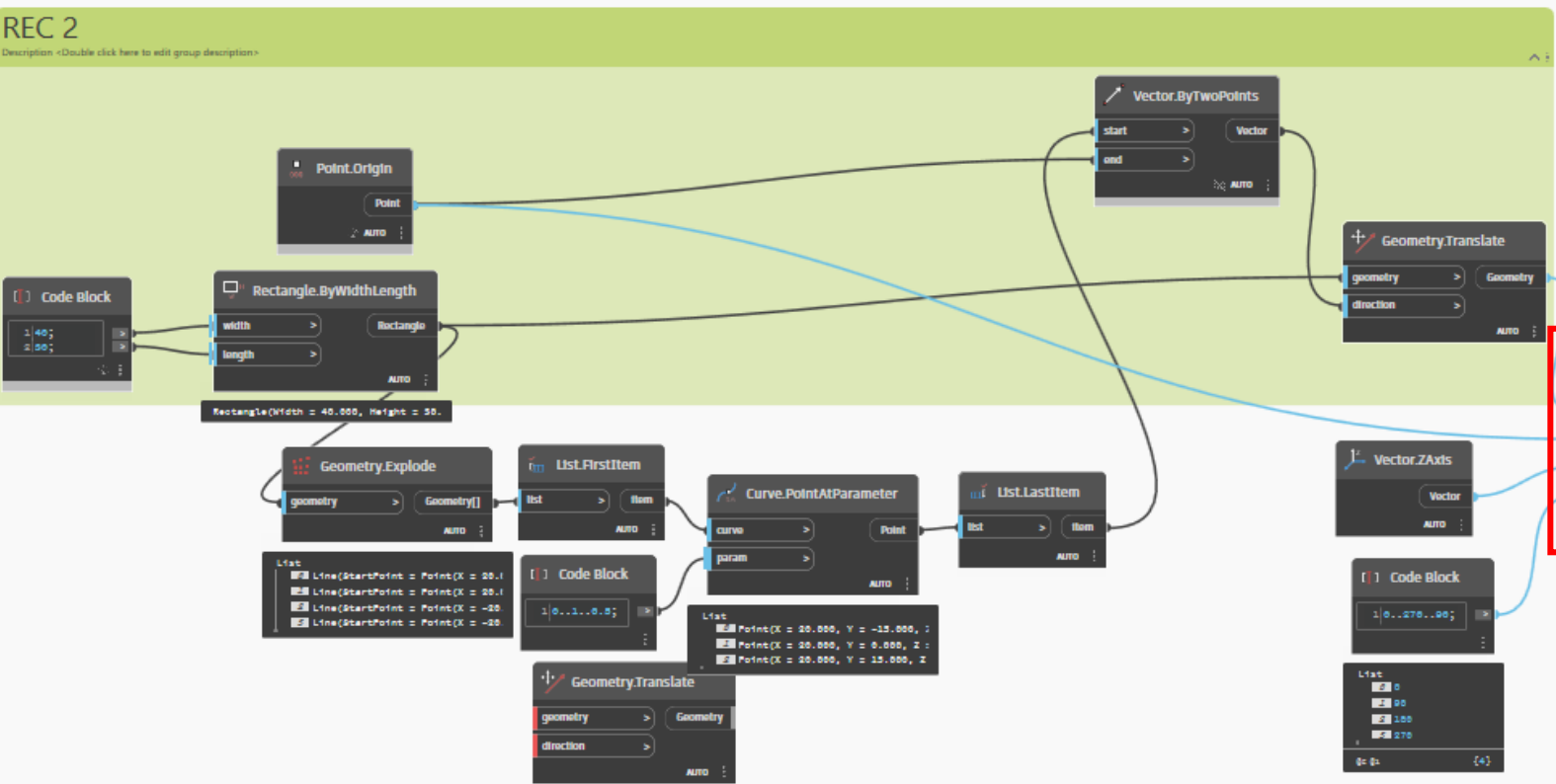
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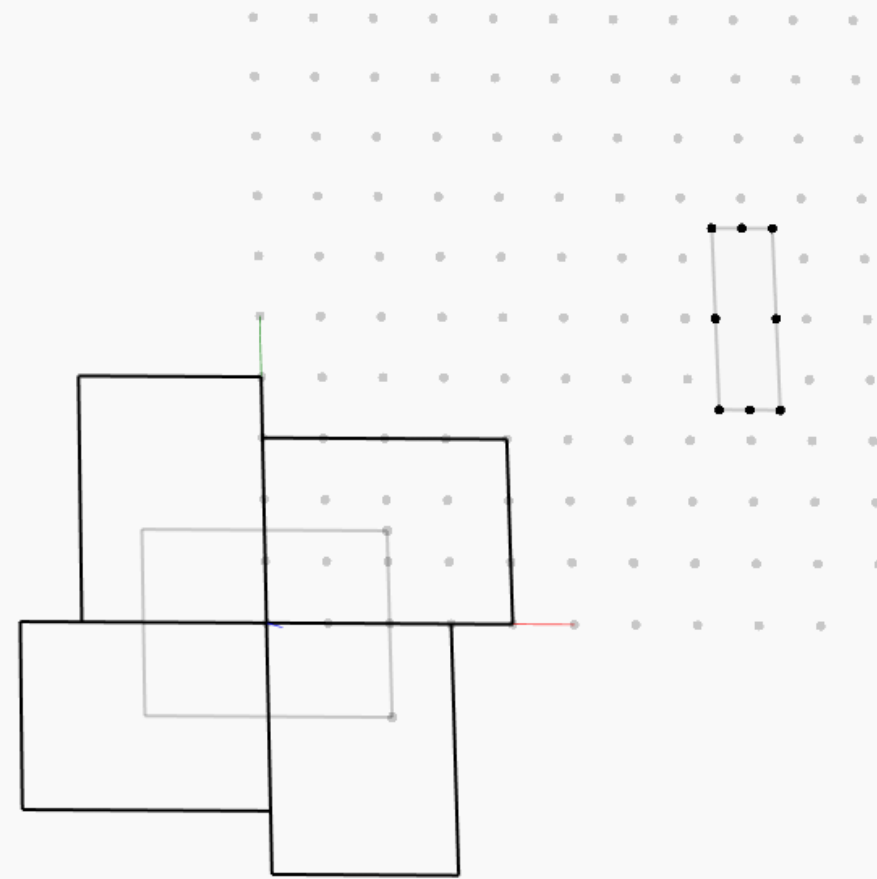
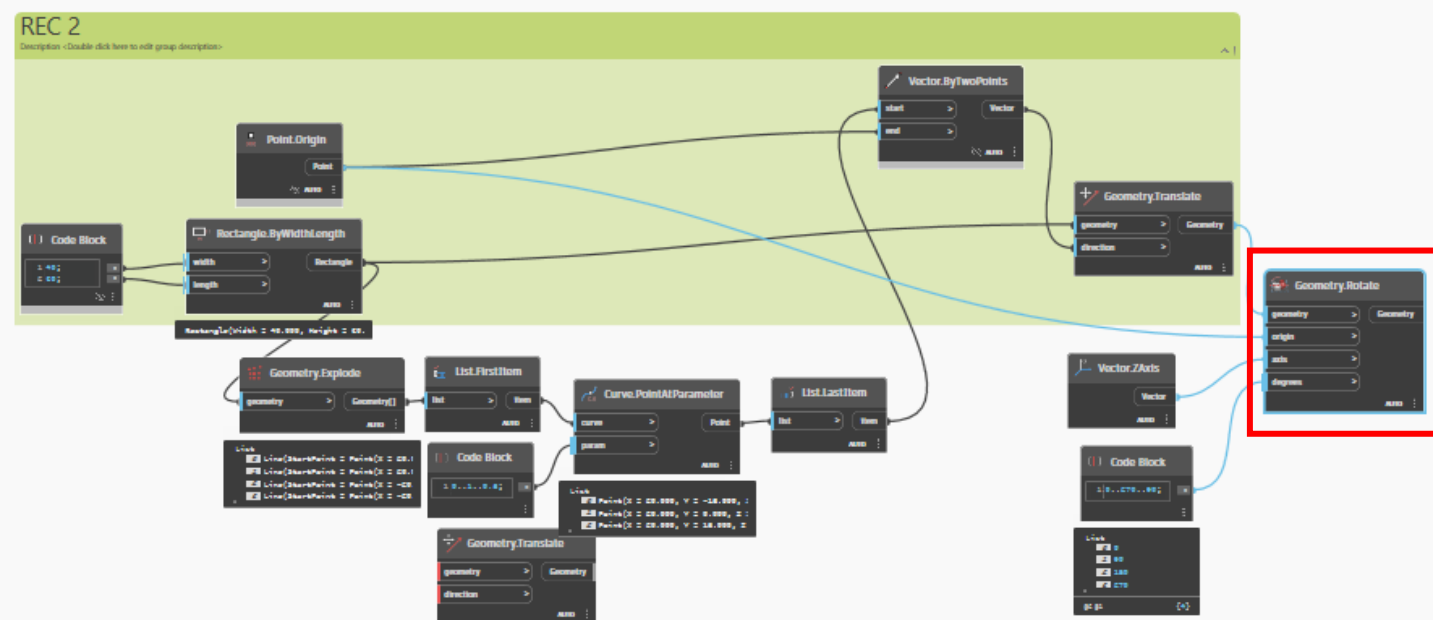
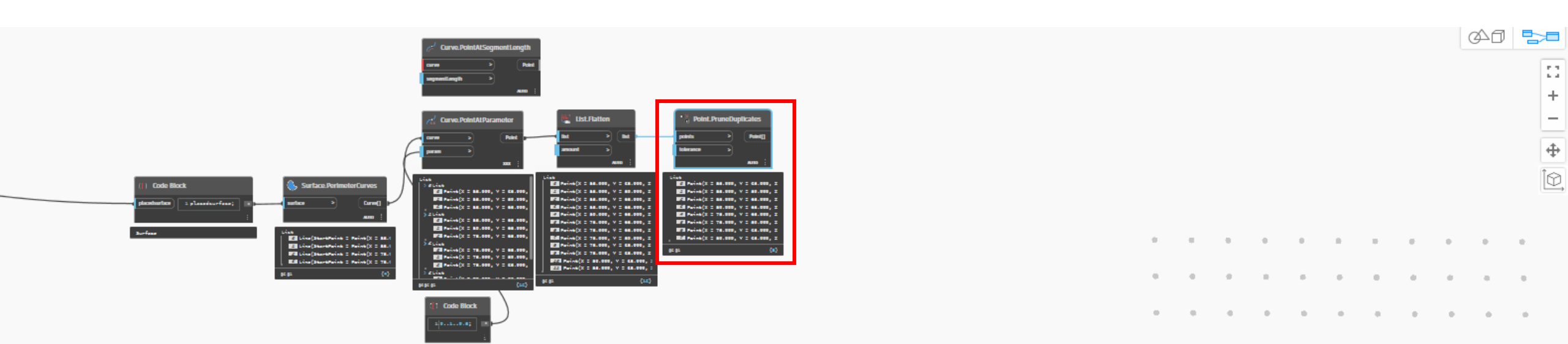


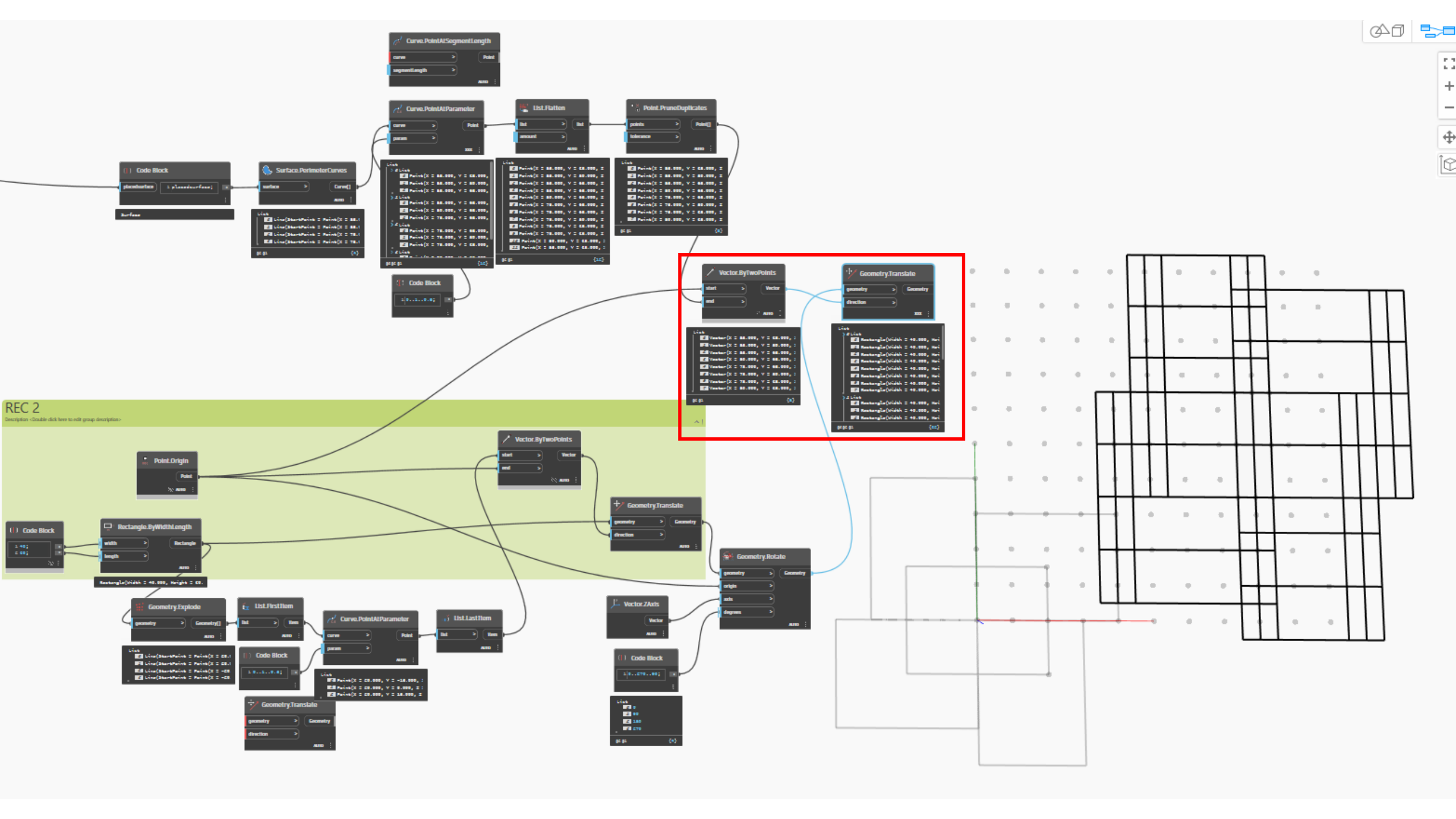








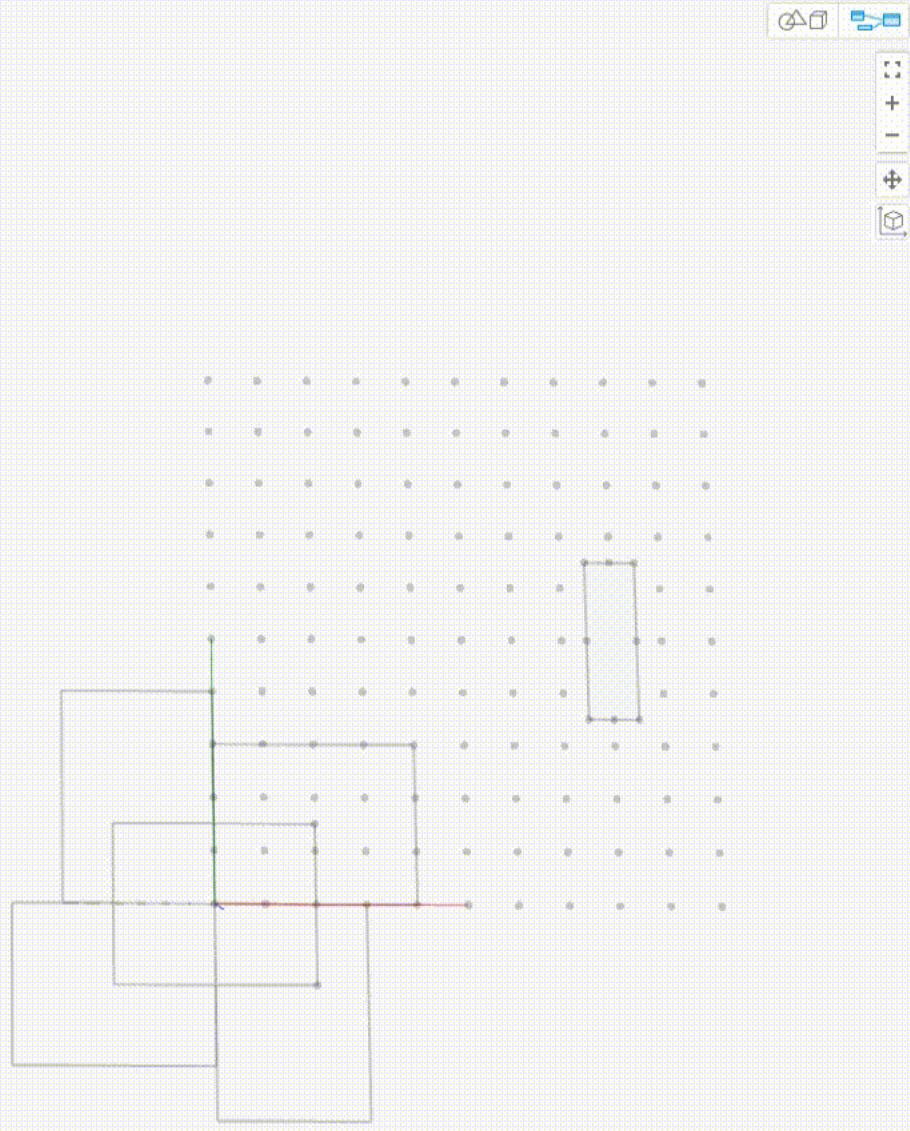
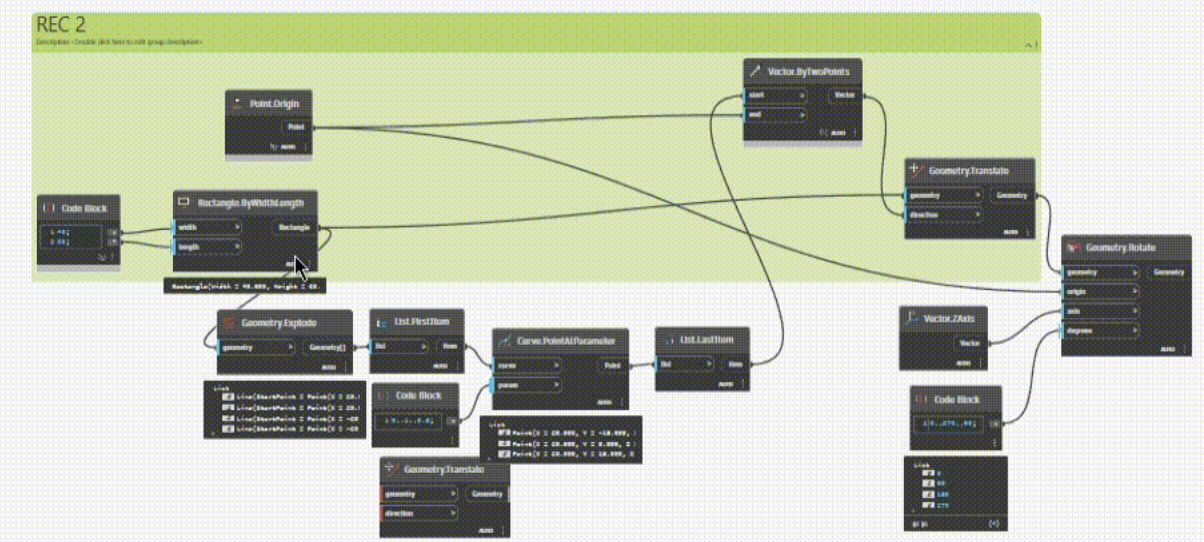


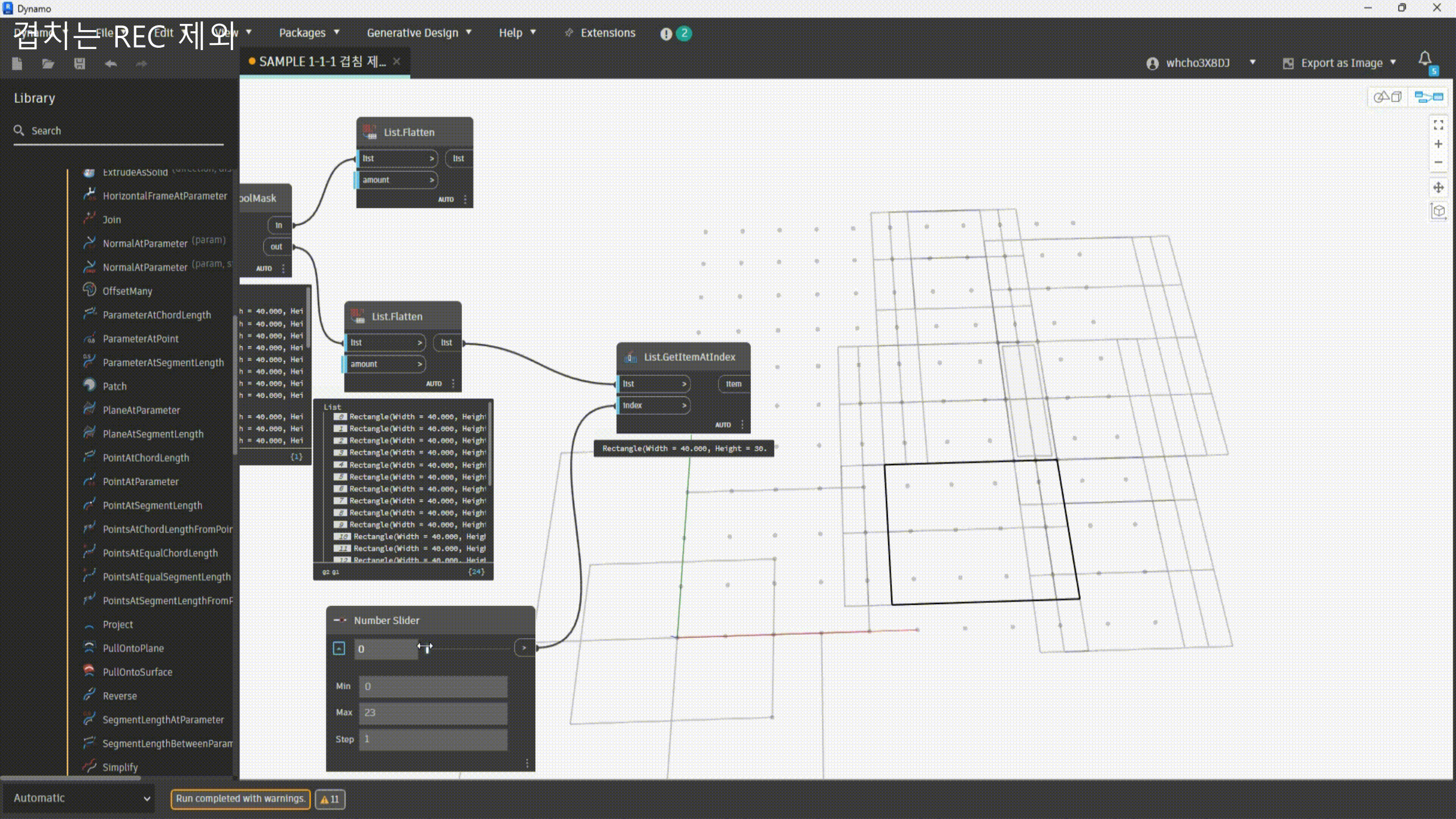


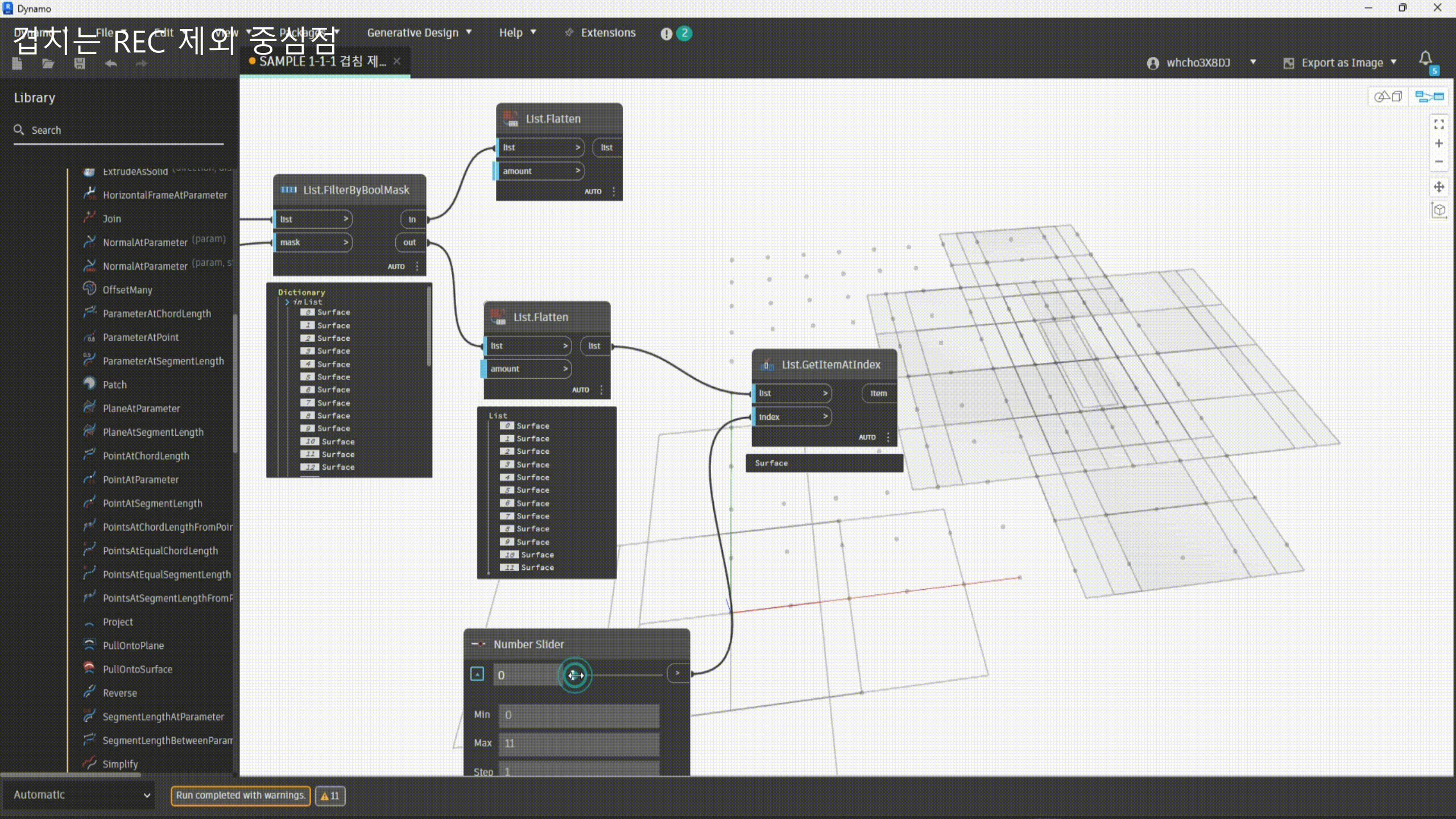
Library

Search

- ExtrudeAsSolid
- HorizontalFrameAtParameter
- Join
- NormalAtParameter (param)
- NormalAtParameter (param, s
- OffsetMany
- ParameterAtChordLength
- ParameterAtPoint
- ParameterAtSegmentLength
- Patch
- PlaneAtParameter
- PlaneAtSegmentLength
- PointAtChordLength
- PointAtParameter
- PointAtSegmentLength
- PointsAtChordLengthFromPoi
- PointsAtEqualChordLength
- PointsAtEqualSegmentLength
- PointsAtSegmentLengthFromF
- Project
- PullOntoPlane
- PullOntoSurface
- Reverse
- SegmentLengthAtParameter
- SegmentLengthBetweenParam
- Simplify



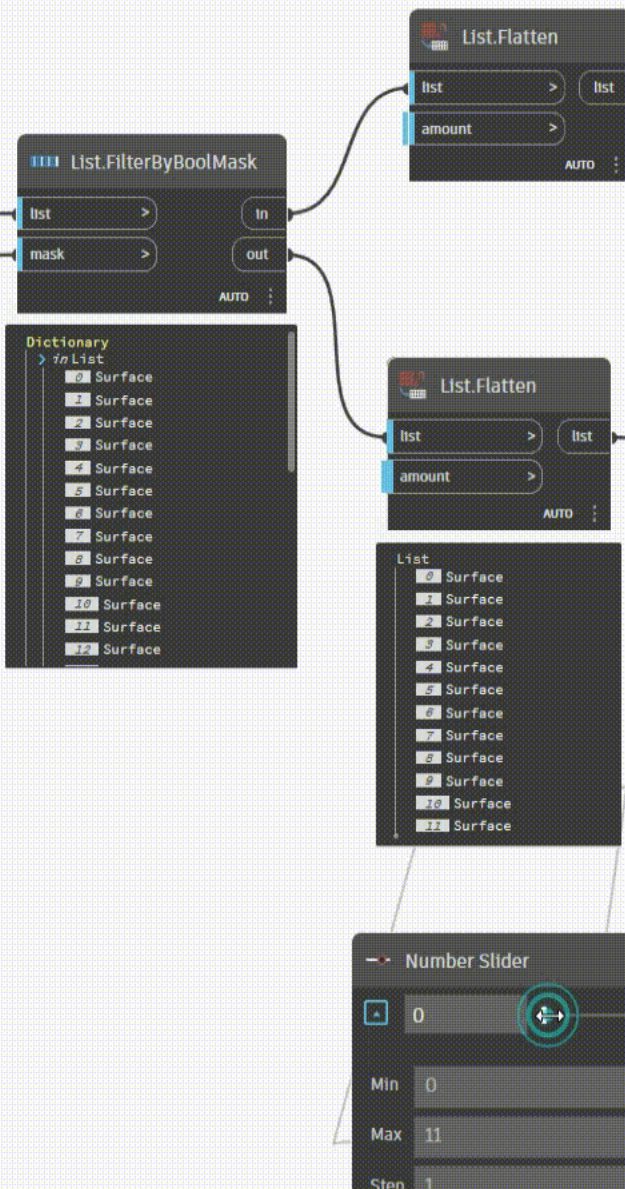




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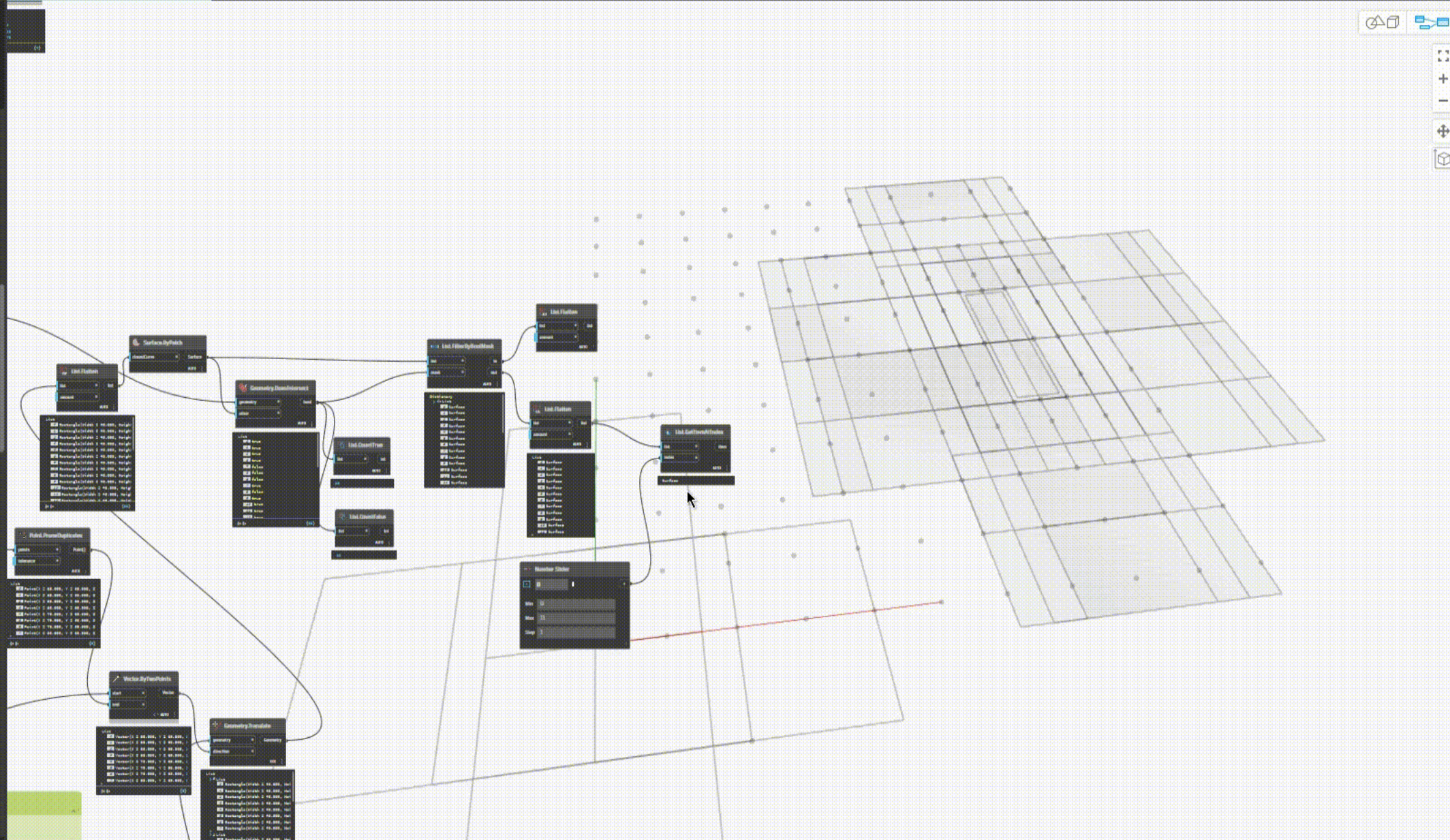
Search

ExtrudeAsSolid (creator, ...)
HorizontalFrameAtParameter
Join
NormalAtParameter (param)
NormalAtParameter (param, S
OffsetMany
ParameterAtChordLength
ParameterAtPoint
ParameterAtSegmentLength
Patch
PlaneAtParameter
PlaneAtSegmentLength
PointAtChordLength
PointAtParameter
PointAtSegmentLength
PointsAtChordLengthFromPoir
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Project
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Reverse
SegmentLengthAtParameter
SegmentLengthBetweenParam
Simplify

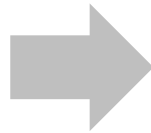
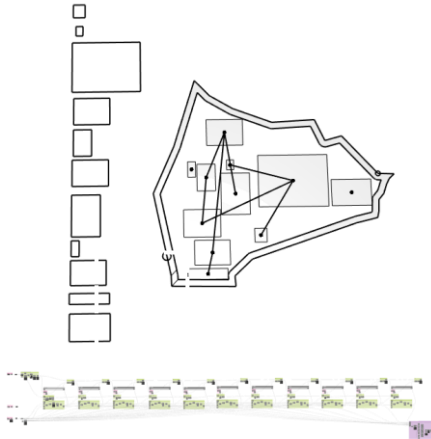


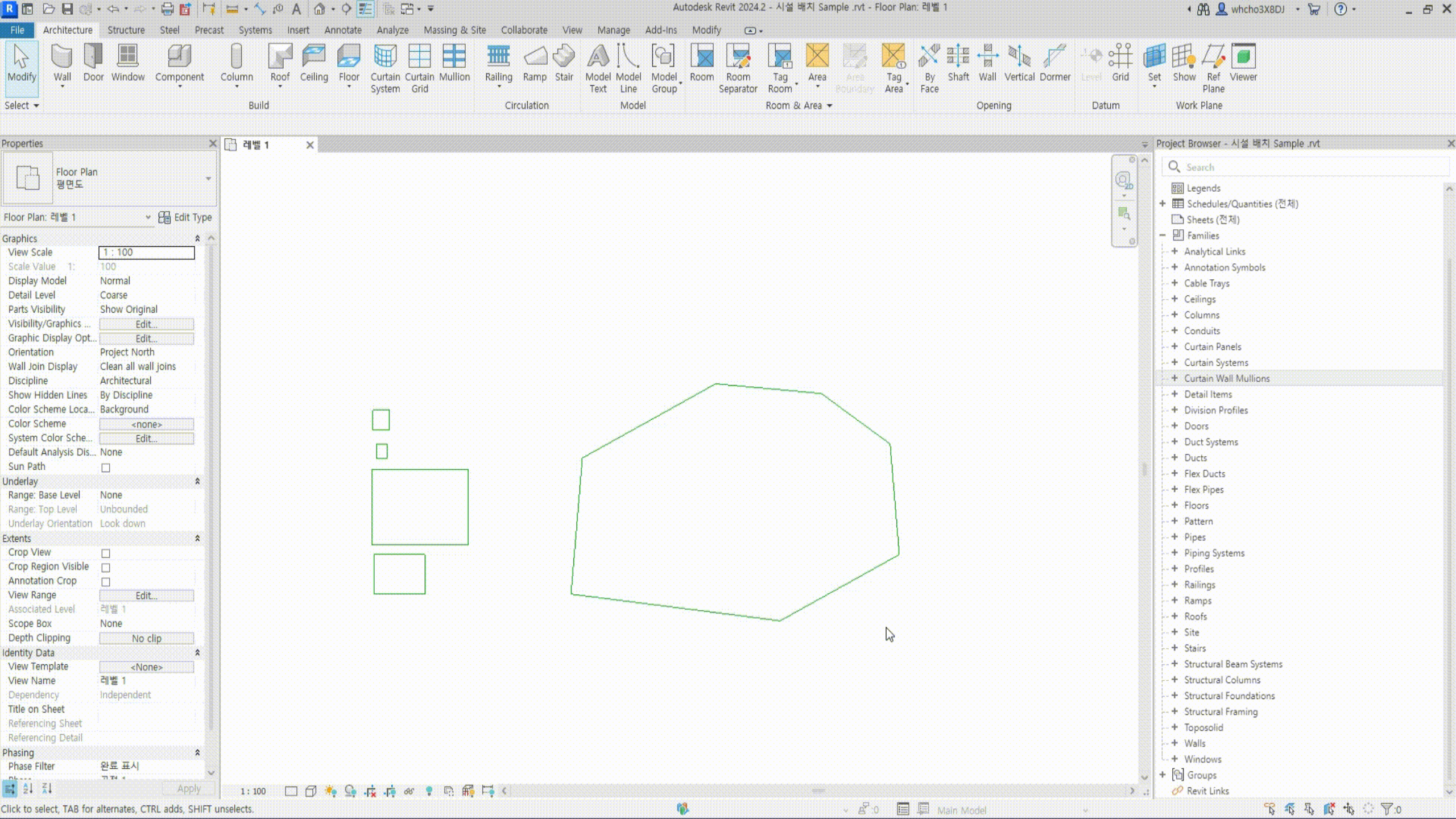
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 Search



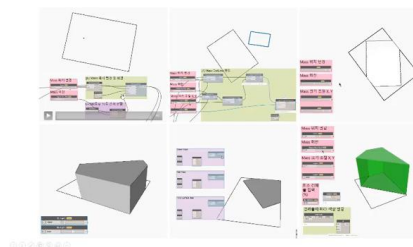
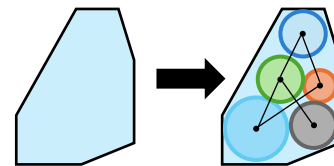
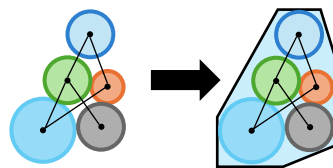
응용 / 샘플파일





02 Space – Out Line

공간 배치를 활용한 형상 생성, 고정된 형상의 공간 배치



공간설계 (space planning) 의미

대한건축학회 건축용어사전

공간설계

[space planning , 空間設計]

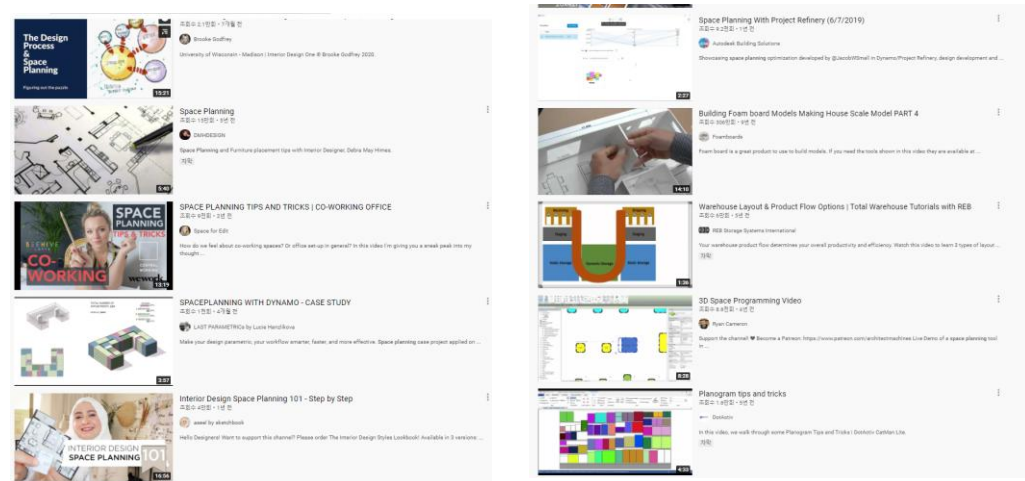
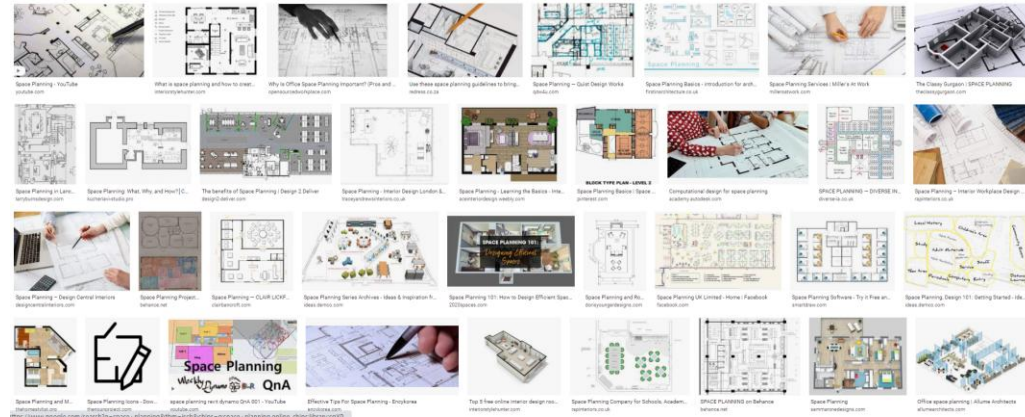
[계획 및 설계]

우리가 감각적으로 인지할 수 있는 공간을 설계자의 의도에 맞게 구성하는 것.

참고 : 건축용어대사전, 기문당

Space planning is the process of analyzing how space in structures and rooms will be used. Good space planning considers possible space uses and ensures that they're used efficiently. The process results in a space plan, a drawing that identifies the needs and required elements of spaces. 2020. 8. 29.

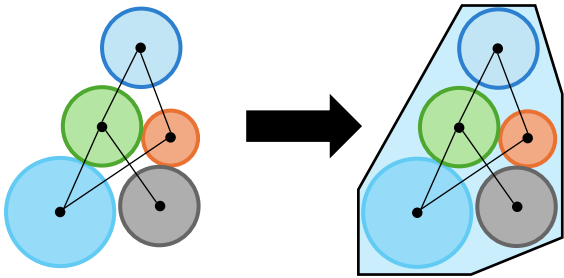
공간계획은 **공간구조와 객실이 어떻게 사용될지를 분석하는 과정이다.** 좋은 공간 계획은 가능한 공간 사용을 고려하고 효율적으로 사용하도록 보장합니다. 이 과정은 공간의 필요와 필요 요소를 식별하는 도면인 공간 계획으로 귀결됩니다. 2020.8.29



공간설계 (space planning)를 위한 공간 배치 방식에 대하여

HOW ?

공간 배치



Space Planning
HOW

방법 1

방법 2

방법 3

방법 4

방법 5



방법
조건
규칙

공간을 규정 짓는 Data > 공간 형선(Geometry) > 공간 계획(배치)



	A	B	C	D	E	F	G	H	I
1	SPACE ID	SPACE NAME	DEPARTMENT	DEPARTMENT ID	QUANTITY	WIDTH	HEIGHT	AREA	TOTAL AREA
2	0	Kitchen	KITCHEN	0	1	80000	80000	640000	640000
3	1	Living Room	LIVING	1	1	100000	100000	1000000	1000000
4	2	Dining	DINING	2	1	100000	100000	1000000	1000000
5	3	Bathroom	BATHROOM	3	1	80000	80000	750000	750000
6	4	Bedroom	BEDROOM	4	1	120000	120000	1320000	1320000
7	5	Closet	CLOSET	5	1	50000	50000	250000	250000
8	6	Entry	CIRCULATION	6	1	40000	40000	320000	320000

Space Planning HOW

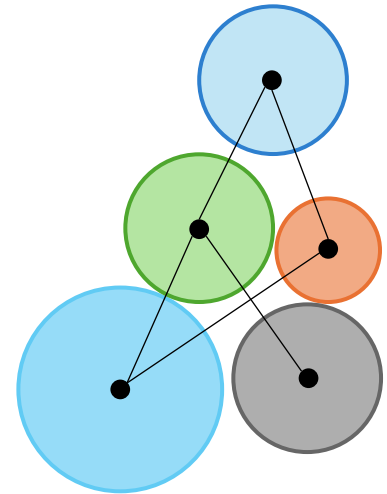
방법 1

방법 2

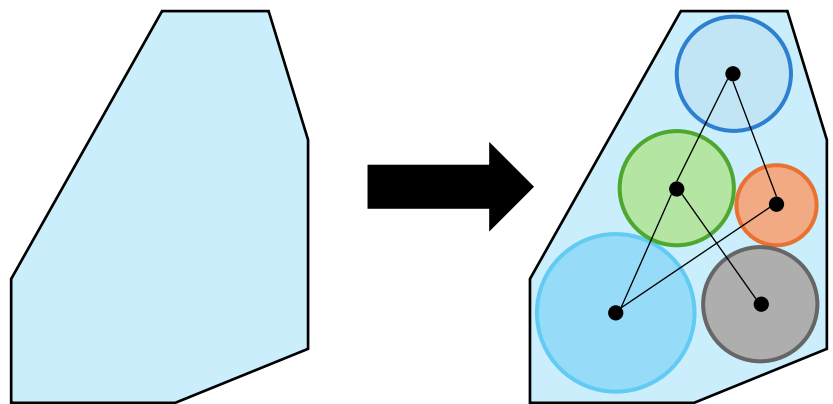
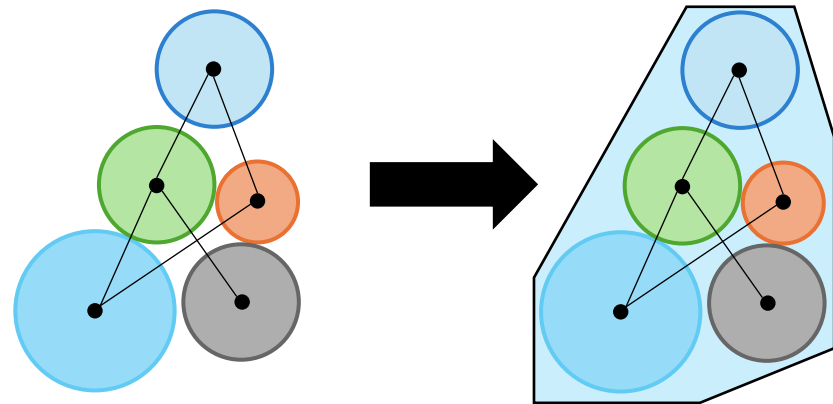
방법 3

방법 4

방법 5

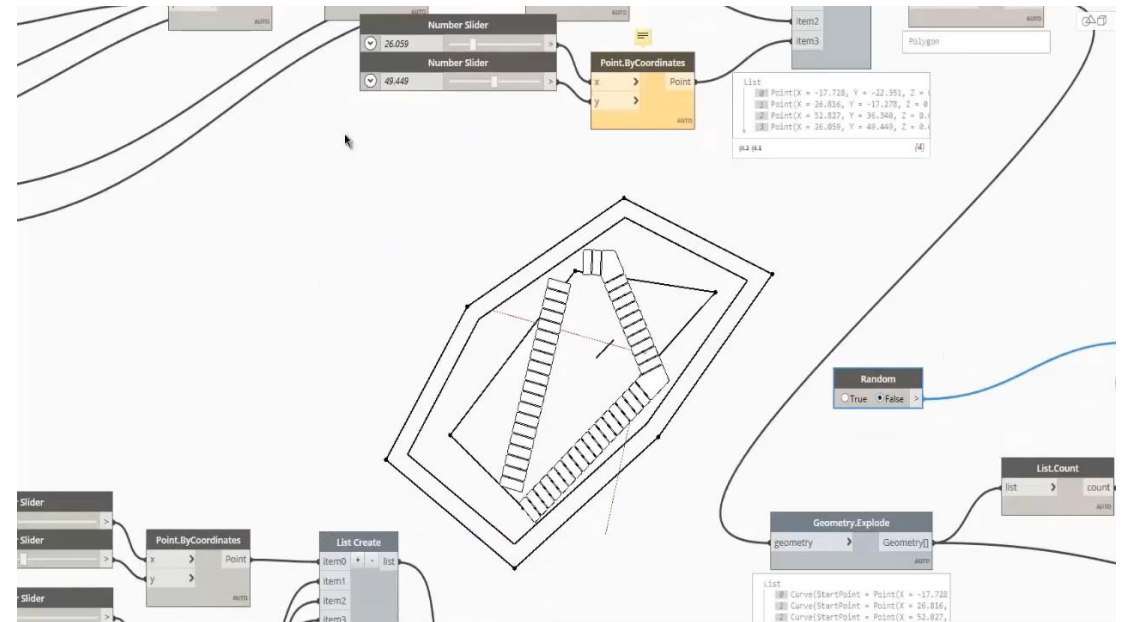
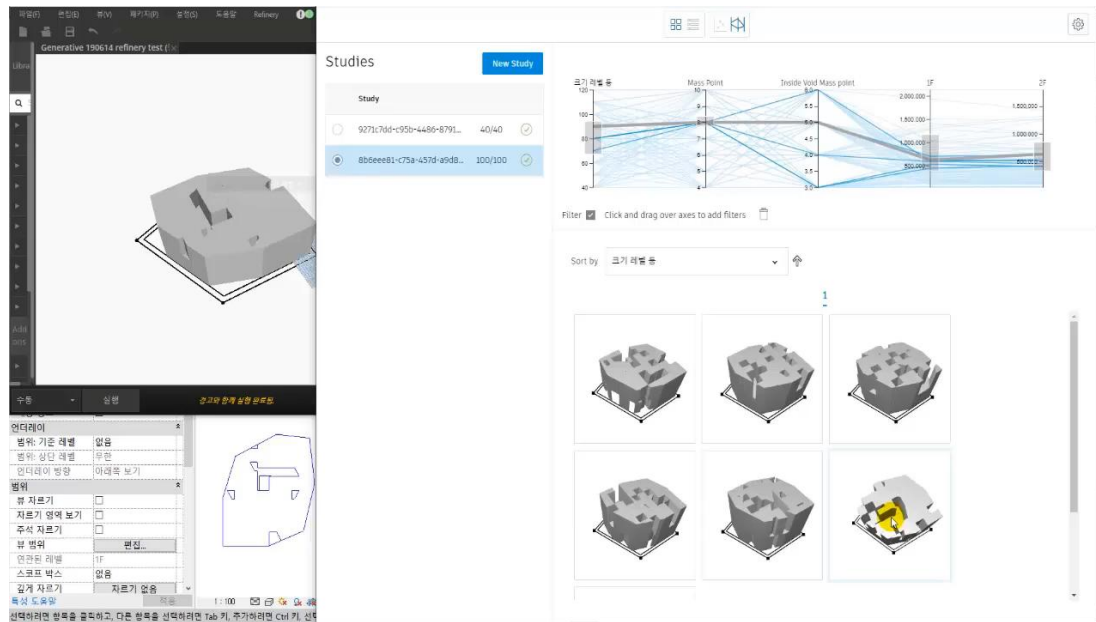
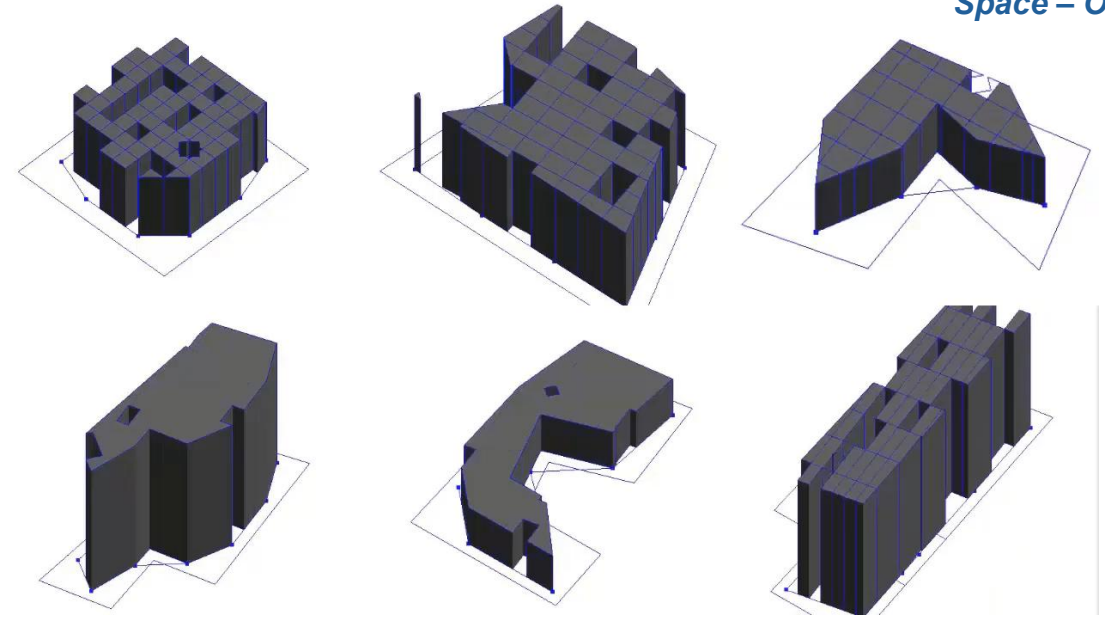
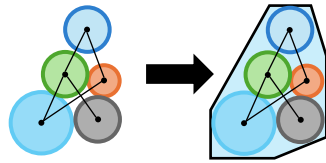


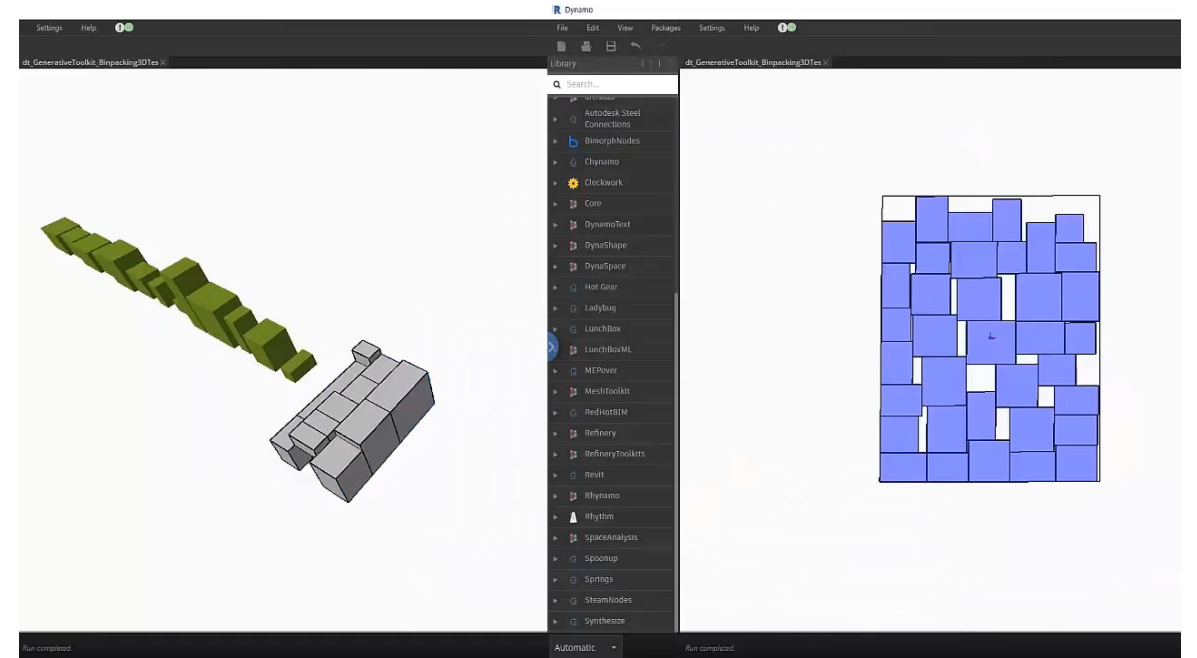
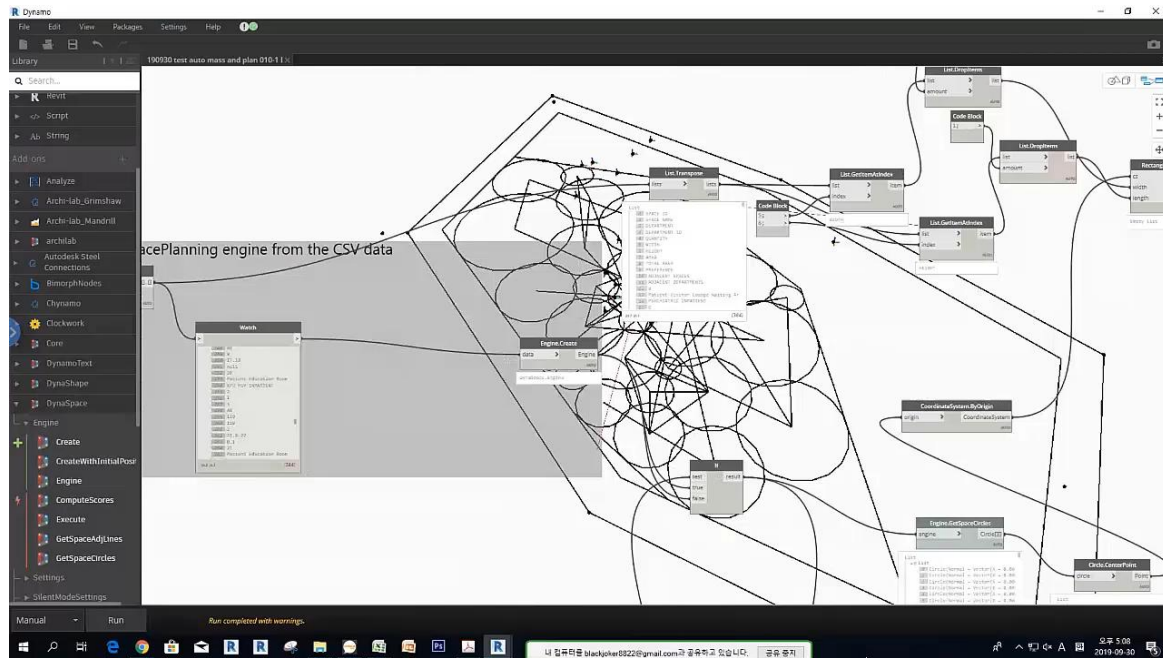
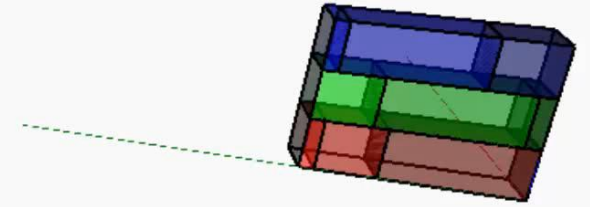
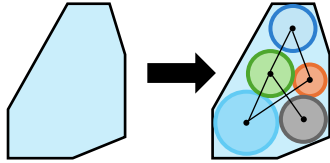
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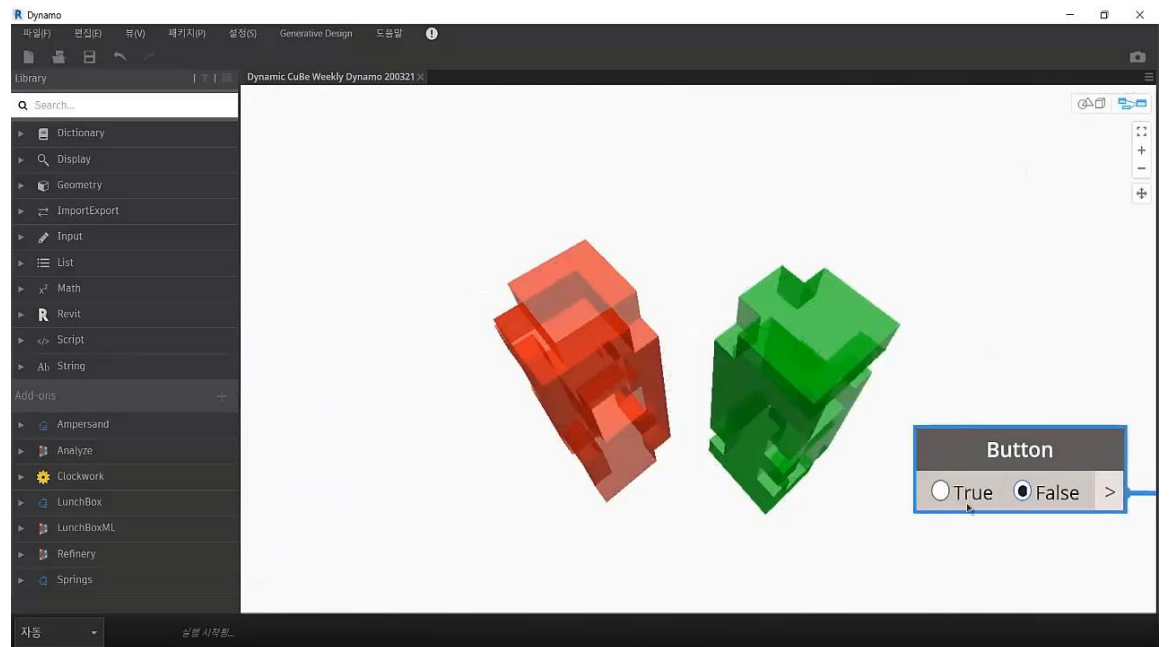
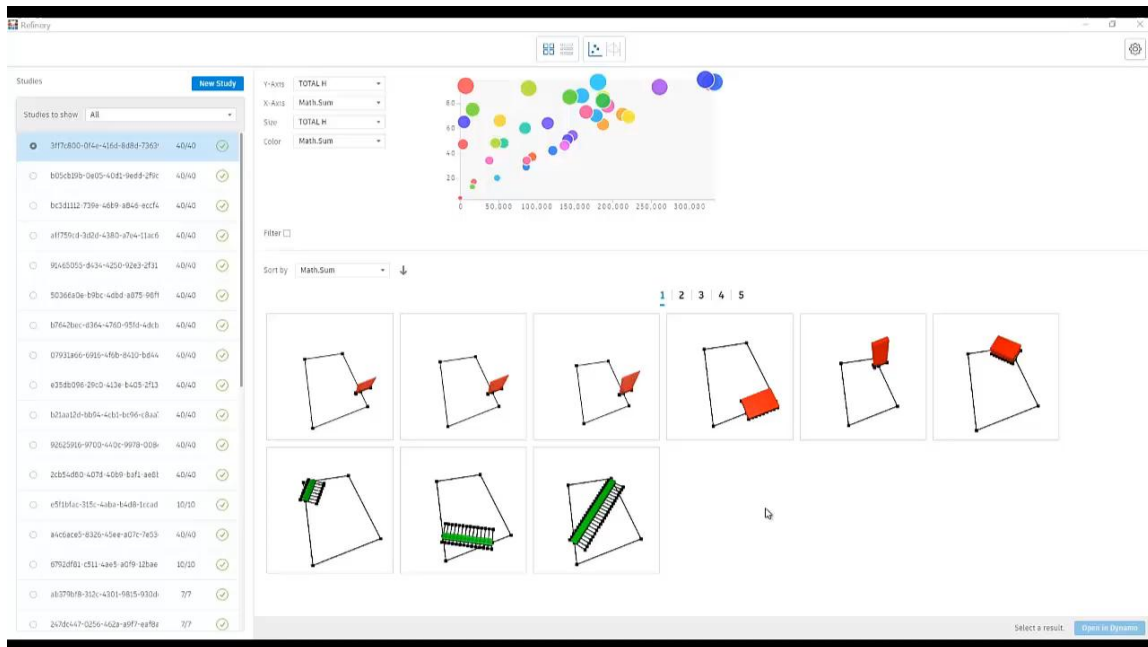
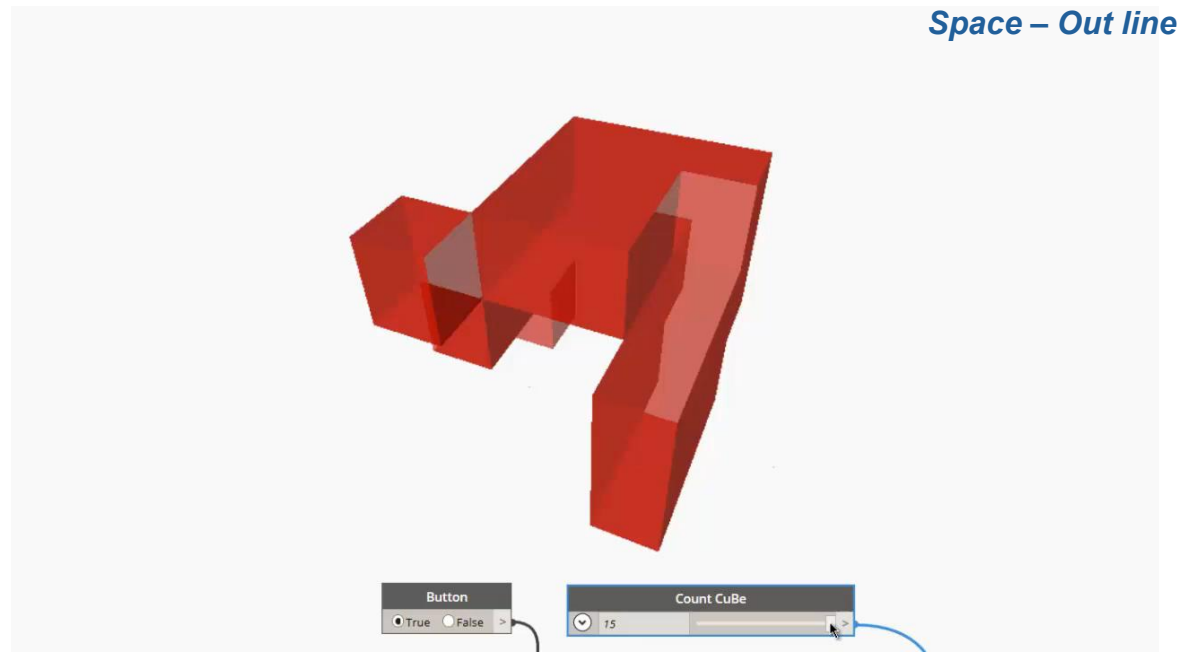
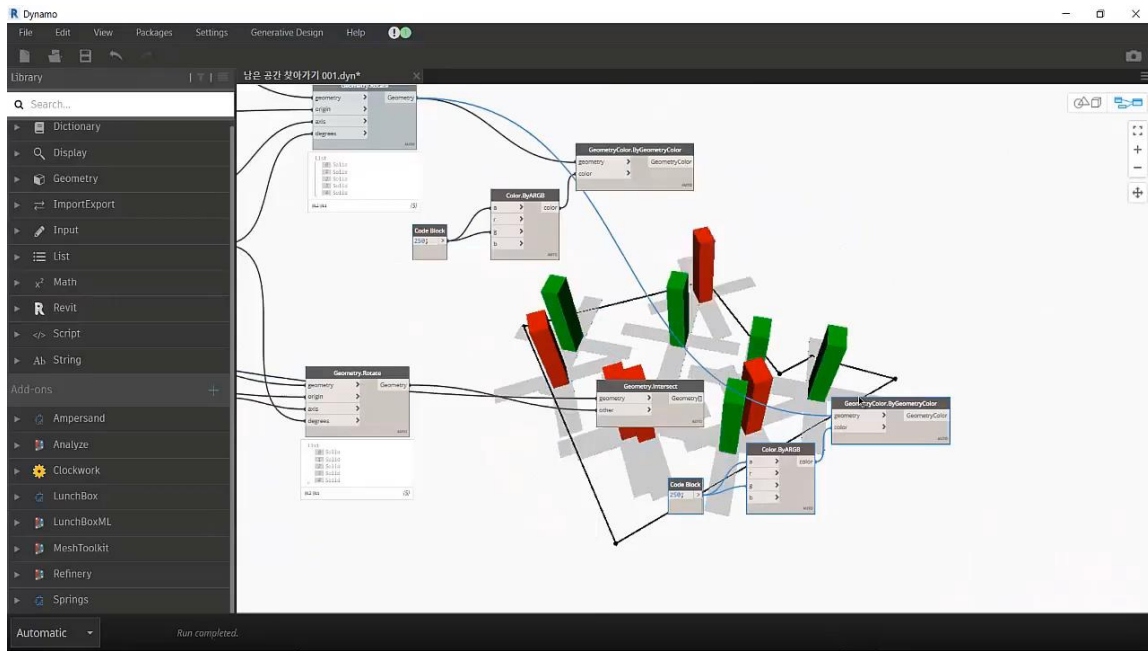


방법 조건 규칙

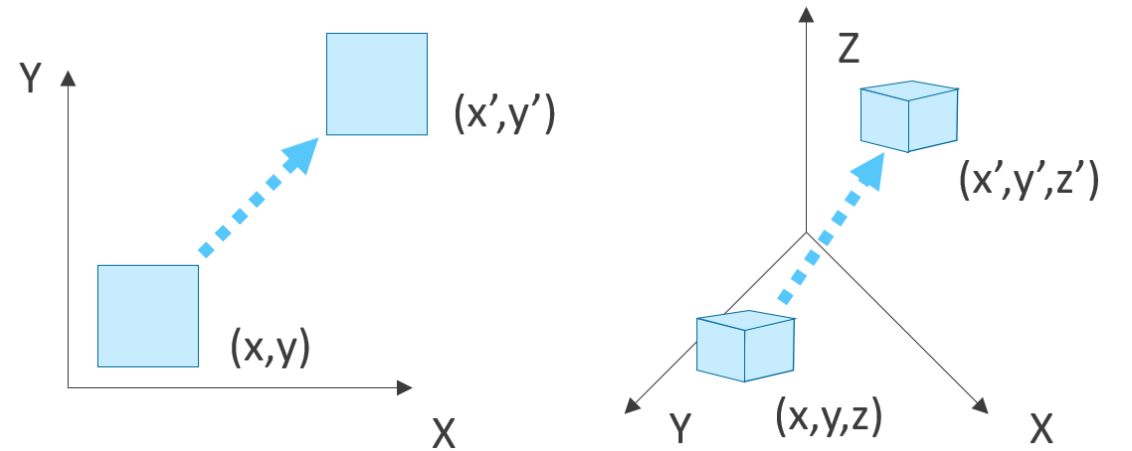
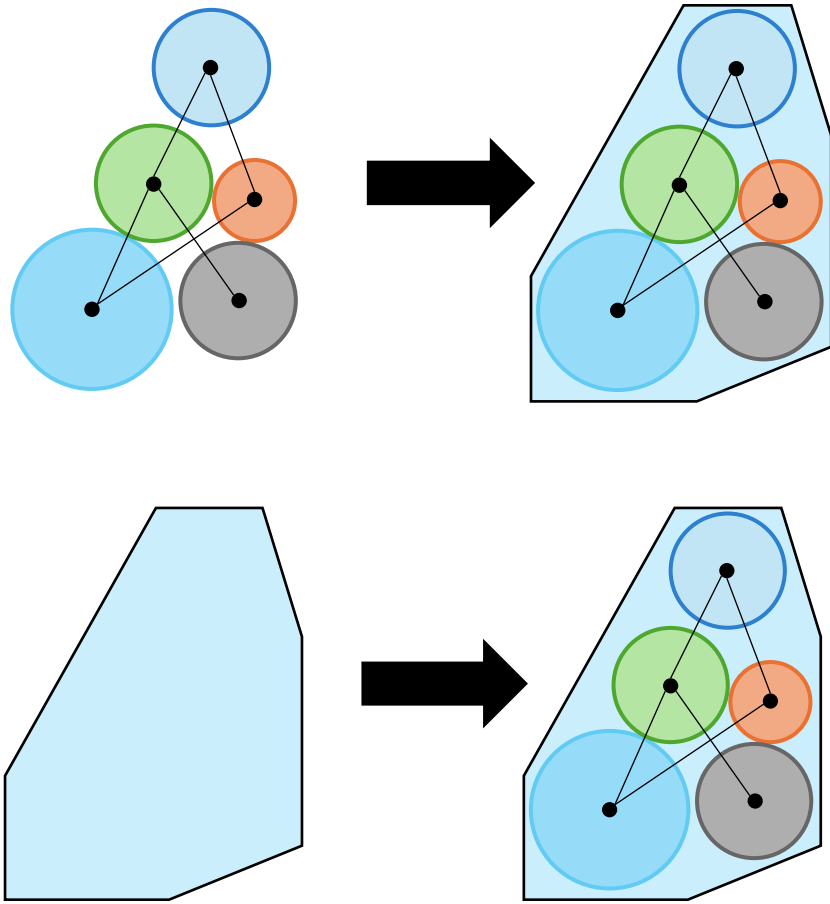
Space – Out line



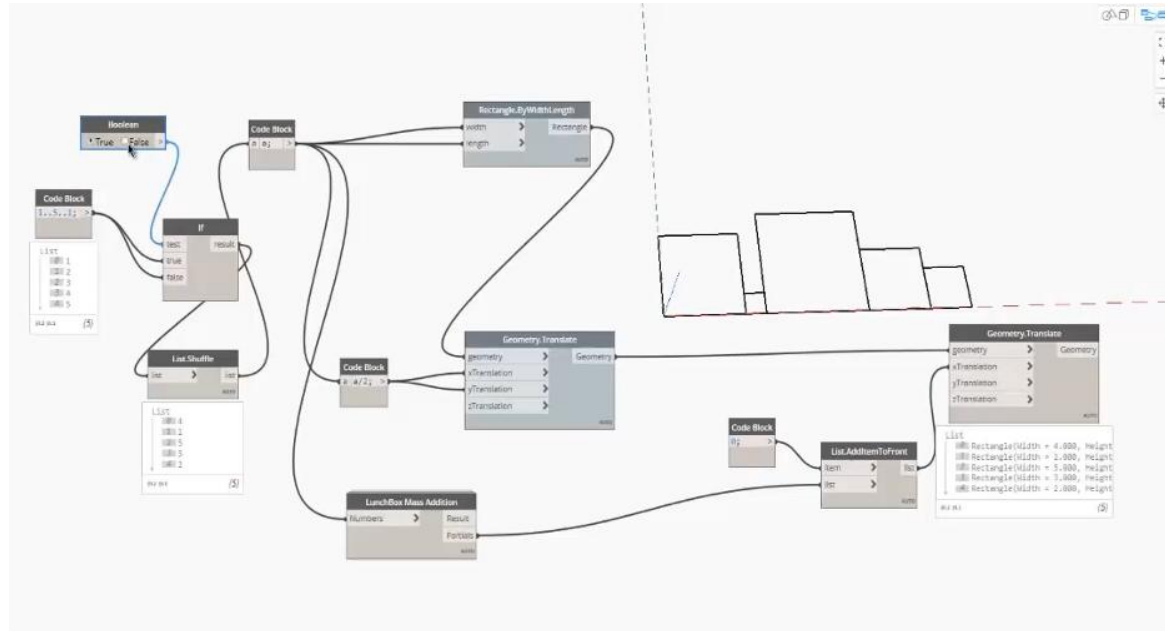
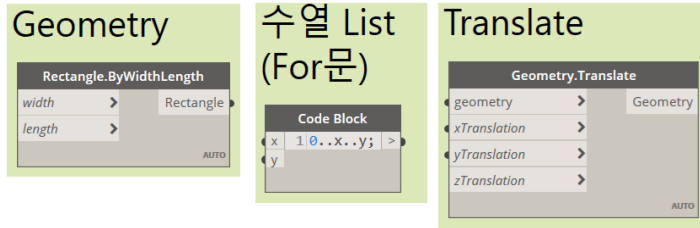




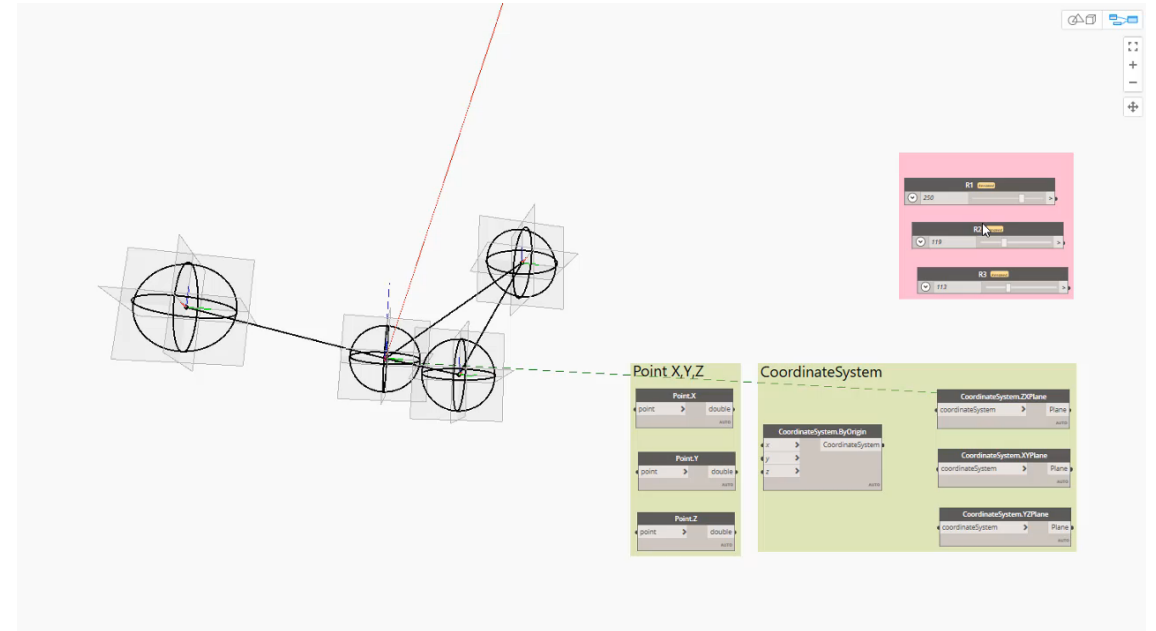
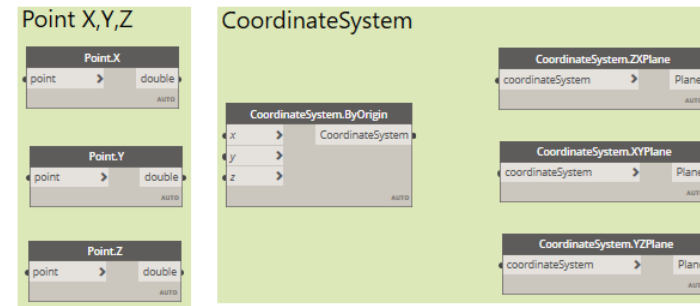
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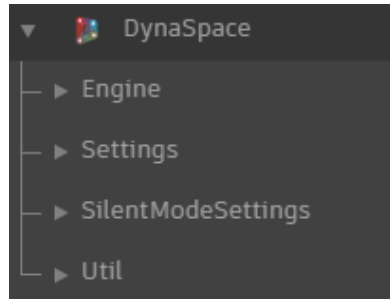
2D : Translate and For문 수열 List



3D : CoordinateSystem and Translate

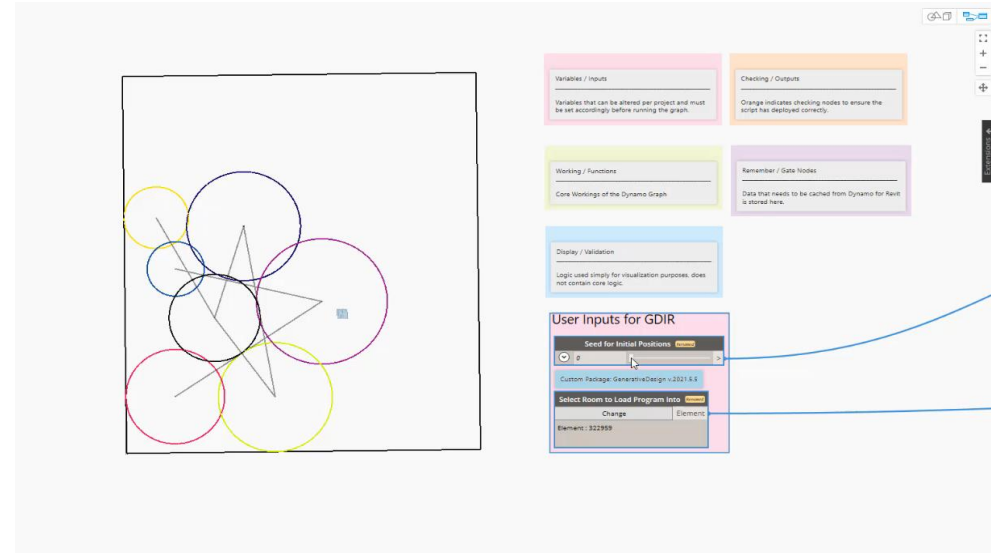


Package 버전 확인 및 Engine Node 업데이트 > 공간 Data 생성



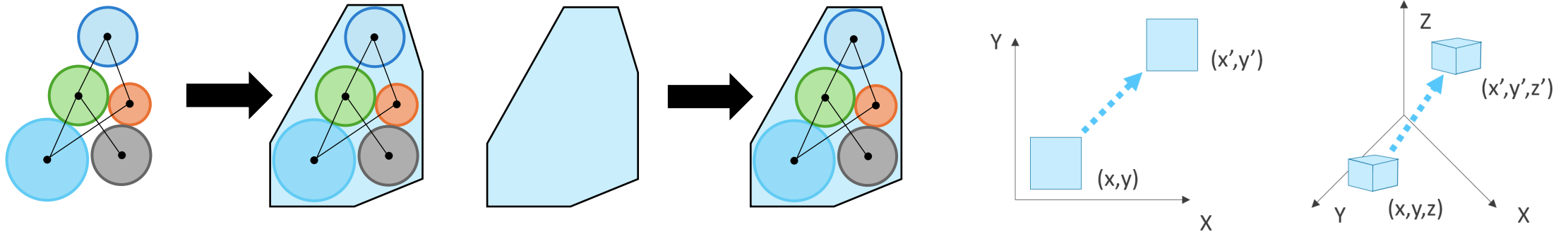
- DynaShape
- GenerativeDesign
- GenerativeDesign.Revit
- GenerativeDesignVisualizer3000™

Engine.Execute	
engine	departmentNames
boundaryVertices	spaceIds
silentModeSettings	spaceNames
reset	spaceCircles
execute	spaceAdjLines
enableManipulation	spaceAdjErrors
showSpaceIds	spaceAdjErrorRatios
showSpaceAdjacency	spaceAdjErrorRatios
showSpaceDepartmentAdjacency	
settings	

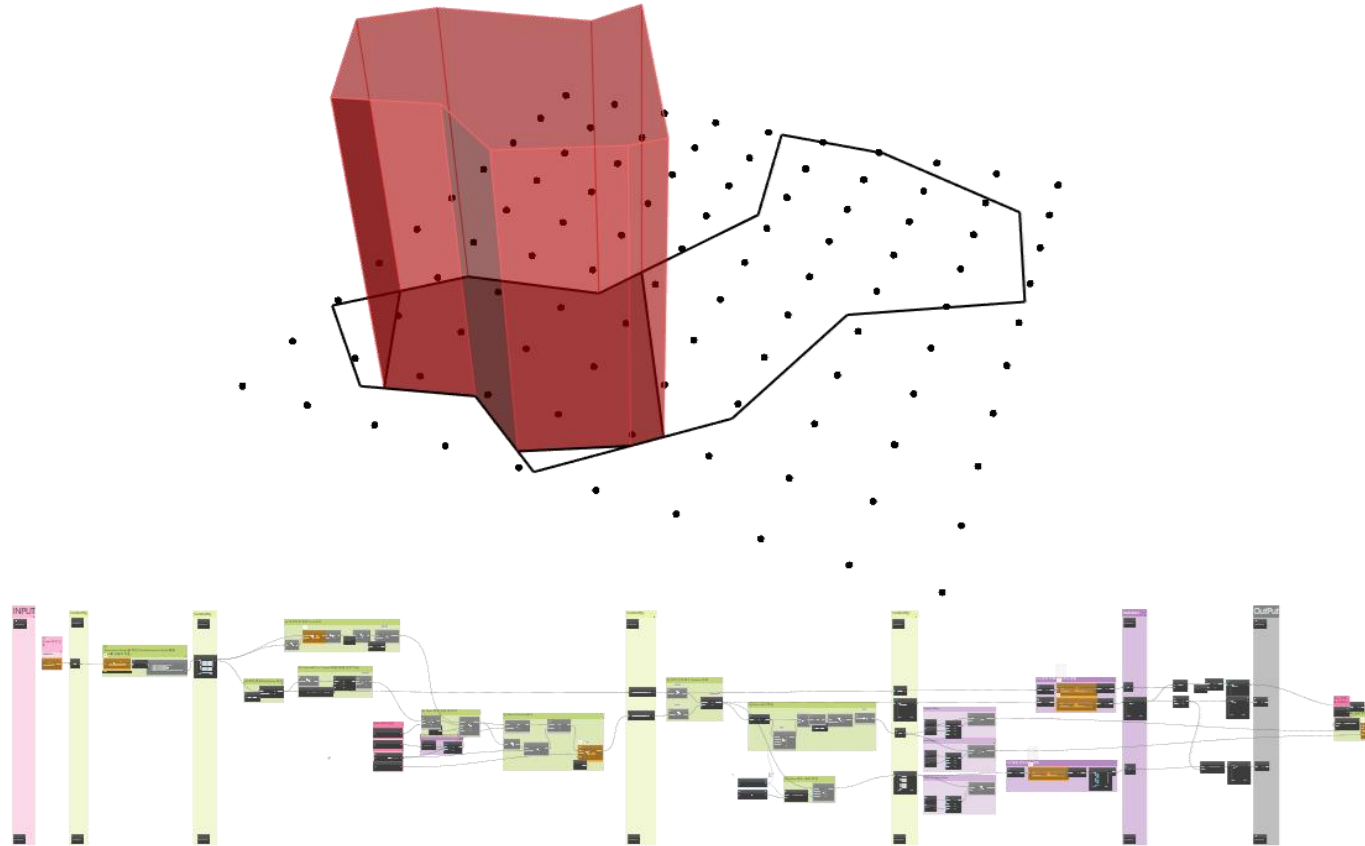


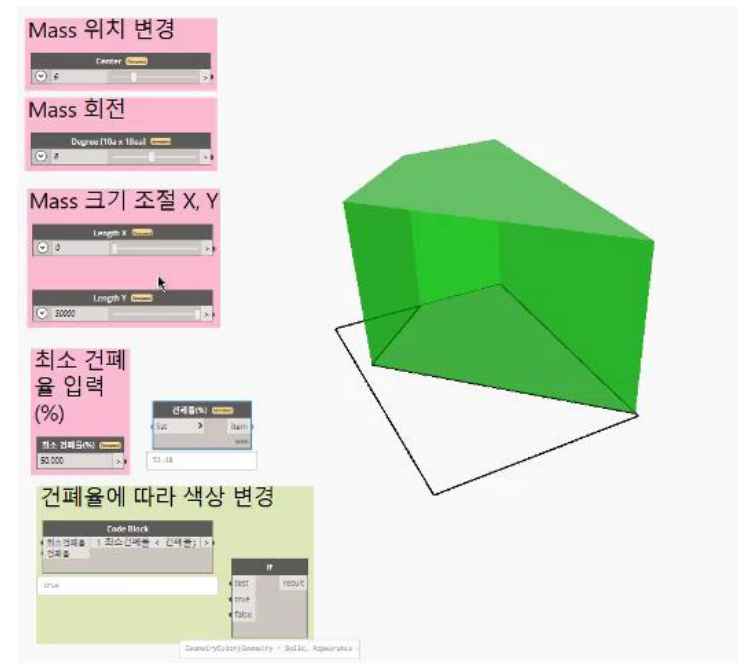
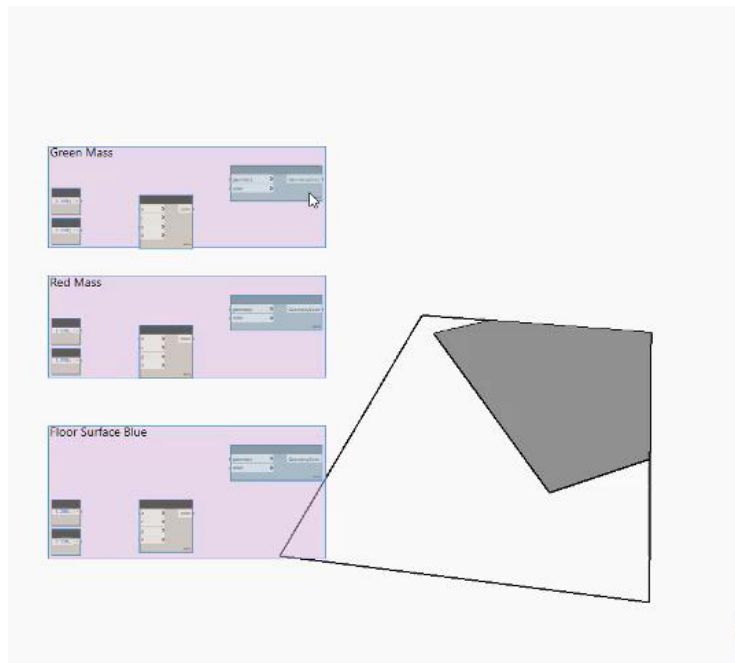
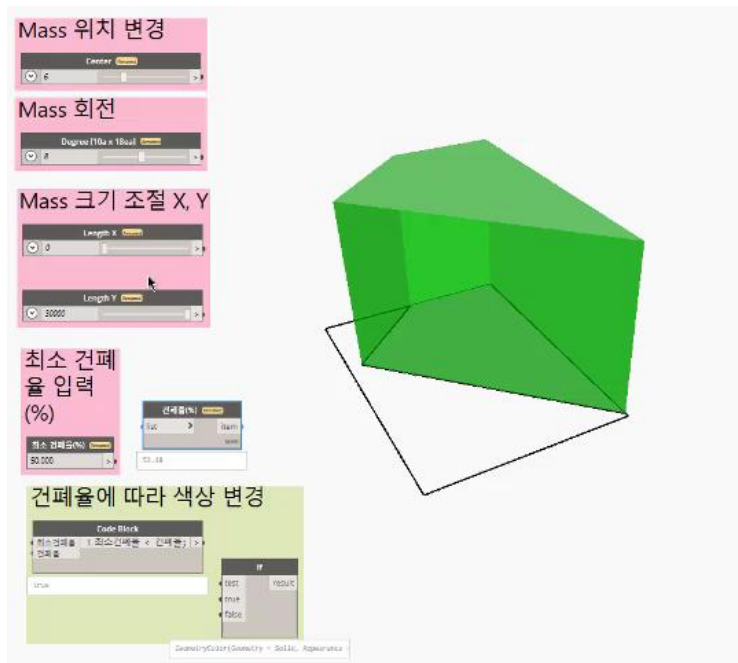
	A	B	C	D	E	F	G	H	I	J	K	L
1	SPACE ID	SPACE NAME	DEPARTMENT	DEPARTMENT ID	QUANTITY	WIDTH	HEIGHT	AREA	TOTAL AREA	PREFERENCE	ADJACENT SPACES	ADJACENT DEPARTMENTS
2	0	Kitchen	KITCHEN	0	1	80000	80000	640000	640000	1	1.2	1.2
3	1	Living Room	LIVING	1	1	100000	100000	1000000	1000000	2	0.2	0.2
4	2	Dining	DINING	2	1	100000	100000	1000000	1000000	3	0.1	0.1
5	3	Bathroom	BATHROOM	3	1	80000	80000	750000	750000	4	4	4
6	4	Bedroom	BEDROOM	4	1	120000	120000	1320000	1320000	5	3.5	3.5
7	5	Closet	CLOSET	5	1	50000	50000	250000	250000	6	4	4
8	6	Entry	CIRCULATION	6	1	40000	40000	320000	320000	7	0	0

형상을 자유롭게 움직이다
공간을 자유롭게 움직이다



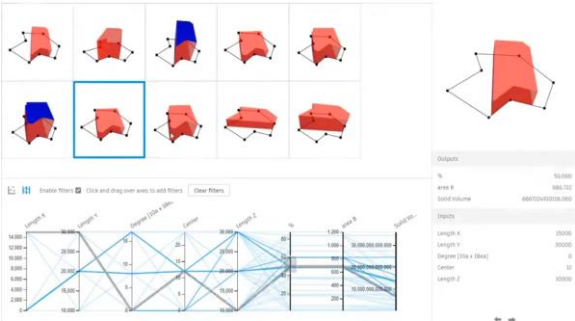
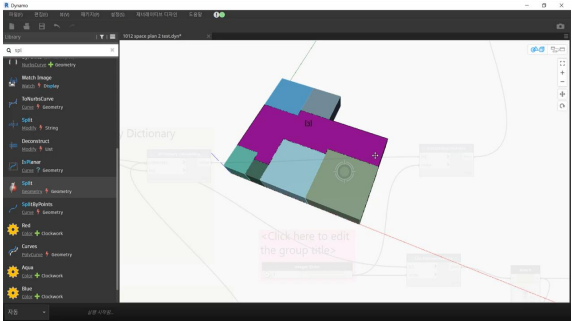
공간에 [방법, 조건, 규칙]을 부여하다

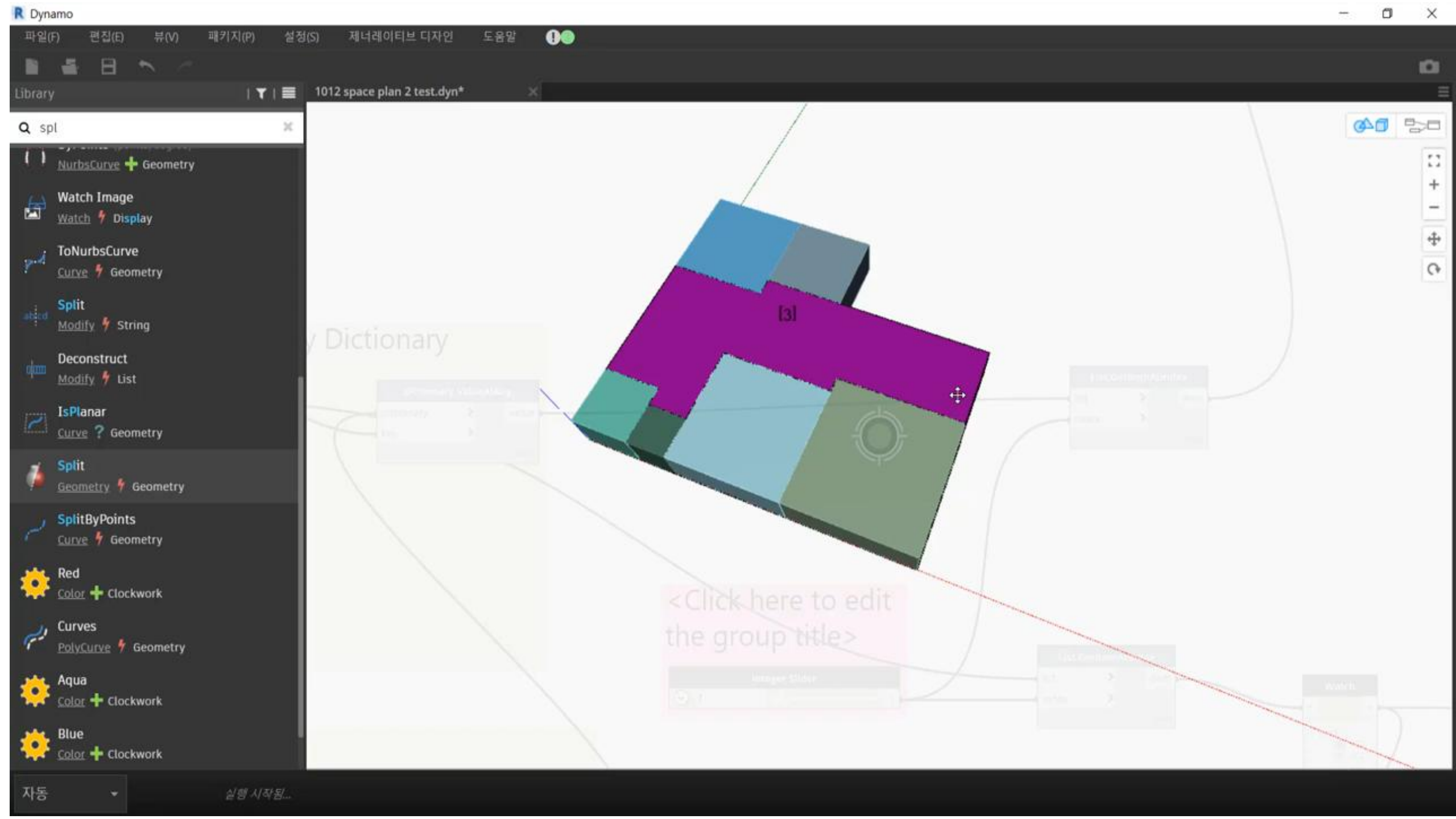


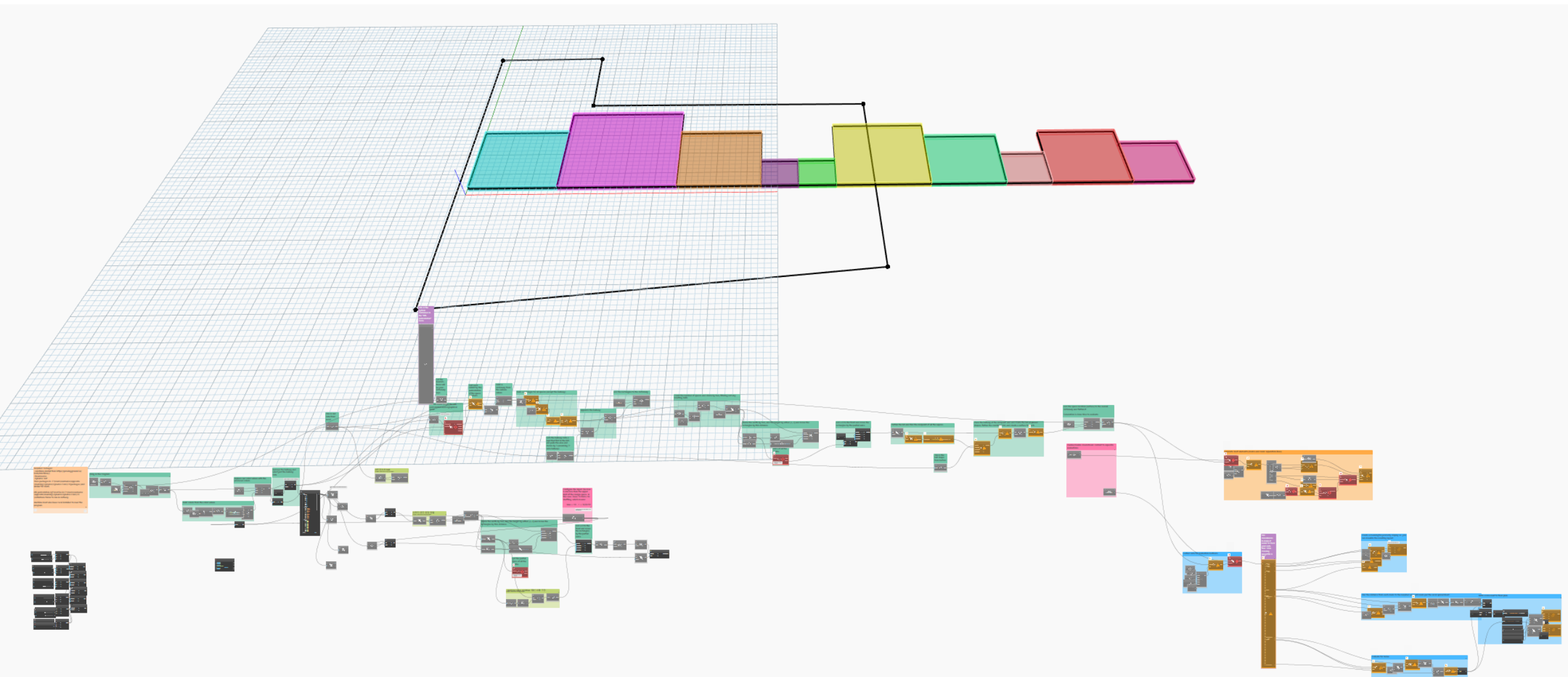


03 Space Planning 01

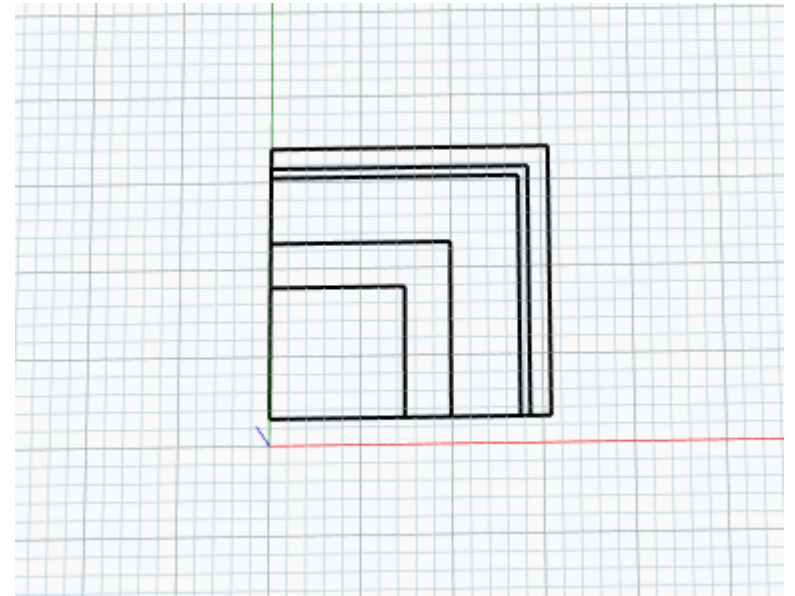
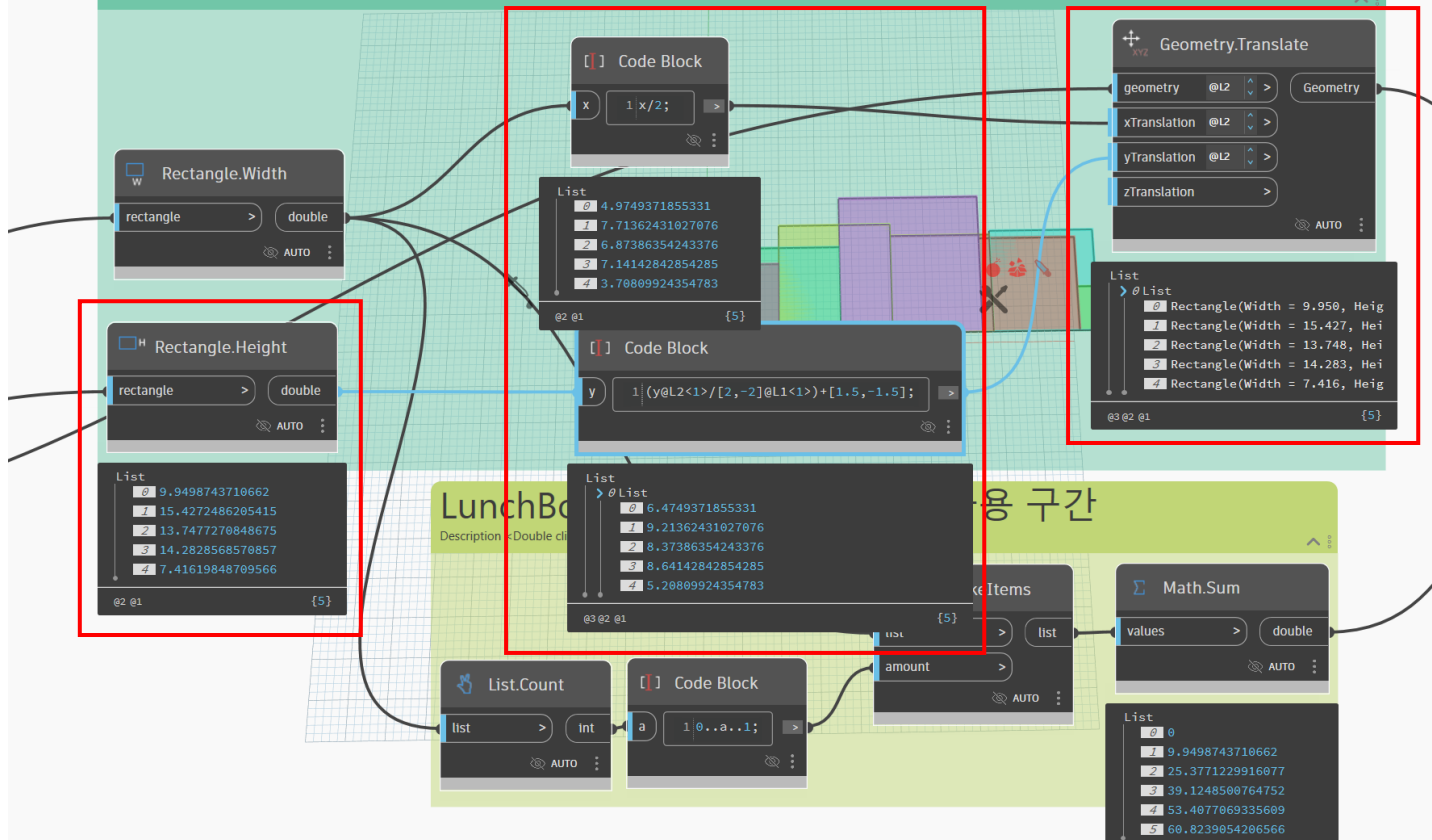
주거 공간 배치 예제

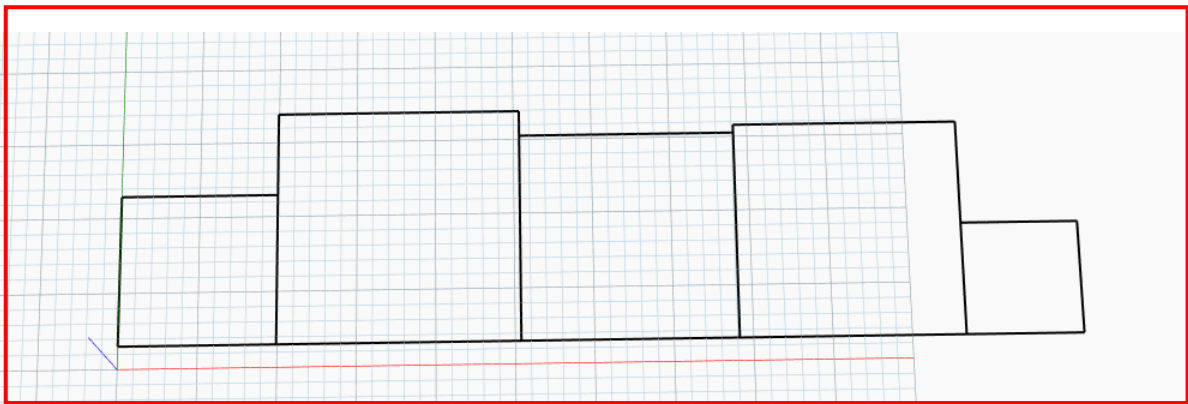




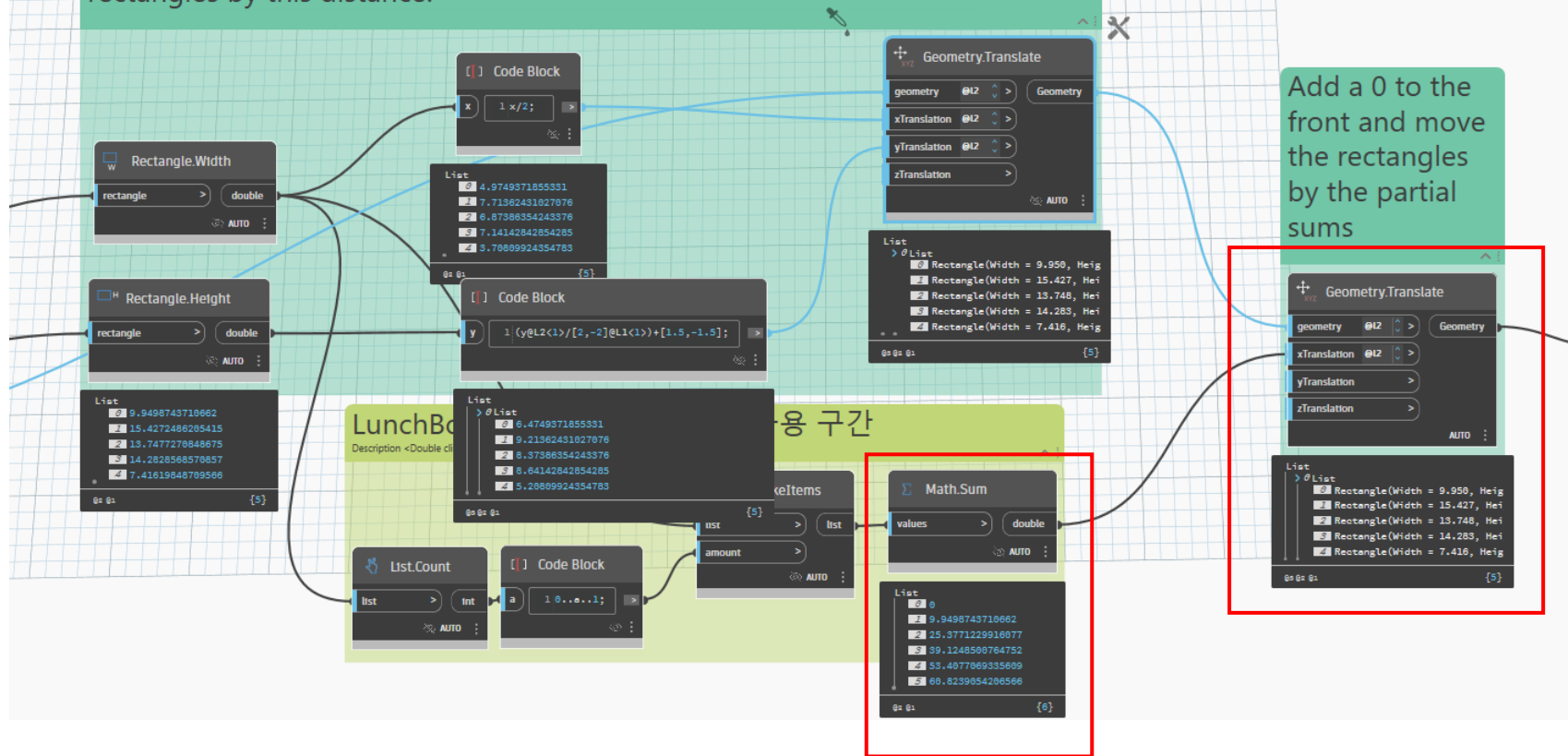


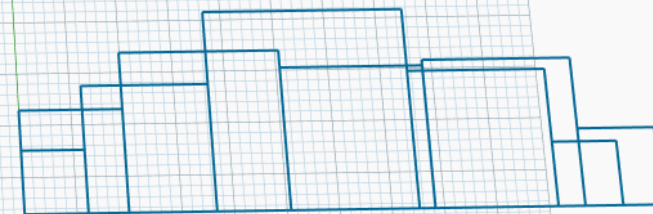
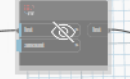
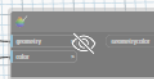
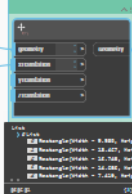
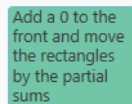
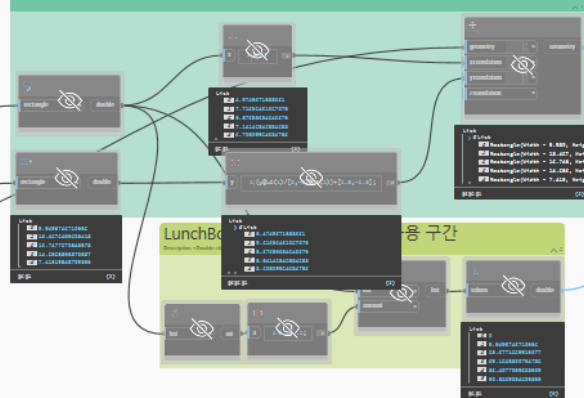
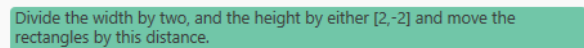
Divide the width by two, and the height by either [2,-2] and move the rectangles by this distance.

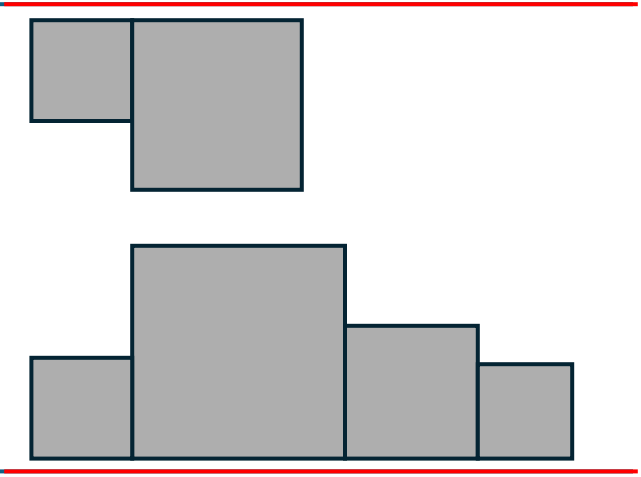
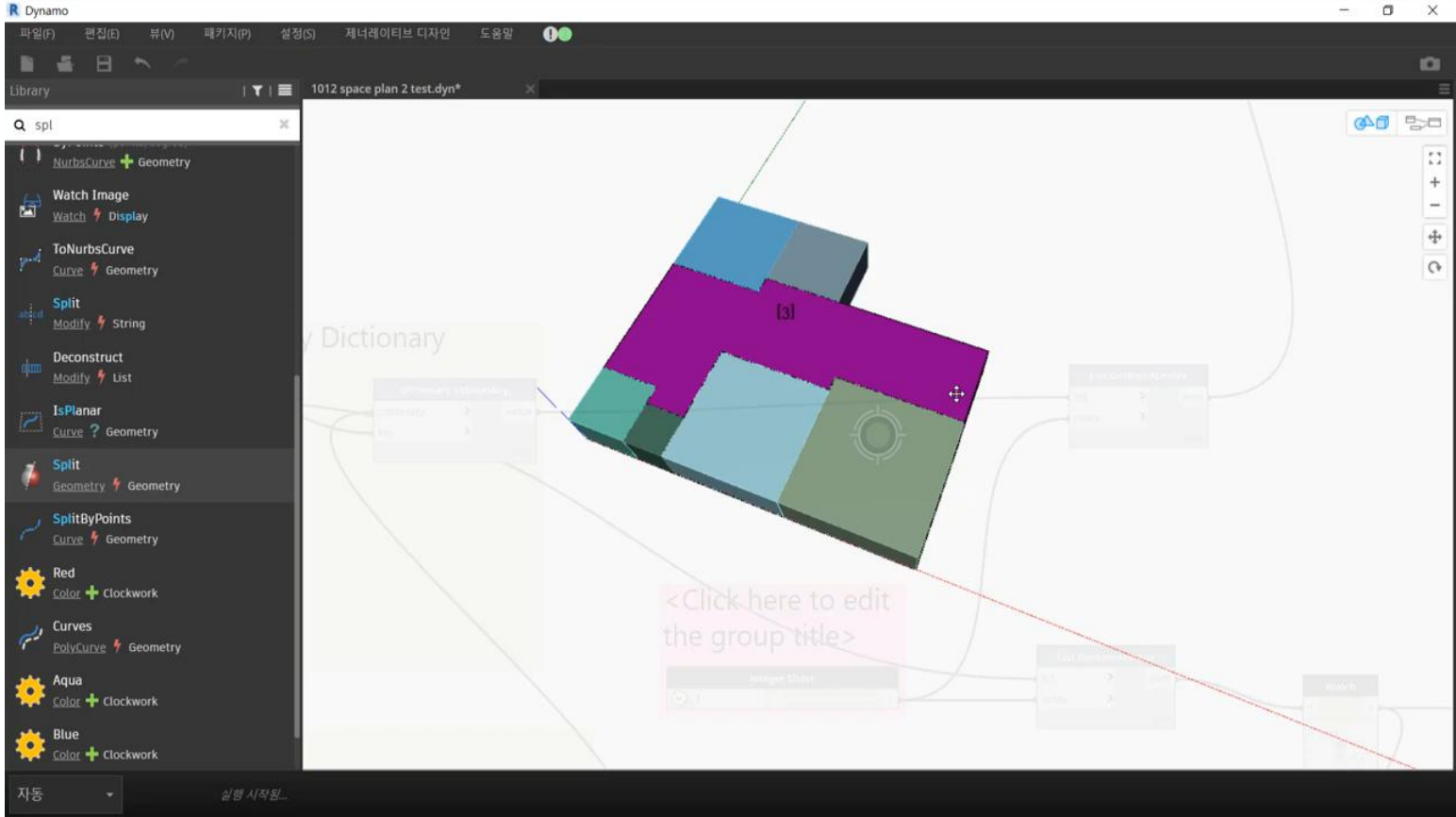




Divide the width by two, and the height by either [2,-2] and move the rectangles by this distance.



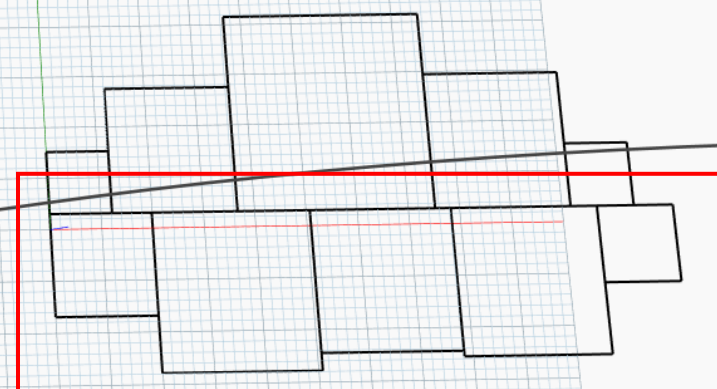
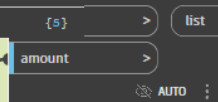
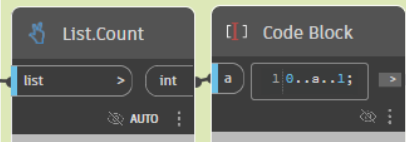
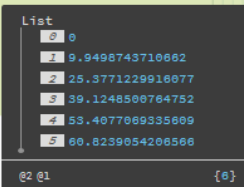
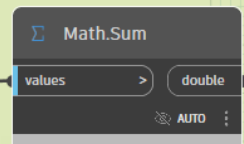
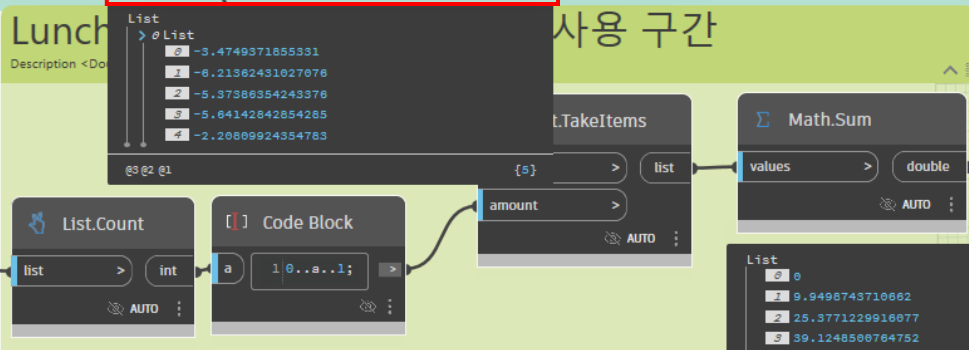
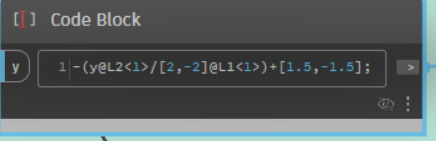
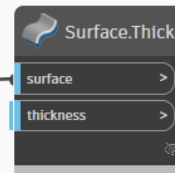
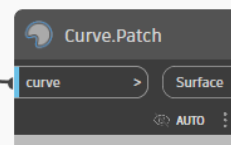
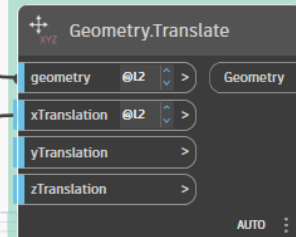




Code Block

$1 - (y @ L2 < 1 > / [2, -2] @ L1 < 1 >) + [1.5, -1.5];$

the rectangles
by the partial
sums



Divide the width by two, and the height by either [2,-2] and move the rectangles by this distance.

Geometry.Translate

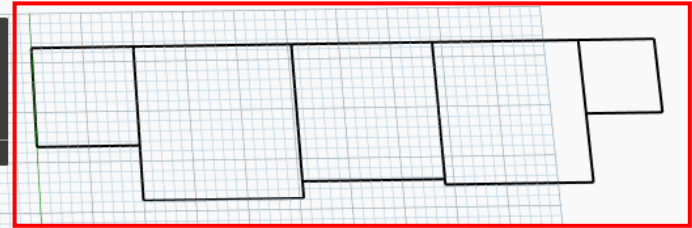
Code Block

$1 - (y @ L2 < 1> / [2, -2] @ L1 < 1>) + [1.5, -1.5] + 45;$

Add a 0 to the front and move the rectangles by the partial sums

Geometry.Translate

Geometry.Translate

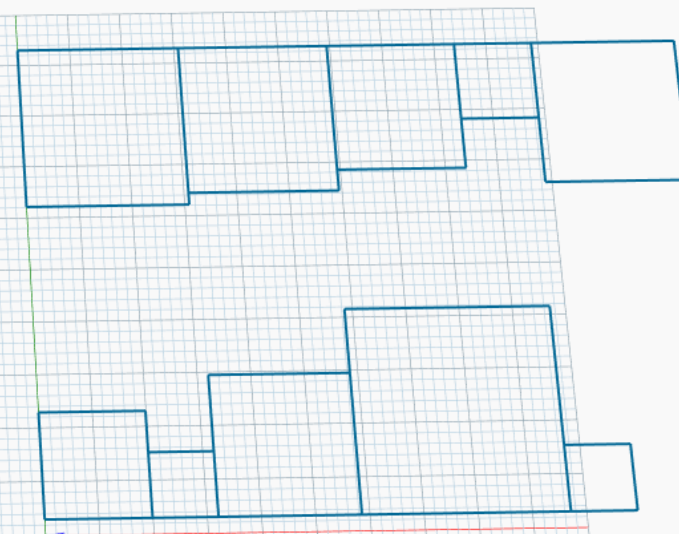
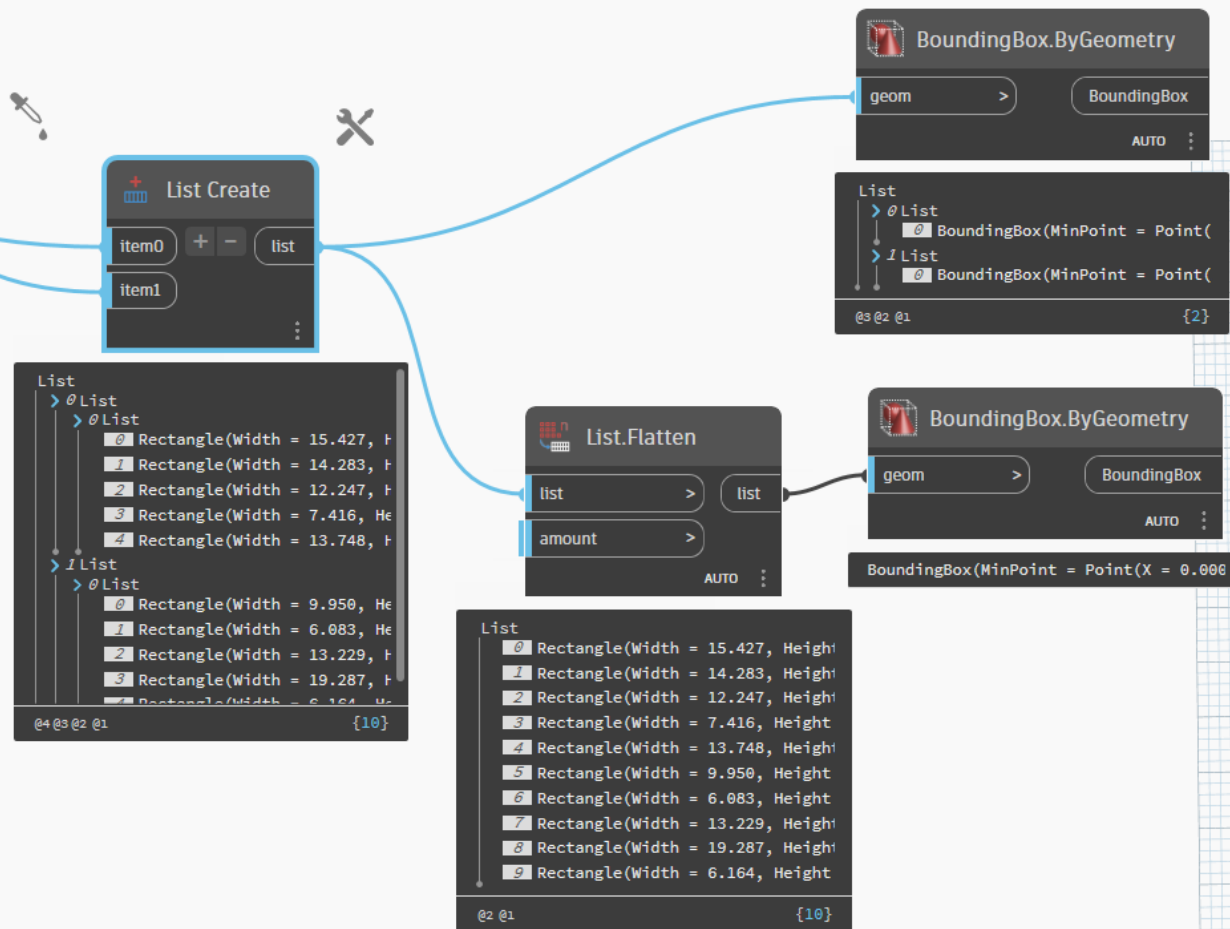


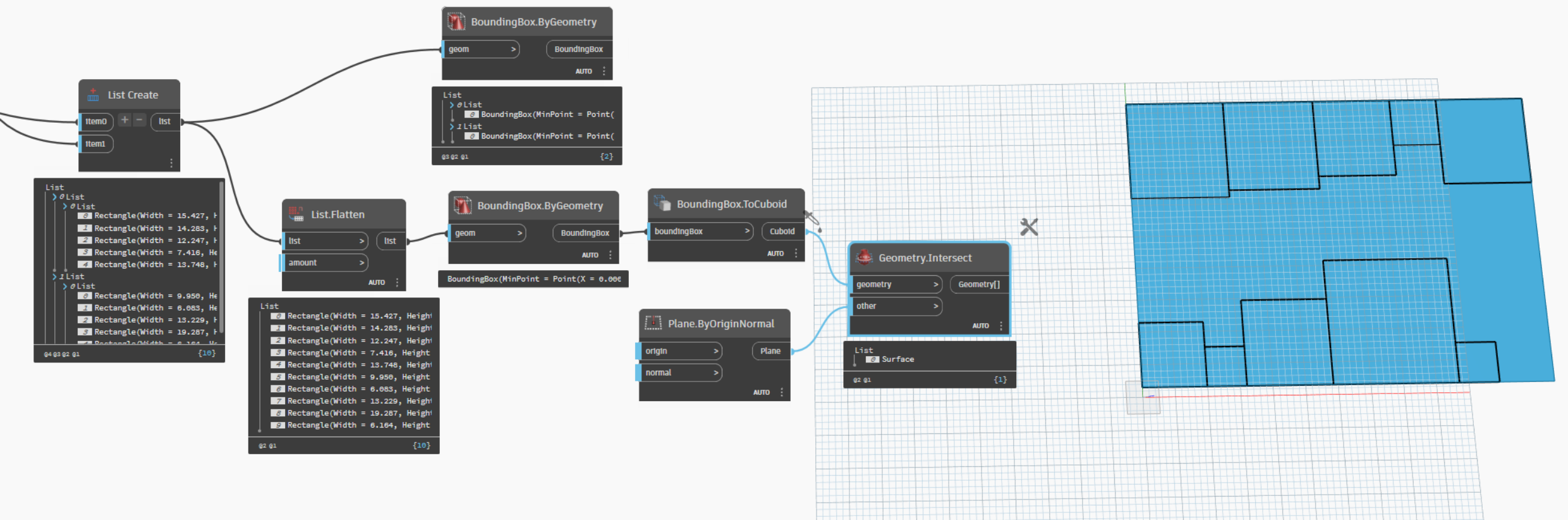
Divide the width by two, and the height by either [2, -2] and move the rectangles by this distance.

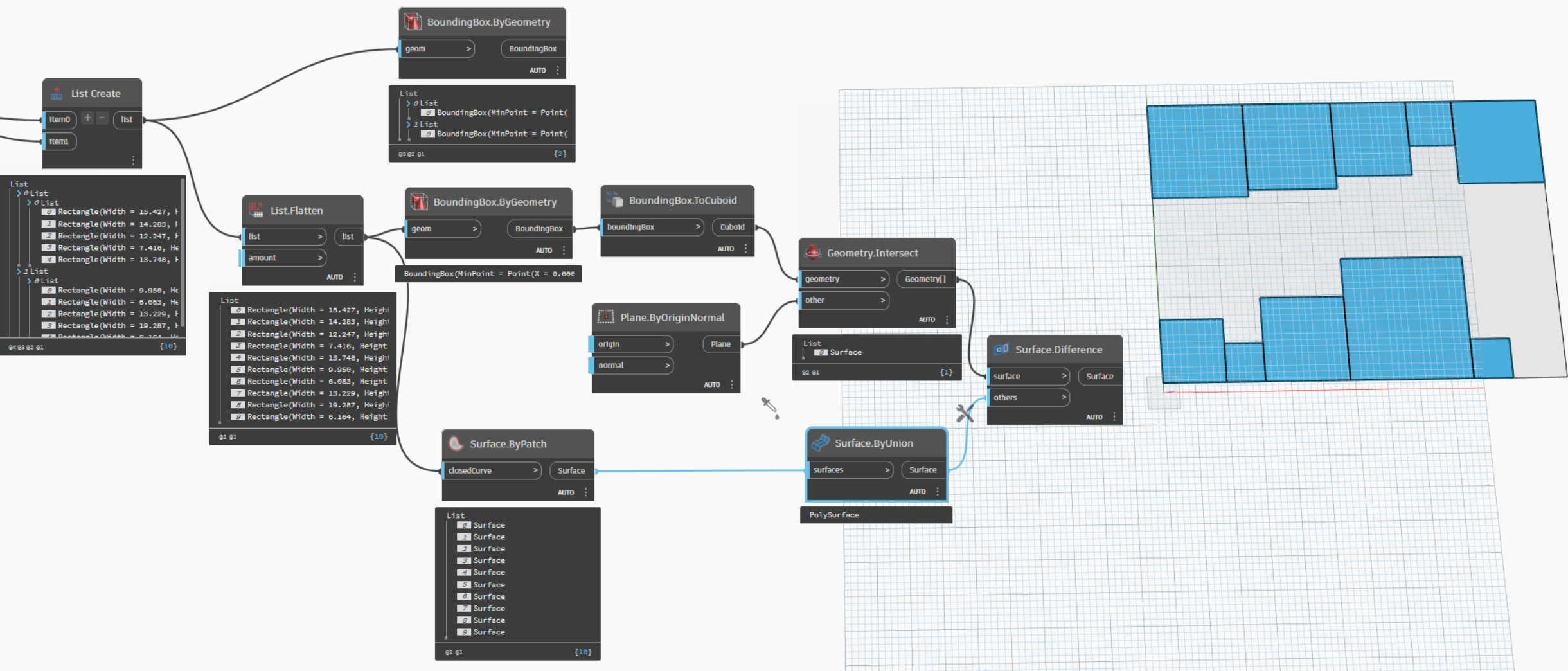
Add a 0 to the front and move the rectangles

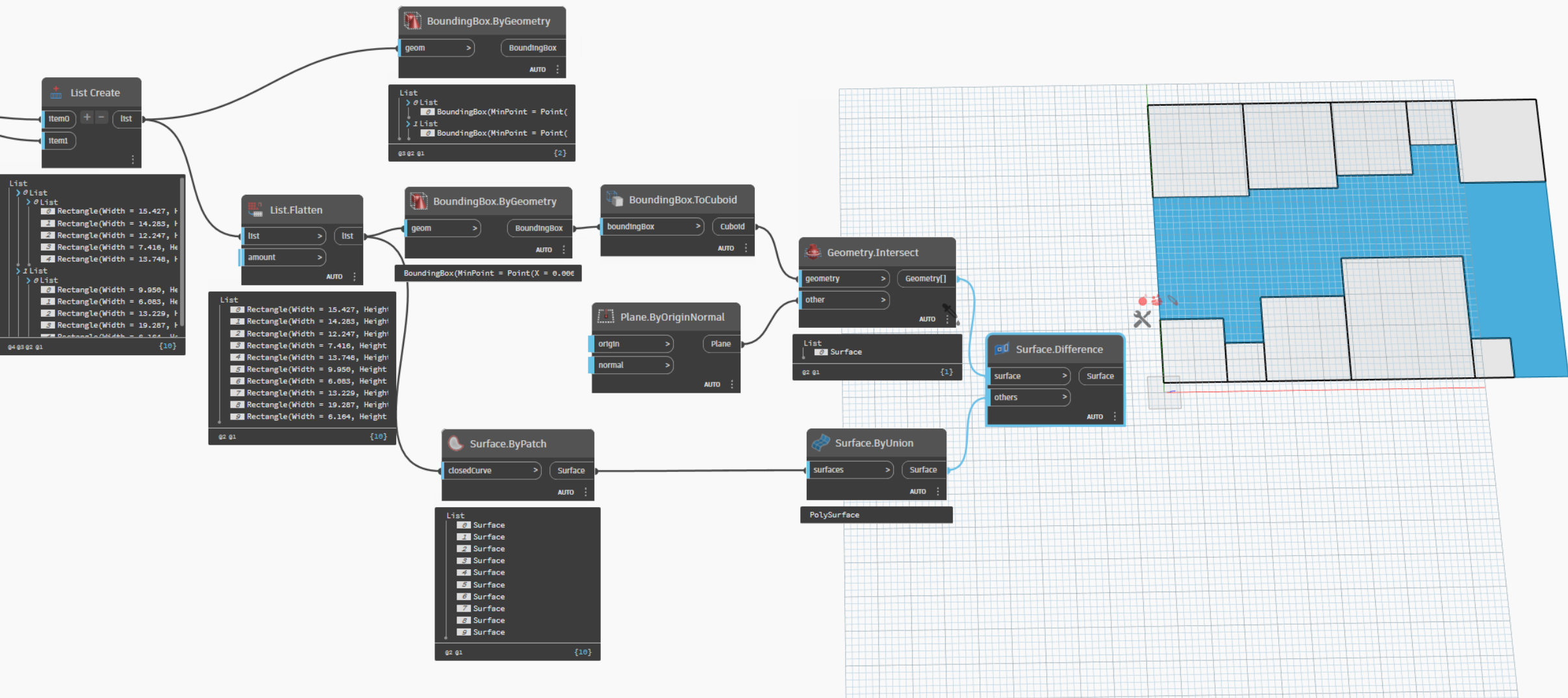
List.Flatten

List





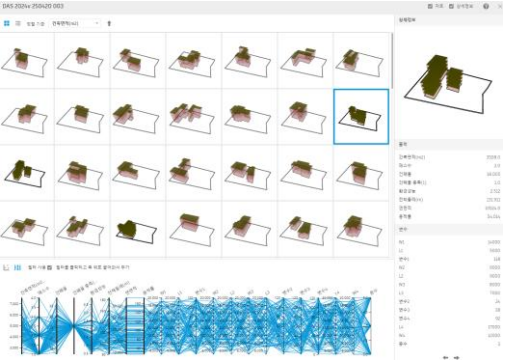
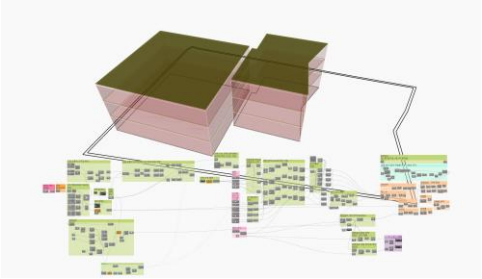




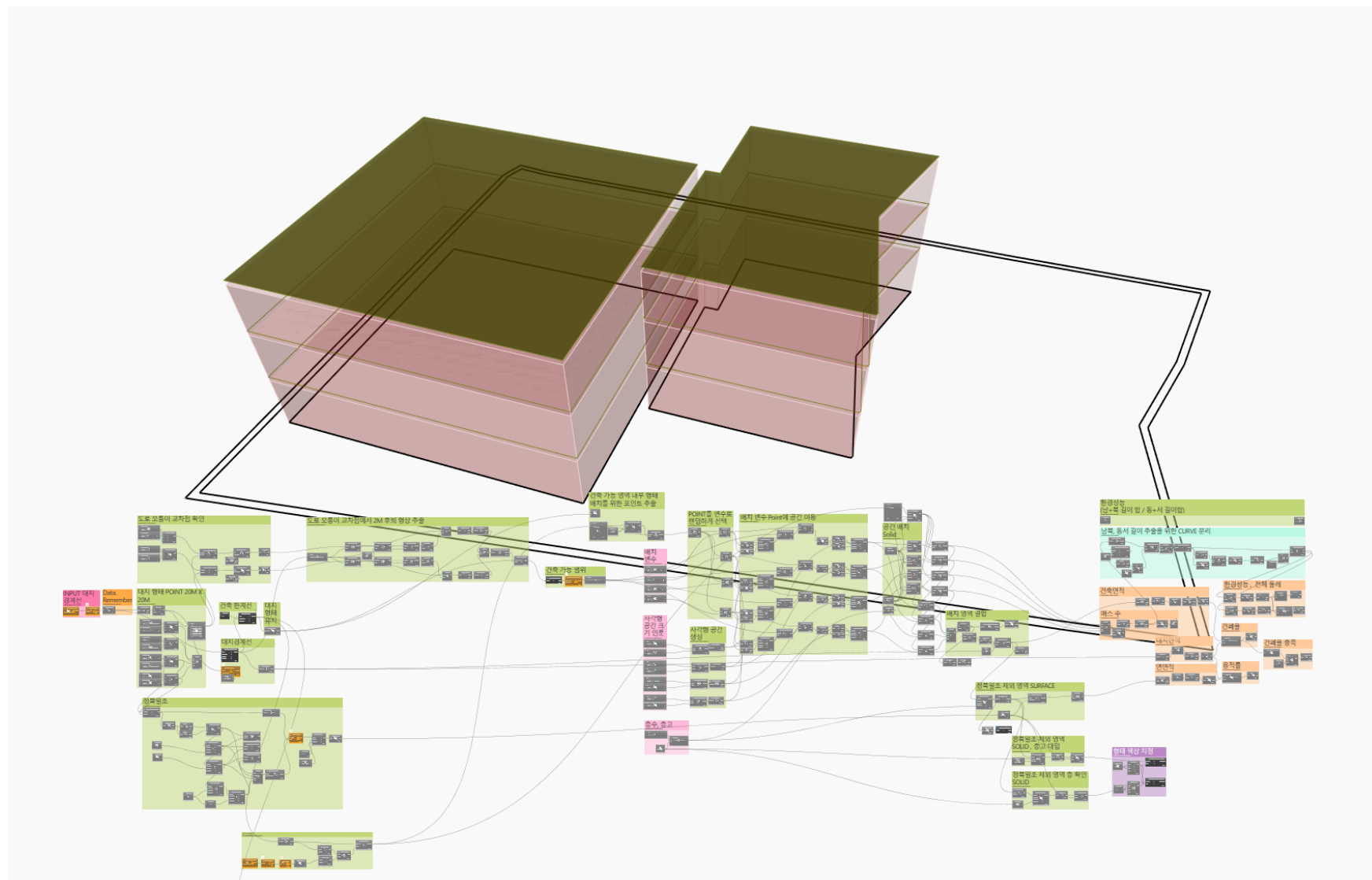


04 Space Planning 02

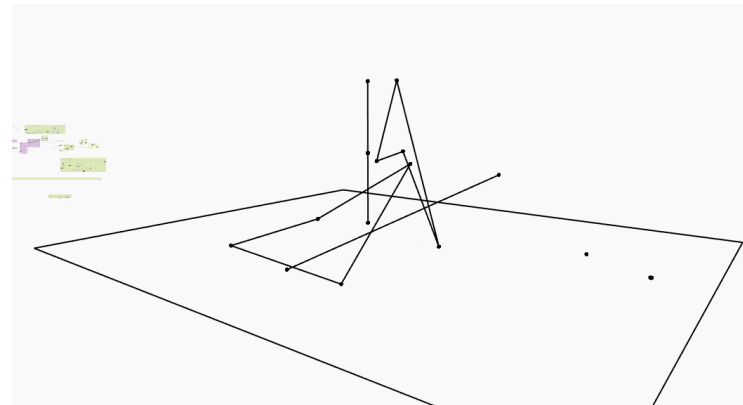
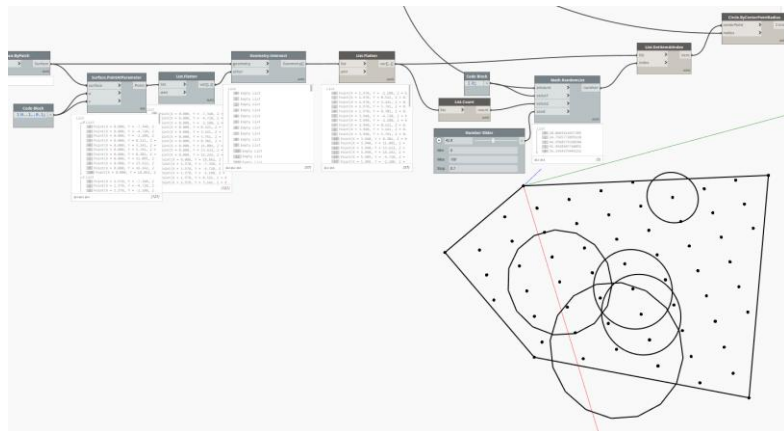
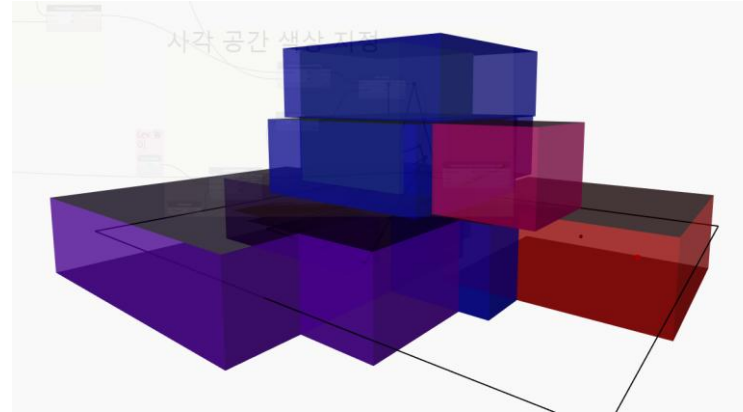
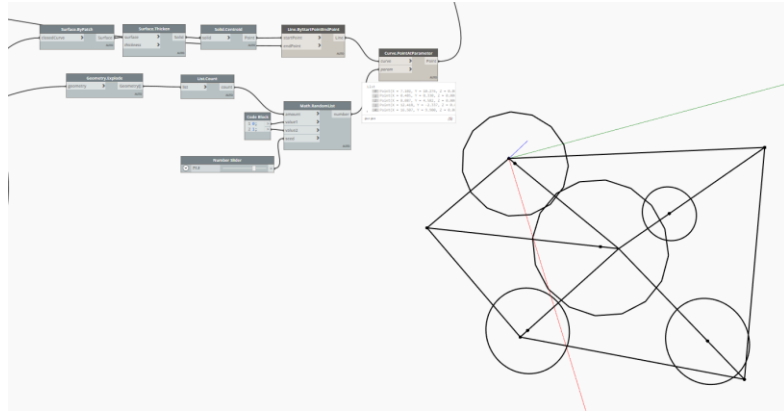
형상 생성을 통한 1~4개 건물 규모 검토 (건폐율, 용적률)



DAS

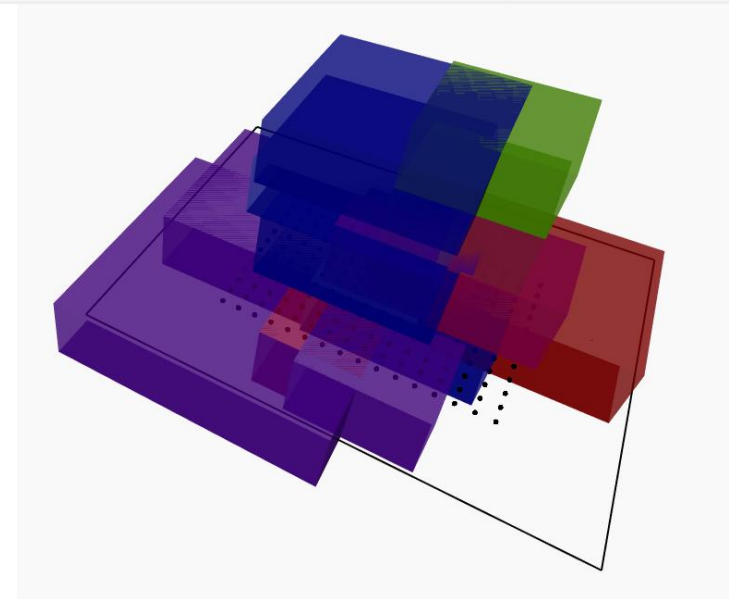
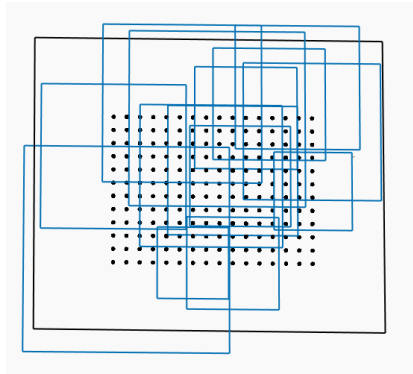
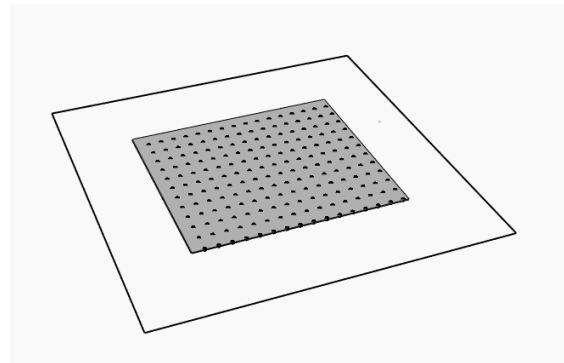
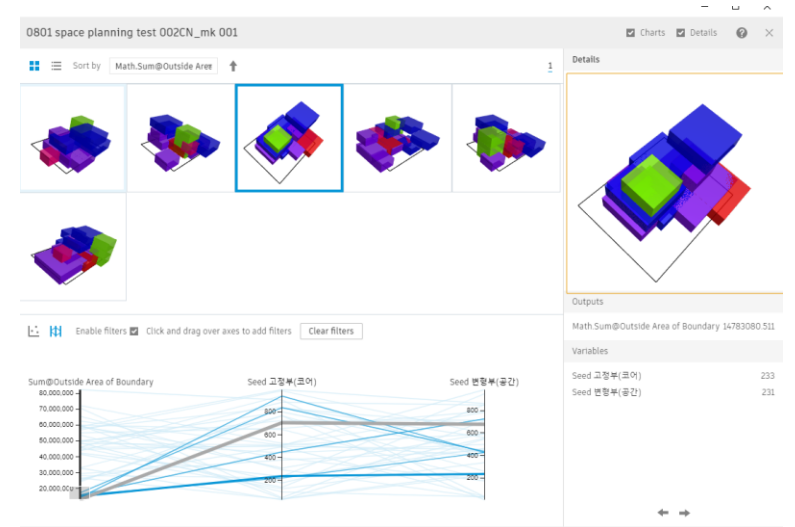
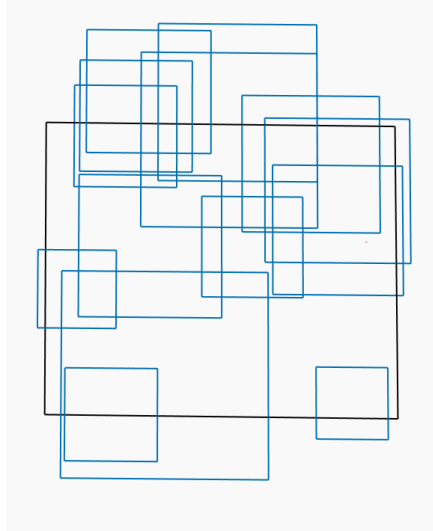
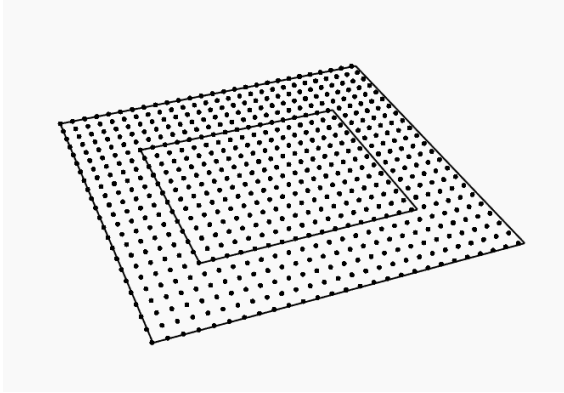


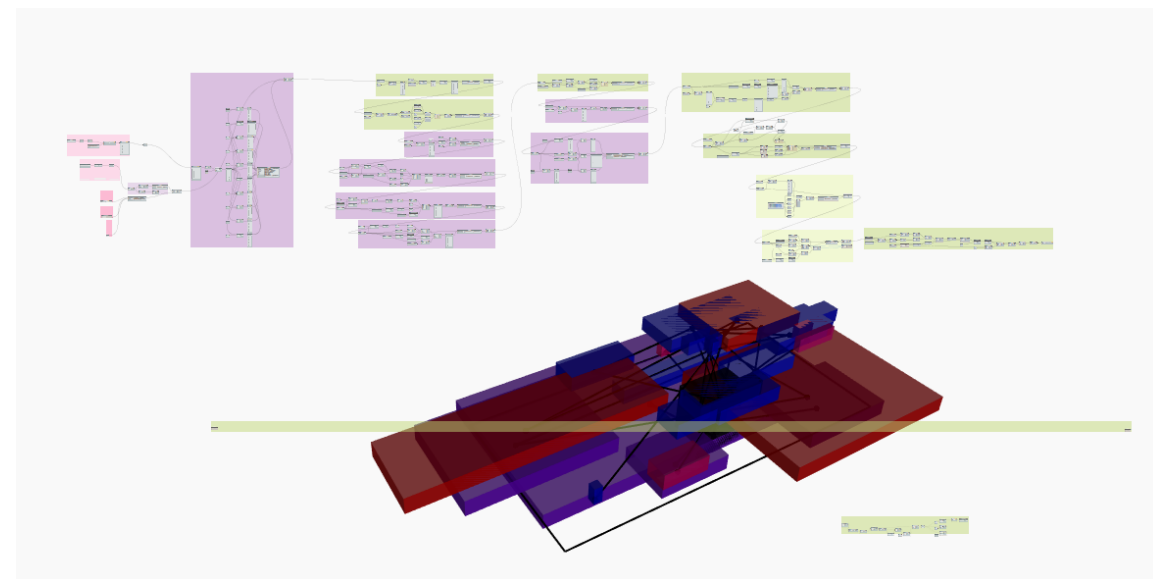
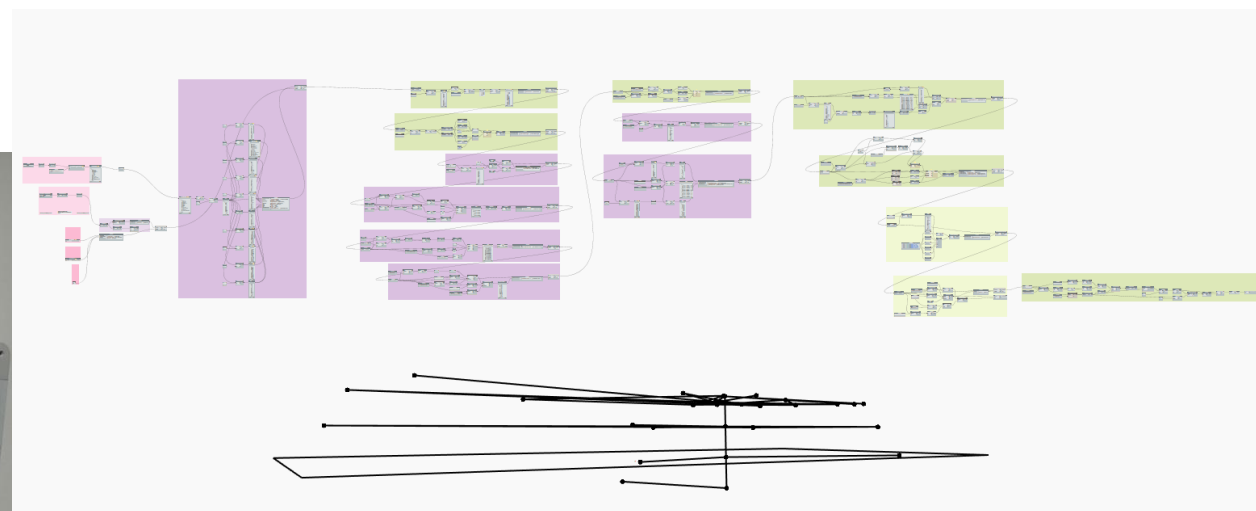
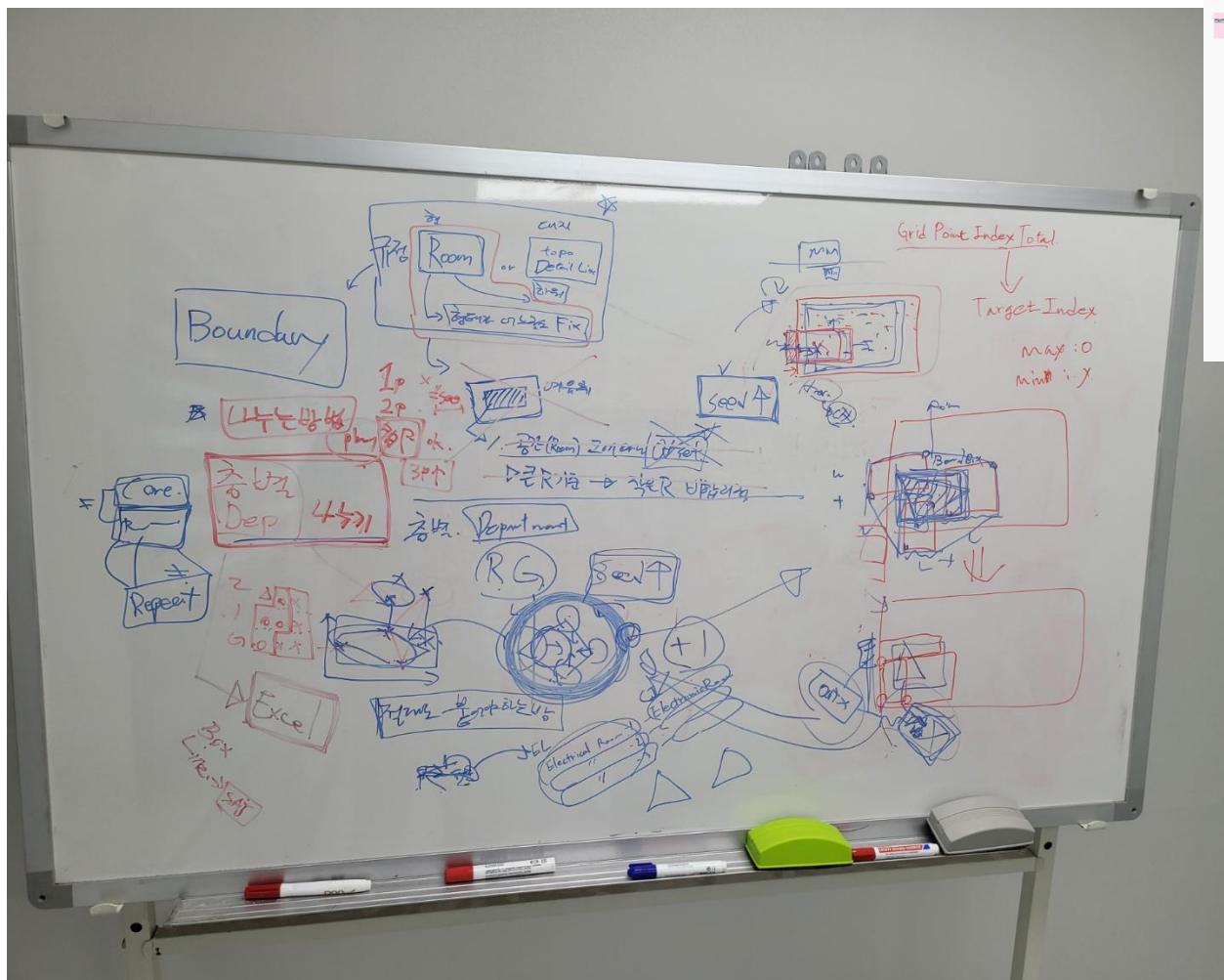
배치 : 선형 / 바운더리 포인트



Department (부서) 층별 배치
> 층별 부서 List 나누어 진행(예상)

배치 : 선형 / 바운더리 포인트





DONE LIST

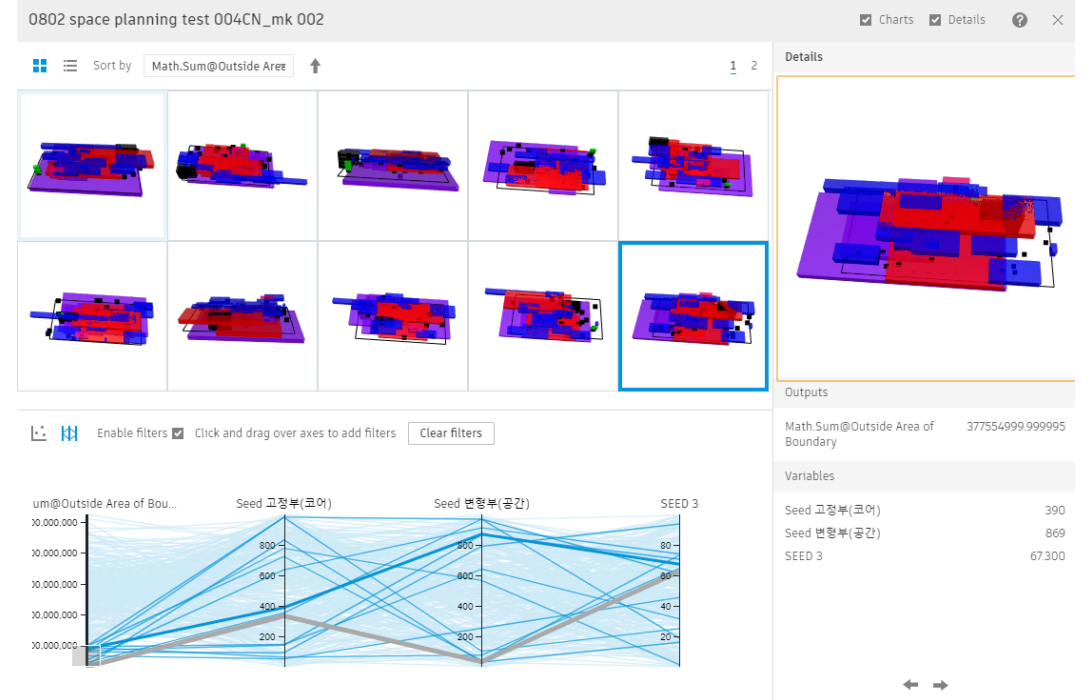
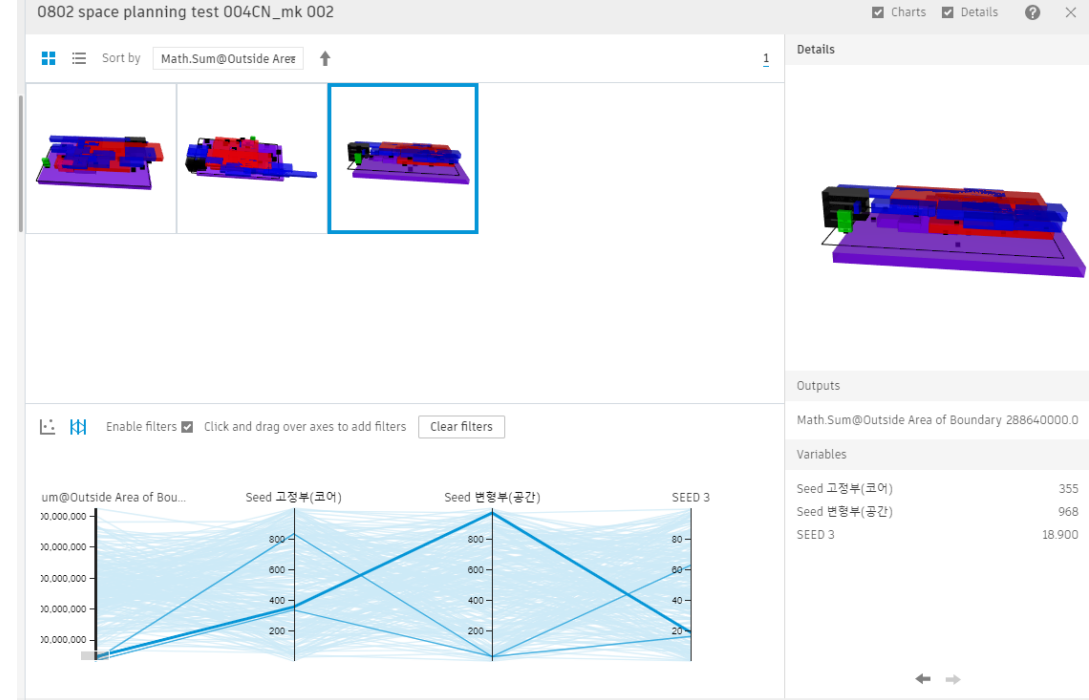
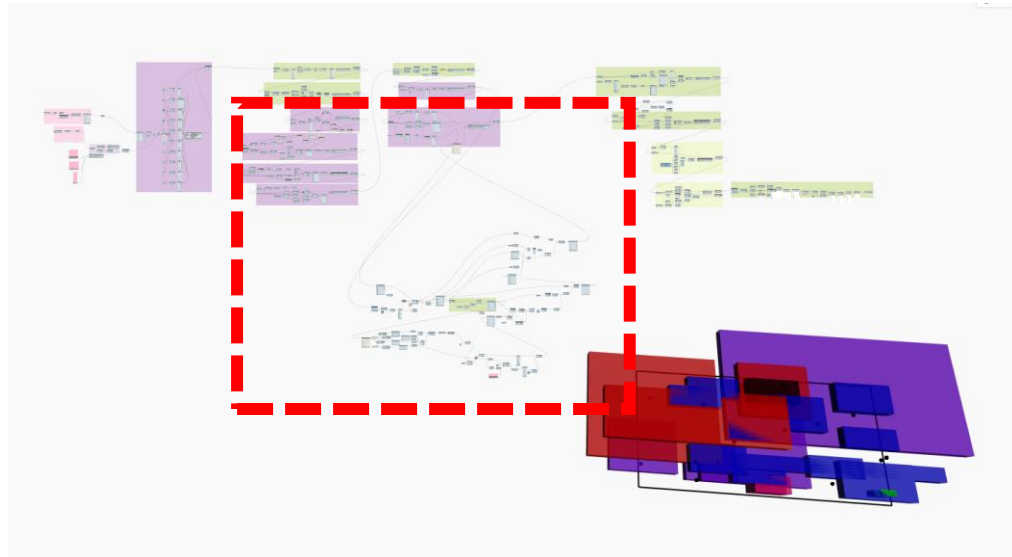
- 고정부 추가(STAIRS, SHAFTS) 다수의 고정부 커버 가능
- LEV별 ROOM 재배치 SEED 추가

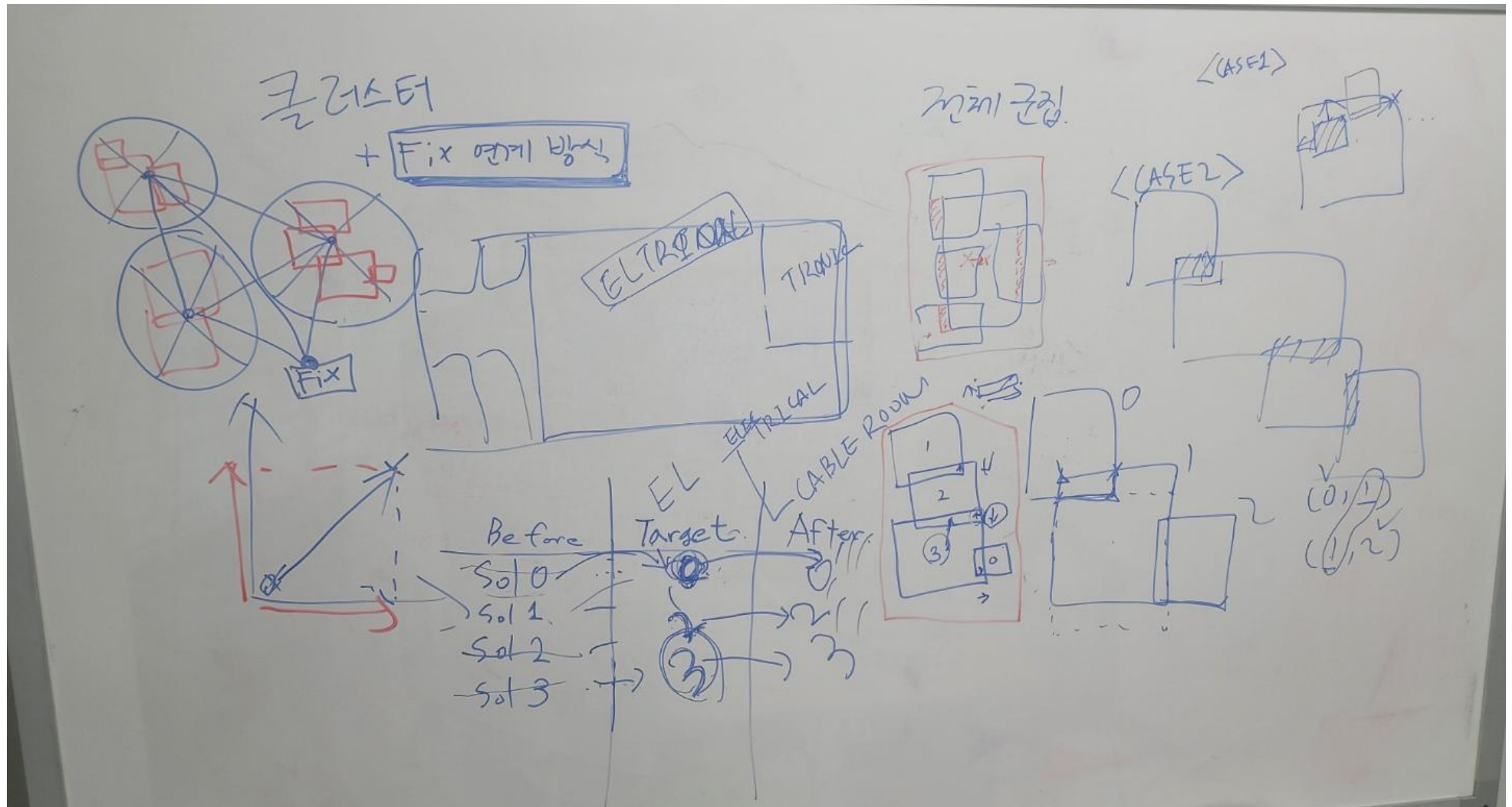
TO DO LIST

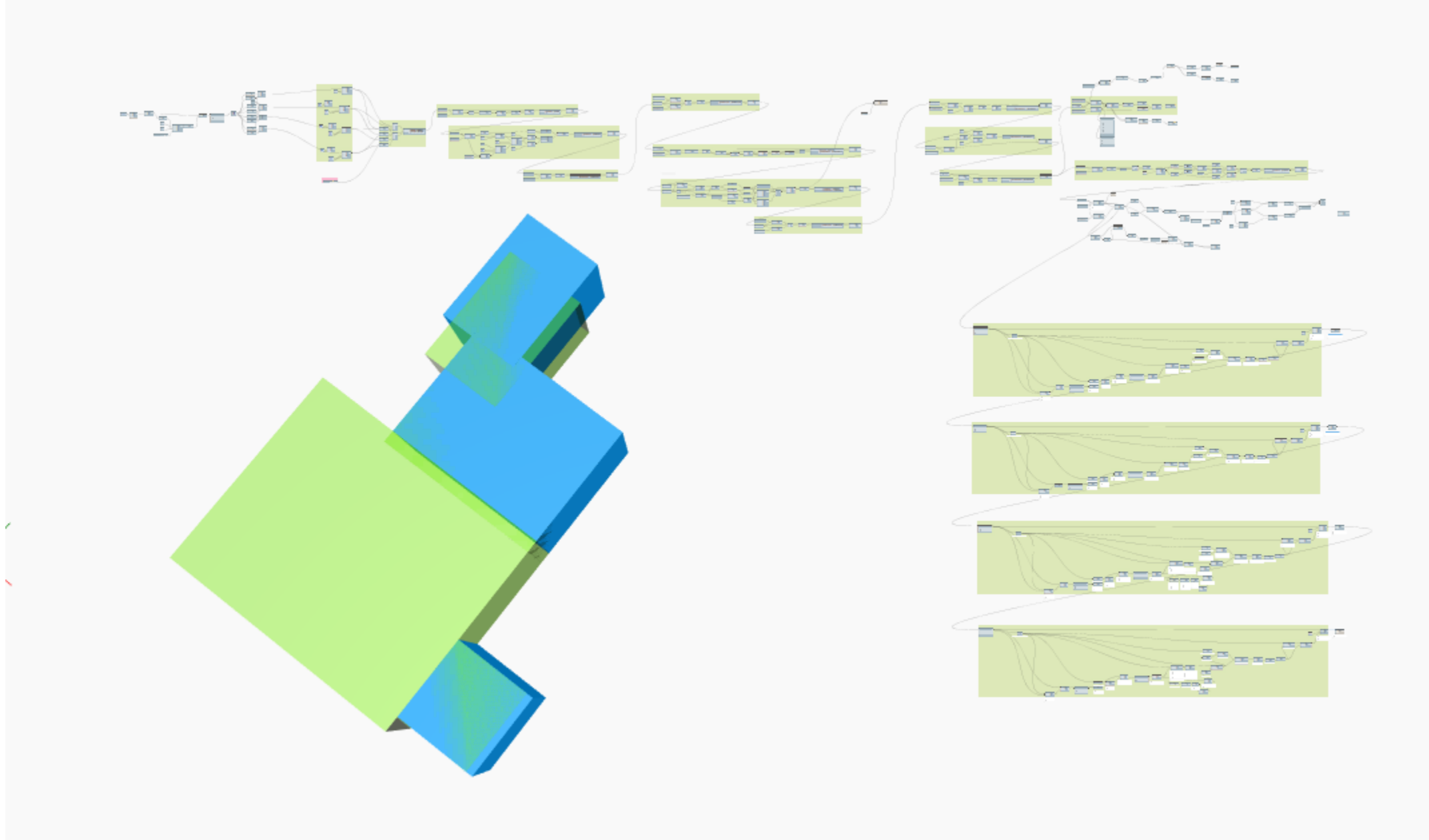
- 기존 평면과 최고 점수의 아웃풋 비교 작업
- ADJ 새로운 접근 방식 알고리즘 추가 필요
(고정부들간의 인접성 / ROOM들간 인접성 - EXCEL +)

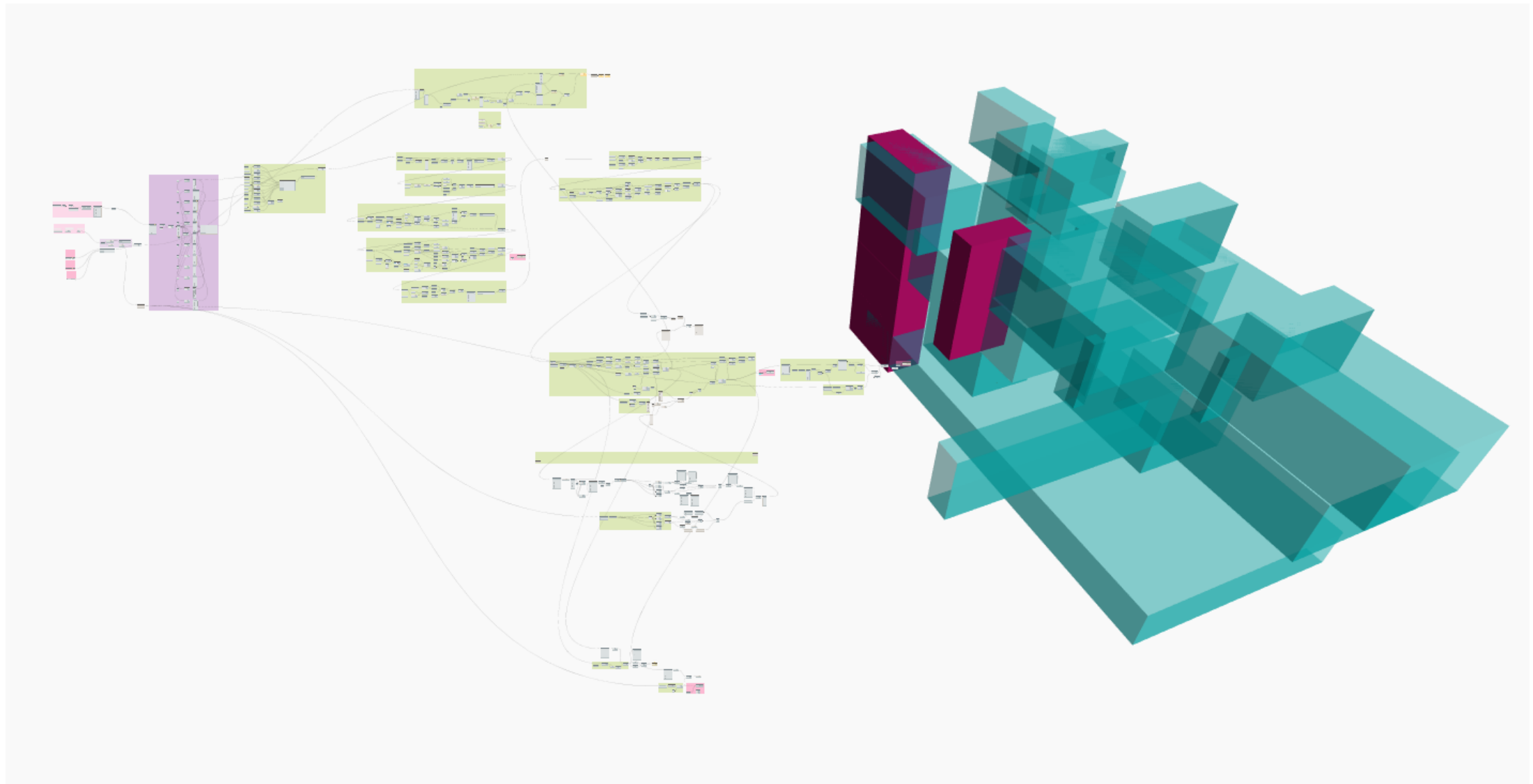
[클러스터(cluster)간 / 클러스터 내 인접성] 고도화

- Dep(부서) 인접성, 수직성에 대한 방법론 고려
- Center point 기준이 아닌 Room 파라미터 값으로 적용 고려
(W,L 값의 Translate)
- Room 간의 이격 거리 좁히기(끌어당김 - 중력)

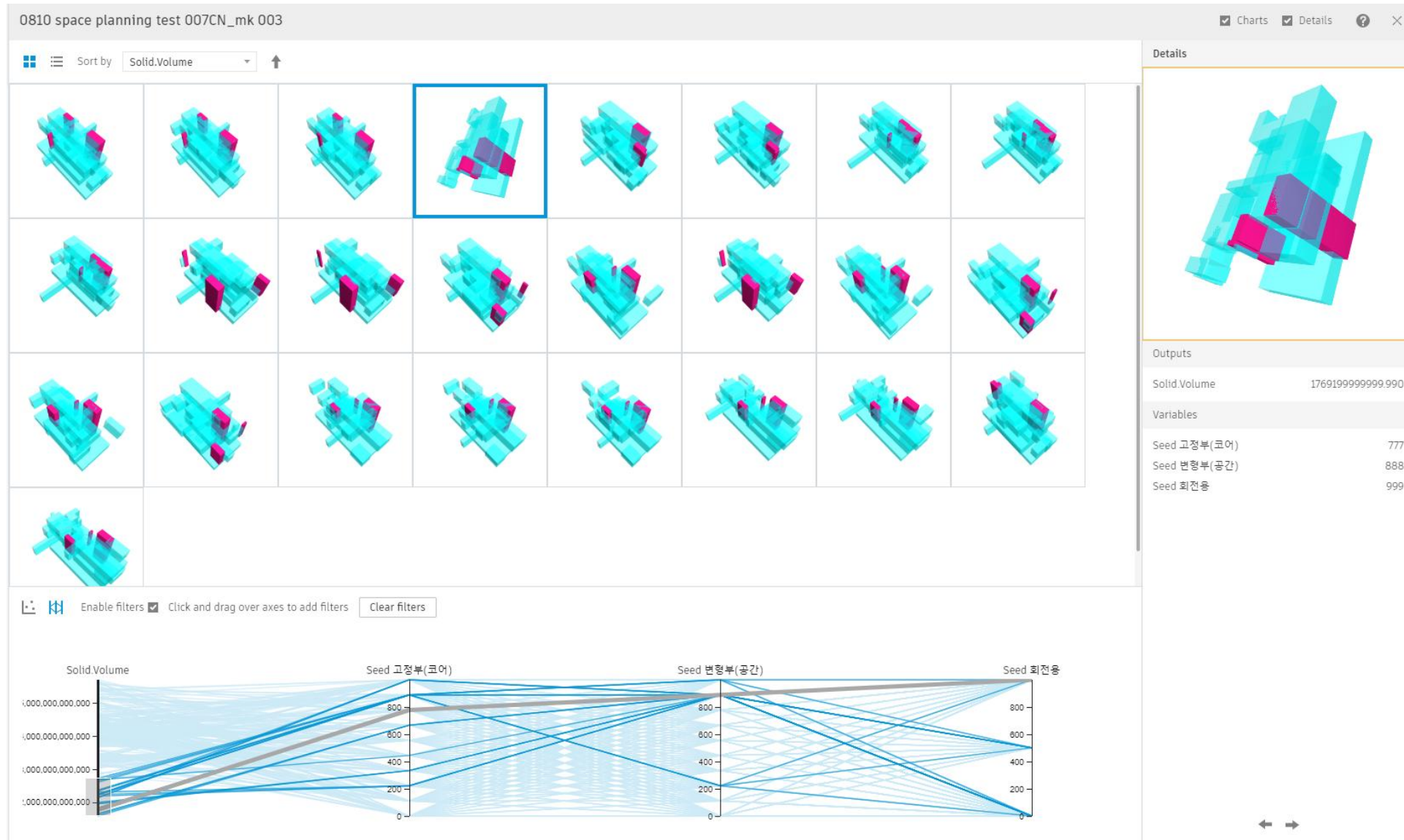


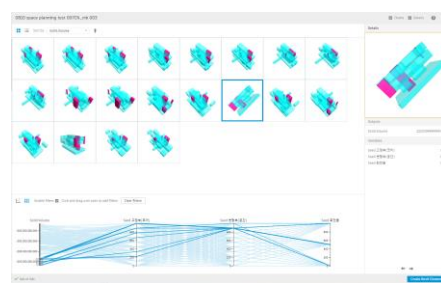
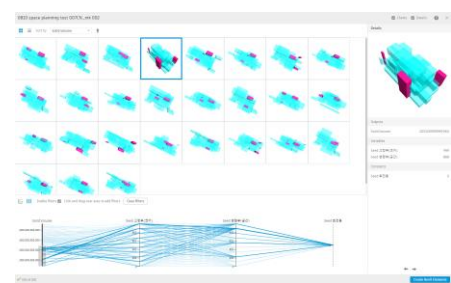
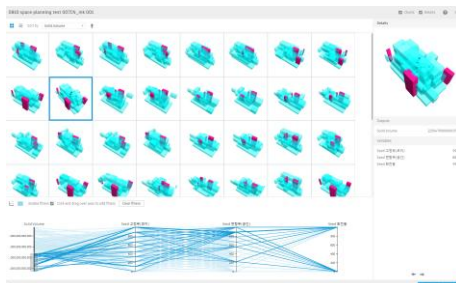
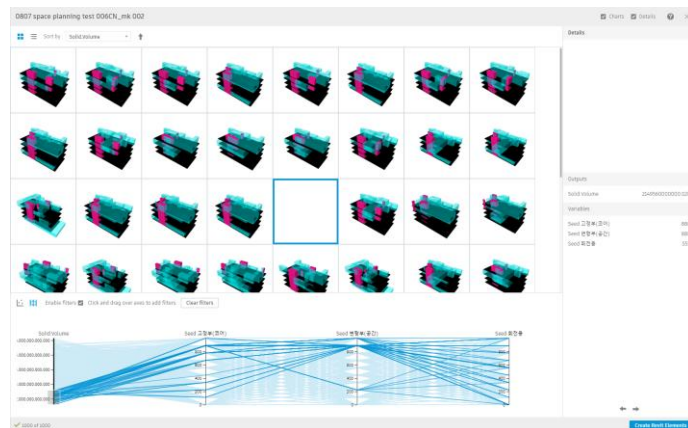
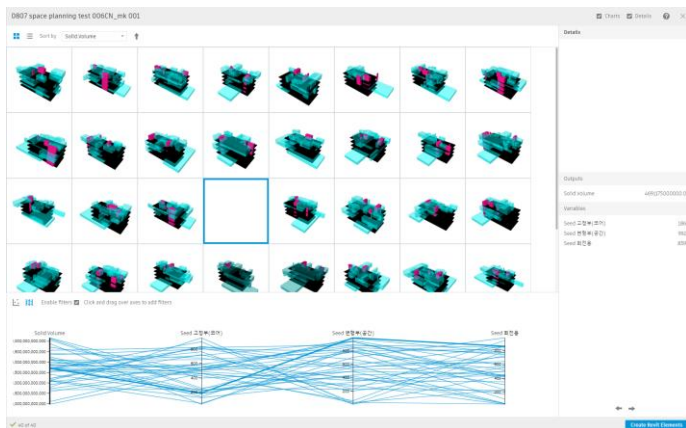
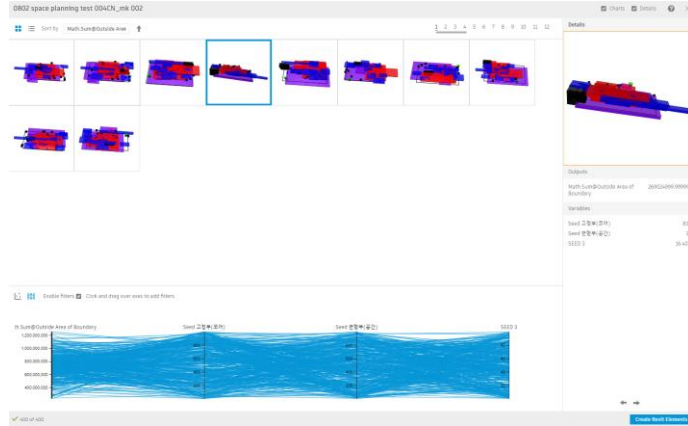
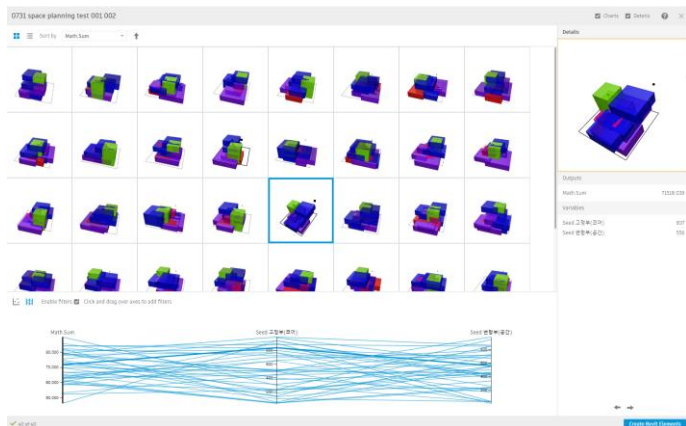




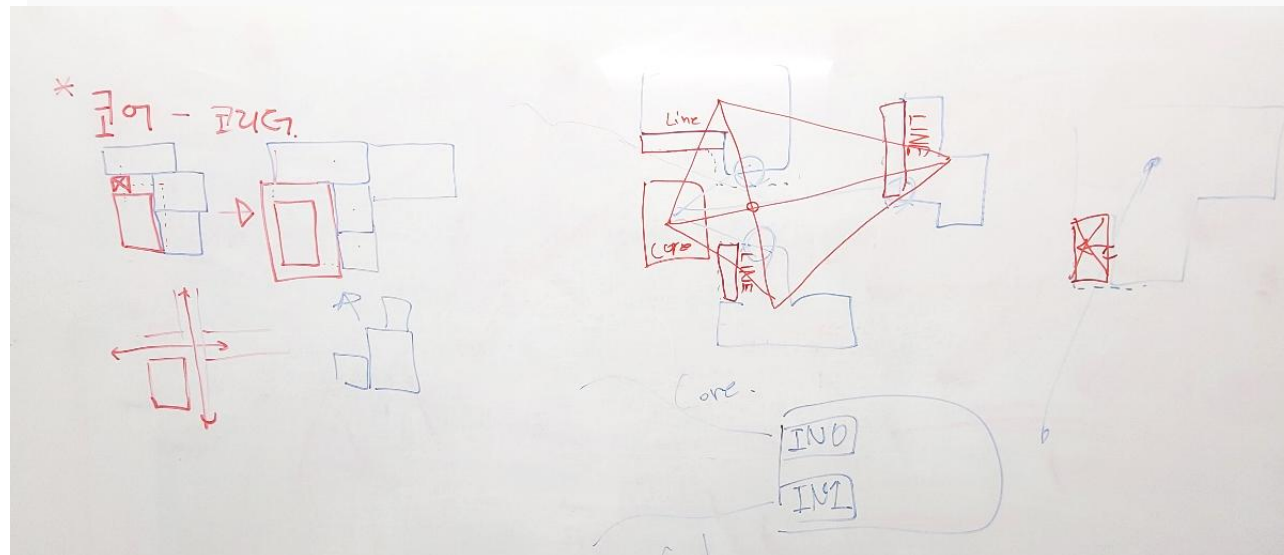
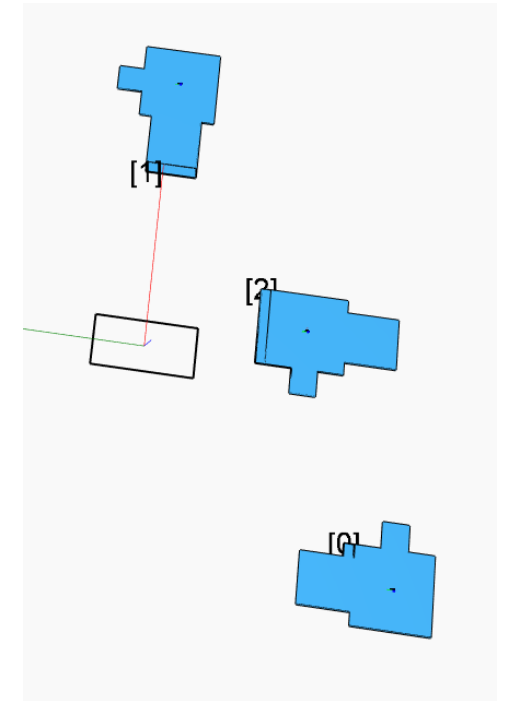
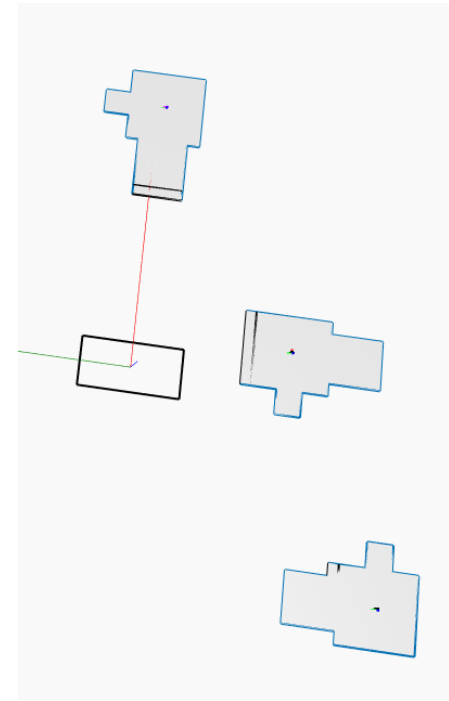
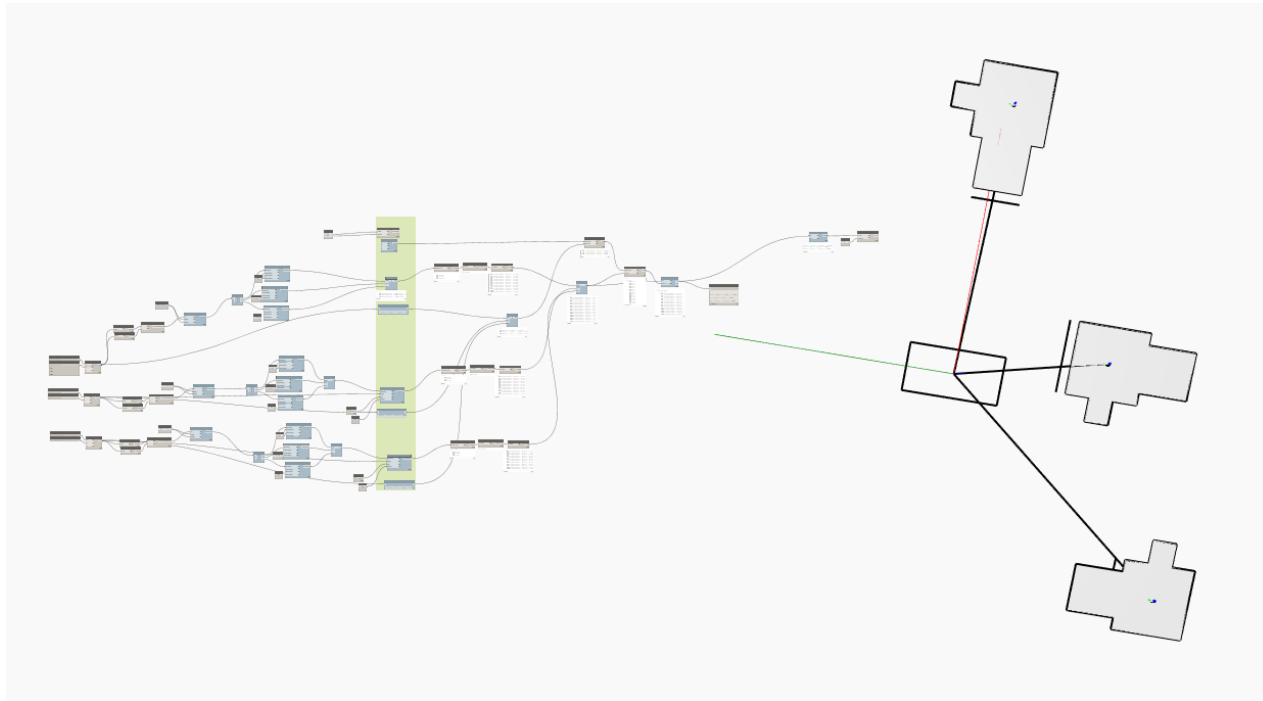


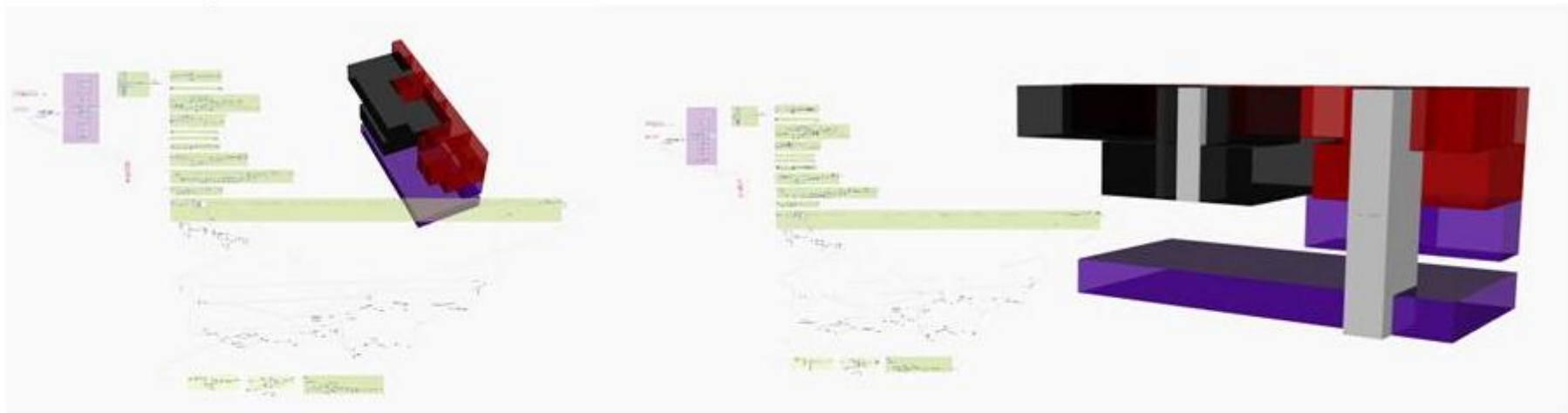
0810 Generative Design Output test



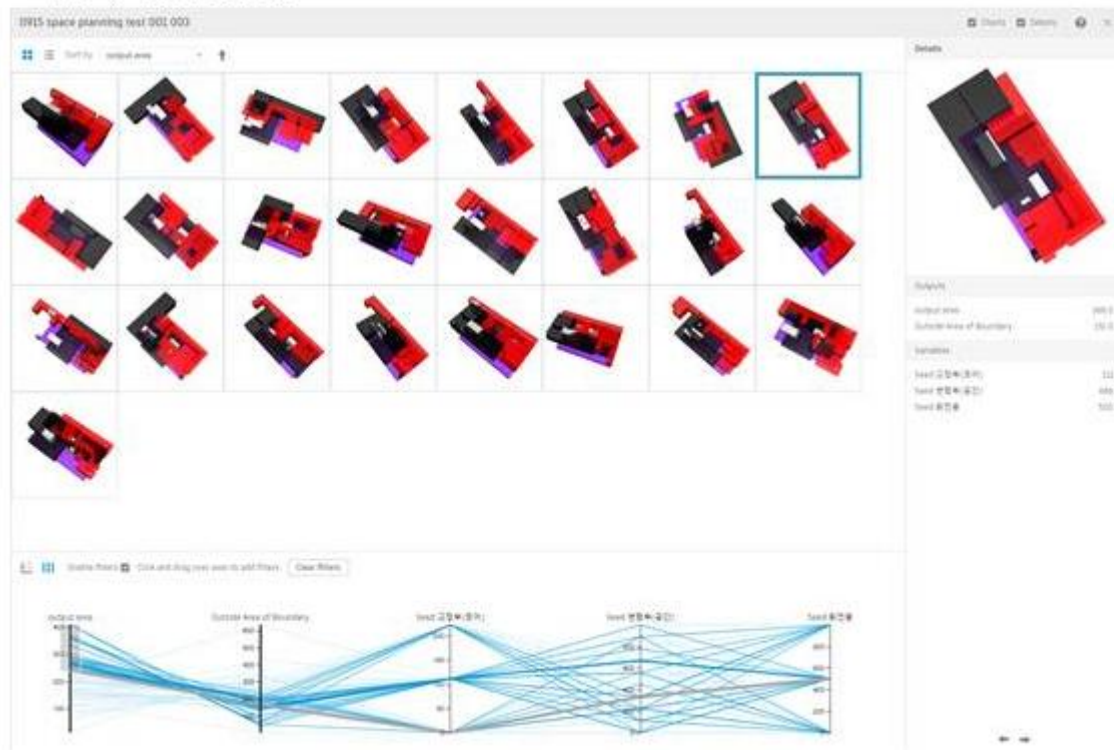


0824 Generative Design Output test





> 확인 후 GD test 진행




```

1
2 "ExcelDataWithNoHead": _ExcelDataWithNoHead,
3 "SpaceID": _SpaceID,
4 "SpaceName": _SpaceName,
5 "AdjacentSpaces": _AdjacentSpaces,
6 "DepartmentID": _DepartmentID,
7 "Department": _Department,
8 "Level": _Level,
9 "Width": _Width,
10 "Length": _Length,
11 "Height": _Height,
12 "Area": _Area,
13 "Rotate": _Rotate,
14 "Cluster": _Cluster,
15 "InteriorFinish": _InteriorFinish,
16 "ExteriorFinish": _ExteriorFinish
17 };

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